Ray Optics and Optical Instruments

Question1

A prism of angle 75° and refractive index $\sqrt{3}$ is coated with thin film of refractive index 1.5 only at the back exit surface. To have total internal reflection at the back exit surface the incident angle must be

$(\sin 15^\circ = 0.25$ **and** $\sin 25^\circ = 0.43)$

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Options:

A.

between 15° and 20°

B.

15°

C.

$< 15^\circ$

D.

$> 25^\circ$

Answer: A

Solution:

Total internal reflection (TIR) occurs at the interface between the prism ($n_p = \sqrt{3}$) and the film ($n_f = 1.5$).

$\sin C = \frac{n_{\text{rarer}}}{n_{\text{denser}}} = \frac{1.5}{\sqrt{3}} = \frac{\sqrt{3}}{2} \Rightarrow C = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60^\circ$

So, the critical angle is $C = 60^{\circ}$.

For TIR to occur at the second surface, the angle of incidence r_2 must satisfy:

$r_2 > C \Rightarrow r_2 > 60^{\circ}$

For a prism with angle $\mathrm{A} = 75^{\circ}$:

$\mathrm{A} = \mathrm{r}_1 + \mathrm{r}_2 \Rightarrow 75^{\circ} = \mathrm{r}_1 + \mathrm{r}_2$

Substituting the condition $\mathrm{r}_2 > 60^{\circ}$:

$r_1 = 75^{\circ} - r_2 < 75^{\circ} - 60^{\circ}$

$\Rightarrow \mathrm{r}_1 < 15^{\circ}$

Apply Snell's Law at the first surface (assuming air to prism) :

$\mathrm{n}_{\text{air}} \quad \sin \mathrm{i} = \mathrm{n}_{\mathrm{p}} \sin \mathrm{r}_1$

$\Rightarrow \sin \mathrm{i} = \sqrt{3} \sin \mathrm{r}_1$

Since $\mathrm{r}_1 < 15^{\circ}$, then $\sin \mathrm{i} < \sqrt{3} \sin 15^{\circ}$:

$\sin i < 1.732 \times 0.25$

$\Rightarrow \sin i < 0.433$

Given that $\sin 25^{\circ} = 0.43$, we can conclude:

$\mathrm{i} < 25^{\circ}$

For the angle r_2 to remain above 60°, i.e., the value of r_1 will be between 0 and 15°. The maximum value of r_2 can be 75° for which $\mathrm{r}_1 = 0$.

$\sin \mathrm{i} = \sqrt{3} \sin 0^{\circ} = 0$

So, the value of i will be between 0° and 25°.

Question2

A collimated beam of light of diameter 2 mm is propagating along x-axis. The beam is required to be expanded in a collimated beam of diameter 14 mm using a system of two convex lenses. If first lens has focal length 40 mm , then the focal length of second lens is _____ mm.

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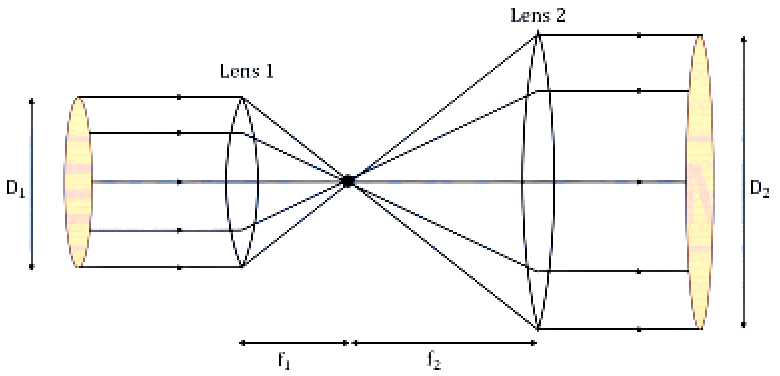
Answer: 280

Solution:

For a beam to enter collimated (parallel rays) and exit collimated with a larger diameter, the two lenses must be placed such that their focal points coincide.

The first lens focuses the parallel rays at its focal point (f_1) and then the second lens is placed so that its focal point (f_2) is at the same location. It then re-collimates the rays diverging from that point.



From similar triangles in the ray diagram, the ratio of the diameters of the beams is directly proportional to the ratio of the focal lengths of the lenses.

$\frac{D_2}{D_1} = \frac{f_2}{f_1}$

Where :

- D_1 = Initial beam diameter $= 2\,\mathrm{mm}$
- D_2 = Final beam diameter $= 14\,\mathrm{mm}$
- f_1 = Focal length of the first lens $= 40\,\mathrm{mm}$
- f_2 = Focal length of the second lens (need to be calculated)

$\Rightarrow \mathrm{f}_2 = \mathrm{f}_1 \times \left(\frac{\mathrm{D}_2}{\mathrm{D}_1}\right)$

$\Rightarrow \mathrm{f}_2 = 40 \times \left(\frac{14}{2}\right)$

$\Rightarrow \mathrm{f}_2 = 280\,\mathrm{mm}$

Therefore, to expand the beam from a diameter of 2 mm to 14 mm using a first lens of 40 mm focal length, the second lens must have a focal length of 280 mm .

Hence, the correct answer is 280 .

Question3

In a microscope the objective is having focal length $f_o = 2$ cm and eye-piece is having focal length $f_e = 4$ cm. The tube length is 32 cm . The magnification produced by this microscope for normal adjustment is ____ .

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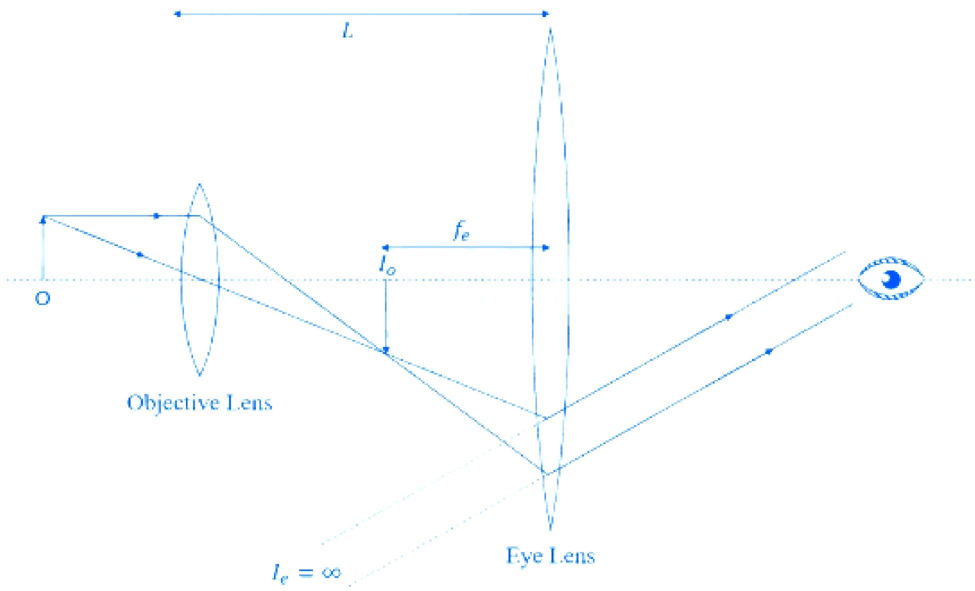
Answer: 100

Solution:



In microscope normal adjustment refers to the condition where the final image is formed at infinity.

The total magnifying power (M) of a compound microscope in normal adjustment is the product of the linear magnification of the objective ($\mathrm{m_o}$) and the angular magnification of the eyepiece ($\mathrm{m_e}$):

$\mathrm{M = m_o \times m_e}$

For an image at infinity:

Objective magnification $\mathrm{m_o} = \frac{\mathrm{L}}{\mathrm{f_O}}$, where L is the tube length.

Eyepiece magnification $\mathrm{m_e} = \frac{\mathrm{D}}{\mathrm{f_e}}$, where D is the least distance of distinct vision (standard value is 25 cm).

Thus, the total magnification formula is:

$M = \frac{L}{f_o} \times \frac{D}{f_e}$

The focal length of objective is $\mathrm{f_o} = 2$ cm

The focal length of eyepiece is $\mathrm{f_e} = 4$ cm

The tube length is $\mathrm{L} = 32$ cm

Substituting the values,

$M=\frac{32}{2}\times\frac{25}{4}100$

Therefore, the total magnification of given microscope is 100 . Hence, the correct answer is $\mathbf{100}$.

Question4

A parallel beam of light travelling in air (refractive index 1.0) is incident on a convex spherical glass surface of radius of curvature 50 cm . Refractive index of glass is 1.5 . The rays converge to a point at a distance x cm from the centre of the curvature of the spherical surface. The value of x is _____ cm .

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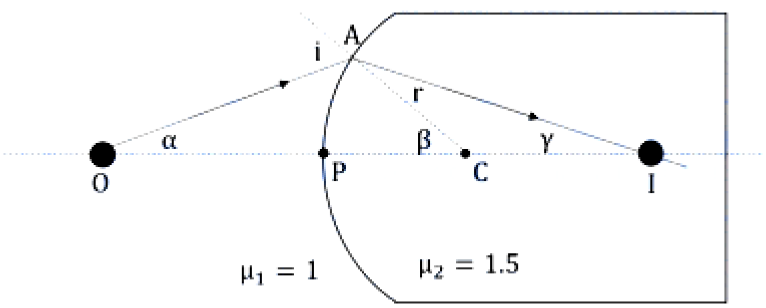
Answer: 100

Solution:

Imagine a point object O placed on the principal axis of a convex spherical surface that separates two media of refractive indices μ_1 and μ_2.

Here the pole of the surface be P and its centre of curvature be C.



Using the formula for refraction at a spherical surface :

$\frac{\mu_2}{\mathrm{v}}-\frac{\mu_1}{\mathrm{u}}=\frac{\mu_2-\mu_1}{\mathrm{R}}$

The incident beam is parallel, which means the light is coming from an object at infinity. Therefore, the object distance $u=-\infty$.

The refractive index of the first medium (air) is $\mu_1=1.0$.

The refractive index of the second medium (glass) is $\mu_2 = 1.5$.

The surface is convex facing the air. This means its centre of curvature lies inside the glass, in the direction of the traveling light. Thus, the radius of curvature is positive: $\mathrm{R} = +50\,\mathrm{cm}$.

Let's plug these into our derived equation to find the image position (v) :

$\frac{1.5}{v} - \frac{1.0}{-\infty} = \frac{1.5-1.0}{+50}$

$\Rightarrow \frac{1.5}{v} - 0 = \frac{0.5}{50}$

$\Rightarrow \frac{1.5}{v} = \frac{1}{100}$

$\Rightarrow \mathrm{v} = 150\,\mathrm{cm}$

So, the parallel rays will converge to form an image at a distance of 150 cm to the right of the pole of the spherical surface.

Pole (P) is at 0 cm . and the centre of curvature (C) is at +50 cm .

Image point / Convergence point (I) is at +150 cm .

So, the distance between C and I is

$x = v - R$

$\Rightarrow \mathrm{x} = 150\,\mathrm{cm} - 50\,\mathrm{cm} = 100\,\mathrm{cm}$

Therefore, the distance from the centre of curvature is 100 cm .

Hence, the correct answer is $\mathbf{100}$.

Question5

The size of the images of an object, formed by a thin lens are equal when the object is placed at two different positions 8 cm and 24 cm from the lens. The focal length of the lens is _____ cm.

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Answer: 16

Solution:

For a thin lens to form images of the same size at two different object positions, one image must be real and the other must be virtual. This typically occurs with a convex lens.

If the object distance is u, focal length of lens is f then lens formula is given as,

$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{1}{v}=\frac{u+f}{uf}$

The magnification is given as $m=\frac{v}{u} \Rightarrow m=\frac{1}{u} \times\left[\frac{uf}{u+f}\right]=\frac{f}{f+u}$

Where:

- f is the focal length.
- u is the object distance (using sign convention, u is negative).

Since the sizes (magnitudes of magnification) are equal but the nature of the images is different:

$|\mathrm{m}_1|=|\mathrm{m}_2| \Rightarrow \mathrm{m}_1=-\mathrm{m}_2$

Given:

- 1. Position $1=\mathrm{u}_1=-8 \mathrm{~cm}$
- 2. Position $2=\mathrm{u}_2=-24 \mathrm{~cm}$

$\frac{f}{f+(-8)}=-\left(\frac{f}{f+(-24)}\right)$

$\Rightarrow \frac{1}{f-8}=-\frac{1}{f-24}$

$\Rightarrow f-24=-(f-8)$

$\Rightarrow f-24=-f+8$

$\Rightarrow 2\mathrm{f}=32$

$\Rightarrow \mathrm{f}=16 \mathrm{~cm}$

The focal length of the lens that produces images of equal size at object distances of 8 cm and 24 cm is 16 cm.

Hence, the correct answer is 16.

Question6

A convex lens of refractive index 1.5 and focal length $f=18 \mathrm{~cm}$ is immersed in water. The difference in focal lengths of the given lens when it is in water and in air is $\alpha \times \mathrm{f}$. The value of α is _____ .

(refractive index of water $=4/3$)

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Answer: 3

Solution:

The Lens Maker's Formula is

$\frac{1}{\mathrm{f}}=\left(\frac{\mu_1}{\mu_{\mathrm{m}}}-1\right)\left(\frac{1}{\mathrm{R}_1}-\frac{1}{\mathrm{R}_2}\right)$

where μ_1 is the refractive index of the lens and μ_m is the refractive index of the surrounding medium.

Case-1 Focal length in Air ($\mathrm{f_a}$)

In air, $\mu_{\mathrm{m}}=1$.

$\frac{1}{\mathrm{f_a}}=\left(\frac{\mu_1}{1}-1\right)\left(\frac{1}{\mathrm{R}_1}-\frac{1}{\mathrm{R}_2}\right)$

Let $\mathrm{K}=\left(\frac{1}{\mathrm{R}_1}-\frac{1}{\mathrm{R}_2}\right) \Rightarrow \frac{1}{\mathrm{f_a}}=(\mu_1-1)\mathrm{K}$

Given $\mu_1=1.5$ and $\mathrm{f_a}=\mathrm{f}=18$ cm :

$\frac{1}{\mathrm{f}}=(1.5-1)\mathrm{K}=0.5\ \mathrm{K} \Rightarrow\ \mathrm{K}=\frac{2}{\mathrm{f}}$

Case-2 Focal length in Water ($\mathrm{f_w}$)

In water, $\mu_{\mathrm{m}}=\mu_{\mathrm{w}}=4/3$:

$\frac{1}{\mathrm{f_w}}=\left(\frac{\mu_1}{\mu_{\mathrm{w}}}-1\right)\mathrm{K}$

$\Rightarrow \frac{1}{\mathrm{f_w}}=\left(\frac{1.5}{4/3}-1\right)\mathrm{K}=\left(\frac{4.5}{4}-1\right)\mathrm{K}=\left(\frac{9}{8}-1\right)\mathrm{K}$

$\frac{1}{\mathrm{f_w}}=\frac{1}{8}\ \mathrm{K}=\frac{1}{8}\times(2\mathrm{f})=\frac{1}{4}\mathrm{f}$

Therefore, $\mathrm{f_w}=4\mathrm{f}$.

The difference in focal lengths is given as :

$\Delta\mathrm{f}=\mathrm{f_w}-\mathrm{f_a}$

$\Rightarrow \Delta\mathrm{f}=4\mathrm{f}-\mathrm{f}=3\mathrm{f}=\alpha\times\mathrm{f}\Rightarrow\alpha=3$

Therefore, the value of α is 3.

Hence, the correct answer is 3.

Question7

If sunlight is focused on a paper using convex lens, it starts burning the paper in shortest time when the lens is kept at 30 cm above the paper. If the radius of curvature of the lens is 60 cm then the refractive index of the lens material is $\frac{\alpha}{10}$. The value of α is ________.

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Answer: 20

Solution:

Since the source (sun) is at infinity, sunlight consists of parallel rays. A convex lens converges parallel rays at its principal focus. Therefore, if the paper burns at a distance of 30 cm, this distance represents the focal length of the lens.

$f = 30$ cm

For a thin lens with radii of curvature R_1 and R_2, the focal length and refractive index μ are related by Lens makers formula:

$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

Assuming the lens is a equi-convex, then $R_1 = +R$ and $R_2 = -R$.

$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R} - \frac{1}{-R}\right) = (\mu - 1)\left(\frac{2}{R}\right)$

$\mu - 1 = \frac{R}{2f}$

The radius of curvature is given as $R = 60$ cm and $f = 30$ cm:

$\mu - 1 = \frac{60}{2\times 30} = \frac{60}{60} = 1$

$\mu = 1 + 1 = 2$

$\Rightarrow 2 = \frac{\alpha}{10}$

$\Rightarrow \alpha = 20$

Therefore, the value of α is 20. Thus, the final answer is 20.

Question8

A concave mirror of focal length 10 cm forms an image which is double the size of object when the object is placed at two different positions. The distance between the two positions of the object is

_____ cm.

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Answer: 10

Solution:

For a mirror, magnification is given as $m = -\frac{v}{u}$, where v is the image distance and u is the object distance.

From the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$:

$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{u-f}{uf} \Rightarrow v = \frac{uf}{u-f}$

Substituting v into the magnification formula:

$m = -\frac{f}{u-f} = \frac{f}{f-u}$

A concave mirror ($f = -10$ cm) produces a magnification $|m| = 2$ in two cases:

Case 1 : Real Image ($m = -2$)

$-2 = -\frac{10}{-10-u_1}$

$\Rightarrow 20 + 2u_1 = -10 \Rightarrow 2u_1 = -30 \Rightarrow u_1 = -15$ cm

Case 2 : Virtual Image ($m = +2$)

$2 = -\frac{10}{-10-u_2}$

$\Rightarrow -20 - 2u_2 = -10 \Rightarrow -2u_2 = 10 \Rightarrow u_2 = -5$ cm

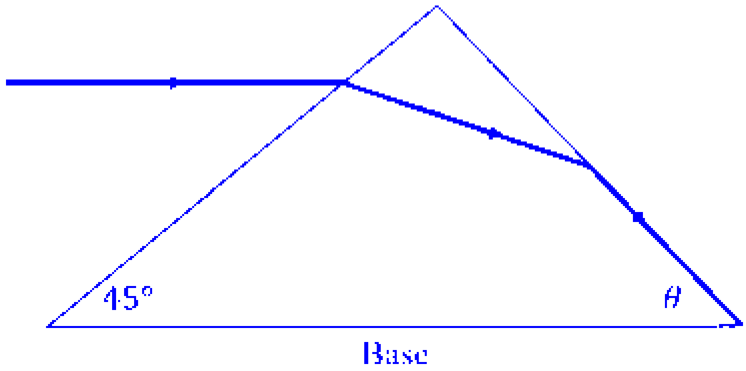
The distance between the two object positions is:

$\Delta u = |u_1 - u_2| = |-15 - (-5)| = |-10| = 10$ cm

Therefore, the distance between the two positions is 10 cm .

Question9

As shown in the diagram, when the incident ray is parallel to base of the prism, the emergent ray grazes along the second surface.



If refractive index of the material of prism is $\sqrt{2}$, the angle θ of prism is.

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Options:

A.

75°

B.

90°

C.

60°

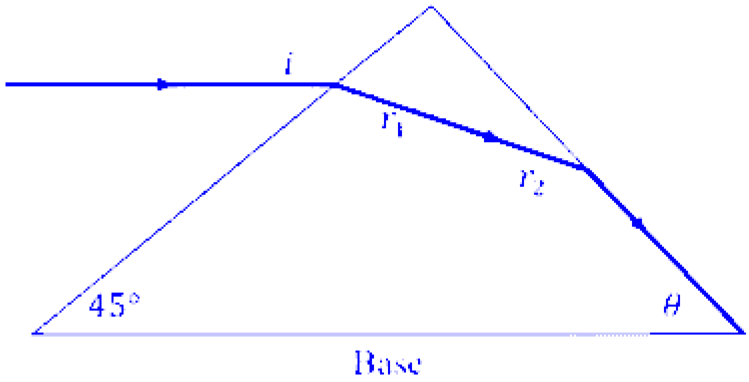
D.

45°

Answer: C

Solution:

The incident ray is parallel to the base where the base angle is 45°.



If the ray is parallel to the base, the angle of incidence i with the normal to the first surface is also 45°

Let the refractive index of the prism be $\mu=\sqrt{2}$ and air be $\mu_{\text{air}}=1$.

Applying Snell's Law at the first surface :

$\mu_{\text{air}} \sin (\mathrm{i})=\mu \sin \left(\mathrm{r}_1\right)$

$\Rightarrow 1 \times \sin \left(45^{\circ}\right)=\sqrt{2} \times \sin \left(\mathrm{r}_1\right)$

$\Rightarrow \frac{1}{\sqrt{2}}=\sqrt{2} \sin \left(\mathrm{r}_1\right)$

$\Rightarrow \sin \left(r_1\right)=\frac{1}{2}$

$\Rightarrow r_1=30^{\circ}$

It is given that the emergent ray grazes along the second surface. This means the angle of emergence e is 90°.

This occurs when the angle of incidence inside the prism at the second surface $\left(\mathrm{r}_2\right)$ is equal to the critical angle (C).

Applying Snell's Law at the second surface :

$\mu \sin \left(\mathrm{r}_2\right)=\mu_{\text{air}} \sin \left(90^{\circ}\right)$

$\Rightarrow \sqrt{2} \times \sin \left(\mathrm{r}_2\right)=1 \times 1$

$\Rightarrow \sin \left(r_2\right)=\frac{1}{\sqrt{2}}$

$\Rightarrow \mathrm{r}_2=45^{\circ}$

For any prism, the relationship between the apex angle A and the internal refracted angles is :

$\mathrm{A}=\mathrm{r}_1+\mathrm{r}_2$

$\Rightarrow \mathrm{A}=30^{\circ}+45^{\circ}=75^{\circ}$

Sum of angles in a triangle is 180° :

$45^{\circ}+\theta+\mathrm{A}=180^{\circ}$

$\Rightarrow 45^{\circ}+\theta+75^{\circ}=180^{\circ}$

$\Rightarrow 120^{\circ}+\theta=180^{\circ}$

$\Rightarrow \theta=60^{\circ}$

So, the angle θ of prism is 60°.

Hence, the correct option is (C).

Question10

Consider an equilateral prism (refractive index $\sqrt{2}$). A ray of light is incident on its one surface at a certain angle i. If the emergent ray is found to graze along the other surface then the angle of refraction at the incident surface is close to ____

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Options:

A.

30°

B.

20°

C.

40°

D.

15°

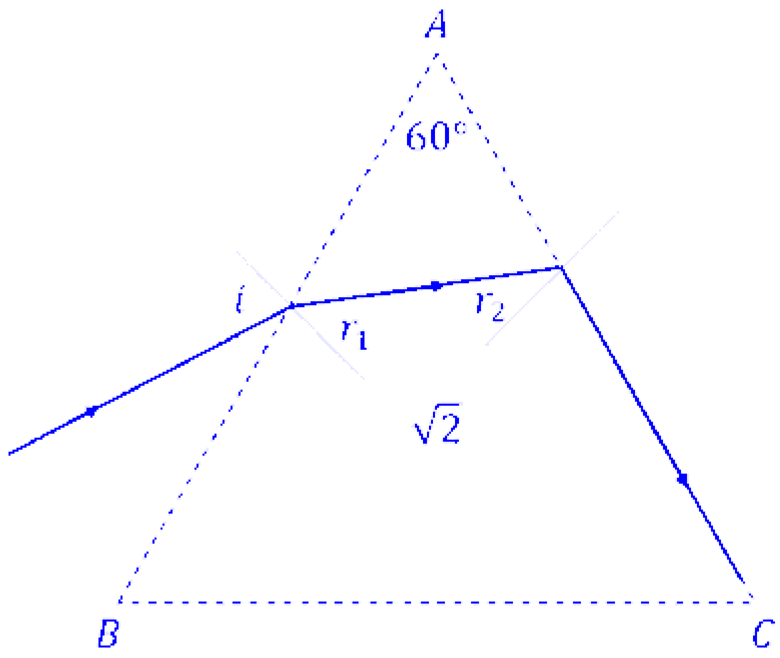
Answer: D

Solution:

The given prism is an equilateral, that means the angle of the prism (A) is 60°.

Refractive index of the prism is $\mu = \sqrt{2}$

The ray grazes the second surface. This means the angle of emergence (e) is 90°.



The angle of incidence at the first surface of prism is i .

Let the angle of refraction at first surface is r_1.

The ray travels straight through the prism and hits the second face AC.

Let the angle of incidence at the second surface be r_2.

The ray bends away from the normal as it exits into the air. Let this angle be e (angle of emergence).

The sum of angles r_1 and r_2 is equal to the angle of prism.

$\mathrm{A} = \mathrm{r}_1 + \mathrm{r}_2$

Using Snell's law at the emergent surface (Face AC),

$\sqrt{2} \cdot \sin(r_2) = 1 \cdot \sin(90^\circ)$

$\Rightarrow \sqrt{2} \cdot \sin(r_2) = 1$

$\Rightarrow \sin(r_2) = \frac{1}{\sqrt{2}} \Rightarrow r_2 = 45^\circ$

Using $\mathrm{A} = \mathrm{r}_1 + \mathrm{r}_2$

$60^\circ = \mathrm{r}_1 + 45^\circ$

$\Rightarrow r_1 = 60^\circ - 45^\circ$

$\Rightarrow r_1 = 15^\circ$

So, the angle of refraction at the incident surface is 15°.

Hence, the correct option is (D).

Question11

A thin convex lens of focal length 5 cm and a thin concave lens of focal length 4 cm are combined together (without any gap) and this combination has magnification m_1 when an object is placed 10 cm before the convex lens. Keeping the positions of convex lens and object undisturbed a gap of 1 cm is introduced between the lenses by moving the concave lens away, which lead to a change in magnification of total lens system to m_2. The value of $\left|\frac{m_1}{m_2}\right|$ is _____ .

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Options:

A.

$\frac{25}{27}$

B.

$\frac{5}{6}$

C.

$\frac{5}{27}$

D.

$\frac{3}{2}$

Answer: B

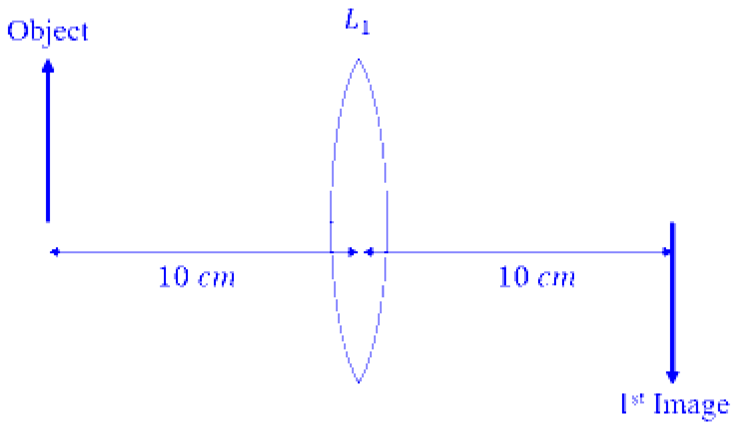
Solution:

The focal length of convex lens $= \mathrm{f}_1 = +5$ cm

The focal length of concave lens $= \mathrm{f}_2 = -4$ cm

Initial object distance: $u = -10$ cm with respect to convex lens.

Using lens formula for first lens,



$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} \Rightarrow \frac{1}{v_1} - \frac{1}{-10} = \frac{1}{5}$

$\Rightarrow \frac{1}{v_1} = \frac{1}{5} - \frac{1}{10} = \frac{1}{10}$

$\mathrm{v}_1 = 10\ \mathrm{cm}$

So, the image by first lens is formed at a distance of 10 cm from the first lens. The positive sign shows that the first image is formed right of first lens.

Case 1 : when lenses are combined together without any gap ($\mathrm{d} = 0$)

When an object is placed at distance u, the first lens forms an image at distance v_1.

Using lens formula:

$\frac{1}{\mathrm{v}_1} - \frac{1}{\mathrm{u}} = \frac{1}{\mathrm{f}_1} \ldots$ (i)

This image acts as a virtual object for the second lens. Since there is no gap, the object distance for the second lens is simply v_1. It forms the final image at distance v .

$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} \ldots (ii)$

From both the equations,

$\left(\frac{1}{v_1} - \frac{1}{u}\right) + \left(\frac{1}{v} - \frac{1}{v_1}\right) = \frac{1}{f_1} + \frac{1}{f_2}$

$\Rightarrow \frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{f_{eq}}$

$\Rightarrow \frac{1}{\mathrm{f_{eq}}} = \frac{1}{\mathrm{f}_1} + \frac{1}{\mathrm{f}_2}$

$\Rightarrow \frac{1}{\mathrm{f_{eq}}} = \frac{1}{5} + \frac{1}{-4} = \frac{4-5}{20} = -\frac{1}{20}$

$\Rightarrow \mathrm{f_{eq}} = -20\ \mathrm{cm}$

Using the lens formula for equivalent lens :

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_{eq}}$

$\Rightarrow \frac{1}{v} - \frac{1}{-10} = -\frac{1}{20}$

$\Rightarrow \frac{1}{v} = -\frac{1}{20} - \frac{1}{10}$

$\Rightarrow \frac{1}{v} = \frac{-1-2}{20} = -\frac{3}{20}$

$\Rightarrow \mathrm{v}=-\frac{20}{3}$ cm

The magnification m_1 for the combined system is the ratio of image distance to object distance:

$m_1=\frac{v}{u}=\frac{-20/3}{-10}=\frac{2}{3}$

Case 2 : When lenses are separated by a gap of d = 1 cm

Refraction through the first lens (Convex, $f_1=+5$ cm)

The object is still at $u_1=-10$ cm.

$\frac{1}{v_1}-\frac{1}{u_1}=\frac{1}{f_1}$

$\Rightarrow \frac{1}{v_1}-\frac{1}{-10}=\frac{1}{5}$

$\Rightarrow \frac{1}{v_1}+\frac{1}{10}=\frac{1}{5}$

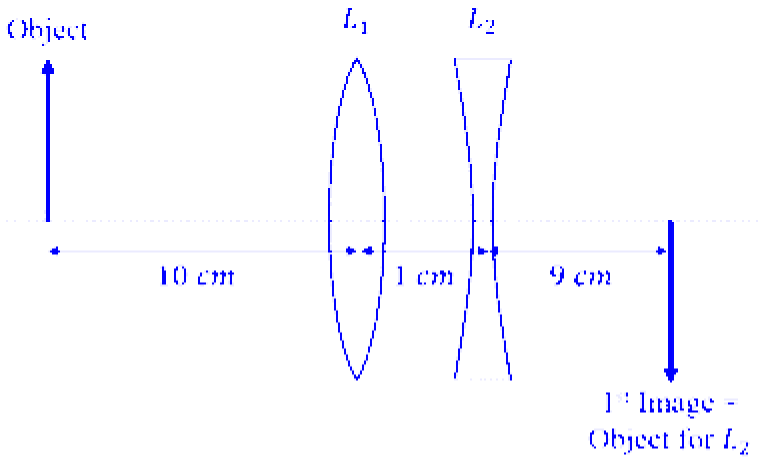
$\Rightarrow \frac{1}{v_1}=\frac{1}{5}-\frac{1}{10}=\frac{2-1}{10}=\frac{1}{10}$

$\Rightarrow v_1=+10$ cm

The lateral magnification of the first lens is :

$m_{L1}=\frac{v_1}{u_1}=\frac{10}{-10}=-1$

Refraction through the second lens (Concave, $f_2=-4$ cm)



The image formed by the first lens acts as the object for the second lens. Since the first image is 10 cm to the right of L_1, and L_2 is placed 1 cm to the right of L_1, the distance of this image from L_2 is:

$u_2=v_1-d=10-1=+9$ cm

It is a virtual object for the second lens, hence the positive sign.

Now, applying the lens formula for L_2 :

$\frac{1}{v_2}-\frac{1}{u_2}=\frac{1}{f_2}$

$\Rightarrow \frac{1}{v_2}-\frac{1}{9}=\frac{1}{-4}$

$\Rightarrow \frac{1}{v_2}=-\frac{1}{4}+\frac{1}{9}=\frac{-9+4}{36}=-\frac{5}{36}$

$\Rightarrow v_2 = -\frac{36}{5}$ cm

The lateral magnification of the second lens is:

$m_{L2} = \frac{v_2}{u_2} = \frac{-36/5}{9} = -\frac{4}{5}$

So, the total magnification in second case is

$m_2 = m_{L1} \times m_{L2} = (-1) \times \left(-\frac{4}{5}\right) = \frac{4}{5}$

So, the required ratio of magnification is $\left|\frac{m_1}{m_2}\right| = \frac{2/3}{4/5} = \frac{5}{6}$.

Question12

In parallax method for the determination of focal length of a concave mirror, the object should always be placed:

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Options:

A.

between the focus (F) and the centre of curvature (C) of the mirror ONLY

B.

between the pole (P) and the focus (F) of the concave mirror ONLY

C.

at any point beyond the focus (F) of the mirror

D.

beyond the centre of the curvature (C) of the mirror ONLY

Answer: C

Solution:

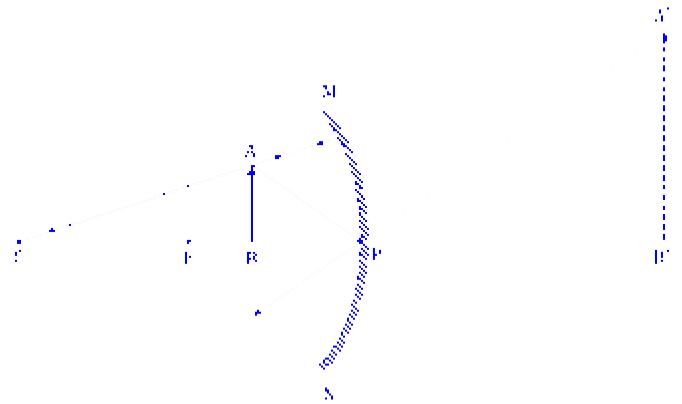
The parallax method is an experimental technique used to locate the exact position of an image. It involves using an object pin and an image pin. The position of the image is found by placing the image pin such that there is no relative shift (no parallax) between the image pin and the actual image of the object pin when you move your eye side to side.

This technique requires the image to be a real image that is formed in the actual space in front of the mirror. So, to use the parallax method, the concave mirror is required because it produces a real image.

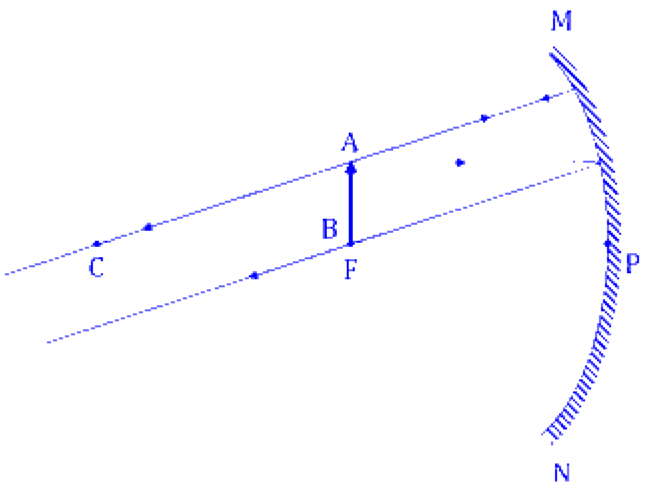
The image formation cases for a concave mirror based on the object's position :

- Between the Pole (P) and Focus (F) :

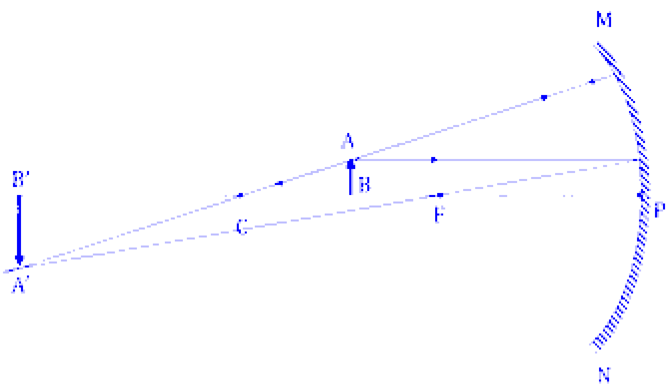
The mirror forms a virtual, erect, and magnified image behind the mirror. A virtual image cannot be located using a physical image pin in standard parallax methods because the light rays do not actually converge at the image location.



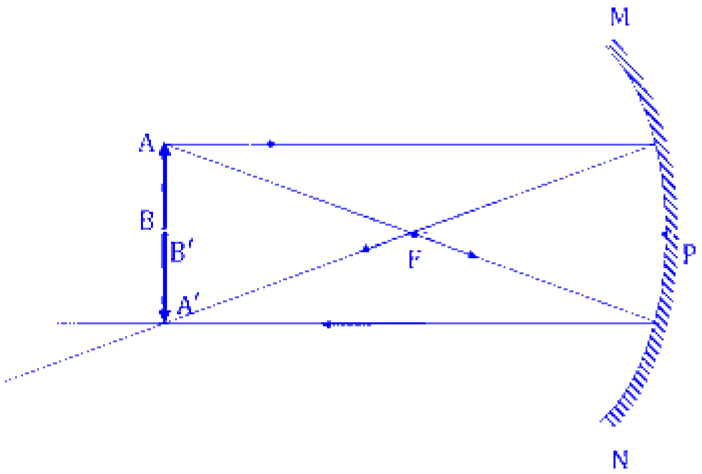
At the Focus (F) : The image is formed at infinity.



Between F and C : The image so formed is real and inverted in front of mirror.



At C : The mirror forms a real and inverted image in front of the mirror.



While placing the object between F and C does form a real image, stating it ONLY works here is incorrect because placing it beyond C also works. Hence, option (A) is incorrect.

Placing the object between P and F forms a virtual image, making the standard parallax method impossible. Hence, option (B) is incorrect.

Placing the object at any point beyond the focus (F) the nature of image formed is real (whether it is formed beyond C , at C , or between C and F). This allows the image pin to be used to remove parallax. Hence,

While placing it beyond C forms a real image, restricting it to ONLY beyond C is incorrect because placing it between F and C also works. Hence, the option (D) is incorrect.

Therefore, the correct option is (C).

Question13

Consider light travelling from a medium A to medium B separated by a plane interface. If the light undergoes total internal reflection during its travel from medium A to B and the speed of light in media A and B are $2.4 \times 10^8 \mathrm{~m/s}$ and $2.7 \times 10^8 \mathrm{~m/s}$, respectively, then the value of critical angle is :

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Options:

A.

$\cos^{-1}\left(\frac{8}{9}\right)$

B.

$\sin^{-1}\left(\frac{9}{8}\right)$

C. $\cot^{-1}\left(\frac{3}{\sqrt{13}}\right)$

D. $\tan^{-1}\left(\frac{8}{\sqrt{17}}\right)$

Answer: D

Solution:

For total internal reflection to occur, light must travel from a denser medium (lower speed) to a rarer medium (higher speed).

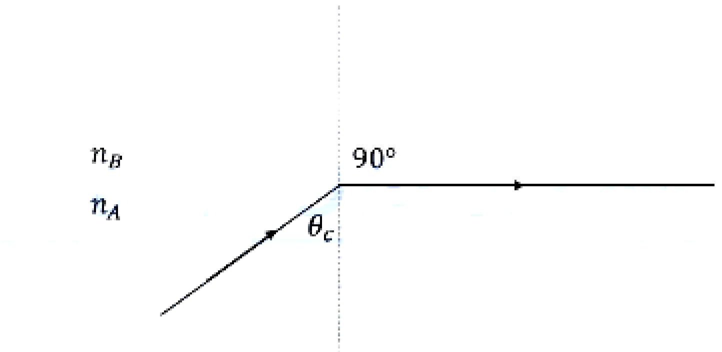
The speed of light in medium A (Denser) is $\mathrm{v_A} = 2.4 \times 10^8\ \mathrm{m/s}$

The speed of light in medium B (rarer) is $v_B = 2.7 \times 10^8\ \mathrm{m/s}$

Since $\mathrm{v_A} < \mathrm{v_B}$, medium A is indeed optically denser than medium B , satisfying the condition for TIR.

The critical angle is determined by Snell's Law at the boundary where the angle of refraction is 90° :

$n_A \sin(\theta_c) = n_B \sin(90^\circ) \Rightarrow \sin(\theta_c) = \frac{n_B}{n_A}$



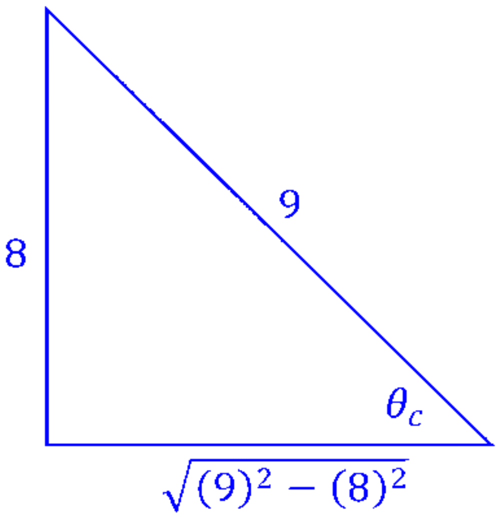
Since the refractive index n is inversely proportional to the speed of light $\left(\mathrm{n} = \frac{\mathrm{c}}{\mathrm{v}}\right)$,

$\frac{n_B}{n_A} = \frac{c/v_B}{c/v_A} = \frac{v_A}{v_B}$

So,

$\sin(\theta_c) = \frac{v_A}{v_B}$

$\Rightarrow \sin(\theta_C) = \frac{2.4\times10^8}{2.7\times10^8} = \frac{8}{9}$



$\tan(\theta_c) = \frac{8}{\sqrt{9^2-8^2}} = \frac{8}{17} \Rightarrow \theta_c = \tan^{-1}\left(\frac{8}{\sqrt{17}}\right)$

Therefore, the critical angle is $\tan^{-1}\left(\frac{8}{\sqrt{17}}\right)$.

Hence, the correct option is (D).

Question14

A thin prism with angle 5° of refractive index 1.72 is combined with another prism of refractive index 1.9 to produce dispersion without deviation. The angle of second prism is _____ .

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Options:

A.

5°

B.

4°

C.

4.5°

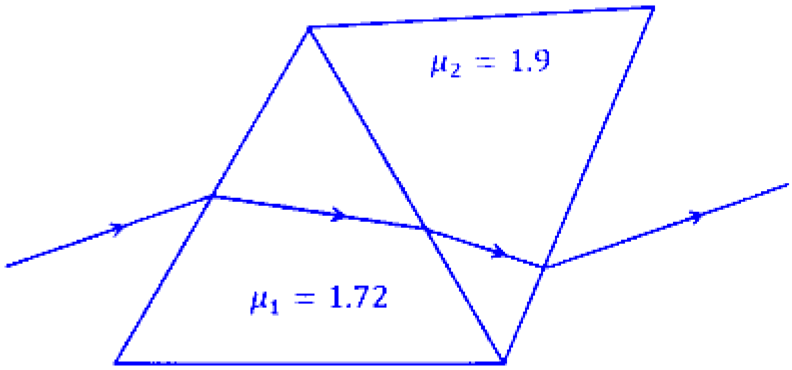
D.

6°

Answer: B

Solution:

When two thin prisms are combined to produce dispersion without deviation, they are arranged such that their refracting angles are inverted relative to each other. The net deviation produced by the combination is zero.



$\delta_{\text{net}} \quad = \delta_1 - \delta_2 = 0$

$\Rightarrow \delta_1 = \delta_2$

The deviation (δ) produced by a thin prism is given by the formula :

$\delta = (\mu - 1)\mathrm{A}$

where :

- μ is the refractive index of the material.
- A is the angle of the prism.

Since the deviations must be equal and opposite :

$(\mu_1 - 1)\mathrm{A}_1 = (\mu_2 - 1)\mathrm{A}_2$

The angle of prism for first prism is $\mathrm{A}_1 = 5^{\circ}$ and its refractive index is $\mu_1 = 1.72$

The refractive index of the second prism is $\mu_2 = 1.9$

$(1.72 - 1) \times 5^{\circ} = (1.9 - 1) \times \mathrm{A}_2$

$\Rightarrow 3.6 = 0.9 \times \mathrm{A}_2 \Rightarrow \ \mathrm{A}_2 = \frac{3.6}{0.9} = 4^{\circ}$

Therefore, the angle of the second prism is 4°. Hence, the correct option is (B).

Question15

In a microscope of tube length 10 cm two convex lenses are arranged with focal length of 2 cm and 5 cm . Total magnification obtained with this system for normal adjustment is $(5)^k$. The value of k is _____ .

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Options:

A.

4

B.

5

C.

2

D.

3.5

Answer: C

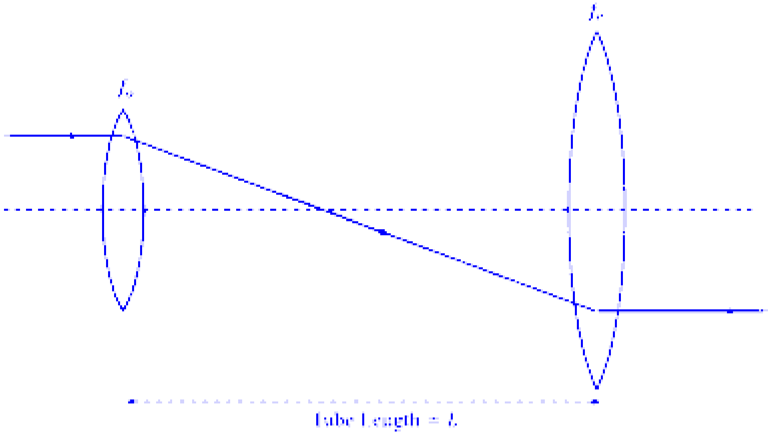
Solution:

A compound microscope consists of two lenses objective lens (lens near the object) and the eyepiece (lens near the eye.). To achieve high magnification, objective lens must have a small focal length compared to eyepiece.

So,

- 1. Focal length of Objective $= f_0 = 2$ cm
- 2. Focal length of Eyepiece $= \mathrm{f_e} = 5$ cm

During normal adjustment in optical instruments the final image is formed at infinity. This allows the observer to view the image with a relaxed eye.



The total magnification (M) of a microscope is the product of the magnification of the objective ($\mathrm{m_o}$) and the magnification of the eyepiece ($\mathrm{m_e}$).

$\mathrm{M = m_o \times m_e}$

The eyepiece acts like a simple magnifying glass. For an image formed at infinity, the angular magnification is the ratio of the distinct vision distance (D) to the focal length ($\mathrm{f_e}$).

$m_e = \frac{D}{f_e}$

Where Least Distance of Distinct Vision $\mathrm{D = 25\ cm}$.

The objective forms a real, inverted image.

The tube length (L) is defined as the distance between the second focal point of the objective and the first focal point of the eyepiece.

$m_o \approx \frac{L}{f_o}$

So, the total magnification of the microscope is,

$\mathrm{M} = \left(\frac{\mathrm{L}}{\mathrm{f_o}}\right) \times \left(\frac{\mathrm{D}}{\mathrm{f_e}}\right)$

Putting the values,

$M = \left(\frac{10\ \mathrm{cm}}{2\ \mathrm{cm}}\right) \times \left(\frac{25}{5}\right) = 25 = 5^2$

$\Rightarrow 5^2 = 5^k \Rightarrow k = 2$

Therefore, the value of k is 2.

Hence, the correct option is (C).

Question16

An unpolarised light is incident at an interface of two dielectric media having refractive indices of 2 (incident medium) and $2\sqrt{3}$ (medium) respectively. To satisfy the condition that reflected and

refracted rays are perpendicular to each other, the angle of incidence is ____

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Options:

A.

45°

B.

60°

C.

10°

D.

30°

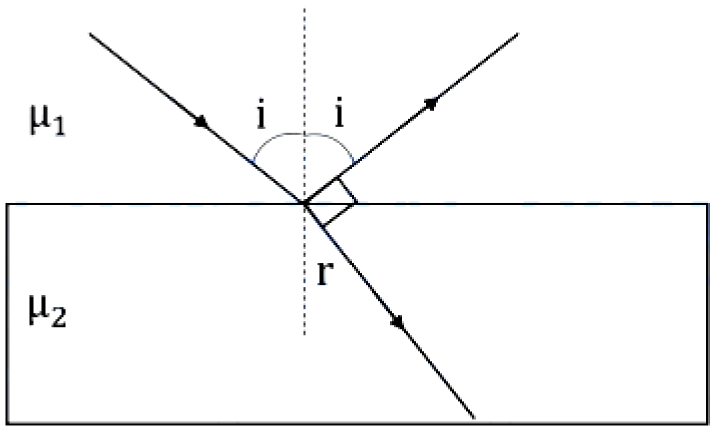
Answer: B

Solution:

Light hits the surface at an angle of incidence i.

Part of the light bounces back. According to the Law of Reflection, the angle of reflection is equal to the angle of incidence (i).

Part of the light enters the second medium. It bends at an angle of refraction r.



The reflected and refracted rays are perpendicular to each other. This means the angle between the reflected ray and the refracted ray is 90°.

The sum of angles on a straight line is 180°.

$\mathrm{i}+90^\circ+\mathrm{r}=180^\circ \Rightarrow \mathrm{i}+\mathrm{r}=90^\circ$

$\Rightarrow \mathrm{r}=90^\circ-\mathrm{i}$

Now using the fundamental law of refraction, Snell's Law :

$\mu_1 \sin(\mathrm{i})=\mathrm{n}_2 \sin(\mathrm{r}) \Rightarrow \mu_1 \sin(\mathrm{i})=\mathrm{n}_2 \sin\left(90^\circ-\mathrm{i}\right)$

$\Rightarrow \mu_1 \sin(\mathrm{i})=\mathrm{n}_2 \cos(\mathrm{i})$

$\Rightarrow \tan(\mathrm{i})=\frac{\mu_2}{\mu_1}$

The refractive index of first medium is $\mu_1=2$ and the refractive index of second medium is $\mu_2=2\sqrt{3}$

$\tan(i)=\frac{2\sqrt{3}}{2}=\sqrt{3} \Rightarrow i=60^\circ$

Therefore, the angle of incidence is 60°.

Hence, the correct option is (B).

Question17

The exit surface of a prism with refractive index n is coated with a material having refractive index $\frac{n}{2}$. When this prism is set for minimum angle of deviation, it exactly meets the condition of critical angle. The prism angle is _____ .

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Options:

A.

60°

B.

30°

C.

15°

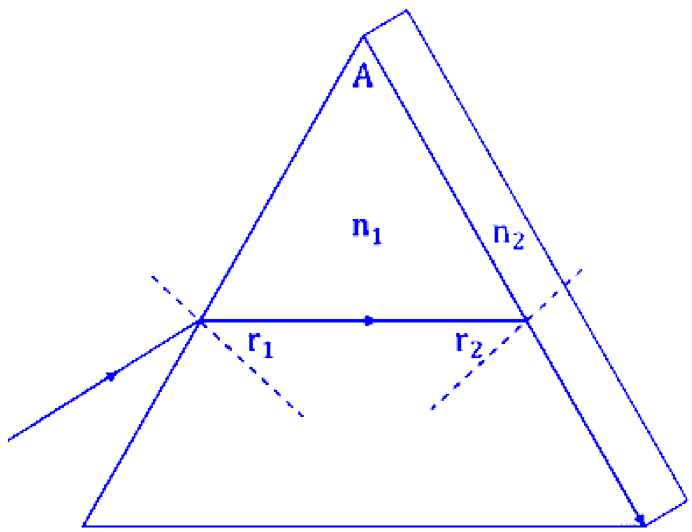
D.

45°

Answer: A

Solution:

Light is traveling from the prism material (denser medium) to the coating material (rarer medium).



Refractive index of Prism $\mathrm{n}_1 = \mathrm{n}$

Refractive index of Coating $\mathrm{n}_2 = \frac{\mathrm{n}}{2}$

According to question at the exit surface, the light exactly meets the condition of critical angle.

The formula for the critical angle is :

$\sin\theta_{\mathrm{C}} = \frac{\mathrm{n}_{\text{rare}}}{\mathrm{n}_{\text{denser}}}$

$\Rightarrow \sin\theta_C = \frac{n/2}{n} = \frac{1}{2}$

$\theta_C = \sin^{-1}(1/2) = 30^\circ$

So, the angle of incidence at the second face (r_2) is equal to the critical angle :

$r_2 = 30^\circ$

Applying the condition for minimum deviation, the light ray passes symmetrically through the prism.

$\mathrm{r}_1 = \mathrm{r}_2 = 30^\circ$

The light ray inside the prism is parallel to the base (for an equilateral prism).

For any prism, the relationship between the Prism Angle A and the internal angles is:

$$\begin{gathered} \mathrm{A} = \mathrm{r}_1 + \mathrm{r}_2 \\ \mathrm{A} = 30^\circ + 30^\circ \\ \mathrm{A} = 60^\circ \end{gathered}$$

Hence, the correct option is (A).

Question18

Five persons P_1, P_2, P_3, P_4 and P_5 recorded object distance (u) and image distance (v) using same convex lens having power +5 D as $(25, 96), (30, 62), (35, 37), (45, 35)$ and $(50, 32)$ respectively. Identify correct statement

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Options:

A.

Readings recorded by P_3 and P_2 persons are incorrect

B.

Readings recorded by all persons are correct

C.

Readings recorded by P_3 person are incorrect

D.

Readings recorded by P_4 and P_5 persons are incorrect

Answer: C

Solution:

The power (P) of the lens is given as +5 D .

The relationship between power and focal length (f) in meters is $\mathrm{P}=\frac{1}{\mathrm{f}}$.

$\mathrm{f}=\frac{1}{\mathrm{P}}=\frac{1}{5} \mathrm{~m}=20 \mathrm{~cm}$

For a convex lens forming a real image, the lens formula is:

$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$

Using sign convention for a real object (u is negative) and a real image (v is positive):

$\frac{1}{20}=\frac{1}{v}-\frac{1}{-u} \Rightarrow \frac{1}{20}=\frac{1}{v}+\frac{1}{u}$

The theoretical image distance $\left(v_{th}\right)$ for each person can be determined for given object distance (u) using $v_{th} = \frac{uf}{u-f}$ (where u is the magnitude).

For person P_1 the object distance

Person	u	v_{th}	Measured v	Absolute Error
P_1 (25,96)	25	$\frac{25 \times 20}{25-20} = 100$ cm	96	$\|100-96\| = 4$ cm
P_2 (30,62)	30	$\frac{30 \times 20}{30-20} = 60$ cm	62	$\|60-62\| = 2$ cm
P_3 (35,37)	35	$\frac{35 \times 20}{35-20} = 46.6$ cm	37	$\|46.6-37\| = 9.6$ cm
P_4 (45,35)	45	$\frac{45 \times 20}{45-20} = 36$ cm	35	$\|36-35\| = 1$ cm
P_5 (50,32)	50	$\frac{50 \times 20}{50-20} = 33.3$ cm	32	$\|33.3-32\| = 1.3$ cm

Most readings have slight experimental errors (within 2-4 cm). However, for Person P_3, the recorded value of 37 cm is significantly different from the theoretical value of 46.6 cm (an error of 9.6 cm). This makes P_3 's reading distinctly incorrect compared to the others.

Therefore, the data recorded by person P_3 is mathematically inconsistent with the properties of a lens with 20 cm focal length.

Hence, the correct option is (C).

Question19

Distance between an object and three times magnified real image is 40 cm . The focal length of the mirror used is _____ cm .

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Options:

A.

-20

B.

$-15/2$

C.

-10

D.

-15

Answer: D

Solution:

Since the image is real, the mirror must be a concave mirror.

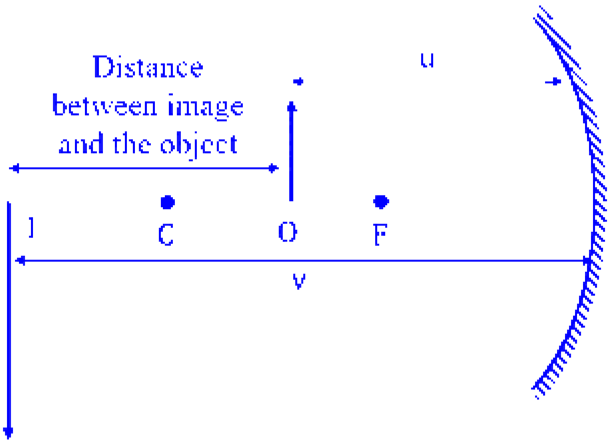
For a real image formed by a concave mirror, the magnification is negative.

$\mathrm{m}=-3$

We know that $m=-\frac{v}{u}$, where m is the magnification of the image, v is the image distance and u is the object distance.

$\Rightarrow -3=-\frac{v}{u} \Rightarrow v=3u$

The distance between the object and the image is 40 cm . In a concave mirror, for a real magnified image, both the object (u) and the image (v) are on the same side of the mirror.



The distance between them is $|\mathrm{v}|-|\mathrm{u}|$.

$|\mathrm{v}|-|\mathrm{u}|=40 \mathrm{~cm}$

Substituting $\mathrm{v}=3\mathrm{u}$,

$3|\mathrm{u}|-|\mathrm{u}|=40$

$\Rightarrow 2|\mathrm{u}|=40 \Rightarrow |\mathrm{u}|=20 \mathrm{~cm}$

Using sign convention, the object distance is negative; u $= -20$ cm.

Therefore, the image distance is :

v $= 3 \times (-20) = -60$ cm

Using the mirror formula :

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

Substituting the values :

$\frac{1}{f} = \frac{1}{-60} + \frac{1}{-20}$

$\Rightarrow \frac{1}{f} = \frac{-1-3}{60} \Rightarrow f = \frac{60}{-4} = -15$ cm

Therefore, the focal length of the concave mirror is -15 cm . The negative sign indicates it is a concave mirror.

Hence, the correct option is (D)

Question20

The magnitudes of power of a biconvex lens (refractive index 1.5) and that of a plano-concave lens (refractive index $= 1.7$) are same. If the curvature of planoconcave lens exactly matches with the curvature of back surface of the biconvex lens, then ratio of radius of curvature of front and back surface of the biconvex lens is ____

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Options:

A.

2 : 5

B.

5 : 2

C.

12 : 5

D.

5 : 12

Answer: B

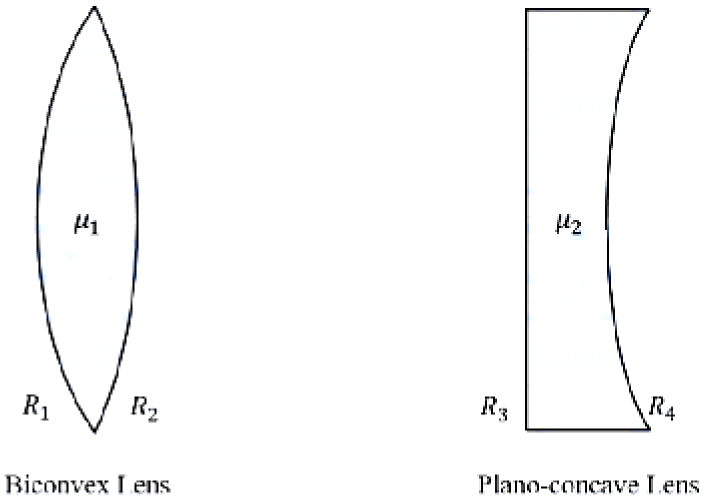
Solution:

If the focal length of a lens is f (in meters) then its power is given as $P=\frac{1}{f} \ldots$ (i)

The Lens Maker's formula is given as $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$, where μ is the refractive index of lens with respect to the medium, R_1 and R_2 are the radii of curvatures (measured using sign convention).

Using the Lens Maker's Formula and power formula :

$$\mathrm{P}=\frac{1}{\mathrm{f}}=(\mu-1)\left(\frac{1}{\mathrm{R}_1}-\frac{1}{\mathrm{R}_2}\right)$$



Here, $\mu_1=$ refractive index of biconvex lens, $R_1 \& R_2$ are the radii of curvatures of front and back curvature of biconvex lens.

Then power of biconvex lens is,

$$P_{\text{convex }}=(1.5-1)\left(\frac{1}{R_1}-\left(\frac{1}{-R_2}\right)\right)=0.5\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$$

For the Plano-Concave Lens

Refractive Index $\mu_2=1.7$

The curvature of the plano-concave lens matches the back surface of the biconvex lens. Thus, its curved surface has radius R_2.

For a plano-concave lens, one surface is flat $R_3=\infty$ and the other is concave (radius $R_4=R_2$).

So, the power of Plano-concave lens is,

$$P_{\text{concave }}=(1.7-1)\left(\frac{1}{R_3}-\frac{1}{R_4}\right)=0.7\left(\frac{1}{\infty}-\frac{1}{R_2}\right)=-\frac{0.7}{R_2}$$

The magnitudes of power are the same

$$\left|P_{\text{convex }}\right|=\left|P_{\text{concave }}\right|$$

$$\Rightarrow 0.5\left(\frac{1}{\mathrm{R}_1}+\frac{1}{\mathrm{R}_2}\right)=\frac{0.7}{\mathrm{R}_2}$$

$$\Rightarrow \frac{5}{\mathrm{R}_1}=\frac{7}{\mathrm{R}_2}-\frac{5}{\mathrm{R}_2}$$

$$\Rightarrow \frac{5}{\mathrm{R}_1}=\frac{2}{\mathrm{R}_2}$$

$$\Rightarrow \frac{\mathrm{R}_1}{\mathrm{R}_2}=\frac{5}{2}$$

Here, R_1 & R_2 are the radii of curvatures of front and back curvatures whose ratio is $5:2$.

Hence, the correct option is (B).

Question21

For a transparent prism, if the angle of minimum deviation is equal to its refracting angle, the refractive index n of the prism satisfies.

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Options:

A.

$n \geq 2$

B.

$\sqrt{2}<n<2$

C.

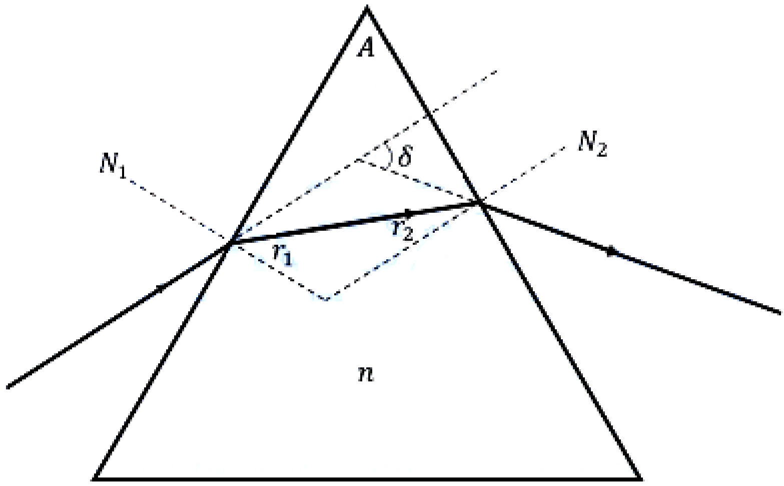
$1<n<2$

D.

$\sqrt{2}<n<2\sqrt{2}$

Answer: B

Solution:



The refractive index n of a prism with refracting angle A and angle of minimum deviation δ_m is given by the formula:

$$\mathrm{n}=\frac{\sin\left(\frac{\mathrm{A}+\delta_{\mathrm{m}}}{2}\right)}{\sin\left(\frac{\mathrm{A}}{2}\right)}$$

According to question the angle of minimum deviation is equal to the refracting angle:

$$\delta_{\mathrm{m}}=\mathrm{A}$$

Putting in the formula,

$$\mathrm{n}=\frac{\sin\left(\frac{A+A}{2}\right)}{\sin\left(\frac{A}{2}\right)}=\frac{\sin(A)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \mathrm{n}=\frac{2\sin\left(\frac{\mathrm{A}}{2}\right)\cos\left(\frac{\mathrm{A}}{2}\right)}{\sin\left(\frac{\mathrm{A}}{2}\right)}=2\cos\left(\frac{\mathrm{A}}{2}\right)$$

The value of the refractive index depends on the possible values for the refracting angle A.

As A approaches 0°, $\cos\left(\frac{\mathrm{A}}{2}\right)\rightarrow\cos\left(0^{\circ}\right)=1$.

Therefore, n approaches $2\times 1=2$.

For minimum deviation to occur, the internal angle of incidence $\mathrm{r}=\frac{\mathrm{A}}{2}$ must be less than the critical angle C to let the light emerge from prism.

$$\sin\left(\frac{A}{2}\right)<\sin(C)=\frac{1}{n}$$

Using $\mathrm{n}=2\cos\left(\frac{\mathrm{A}}{2}\right)$,

$$2\sin\left(\frac{\mathrm{A}}{2}\right)\cos\left(\frac{\mathrm{A}}{2}\right)<1\Rightarrow\sin(\mathrm{A})<1$$

This condition is satisfied for all $\mathrm{A}<90^{\circ}$.

If $\mathrm{A}=90^{\circ}$, then $\mathrm{n}=2\cos\left(45^{\circ}\right)=2\times\frac{1}{\sqrt{2}}=\sqrt{2}$

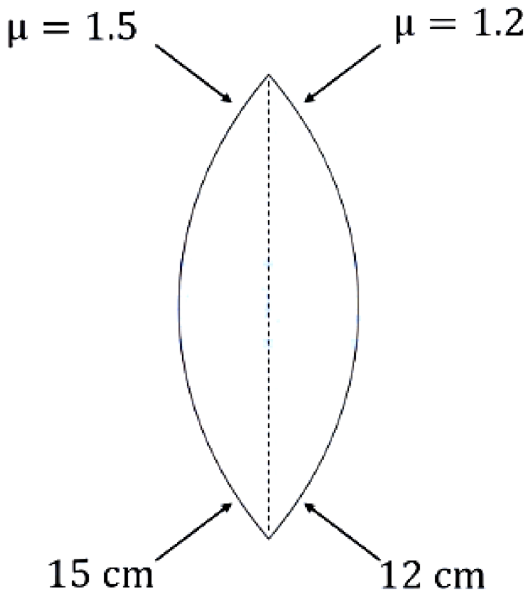
For a practical transparent prism where $0<\mathrm{A}<90^{\circ}$, the refractive index must fall in the range:

$\sqrt{2} < n < 2$

Hence, the correct option is (B).

Question22

A biconvex lens is formed by using two thin planoconvex lenses, as shown in the figure. The refractive index and radius of curved surfaces are also mentioned in figure. When an object is placed on the left side of lens at a distance of 30 cm from the biconvex lens, the magnification of the image will be :



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Options:

A.

-2

B.

-2.5

C.

$+2.5$

D.

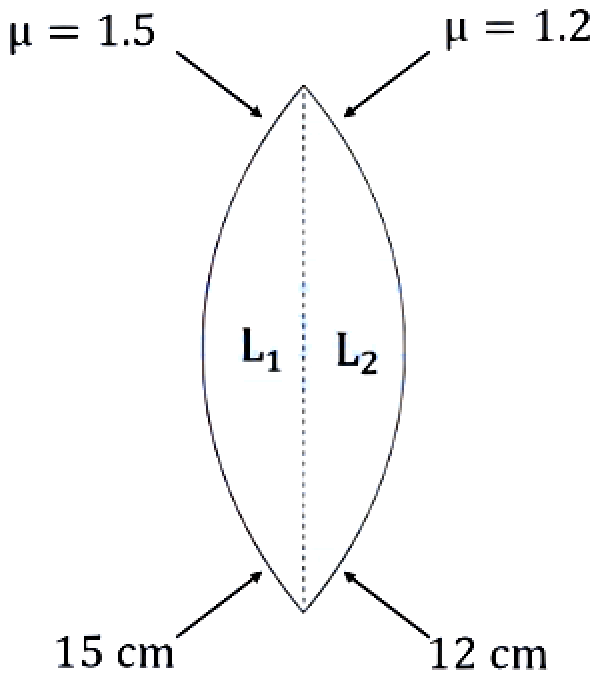
+2

Answer: A

Solution:

We use the Lens Maker's Formula for each planoconvex lens :

$\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$



For left Plano-convex lens $\left(\mathrm{L}_1\right)$:

The refractive index of the lens is $\mu_1=1.5$

Radii of curvatures are $\mathrm{R}_1=15 \mathrm{~cm}$ (curved surface), $\mathrm{R}_2=\infty$ (plane surface)

$\frac{1}{f_1}=(1.5-1)\left(\frac{1}{15}-\frac{1}{\infty}\right)=0.5 \times \frac{1}{15}=\frac{1}{30}$

$\Rightarrow \mathrm{f}_1=+30 \mathrm{~cm}$

For right Plano-convex lens $\left(\mathrm{L}_2\right)$:

The refractive index of lens $\mu_2=1.2$

Radii of curvatures are $\mathrm{R}_1=\infty$, (Plane surface) and $\mathrm{R}_2=-12 \mathrm{~cm}$ ((curved surface)

$\frac{1}{f_2}=(1.2-1)\left(\frac{1}{\infty}-\frac{1}{-12}\right)=0.2 \times \frac{1}{12}=\frac{1}{60}$

$\Rightarrow \mathrm{f}_1=+60 \mathrm{~cm}$

Since the lenses are thin and in contact, the total power is the sum of individual powers :

$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

$\Rightarrow \frac{1}{F} = \frac{1}{30} + \frac{1}{60} = \frac{2+1}{60} = \frac{3}{60} = \frac{1}{20}$

$\Rightarrow \mathrm{F} = +20\,\mathrm{cm}$

Given object distance $\mathrm{u} = -30\,\mathrm{cm}$

Using the Lens Formula: $\frac{1}{\mathrm{v}} - \frac{1}{\mathrm{u}} = \frac{1}{\mathrm{F}}$

$\frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$

$\Rightarrow \frac{1}{v} + \frac{1}{30} = \frac{1}{20}$

$\Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{3-2}{60} = \frac{1}{60}$

$\mathrm{v} = +60\,\mathrm{cm}$

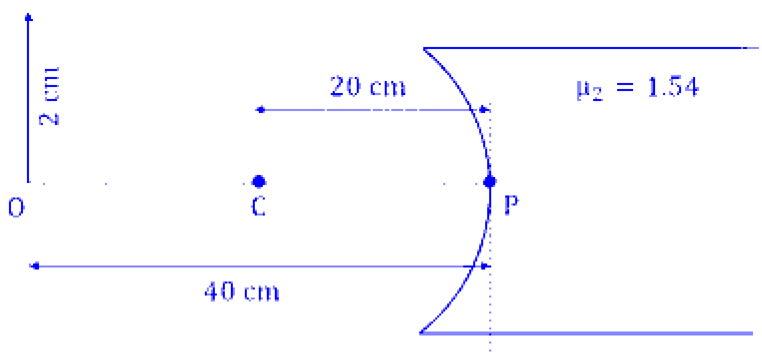
Magnification for a lens is given by : $\mathrm{m} = \mathrm{v}/\mathrm{u}$

$m = \frac{60}{-30} = -2$

The negative sign indicates that the image is real and inverted, and its size is twice the size of the object. Hence, the correct option is (A).

Question23

Refer the figure given below. μ_1 and μ_2 are refractive indices of air and lens material. The height of image will be ________ cm.



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Options:

A.

1

B.

0.5

C.

1.2

D.

0.25

Answer: A

Solution:

Using the Cartesian sign convention:

Refractive index of first medium (air), $\mu_1 = 1$

Refractive index of second medium, $\mu_2 = 1.54$

Object distance, $u = -40$ cm

Radius of curvature, $R = -20$ cm

Height of object, $h_o = 2$ cm

Using the spherical surface refraction formula

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

Substituting the values:

$\frac{1.54}{v} - \frac{1}{-40} = \frac{1.54-1}{-20}$

$\frac{1.54}{v} + \frac{1}{40} = \frac{0.54}{-20}$

$\frac{1.54}{v} = \frac{-1-3.08}{40} = \frac{-4.08}{40}$

$v = \frac{1.54 \times 40}{-4.08}$ cm ≈ 29.6 cm

For refraction at a single spherical surface, magnification is given by:

$m = \frac{h_i}{h_o} = \frac{\mu_1 v}{\mu_2 u}$

$m = \frac{1 \times (-29.61)}{1.54 \times (-40)} = \frac{29.61}{61.6} \approx 0.48$

$h_i = m \times h_o = 0.48 \times 2 \approx 0.96$ cm

Rounding to the nearest significant value, the height is approximately 1 cm.

Therefore, the correct option is (A).

Question24

For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle and minimum deviation will be _______.

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Options:

A.

3 : 4

B.

3 : 2

C.

2 : 1

D.

1 : 2

Answer: B

Solution:

For a thin prism with a small prism angle (A), the angle of minimum deviation is given by,

$\delta_m = (\mu - 1)A$

Where μ is the refractive index of the glass, $\mu = 1.5$:

Putting the value of refractive index, $\delta_m = (1.5 - 1)A = 0.5A = \frac{A}{2}$

At the position of minimum deviation, the angle of incidence (i) is related to the prism angle (A) and the minimum deviation (δ_m) by

$i = \frac{A+\delta_m}{2}$

Substituting the value of $\delta_m = \frac{A}{2}$

$i = \frac{A+\frac{A}{2}}{2} = \frac{3A/2}{2} = \frac{3A}{4}$

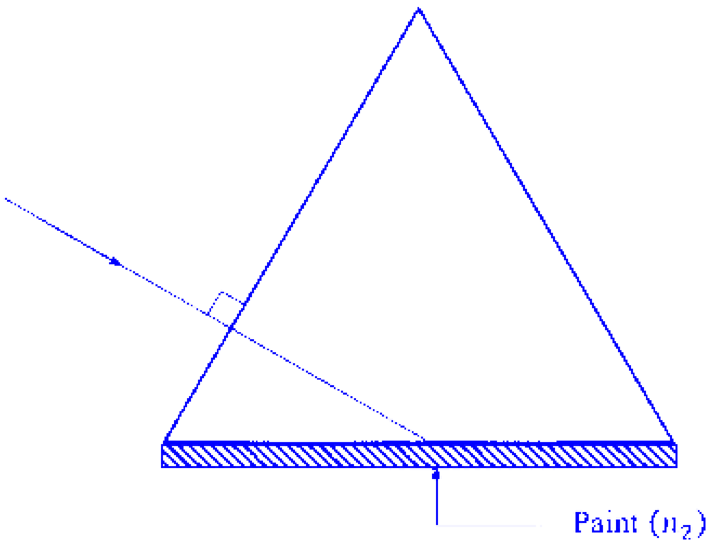
So, the ratio of the incident angle (i) to the minimum deviation (δ_m) is,

$\frac{i}{\delta_m} = \frac{3A/4}{A/2} = \frac{6}{4} = \frac{3}{2}$

Therefore, the ratio of the incident angle and minimum deviation is 3 : 2. Thus, the correct option is (B).

Question25

One side of an equilateral prism is painted by a transparent material of refractive index n_2. The refractive index of prism is 1.6. The minimum value of n_2 required for total internal reflection from painted face is ________.



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Options:

A.

$\frac{3\sqrt{3}}{1.6}$

B.

$\sqrt{3}$

C.

$$\frac{3.2}{\sqrt{3}}$$

D.

$$\frac{4\sqrt{3}}{5}$$

Answer: D

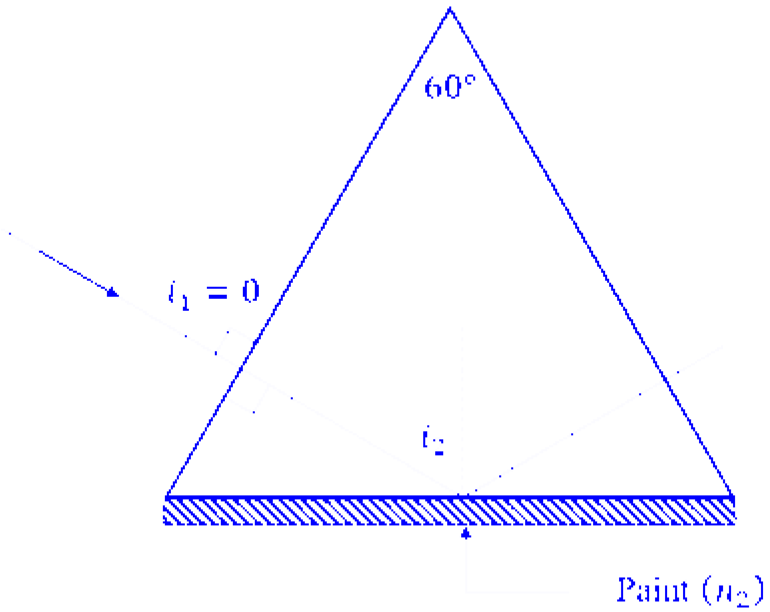
Solution:

An equilateral prism has interior angles of $\mathrm{A} = 60^{\circ}$.

The light ray enters the first face at a normal incidence (perpendicularly).

Angle of incidence at the first face $\mathrm{i}_1 = 0^{\circ}$.

By Snell's Law $\mathrm{n}_1 \sin \mathrm{i}_1 = \mathrm{n}_2 \sin \mathrm{r}_1$, the angle of refraction $\mathrm{r}_1 = 0^{\circ}$.



In a prism, the relation between the apex angle A and the internal angles is $A = r_1 + r_2$.

$60^{\circ} = 0^{\circ} + \mathrm{i}_2 \Rightarrow \mathrm{i}_2 = 60^{\circ}$

The light ray strikes the painted face at an angle of 60° relative to the normal.

For TIR to occur at the interface between the prism ($\mathrm{n}_{\text{prism}} = 1.6$) and the paint (n_2), the angle of incidence i_2 must be greater than or equal to the critical angle θ_{c}.

$\sin i_2 \geq \sin \theta_c$

The critical angle is defined by $\sin \theta_{\mathrm{c}} = \frac{\mathrm{n}_{\text{rarer}}}{\mathrm{n}_{\text{denser}}} = \frac{\mathrm{n}_2}{\mathrm{n}_{\text{prism}}}$

$\sin 60^{\circ} \geq \frac{n_2}{1.6}$

Substituting the value of $\sin 60^{\circ} = \frac{\sqrt{3}}{2}$:

$\frac{\sqrt{3}}{2} \geq \frac{n_2}{1.6}$

$\Rightarrow n_2 \leq \frac{1.6 \times \sqrt{3}}{2}$

$\Rightarrow n_2 \leq 0.8\sqrt{3}$

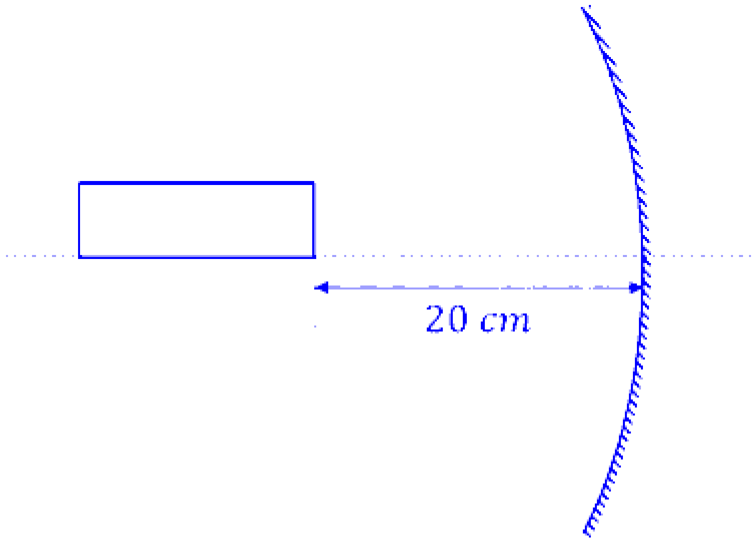
$\Rightarrow n_2 \leq \frac{4\sqrt{3}}{5}$

The maximum value n_2 can take to ensure TIR is $\frac{4\sqrt{3}}{5}$.

Hence, the correct option is (D).

Question26

A rod of length 10 cm lies along the principle axis of a concave mirror of focal length 10 cm as shown in figure. The length of the image is _____ cm.



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Options:

A.

2.5

B.

5

C.

7.5

D.

7

Answer: B

Solution:

Using mirror formula,

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

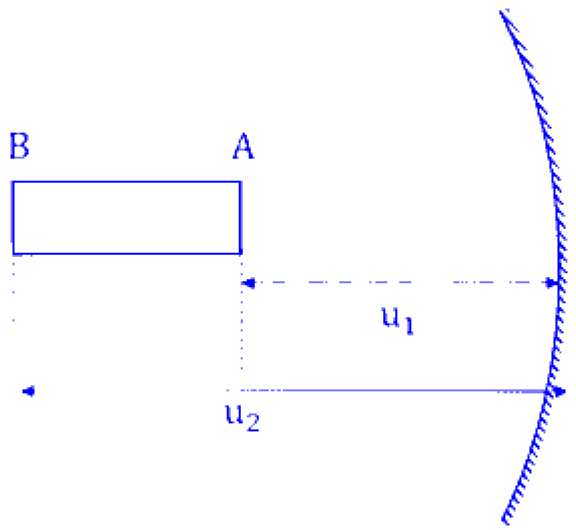
Where, the focal length is f , object distance is u , and image distance is v .

According to the sign convention for a concave mirror:

Focal length $f = -10$ cm

Distances measured in the direction of incident light are positive; opposite is negative.

The end of the rod closer to the mirror is at a distance of 20 cm from the pole.



$u_1 = -20$ cm

Using mirror formula for point A,

$\frac{1}{-10} = \frac{1}{v_1} + \frac{1}{-20}$

$\Rightarrow \frac{1}{v_1} = \frac{1}{20} - \frac{1}{10} = \frac{1-2}{20} = -\frac{1}{20}$

$\Rightarrow v_1 = -20$ cm

Since $u = 2f = R$ (the center of curvature), the image is formed at the same location ($v = -20$ cm).

The rod has a length of 10 cm and extends away from the mirror. Therefore, the far end (B) is at :

$u_2 = -(20 + 10) = -30 \text{ cm}$

Using mirror formula for point B ,

$\frac{1}{-10} = \frac{1}{v_2} + \frac{1}{-30}$

$\Rightarrow \frac{1}{v_2} = \frac{1}{30} - \frac{1}{10} = \frac{1-3}{30} = -\frac{2}{30} = -\frac{1}{15}$

$\Rightarrow v_2 = -15 \text{ cm}$

The length of the image is the magnitude of the difference between the two image positions:

$\text{Length} = |v_1 - v_2|$

$\text{Length} = |-20 - (-15)| = |-5| = 5 \text{ cm}$

Therefore, the length of image of the rod is 5 cm .

Thus, the correct option is (B).

Question27

A convex lens is made from glass material having refractive index of 1.4 with same radius of curvature on both sides. The ratio of its focal length and radius of curvature is _____ .

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Options:

A.

0.5

B.

2.5

C.

0.8

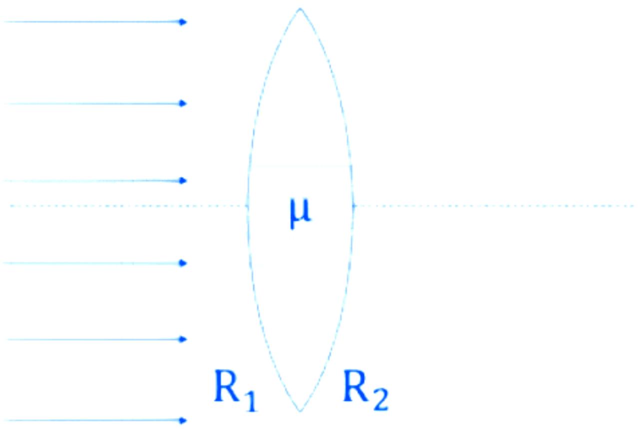
D.

1.25

Answer: D

Solution:

For a thin lens with two curved surfaces, the focal length (f) is related to the refractive index (n) and the radii of curvature (R_1 and R_2) by Lens Makers' formula,



$\frac{1}{f} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

The refractive index of the material, $n = 1.4$.

The lens is convex with the same radius of curvature on both sides (R).

According to the Cartesian sign convention:

The first surface is convex towards the incident light, so $R_1 = +R$.

The second surface is concave towards the incident light, so $R_2 = -R$.

Substituting the values into the formula :

$\frac{1}{f} = (1.4 - 1)\left(\frac{1}{R} - \frac{1}{-R}\right)$

$\Rightarrow \frac{1}{f} = (0.4)\left(\frac{1}{R} + \frac{1}{R}\right)$

$\Rightarrow \frac{1}{f} = 0.4 \times \frac{2}{R}$

$\Rightarrow \frac{1}{f} = \frac{0.8}{R}$

$\Rightarrow f = \frac{R}{0.8}$

$\Rightarrow \frac{f}{R} = \frac{1}{0.8} = 1.25$

Therefore, the ratio of its focal length and radius of curvature is 1.25 .

Thus, the correct option is (D).

Question28

A compound microscope is designed with two symmetric biconvex lenses. The objective lens is cut vertically, creating two identical plano-convex lenses. One of them is used in place of original objective lens. To retain same magnification keeping the object distance unchanged, the tube length has to be

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Options:

A.

increased two times

B.

increased $\frac{3}{2}$ times

C.

decreased two times

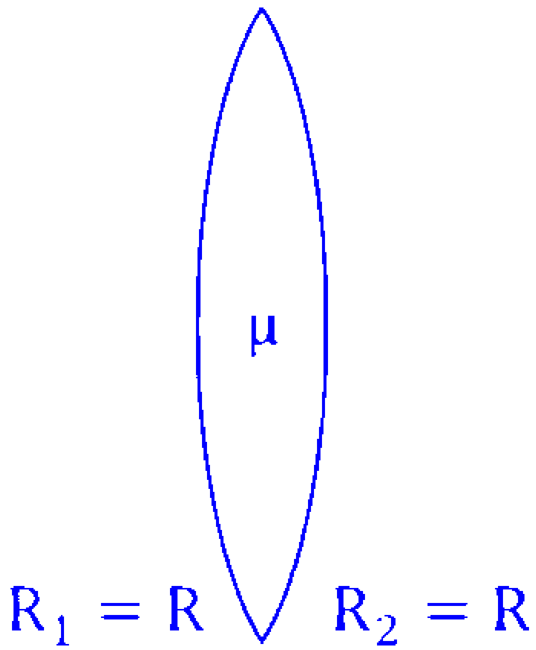
D.

decreased $\frac{3}{2}$ times

Answer: A

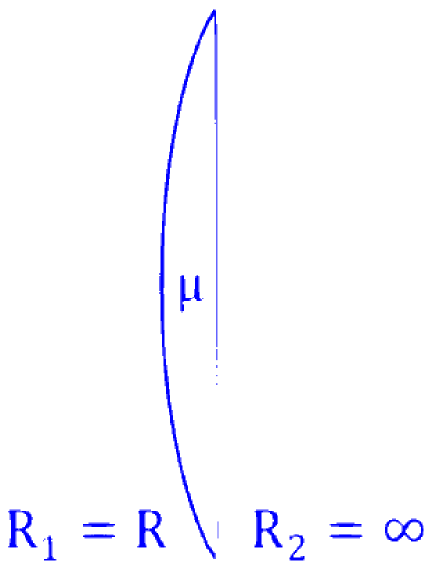
Solution:

According to the lens maker's formula, the focal length (f) of a symmetric biconvex lens with radius of curvature R and refractive index μ is :



$\frac{1}{f}=(\mu-1)\left(\frac{1}{R}-\frac{1}{-R}\right)=\frac{2(\mu-1)}{R}$

When the lens is cut vertically, the new lens is plano-convex with one flat surface ($R_2=\infty$). Its focal length (f') is :



$\frac{1}{f'}=(\mu-1)\left(\frac{1}{R}-\frac{1}{\infty}\right)=\frac{\mu-1}{R}$

Comparing the two equations, $\frac{1}{f'}=\frac{1}{2f} \Rightarrow f'=2f$, i.e., the focal length of the objective lens doubles.

The total magnification (M) for a compound microscope is given by:

$$M = m_o \times m_e \approx \frac{L}{f_o} \times \frac{D}{f_e}$$

Where L is the tube length, f_o is the focal length of the objective, and f_e is the focal length of the eyepiece.

It is given that magnification (M) and the eyepiece (f_e) remain constant. To keep M the same while f_o becomes $2f_o$:

$$M = \frac{L}{f_o} = \frac{L'}{2f_o}$$

$\Rightarrow L' = 2\,L$

Therefore, the tube length has to be increased two times to keep the magnification of compound microscope unchanged.

Hence, the correct option is (A).

Question29

A ray of light passing through an equilateral prism is having velocity $2.12 \times 10^8\ \mathrm{m/s}$ in the prism material, then the minimum angle of deviation is

_____ degrees.

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Options:

A.

45

B.

30

C.

28

D.

58

Answer: B

Solution:

The refractive index is the ratio of the speed of light in a vacuum (c) to the speed of light in the medium (v).

$\mu=\frac{c}{v}$

The speed of light in vacuum is $\mathrm{c}=3\times10^{8}\ \mathrm{m/s}$ and the speed of light in material of prism is given as $\mathrm{v}=2.12\times10^{8}\ \mathrm{m/s}$.

So, the refractive index of material of prism is;

$\mu=\frac{3\times10^{8}}{2.12\times10^{8}}\approx1.414\approx\sqrt{2}$

As the prism is equilateral, so the angle of prism is $\mathrm{A}=60^{\circ}$.

The the angle of minimum deviation through prism is given as,

$\mu=\frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

Substituting the refractive index $\mu=\sqrt{2}$ and prism angle $\mathrm{A}=60^{\circ}$:

$\sqrt{2}=\frac{\sin\left(\frac{60+\delta_{\mathrm{m}}}{2}\right)}{\sin\left(\frac{60}{2}\right)}$

$\Rightarrow\sqrt{2}=\frac{\sin\left(\frac{60+\delta m}{2}\right)}{\sin\left(30^{\circ}\right)}$

$\Rightarrow\sqrt{2}\times0.5=\sin\left(\frac{60+\delta_{m}}{2}\right)$

$\Rightarrow\frac{\sqrt{2}}{2}=\sin\left(\frac{60+\delta_{m}}{2}\right)\Rightarrow\frac{1}{\sqrt{2}}=\sin\left(\frac{60+\delta_{m}}{2}\right)=\sin45^{\circ}$

$\Rightarrow45^{\circ}=\frac{60+\delta_{\mathrm{m}}}{2}$

$\Rightarrow90^{\circ}=60^{\circ}+\delta_{\mathrm{m}}\Rightarrow\delta_{\mathrm{m}}=30^{\circ}$

Therefore, the correct option is (B).

Question30

An object AB is placed 15 cm on the left of a convex lens P of focal length 10 cm . Another convex lens Q is now placed 15 cm right of lens P. If the focal length of lens Q is 15 cm , the final image is _____ .

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Options:

A.

virtual, formed at 7.5 cm right of lens Q, with a size bigger than that of AB real, formed at 7.5 cm right of lens

B.

Q, with a size same as that of AB

C.

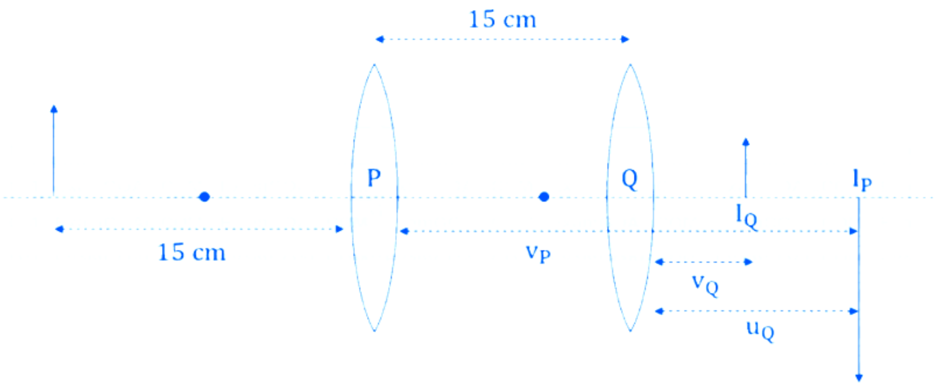
formed at infinity.

D.

real, formed at 7 cm right of lens Q, with a size smaller than that of AB

Answer: B

Solution:



For Lens P:

Focal length is $f_P = +10$ cm

Object distance is $u_P = -15$ cm

Using the lens formula.

$\frac{1}{v_P} - \frac{1}{u_P} = \frac{1}{f_P}$

$\Rightarrow \frac{1}{v_P} - \frac{1}{-15} = \frac{1}{10}$

$\Rightarrow \frac{1}{v_P} = \frac{1}{10} - \frac{1}{15} = \frac{3-2}{30} = \frac{1}{30}$

$v_P = +30$ cm

The image formed by lens P is 30 cm to the right of lens P .

So, the magnification of image formed by lens P is,

$m_P = \frac{v_P}{u_P} = \frac{30}{-15} = -2$

Lens Q is placed 15 cm to the right of lens P and the image from lens P acts as the object for lens Q .

Since the image from P is at 30 cm and lens Q is at 15 cm , the object for lens Q is $30 - 15 = 15$ cm to the right of lens Q.

The image formed by P acts as virtual object for the lens Q .

For lens Q :

Object distance is $u_Q = +15$ cm

Focal length is $f_Q = +15$ cm

Using the lens formula,

$\frac{1}{v_Q} - \frac{1}{u_Q} = \frac{1}{f_Q}$

$\Rightarrow \frac{1}{v_Q} - \frac{1}{15} = \frac{1}{15}$

$\Rightarrow \frac{1}{v_Q} = \frac{1}{15} + \frac{1}{15} = \frac{2}{15}$

$\Rightarrow v_Q = +7.5$ cm

If the object is real and the image is formed on opposite side of the convex lens, then image so formed is real but if the object is virtual and image is formed on the same side of the convex lens, then the image so formed is real.

The final image formed by lens Q is real and formed 7.5 cm to the right of lens Q .

So, the magnification of image formed by lens P is,

$m_Q = \frac{v_Q}{u_Q} = \frac{7.5}{15} = 0.5$

Total magnification $M = m_P \times m_Q$

$M = (-2) \times (0.5) = -1$

Since $|M| = 1$, the size of the final image is the same as that of object AB .

Therefore, the final image is real, formed at 7.5 cm right of lens Q , with a size same as that of AB .

Thus, the correct option is (B).

Question31

A thin convex lens and a thin concave lens are kept in contact and are co-axial. Which of the following statements is correct for this

combination of two lenses ?

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Options:

A.

behaves as concave lens if $|f_{\text{convex}}| > |f_{\text{concave}}|$

B.

behaves as concave lens if $|f_{\text{convex}}| < |f_{\text{concave}}|$

C.

behaves as convex lens if $|f_{\text{convex}}| > |f_{\text{concave}}|$

D.

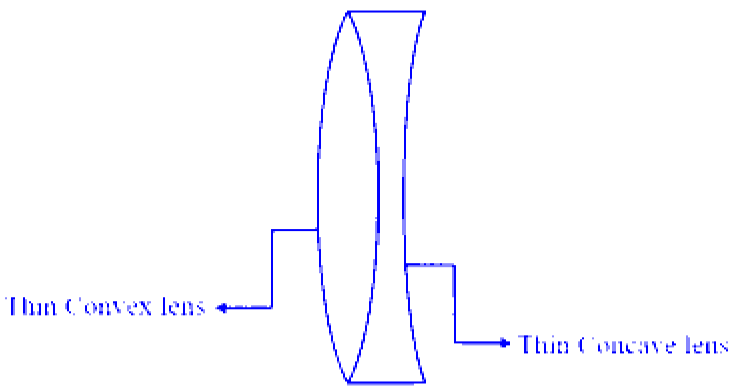
Focal length of the lens system will change if the positions of two lenses are interchanged

Answer: A

Solution:

For two thin lenses in contact, the total power P is the sum of their individual powers :

$\mathrm{P} = \mathrm{P}_1 + \mathrm{P}_2$



The power of lens is given as $P = \frac{1}{f}$, where f is the focal length of lens in meters.

$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

For a convex lens, the focal length is positive $\Rightarrow \mathrm{f}_1 = +|\mathrm{f}_{\text{convex}}|$.

For a concave lens, the focal length is negative $\Rightarrow f_2 = -|f_{\text{concave}}|$.

Substituting these into the formula of equivalent power:

$$\frac{1}{F} = \frac{1}{|f_{\text{convex}}|} - \frac{1}{|f_{\text{concave}}|}$$

$$\Rightarrow \frac{1}{F} = \frac{|f_{\text{concave}}| - |f_{\text{convex}}|}{|f_{\text{convex}}| \cdot |f_{\text{concave}}|}$$

The combination behaves as a concave lens if the equivalent focal length F is negative ($F < 0$).

As denominator of the expression is always positive in this case, so for the focal length (F) to be true, the numerator must be negative.

$$|f_{\text{concave}}| - |f_{\text{convex}}| < 0$$
$$|f_{\text{convex}}| > |f_{\text{concave}}|$$

Conversely, it behaves as a convex lens if F is positive, which happens when

$|f_{\text{concave}}| > |f_{\text{convex}}|$.

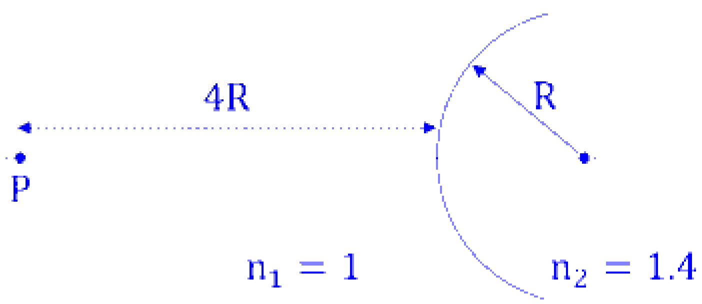
For thin lenses in contact, the formula $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$ is independent of the order of the lenses. Therefore, interchanging them does not change the focal length.

Therefore, the system behaves as a concave lens if $|f_{\text{convex}}| > |f_{\text{concave}}|$.

Thus, the correct option is (A).

Question32

A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A point source is placed at a distance of $4R$ in front of spherical interface. The magnitude of the magnification of point source image is _____ .



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Options:

A.

1.66

B.

2.33

C.

2.66

D.

1.33

Answer: A

Solution:

When light travels from one medium to another through a curved surface, the positions of the object (u) and the image (v) are related by the formula:

$\frac{\mathrm{n}_2}{\mathrm{v}}-\frac{\mathrm{n}_1}{\mathrm{u}}=\frac{\mathrm{n}_2-\mathrm{n}_1}{\mathrm{R}}$

Where n_1 is the refractive index of the medium where the object is placed, n_2 is the refractive index of the medium where light enters, R is the radius of curvature.

The refractive index of medium in which object is $\mathrm{n}_1=1$, and that for the image $\mathrm{n}_2=1.4$.

The object distance is $u=-4R$ (as per sign convention), and radius is $+R$ (convex surface).

$\frac{1.4}{v}-\frac{1}{-4R}=\frac{1.4-1}{R}$

$\Rightarrow \frac{1.4}{v}+\frac{1}{4R}=\frac{0.4}{R}$

$\Rightarrow \frac{1.4}{v}=\frac{0.4}{R}-\frac{0.25}{R}=\frac{0.15}{R}$

$\Rightarrow \mathrm{v}=\frac{1.4\mathrm{R}}{0.15}=\frac{140\mathrm{R}}{15}=\frac{28}{3}\mathrm{R}$

For a spherical refracting interface, the magnification is given by :

$\mathrm{m}=\frac{\mathrm{n}_1\mathrm{v}}{\mathrm{n}_2\mathrm{u}}$

$\Rightarrow \mathrm{m}=\frac{1\times\left(\frac{28}{3}\mathrm{R}\right)}{1.4\times(-4\mathrm{R})}$

$\Rightarrow \mathrm{m}=\frac{28\mathrm{R}/3}{-5.6\mathrm{R}}=\frac{28}{3\times(-5.6)}=\frac{28}{-16.8}$

$\Rightarrow |\mathrm{m}|=\frac{28}{16.8}=1.666\ldots$

Therefore, the magnitude of the magnification of point source image is 1.66 .

Thus, the correct option is (A).

Question33

Angle of minimum deviation is equal to the half of the angle of prism in an equilateral prism. The refractive index of the prism is ____

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Options:

A.

1.5

B.

$\sqrt{3}$

C.

$\sqrt{2}$

D.

1.65

Answer: C

Solution:

For a prism with angle A and refractive index μ, the relationship between the angle of minimum deviation (δ_{m}) and refractive index is given by:

$$\mu=\frac{\sin \left(\frac{A+\delta_m}{2}\right)}{\sin \left(\frac{A}{2}\right)}$$

For equilateral prism the angle of the prism is $\mathrm{A}=60^{\circ}$.

The angle of minimum deviation is half the angle of prism $\left(\delta_m=\frac{A}{2}\right)$.

Therefore, $\delta_m=\frac{60^{\circ}}{2}=30^{\circ}$

Substituting $\mathrm{A}=60^{\circ}$ and $\delta_{\mathrm{m}}=30^{\circ}$ into the formula:

$$\mu=\frac{\sin \left(\frac{60^{\circ}+30^{\circ}}{2}\right)}{\sin \left(\frac{60^{\circ}}{2}\right)}$$

$\Rightarrow \mu = \frac{\sin(45^\circ)}{\sin(30^\circ)}$

$\Rightarrow \mu = \frac{1/\sqrt{2}}{1/2} = \frac{2}{\sqrt{2}}$

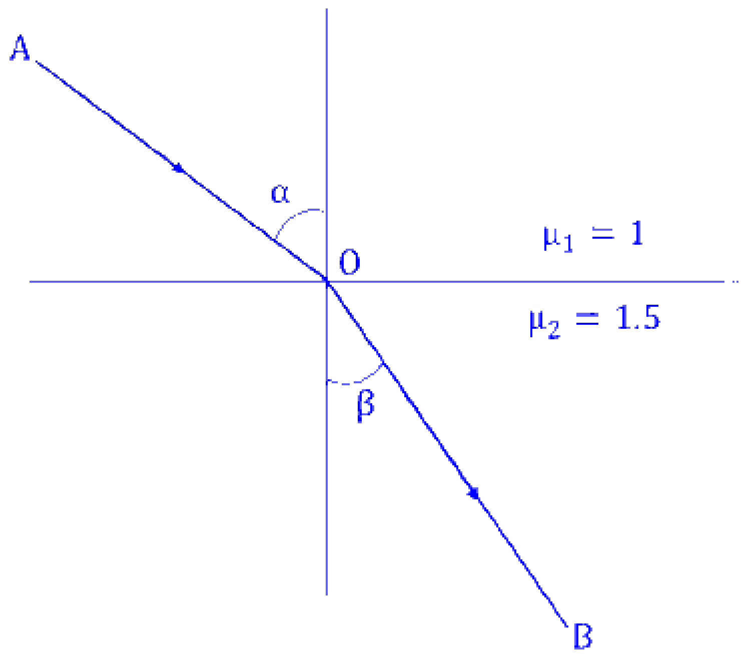
$\Rightarrow \mu = \sqrt{2}$

Therefore, the refractive index of the prism is $\sqrt{2}$.

Thus the correct option is (C).

Question34

Light ray incident along a vector $\overrightarrow{AO}(\overrightarrow{AO} = 2\hat{i} - 3\hat{j})$ emerges out along vector $\overrightarrow{OB}(\overrightarrow{OB} = C\hat{i} - 4\hat{j})$ as shown in the figure below. The value of C is _____ .



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Options:

A.

1.6

B.

0.16

C.

11.6

D.

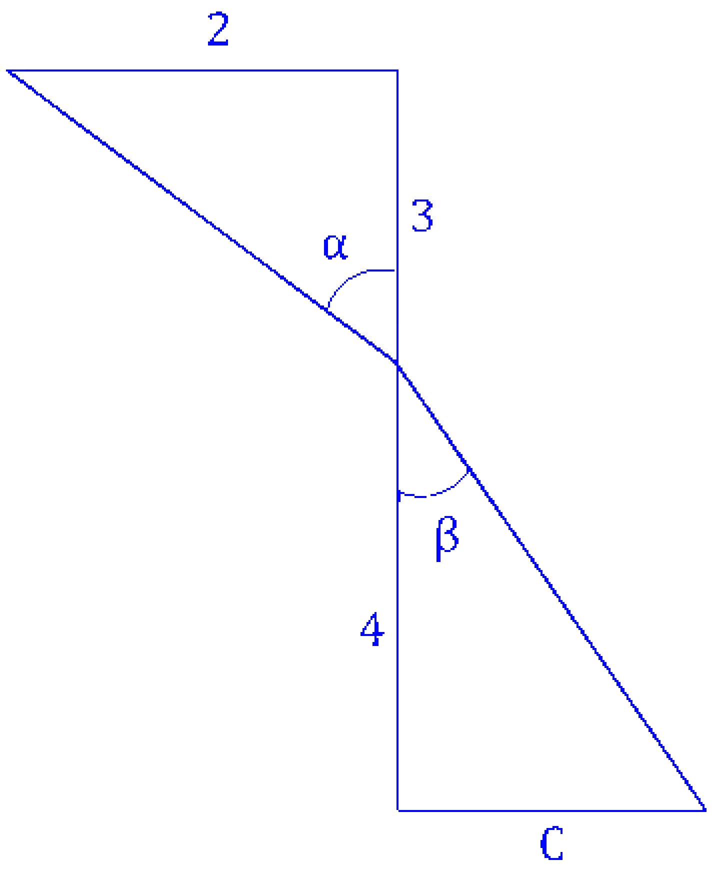
16

Answer: A

Solution:

A light ray travels from a medium with refractive index $\mu_1 = 1$ (air) to a medium with $\mu_2 = 1.5$. The incident vector is $\overrightarrow{\mathrm{AO}} = 2\hat{\imath} - 3\hat{\jmath}$ and the refracted vector is $\overrightarrow{\mathrm{OB}} = C\hat{\imath} - 4\hat{\jmath}$.

To use Snell's Law, we need the angles of incidence and refraction relative to the normal.



The incident ray is along the vector $\overrightarrow{\mathrm{AO}}$ points towards the origin.

$\tan\theta_1 = \frac{\text{Horizontal component}}{\text{Vertical component}} = \frac{2}{3}$

So, using the trigonometric identity,

$\sin\theta_1 = \frac{2}{\sqrt{2^2+3^2}} = \frac{2}{\sqrt{13}}$

The refracted ray is along the vector $\overrightarrow{\mathrm{OB}}$ points away from the origin. The angle with the vertical is:

$\tan\theta_2 = \frac{C}{4}$

From this, $\sin\theta_2$ is:

$\sin\theta_2 = \frac{C}{\sqrt{C^2+4^2}} = \frac{C}{\sqrt{C^2+16}}$

Applying Snell's Law

$\mu_1 \sin\theta_1 = \mu_2 \sin\theta_2$

Substituting the values :

$1 \times \left(\frac{2}{\sqrt{13}}\right) = 1.5 \times \left(\frac{C}{\sqrt{C^2+16}}\right)$

$\Rightarrow \frac{2}{\sqrt{13}} = \frac{3}{2} \times \frac{C}{\sqrt{C^2+16}}$

$\Rightarrow 4\sqrt{C^2+16} = 3C\sqrt{13}$

Squaring both sides,

$16\left(C^2+16\right) = 9C^2(13)$

$\Rightarrow 16C^2 + 256 = 117C^2$

$\Rightarrow 256 = 117C^2 - 16C^2$

$\Rightarrow 256 = 101C^2$

$\Rightarrow C^2 = \frac{256}{101} \approx 2.53$

$\Rightarrow C = \sqrt{2.53} \approx 1.6$

Therefore, the value of C is 1.6 .

Hence, the correct option is (A).

Question35

A thin biconvex lens is prepared from the glass $(\mu = 1.5)$ both curved surfaces of which have equal radii of 20 cm each. Left side surface of the lens is silvered from outside to make it reflecting. To have the position of image and object at the same place, the object should be placed, from the lens at a distance of ____ cm.

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Options:

A.

10

B.

12.5

C.

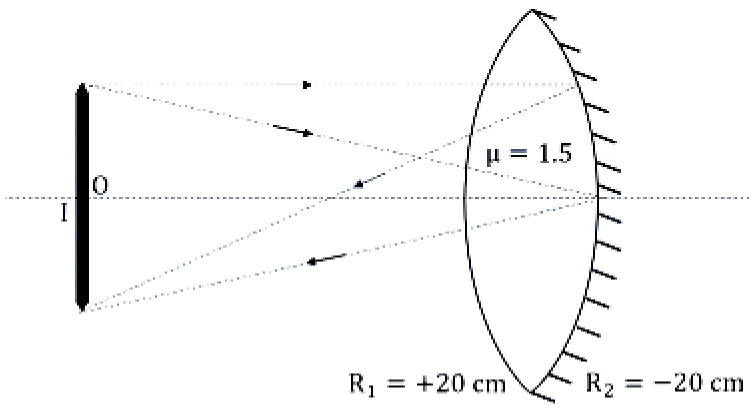
13

D.

13.5

Answer: A

Solution:



When one surface of a lens is silvered, it behaves like a mirror. The effective system consists of a lens, followed by a mirror, and then the lens again (as light passes through the lens, reflects at the silvered surface, and passes back through the lens). The effective power (P_{eq}) of this silvered lens is given by:

$\mathrm{P}_{\mathrm{eq}}=2\mathrm{P}_{\mathrm{l}}+\mathrm{P}_{\mathrm{m}}$

Where, $\mathrm{P}_{\mathrm{l}}=$ Power of the lens and $\mathrm{f}_{\mathrm{m}}=$ Focal length of the silvered surface (mirror).

The power of lens is given as $P=\frac{1}{f}$, where focal length (f) should be in meters.

The power of mirror is given as $P=-\frac{1}{f}$, where focal length (f) should be in meters.

Using the Lens Maker's Formula:

$\frac{1}{\mathrm{f}_{\mathrm{l}}}=(\mu-1)\left(\frac{1}{\mathrm{R}_1}-\frac{1}{\mathrm{R}_2}\right)$

For a biconvex lens, the refractive index of material of lens is $\mu=1.5$, the radius of curvature for the first surface is $\mathrm{R}_1=+20\ \mathrm{cm}$ (convex surface) and $\mathrm{R}_2=-20\ \mathrm{cm}$ is for the second surface (opposite convex surface)

$$\frac{1}{f_1}=(1.5-1)\left(\frac{1}{20}-\left(\frac{1}{-20}\right)\right)$$

$$\Rightarrow \frac{1}{\mathrm{f}_1}=0.5\left(\frac{1}{20}+\frac{1}{20}\right)=0.5\times\frac{2}{20}=\frac{1}{20}$$

So, the focal length of lens is $\mathrm{f}_{\mathrm{l}}=20\,\mathrm{cm}$.

The power of lens is, $\mathrm{P}_{\mathrm{l}}=\frac{1}{(20/100)}=5\mathrm{D}$.

Since the silvered surface is the back surface of a biconvex lens, it acts as a concave mirror for the light inside the lens.

So, the radius of curvature is $\mathrm{R}=-20\,\mathrm{cm}$.

$$\mathrm{f}_{\mathrm{m}}=\frac{\mathrm{R}}{2}=-\frac{20}{2}=-10\,\mathrm{cm}$$

The optical power of mirror is given as $\mathrm{P}_{\mathrm{m}}=-\frac{1}{\mathrm{f}_{\mathrm{m}}}$, where focal length should be in meters.

The power of mirror is, $P_m=-\frac{1}{(-10/100)}=10\mathrm{D}$.

Using the effective power formula,

$$\mathrm{P}_{\mathrm{eq}}=(2\times 5+10)\mathrm{D}=20\mathrm{D}$$

As the effective focal length of the system is +20 D , and the whole system behaves like a mirror,

$$f_{\mathrm{eq}}=-\frac{1}{P_{\mathrm{eq}}}\Rightarrow f_{\mathrm{eq}}=-\frac{1}{20}\,\mathrm{m}=-\frac{100}{5}\,\mathrm{cm}=-5\,\mathrm{cm}$$

When the object is placed at the centre of curvature (C), the position of image and object are at the same place. So, the object distance is equal to the radius of curvature.

$$\mathrm{f}_{\mathrm{eq}}=\frac{\mathrm{R}_{\mathrm{eq}}}{2}\Rightarrow \mathrm{R}_{\mathrm{eq}}=2\times\mathrm{f}_{\mathrm{eq}}=2\times(-5)\mathrm{cm}$$

$$\Rightarrow \mathrm{R}_{\mathrm{eq}}=-10\,\mathrm{cm}$$

Therefore, to have the image and object at the same place, the object should be placed at 10 cm .

Hence, the correct option is (A).

Question36

The driver sitting inside a parked car is watching vehicles approaching from behind with the help of his side view mirror, which is a convex mirror with radius of curvature $\mathrm{R}=2\ \mathrm{m}$. Another car approaches him from behind with a uniform speed of $90\ \mathrm{km/hr}$. When the car is at a distance of 24 m from him, the magnitude of the

acceleration of the image of the car in the side view mirror is ' a '. The value of $100a$ is __________ $\mathrm{m/s^2}$.

JEE Main 2025 (Online) 22nd January Morning Shift

Answer: 8

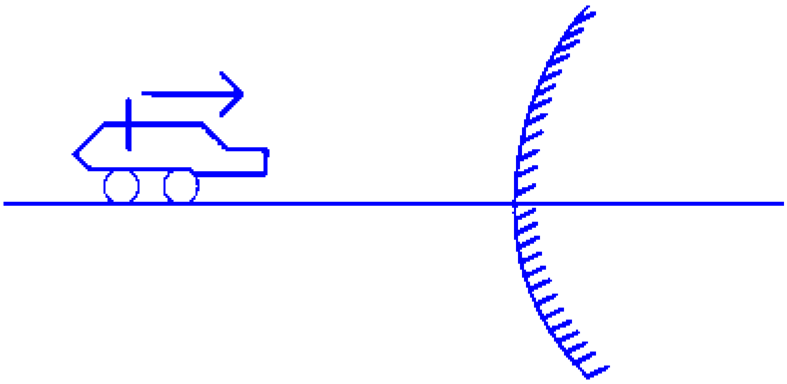
Solution:

Given, R = 2m, So, f = 1m (as $f = \frac{R}{2}$)

$V_0 = 90\,km/hr = 90 \times \frac{1000}{3600} = 25\,m/s$

v_0 is uniform, so $a_0 = 0$

$u = -24\,m$



Using mirror formula,

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \frac{1}{v} + \frac{1}{-24} = \frac{1}{1} \Rightarrow \frac{1}{v} = 1 + \frac{1}{24} = \frac{25}{24}$

$\Rightarrow v = \frac{24}{25}m$

$m = \frac{-v}{u}$

by differentiating mirror formula,

$-\frac{1}{v^2}\frac{dv}{dt} - \frac{1}{u^2}\frac{du}{dt} = 0$

$\Rightarrow -\frac{1}{v^2}v_I - \frac{1}{u^2}v_0 = 0$ ...... (1)

$\Rightarrow v_I = \frac{-v^2}{u^2}v_0 = -\frac{24^2}{25^2}$

$\Rightarrow v_I = \frac{-1}{25}$ m/s

by differentiating (1) w.r.t. time,

$\frac{-1}{v^2}a_i + \frac{2}{v^3}v_I^2 - \frac{1}{u^2}a_0 + \frac{2}{u^3}v_0^2 = 0$

$a_I = \frac{2}{v}v_I^2 + \frac{2v^2}{u^3}v_0^2$

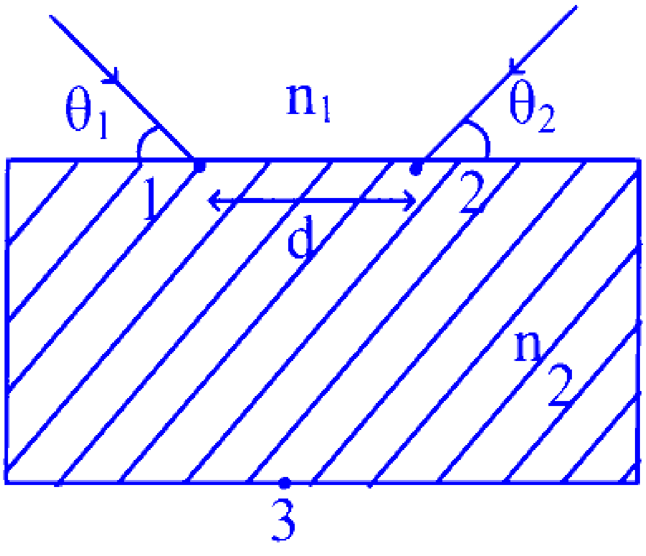
$= \frac{2}{24}\frac{1}{25^2} + \frac{2}{-24^3}\frac{24^2}{25^2}25^2$

$= \frac{1}{12}\left[\frac{1}{25} - 1\right] = \frac{1}{12} \times \frac{-24}{25} = -\frac{2}{25}$

$\Rightarrow 100\, a_I = \frac{-2}{25} \times 100 = 8\, m/s^2$

Question37

Two light beams fall on a transparent material block at point 1 and 2 with angle θ_1 and θ_2, respectively, as shown in figure. After refraction, the beams intersect at point 3 which is exactly on the interface at other end of the block. Given : the distance between 1 and 2, d $= 4\sqrt{3}$ cm and $\theta_1 = \theta_2 = \cos^{-1}\left(\frac{n_2}{2n_1}\right)$. where refractive index of the block $n_2 >$ refractive index of the outside medium n_1, then the thickness of the block is ______ cm .



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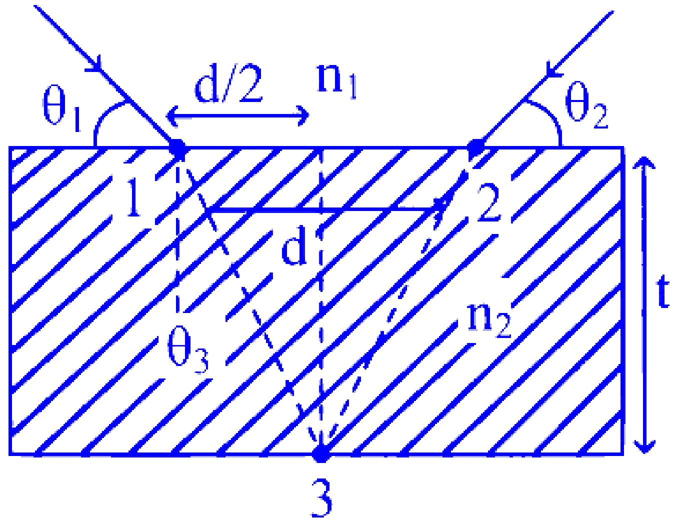
Answer: 6

Solution:



$$n_1 \sin(90 - \theta_1) = n_2 \sin\theta_3$$
$$n_1 \cos\theta_1 = n_2 \sin\theta_3$$
$$n_1 \frac{n_2}{2n_1} = n_2 \sin\theta_3$$
$$\frac{1}{2} = \sin\theta_3, \theta_3 = 30$$
$$\tan 30 = \frac{d}{2(t)}$$
$$t = \frac{d\sqrt{3}}{2} = \frac{4\sqrt{3} \times \sqrt{3}}{2} \text{ cm} = 6 \text{ cm}$$

Question38

A ray of light suffers minimum deviation when incident on a prism having angle of the prism equal to 60°. The refractive index of the prism material is $\sqrt{2}$. The angle of incidence (in degrees) is__________

JEE Main 2025 (Online) 2nd April Evening Shift

Answer: 45

Solution:

To find the angle of incidence when a ray of light experiences minimum deviation in a prism, we use the formula for the refractive index μ:

$$\mu = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Given:

The angle of the prism, $A = 60^\circ$

The refractive index of the prism material, $\mu = \sqrt{2}$

Minimum deviation, $\delta_m = 30^\circ$

At minimum deviation, the angle of incidence i equals the angle of emergence e. Therefore, the relation between the angle of minimum deviation and the angle of the prism is represented as:

$$\delta_m = 2i - A$$

Solving for i:

$$\delta_m = 2i - A \implies 30^\circ = 2i - 60^\circ \implies 2i = 90^\circ \implies i = 45^\circ$$

Thus, the angle of incidence is 45°.

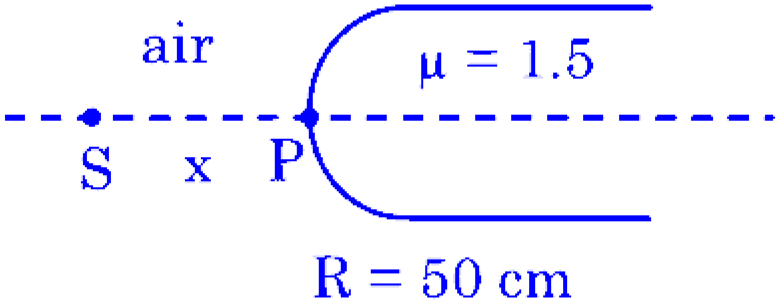
Question39

Light from a point source in air falls on a spherical glass surface (refractive index, $\mu = 1.5$ and radius of curvature $= 50$ cm). The image is formed at a distance of 200 cm from the glass surface inside the glass. The magnitude of distance of the light source from the glass surface is ___________m.

JEE Main 2025 (Online) 3rd April Evening Shift

Answer: 4

Solution:



$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$
$$\frac{1.5}{200}-\frac{1}{-x}=\frac{1.5-1}{50}$$
$$\frac{1}{x}=\frac{1}{100}-\frac{3}{400}$$
$x=400\ \mathrm{cm}$
$x=4\ \mathrm{m}$

Question40

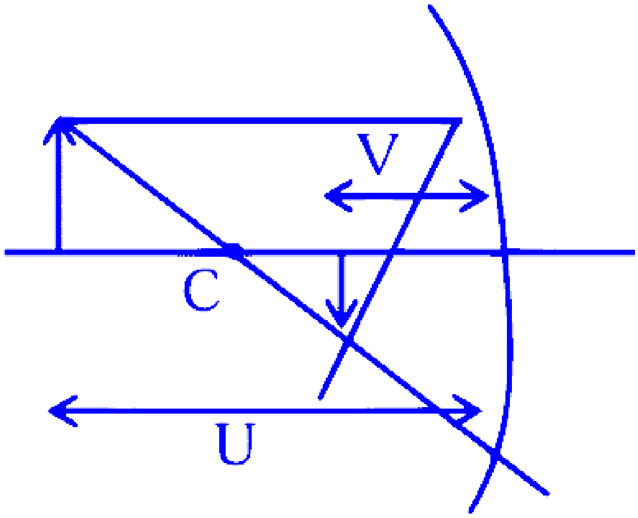
Distance between object and its image (magnified by $-\frac{1}{3}$) is 30 cm . The focal length of the mirror used is $\left(\frac{x}{4}\right)$cm,where magnitude of value of x is __________.

JEE Main 2025 (Online) 4th April Morning Shift

Answer: 45

Solution:

$$M=-\frac{1}{3}$$
$$-\frac{-V}{-U}=\frac{-1}{3}\Rightarrow V=\frac{U}{3}$$
Distance b/w object and image :



$U - V = 30$

$U - \frac{\mu}{3} = 30$

$\Rightarrow U = 45 \quad V = 15$

$\frac{1}{f} = \frac{1}{V} + \frac{1}{U} = -\frac{1}{15} - \frac{1}{45}$

$\Rightarrow F = \frac{45}{4}$

$x = 45$

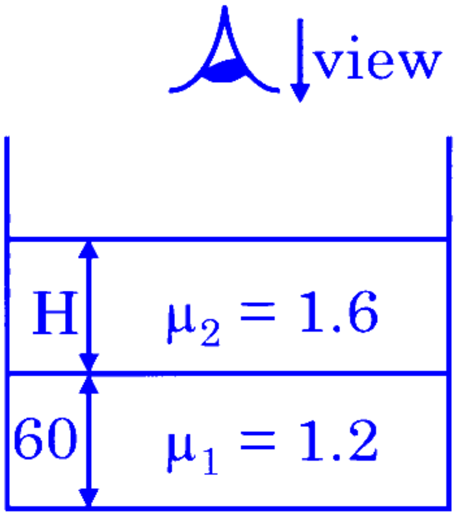
Question41

A container contains a liquid with refractive index of 1.2 up to a height of 60 cm and another liquid having refractive index 1.6 is added to height H above first liquid. If viewed from above, the apparent shift in the position of bottom of container is 40 cm . The value of H is ________ cm . (Consider liquids are immisible)

JEE Main 2025 (Online) 7th April Morning Shift

Answer: 80

Solution:



$y =$ apparent depth of bottom

$$\frac{y}{1} = \frac{H}{1.6} + \frac{60}{1.2}$$

Shift $= 40$

$$H + 60 - y = 40$$

$$H + 60 - \frac{\text{H}}{1.6} - \frac{60}{1.2} = 40$$

$$\frac{6}{16}\text{H} = 30$$

$$H = 80 \text{ cm}$$

Question42

Given is a thin convex lens of glass (refractive index μ) and each side having radius of curvature R. One side is polished for complete reflection. At what distance from the lens, an object be placed on the optic axis so that the image gets formed on the object itself?

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Options:

A. R/μ

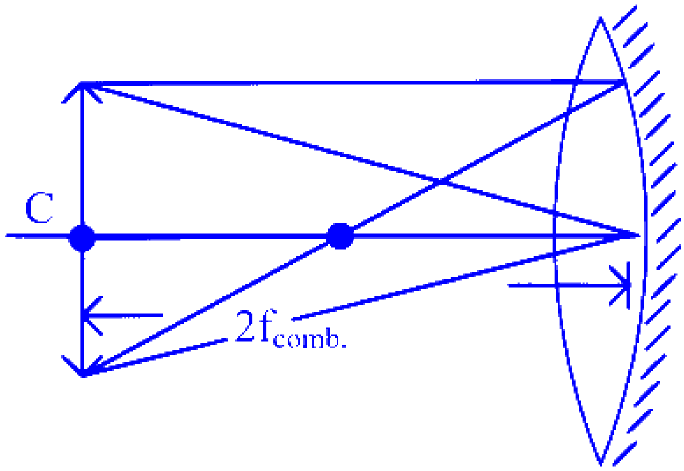
B. $R/(2\mu-1)$

C. $\mathrm{R}/(2\mu-3)$

D. $\mu\mathrm{R}$

Answer: B

Solution:



We know, for a combination of a lens and a mirror,

$P_{comb}=2P_l+P_m$ ...... (1)

Also using lens maker formula,

$\frac{1}{f_l}=(\mu-1)\left(\frac{1}{R}-\frac{1}{-R}\right)=(\mu-1)\frac{2}{R}$

So using eq. (i),

$-\frac{1}{f_{comb}}=2\left(\frac{1}{f_l}\right)-\frac{1}{f_m}$

$\Rightarrow -\frac{1}{f_{comb}}=\frac{4(\mu-1)}{R}-\frac{1}{\left(-\frac{R}{2}\right)}$

$\Rightarrow -\frac{1}{f_{comb}}=\frac{4(\mu-1)}{R}+\frac{2}{R}$

$\Rightarrow -\frac{1}{f_{comb}}=\frac{2}{R}(2\mu-2+1)=\frac{2}{R}(2\mu-1)$

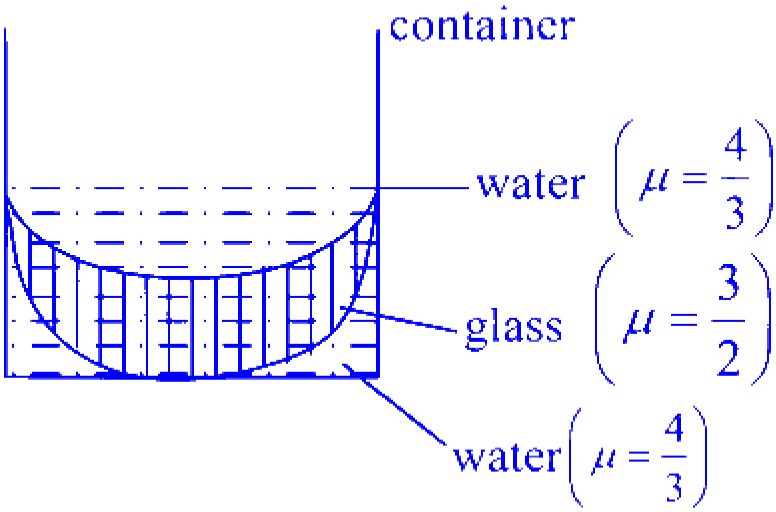
$\Rightarrow f_{comb}=\frac{-R}{2(2\mu-1)}$

$\Rightarrow R_{comb}=2f_{comb}=\frac{-2R}{-2(2\mu-1)}=\frac{-R}{2\mu-1}$

Hence, distance $=\frac{R}{2\mu-1}$

Question43

In the diagram given below, there are three lenses formed. Considering negligible thickness of each of them as compared to $|R_1|$ and $|R_2|$, i.e., the radii of curvature for upper and lower surfaces of the glass lens, the power of the combination is



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Options:

A. $-\frac{1}{6}\left(\frac{1}{|\mathrm{R}_1|}-\frac{1}{|\mathrm{R}_2|}\right)$

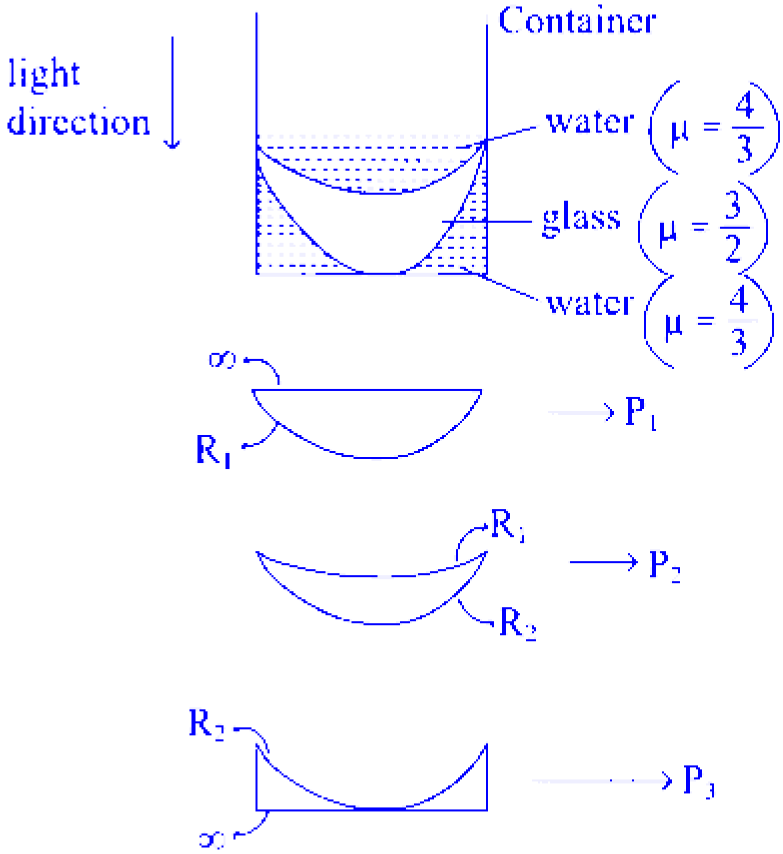
B. $-\frac{1}{6}\left(\frac{1}{|\mathrm{R}_1|}+\frac{1}{|\mathrm{R}_2|}\right)$

C. $\frac{1}{6}\left(\frac{1}{|\mathrm{R}_1|}+\frac{1}{|\mathrm{R}_2|}\right)$

D. $\frac{1}{6}\left(\frac{1}{|\mathrm{R}_1|}-\frac{1}{|\mathrm{R}_2|}\right)$

Answer: A

Solution:



$P_{eq} = P_1 + P_2 + P_3$

Using lens maker formula, $\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

Note : here, we will take air as medium for all three lenses so $\mu_1 = 1$

$P_1 = \frac{1}{f_1} = \left(\frac{4}{3} - 1\right)\left(\frac{1}{\infty} - \frac{1}{-R_1}\right)$

$P_1 = \frac{1}{3}\left(\frac{1}{R_1}\right)$ ..... (1)

$P_2 = \frac{1}{f_2} = \left(\frac{3}{2} - 1\right)\left(\frac{1}{-R_1} - \frac{1}{-R_2}\right)$

$\Rightarrow P_2 = \frac{-1}{2}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ .... (2)

$P_3 = \frac{1}{f_3} = \left(\frac{4}{3} - 1\right)\left(\frac{1}{-R_2} - \frac{1}{\infty}\right)$

$\Rightarrow P_3 = \frac{-1}{3}\frac{1}{R_2}$ ...... (3)

So, $P_{eq} = \frac{1}{3R_1} - \frac{1}{2}\left(\frac{1}{R_1} - \frac{1}{R_2}\right) - \frac{1}{3R_2}$

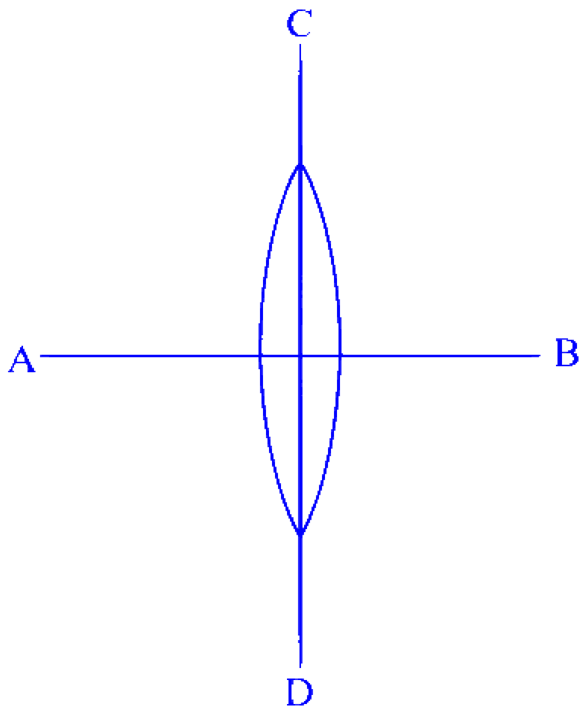
$= \frac{1}{3}\left(\frac{1}{R_1} - \frac{1}{R_2}\right) - \frac{1}{2}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$= \left(\frac{1}{3} - \frac{1}{2}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$P_{eq} = \frac{-1}{6}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

Question44

A symmetric thin biconvex lens is cut into four equal parts by two planes AB and CD as shown in figure. If the power of original lens is 4D then the power of a part of the divided lens is



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Options:

A. 2D

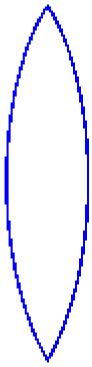
B. 8D

C. 4D

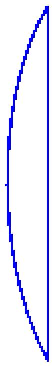
D. D

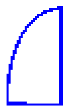
Answer: A

Solution:



$$\frac{1}{f_1} = (\mu - 1)\frac{2}{R} = P = 4D$$



$=$ 

$$\frac{1}{f_2} = (\mu - 1)\frac{1}{R} = \frac{P}{2} = 2D$$

Question45

Given a thin convex lens (refractive index μ_2), kept in a liquid (refractive index $\mu_1, \mu_1 < \mu_2$) having radii of curvatures $|R_1|$ and $|R_2|$. Its second surface is silver polished. Where should an object be placed on the optic axis so that a real and inverted image is formed at the same place?

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Options:

A. $\frac{(\mu_2+\mu_1)|R_1|}{(\mu_2-\mu_1)}$

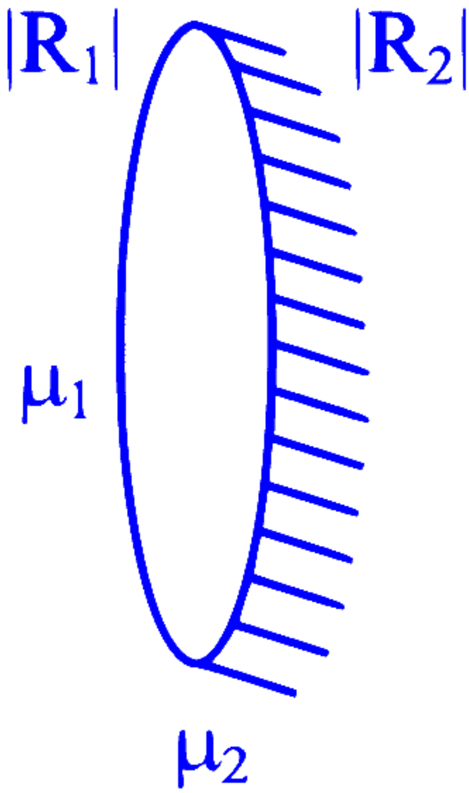
B. $\frac{\mu_1|\mathrm{R}_1|\cdot|\mathrm{R}_2|}{\mu_2(2|\mathrm{R}_1|+|\mathrm{R}_2|)-\mu_1\sqrt{|\mathrm{R}_1|\cdot|\mathrm{R}_2|}}$

C. $\frac{\mu_1|R_1|\cdot|R_2|}{\mu_2(|R_1|+|R_2|)-\mu_1|R_1|}$

D. $\frac{\mu_1|\mathrm{R}_1|\cdot|\mathrm{R}_2|}{\mu_2(|\mathrm{R}_1|+|\mathrm{R}_2|)-\mu_1|\mathrm{R}_2|}$

Answer: D

Solution:



$$\frac{1}{\mathrm{f}_{\mathrm{eq}}}=\frac{2}{\mathrm{f}_{\mathrm{L}}}-\frac{1}{\mathrm{f}_{\mathrm{m}}}$$

$$\mathrm{f}_{\mathrm{m}}=-\frac{|\mathrm{R}_2|}{2}$$

$$\frac{1}{\mathrm{f}_{\mathrm{L}}}=\left(\frac{\mu_2}{\mu_1}-1\right)\left(\frac{1}{\mathrm{R}_1}+\frac{1}{\mathrm{R}_2}\right)$$

$$\frac{1}{f_{eq}}=2\left(\frac{\mu_2-\mu_1}{\mu_1}\right)\left(\frac{R_1+R_2}{R_1R_2}\right)+\frac{2}{R_2}$$

$$=\frac{2}{R_2}\left[\frac{(\mu_2-\mu_1)(R_1+R_2)+\mu_1R_1}{\mu_1R_1}\right]$$

$$=\frac{2}{R_2}\left[\frac{\mu_2R_1+\mu_2R_2-\mu_1R_1-\mu_1R_2+\mu_1R_1}{\mu_1R_1}\right]$$

$$\frac{1}{f_{eq}}=\frac{2\left[\mu_2R_1+\mu_2R_2-\mu_1R_2\right]}{\mu_1R_1R_2}$$

For same size of image

$u=2f$

$$\mathbf{u}=\frac{\mu_1\mathbf{R}_1\mathbf{R}_2}{\mu_2R_1+\mu_2R_2-\mu_1R_2}$$

Question46

A spherical surface of radius of curvature R, separates air from glass (refractive index $=1.5$). The centre of curvature is in the glass medium. A point object ' O ' placed in air on the optic axis of the surface, so that its real image is formed at 'I' inside glass. The line OI intersects the spherical surface at P and $PO=PI$. The distance PO equals to

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Options:

A. 5R

B. 2R

C. 1.5R

D. 3R

Answer: A

Solution:

We wish to find the distance from the point where the ray joining the object (O) and its image (I) meets the spherical surface (P) to the object. The given condition is that

$$PO = PI,$$

meaning that P is the midpoint of OI.

Let's work through the steps:

Setup the coordinate system and geometry:

- Choose the vertex of the spherical surface (the "pole") as the origin (x = 0).
- Let the optical axis be the x-axis.
- Since the object is in air, place O at x = –u (with u > 0).
- Since the real image is formed inside the glass, place I at x = v (with v > 0).
- The spherical surface has radius of curvature R and its center of curvature is in the glass. Thus, we take the center of the sphere as $C = (R, 0)$.
- The vertex (the point on the sphere closest to the object) is then located at x = 0.

Position of P:

- The ray joining O and I is along the x-axis. Its midpoint is

  $$x_P = \frac{-u+v}{2}.$$

- However, since the physical part of the spherical surface is the convex cap met first by the incident light, the ray in fact meets the surface at the vertex, which is at x = 0.
- Equating the midpoint to 0, we have

  $$\frac{-u+v}{2} = 0 \quad \Longrightarrow \quad v = u.$$

- Thus, the object and its image are equidistant from the vertex, and

  $$PO = |0 - (-u)| = u.$$

Relate u and R using the refraction at a spherical surface:

- The formula for refraction at a spherical surface (using the appropriate sign–convention) is

  $$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R},$$

  where for our problem

  – $n_1 = 1$ (air)

  – $n_2 = 1.5$ (glass).

- With $v = u$, substitute in:

  $$\frac{1}{u} + \frac{1.5}{u} = \frac{0.5}{R} \quad \Longrightarrow \quad \frac{2.5}{u} = \frac{0.5}{R}.$$

- Solving for u:

  $u = \frac{2.5\,R}{0.5} = 5R.$

Conclusion:

- Since $PO = u$, we have

  $PO = 5R.$

Thus, the distance PO equals $5R$, which is Option A.

Question47

What is the lateral shift of a ray refracted through a parallel-sided glass slab of thickness ' h ' in terms of the angle of incidence ' i ' and angle of refraction ' r ', if the glass slab is placed in air medium?

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Options:

A. h

B. $\frac{h\cos(i-r)}{\sin r}$

C. $\frac{\mathrm{h}\tan(\mathrm{i}-\mathrm{r})}{\tan \mathrm{r}}$

D. $\frac{h\sin(i-r)}{\cos r}$

Answer: D

Solution:

We can derive the lateral shift (d) by considering the geometry of a ray entering a parallel-sided slab. Here's a step-by-step explanation:

The slab has a thickness h (measured perpendicular to the surfaces). When the ray enters the glass from air at angle i (with respect to the normal), it refracts inside and makes an angle r.

Inside the slab, the ray travels a distance

$L = \frac{h}{\cos r}$

because the vertical component of the path must cover a distance h.

The ray inside the slab is shifted laterally (parallel to the surface). Its lateral displacement over the distance L is

$L \sin r = \frac{h \sin r}{\cos r}$.

However, the emergent ray is parallel to the incident ray but not collinear. By constructing the appropriate geometry (extending the emergent ray backwards until it meets the incident ray's extension at the top interface), you find that the lateral shift between the incident ray and the emergent ray is given by

$d = \frac{h \sin(i-r)}{\cos r}$.

This expression correctly accounts for both the refraction inside the slab and the geometry of the displacement.

Thus, the correct answer is:

Option D: $\frac{h \sin(i-r)}{\cos r}$.

Question48

The refractive index of the material of a glass prism is $\sqrt{3}$. The angle of minimum deviation is equal to the angle of the prism. What is the angle of the prism?

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Options:

A. 60°

B. 50°

C. 58°

D. 48°

Answer: A

Solution:

Step 1: Symmetry at Minimum Deviation

In a prism, when the deviation is minimum, the path of light is symmetric. This means that the angle of incidence (i) is equal to the angle of emergence (i'), and the light inside the prism makes equal angles with the prism faces. If the prism angle is A, then the refracted angle at each interface is:

$r = \frac{A}{2}$

Step 2: Relating Deviation to the Angles

The formula for the deviation (D) in a prism is given by:

$$D = i + i' - A$$

At minimum deviation, $i = i'$, so:

$$D_{\text{min}} = 2i - A$$

The special condition given is that the minimum deviation is equal to the prism angle:

$$D_{\text{min}} = A$$

Thus:

$$A = 2i - A \implies 2i = 2A \implies i = A$$

Step 3: Applying Snell's Law

At the first surface, Snell's law gives:

$$\sin i = \mu \sin r$$

Substitute the values $i = A$ and $r = \frac{A}{2}$:

$$\sin A = \mu \sin \frac{A}{2}$$

Given that the refractive index $\mu = \sqrt{3}$, we have:

$$\sin A = \sqrt{3} \sin \frac{A}{2}$$

Step 4: Solving the Equation

Recall the double-angle formula for sine:

$$\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$$

Substitute this into the previous equation:

$$2 \sin \frac{A}{2} \cos \frac{A}{2} = \sqrt{3} \sin \frac{A}{2}$$

Assuming $\sin \frac{A}{2} \neq 0$, we can divide both sides by $\sin \frac{A}{2}$:

$$2 \cos \frac{A}{2} = \sqrt{3}$$

Solve for $\cos \frac{A}{2}$:

$$\cos \frac{A}{2} = \frac{\sqrt{3}}{2}$$

Since:

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

It follows:

$\frac{A}{2} = 30^\circ \quad \implies \quad A = 60^\circ$

Final Answer:

The angle of the prism is 60°.

Question49

A concave mirror of focal length f in air is dipped in a liquid of refractive index μ. Its focal length in the liquid will be:

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Options:

A. μf

B. f

C. $\frac{f}{\mu}$

D. $\frac{f}{(\mu-1)}$

Answer: B

Solution:

When a concave mirror is dipped in a liquid, its focal length is still determined by the geometry of the mirror. For a spherical concave mirror, the focal length is given by

$f = \frac{R}{2}$

where R is the radius of curvature of the mirror. This derivation comes solely from geometrical considerations and the law of reflection, which states that the angle of incidence equals the angle of reflection. Since reflection is independent of the medium in which the mirror is immersed, the focal length remains unchanged.

Thus, even when the mirror is in a liquid of refractive index μ, its focal length is still

f.

The correct answer is **Option B:** f.

Question50

A plano-convex lens having radius of curvature of first surface 2 cm exhibits focal length of f_1 in air. Another plano-convex lens with first surface radius of curvature 3 cm has focal length of f_2 when it is immersed in a liquid of refractive index 1.2. If both the lenses are made of same glass of refractive index 1.5 , the ratio of f_1 and f_2 will be

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Options:

A. 2 : 3

B. 1 : 3

C. 1 : 2

D. 3 : 5

Answer: B

Solution:

$\frac{1}{f} = \left(\frac{n}{n'} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

For a thin plano-convex lens, one surface is flat, meaning $R_2 = \infty$ and $\frac{1}{R_2} = 0$. The formula reduces to:

$\frac{1}{f} = \left(\frac{n}{n'} - 1\right)\frac{1}{R_1}$

Thus, the focal length is:

$f = \frac{R_1}{\left(\frac{n}{n'} - 1\right)}$

For the First Lens (in Air)

Refractive index of lens, $n = 1.5$

Surrounding medium (air), $n' = 1$

Radius of curvature, $R_1 = 2$ cm

Substitute into the formula:

$f_1 = \frac{2}{\left(\frac{1.5}{1} - 1\right)} = \frac{2}{(1.5-1)} = \frac{2}{0.5} = 4 \text{ cm}$

For the Second Lens (in Liquid)

Refractive index of lens, $n = 1.5$

Surrounding medium (liquid), $n' = 1.2$

Radius of curvature, $R_1 = 3$ cm

Substitute into the formula:

$f_2 = \frac{3}{\left(\frac{1.5}{1.2} - 1\right)}$

First, calculate the term inside the parentheses:

$\frac{1.5}{1.2} = 1.25 \quad \Rightarrow \quad 1.25 - 1 = 0.25$

Now, compute f_2:

$f_2 = \frac{3}{0.25} = 12 \text{ cm}$

Ratio of the Focal Lengths

$\frac{f_1}{f_2} = \frac{4}{12} = \frac{1}{3}$

Thus, the ratio of f_1 to f_2 is:

1 : 3

Question51

A thin plano convex lens made of glass of refractive index 1.5 is immersed in a liquid of refractive index 1.2. When the plane side of the lens is silver coated for complete reflection, the lens immersed in the liquid behaves like a concave mirror of focal length 0.2 m . The radius of curvature of the curved surface of the lens is

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Options:

A. 0.15 m

B. 0.10 m

C. 0.25 m

D. 0.20 m

Answer: B

Solution:

$\frac{1}{f} = \frac{2(n_2 - n_1)}{n_1 R}$

For a thin plano-convex lens made of glass (refractive index $n_2 = 1.5$) immersed in a liquid (refractive index $n_1 = 1.2$), when the plane side is silver-coated, a ray entering the curved surface from the liquid is refracted into the glass, reflects off the coated plane (thereby reversing its direction) and finally refracts back into the liquid at the same curved surface. In the paraxial approximation the net effect is equivalent to a mirror in the liquid with an effective focal length f.

A ray that is incident parallel to the optical axis (at height h) will, after this combined refraction–reflection–refraction process, emerge with an angular deviation corresponding to a mirror having a power given by:

$\frac{h}{f} = \frac{2(n_2 - n_1)}{n_1 R} h\,.$

This gives the relation:

$\frac{1}{f} = \frac{2(n_2 - n_1)}{n_1 R}\,.$

Given that the effective focal length is $f = 0.2\,\text{m}$, substitute the values:

$\frac{1}{0.2} = \frac{2(1.5 - 1.2)}{1.2\,R}\,.$

Simplify the numerator:

$\frac{1}{0.2} = \frac{2(0.3)}{1.2\,R} = \frac{0.6}{1.2\,R}\,.$

Since

$\frac{0.6}{1.2} = 0.5,$

the equation becomes:

$5 = \frac{0.5}{R}\,.$

Solving for R:

$R = \frac{0.5}{5} = 0.10\,\text{m}\,.$

Thus, the radius of curvature of the curved surface of the lens is $0.10\,\text{m}$.

Question52

What is the relative decrease in focal length of a lens for an increase in optical power by 0.1 D from 2.5D ? ['D' stands for dioptre]

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Options:

A. 0.04

B. 0.1

C. 0.40

D. 0.01

Answer: A

Solution:

$P_1 = 2.5\,D, \quad f_1 = \frac{1}{P_1} = \frac{1}{2.5} = 0.4\,m$

After an increase in optical power by 0.1 D:

$P_2 = 2.5\,D + 0.1\,D = 2.6\,D, \quad f_2 = \frac{1}{P_2} = \frac{1}{2.6} \approx 0.3846\,m$

The change in focal length is:

$\Delta f = f_1 - f_2 = 0.4\,m - 0.3846\,m \approx 0.0154\,m$

The relative decrease in focal length is then calculated as:

Relative decrease $= \frac{\Delta f}{f_1} = \frac{0.0154}{0.4} \approx 0.0385 \approx 0.04$

Thus, the relative decrease in focal length is approximately **0.04**.

Question53

A photograph of a landscape is captured by a drone camera at a height of 18 km . The size of the camera film is $2\ \mathrm{cm} \times 2\ \mathrm{cm}$ and the area of the landscape photographed is $400\ \mathrm{km}^2$. The focal length of the lens in the drone camera is :

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Options:

A. 2.8 cm

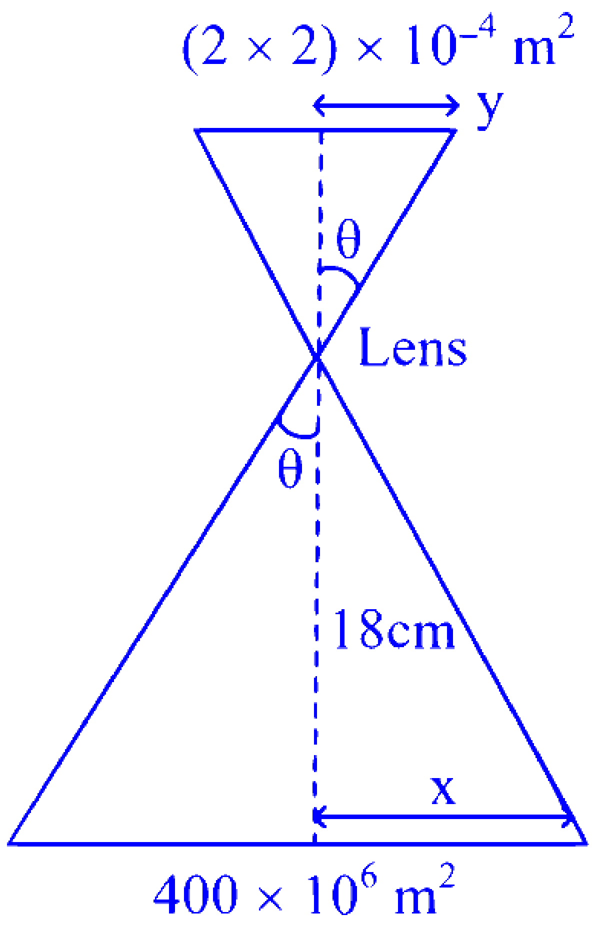
B. 0.9 cm

C. 2.5 cm

D. 1.8 cm

Answer: D

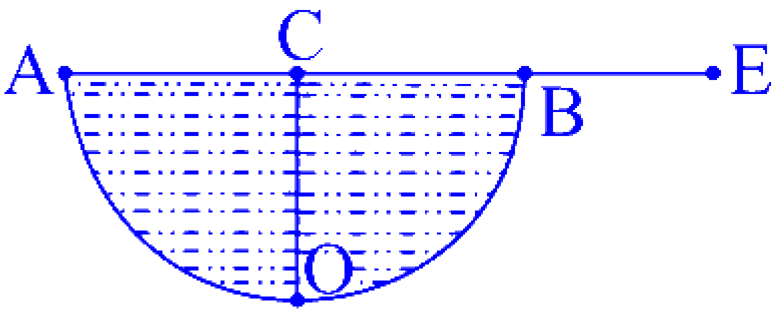
Solution:



$H = 18\ km$
Size of camera film $= 2\ cm \times 2\ cm$
$A_{image} = 400\ km^2$
$x = 20 \times 10^3\ m = 2 \times 10^4\ m$
$y = 2 \times 10^{-2}\ m$
$\frac{x}{y} = 10^6 = \frac{18Km}{f}$
$f = 18 \times 10^{-3}\ m = 18\ mm$
$f = 1.8\ cm$

Question54

A hemispherical vessel is completely filled with a liquid of refractive index μ. A small coin is kept at the lowest point (O) of the vessel as shown in figure. The minimum value of the refractive index of the liquid so that a person can see the coin from point E (at the level of the vessel) is __________.



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Options:

A. $\frac{3}{2}$

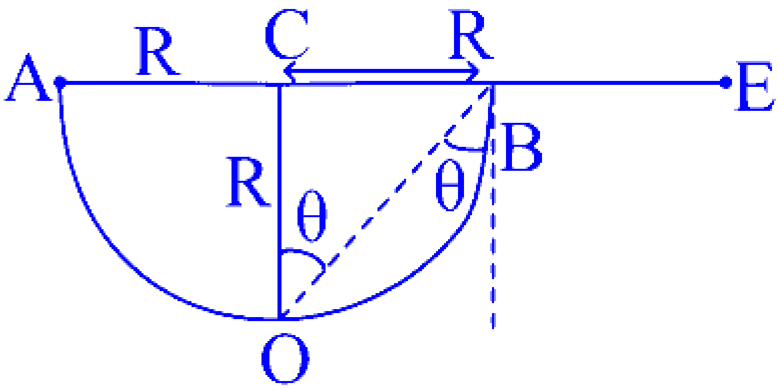
B. $\frac{\sqrt{3}}{2}$

C. $\sqrt{3}$

D. $\sqrt{2}$

Answer: D

Solution:



from ΔOCB, $\tan\theta = \frac{CB}{OC} = \frac{R}{R} = 1$

$\Rightarrow \theta = 45^{\circ}$

We know, $\sin c = \frac{1}{\mu}$

for minimum μ, sinc maximum $\Rightarrow$ C maximum

Here, $c_{\max} = \theta = 45^{\circ}$

So, $\mu = \frac{1}{\sin 45^{\circ}} = \frac{1}{\frac{1}{\sqrt{2}}}$

$\Rightarrow \mu = \sqrt{2}$

Question55

A thin prism P_1 with angle 4° made of glass having refractive index 1.54 , is combined with another thin prism P_2 made of glass having refractive index 1.72 to get dispersion without deviation. The angle of the prism P_2 in degrees is

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Options:

A. 1.5

B. 16/3

C. 3

D. 4

Answer: C

Solution:

We know, for thin prism,

$\mu = \frac{A+\delta}{A}$

where, μ = refractive index

A = Angle of prism

δ = Angle of deviation

We can write, $\mu A = A + \delta$

$\Rightarrow \delta = (\mu - 1)A$

given, $\delta_{net} = 0$

$\Rightarrow (\mu_1 - 1)A_1 - (\mu_2 - 1)A_2 = 0$

$\Rightarrow (1.54 - 1)4 - (1.72 - 1)A_2 = 0$

$\Rightarrow A_2 = \frac{54\times4}{72} \Rightarrow A_2 = 3^\circ$

Question56

In a long glass tube, mixture of two liquids A and B with refractive indices 1.3 and 1.4 respectively, forms a convex refractive meniscus towards A . If an object placed at 13 cm from the vertex of the meniscus in A forms an image with a magnification of $' - 2'$ then the radius of curvature of meniscus is :

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Options:

A.

$\frac{2}{3}$ cm

B.

$\frac{4}{3}$ cm

C.

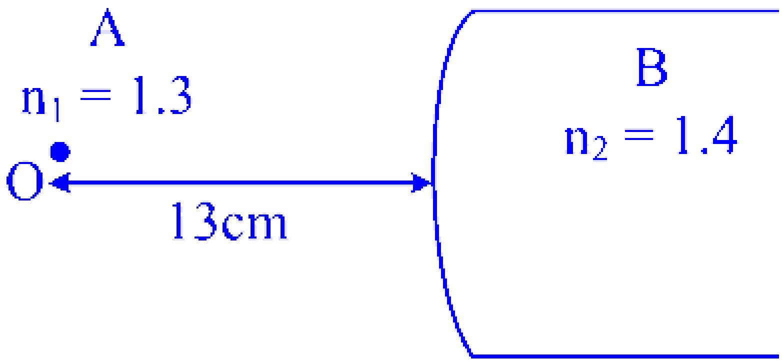
$\frac{1}{3}$ cm

D.

1 cm

Answer: A

Solution:



$$\frac{n_2}{v} - \frac{n_1}{u} \frac{n_2 - n_1}{R}$$
$$\frac{1.4}{v} - \frac{1.3}{-13} = \frac{0.1}{R}$$
$$\frac{1.4}{v} = \frac{1 - R}{10R}$$
$$\frac{1.4}{v} = \frac{1 - R}{10R}$$
$$m = \frac{v/n_2}{u/n_1}$$
$$-2 \times \frac{(-13)}{1.3} = \frac{10R}{1 - R}$$
$$R = \frac{2}{3} \text{ cm}$$

Question57

A concave mirror produces an image of an object such that the distance between the object and image is 20 cm. If the magnification of the image is –3, then the magnitude of the radius of curvature of the mirror is :

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Options:

A.

3.75 cm

B.

15 cm

C.

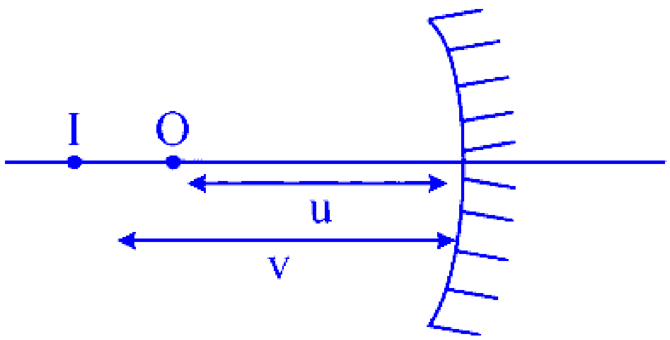
7.5 cm

D.

30 cm

Answer: B

Solution:



$\mathrm{m}=-3=-\frac{\mathrm{v}}{\mathrm{u}}$ and $\mathrm{v}-\mathrm{u}=20\ \mathrm{cm}$

$\mathrm{f}=\frac{\mathrm{vu}}{\mathrm{v}+\mathrm{u}}=\frac{(-30)(-10)}{-30-10}$

$\therefore \mathrm{R}=+15$

Question58

Let u and v be the distances of the object and the image from a lens of focal length f. The correct graphical representation of u and v for a convex lens when $|u| > f$, is

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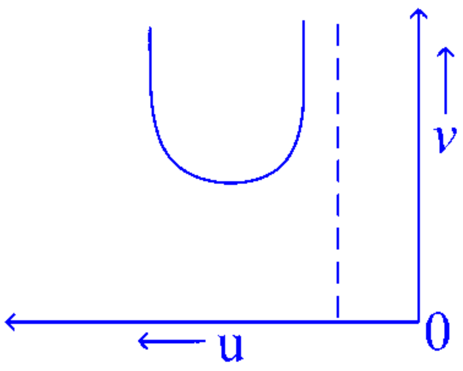
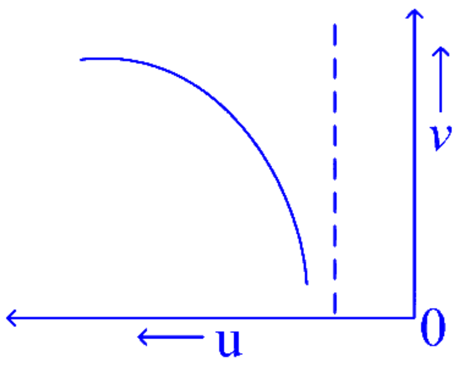
Options:

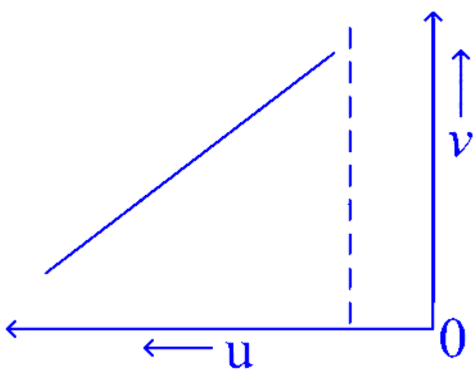
A.

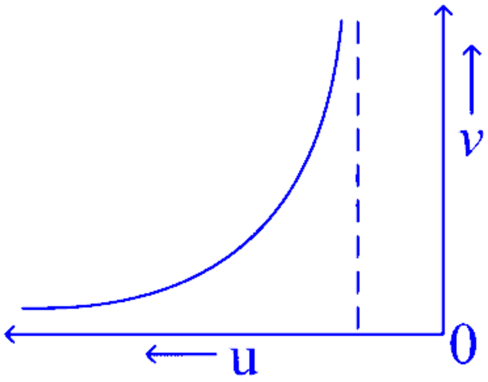


B.



C.

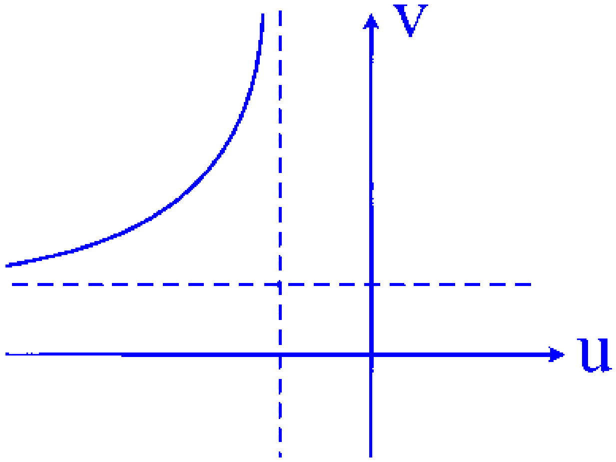


D. 

Answer: D

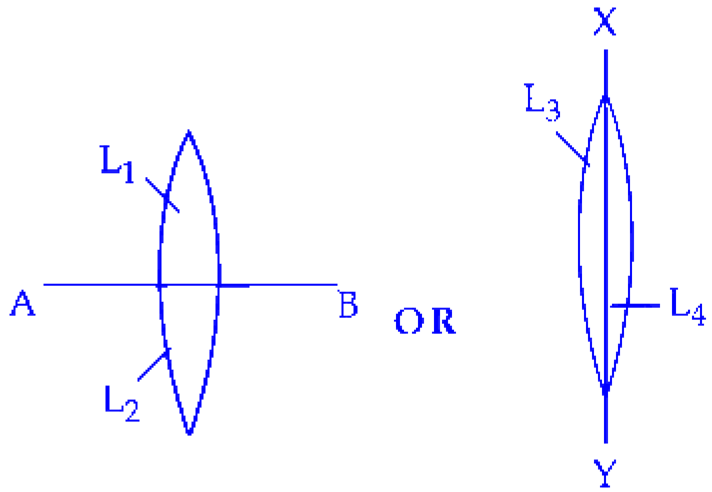
Solution:

$(u+f)(v-f)=f^2$



Question59

Two identical symmetric double convex lenses of focal length f are cut into two equal parts L_1, L_2 by AB plane and L_3, L_4 by XY plane as shown in figure respectively. The ratio of focal lengths of lenses L_1 and L_3 is



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Options:

A.

1 : 2

B.

1 : 1

C.

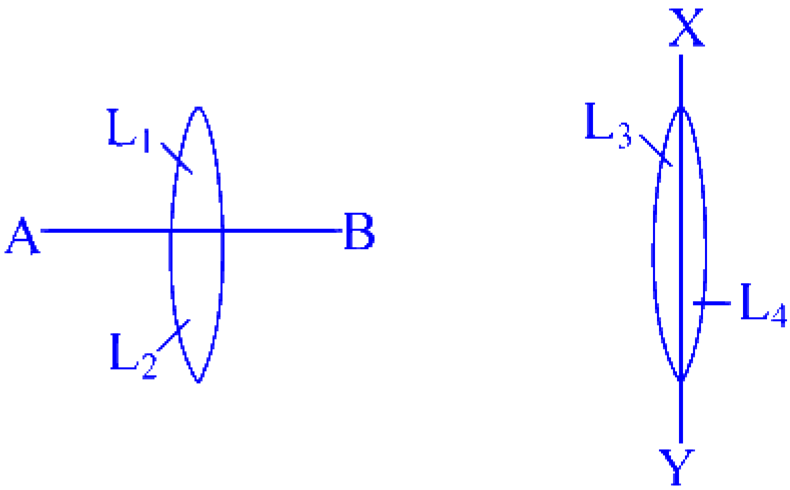
2 : 1

D.

1 : 4

Answer: A

Solution:



Here,

we will use lens maker formula,

$\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

So, the focal length of L_1 remains the same as the original lens because the curvature of the surfaces does not change

So, $(f)_{L_1}=f$ .... (1)

Now, for L_3, L_3 is plano-convex lens

$R_1=R,\ R_2=\infty$

So, $\frac{1}{(f)_{L_3}}=(\mu-1)\left(\frac{1}{R}-\frac{1}{\infty}\right)$

$\Rightarrow \frac{1}{(f)_{L_3}}=(\mu-1)\frac{1}{R}$ ..... (2)

For whole double convex lens,

$\frac{1}{f}=(\mu-1)\left(\frac{1}{R}-\frac{1}{-R}\right)$ (As $R_1=R$, $R_2=-R$)

$\Rightarrow \frac{1}{f}=(\mu-1)\left(\frac{1}{R}+\frac{1}{R}\right)$

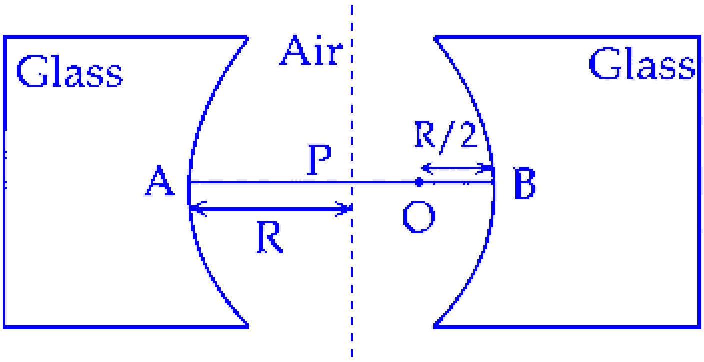
$\Rightarrow \frac{1}{f}=(\mu-1)\left(\frac{2}{R}\right)$ .... (3)

From (2) and (3),

$\frac{1}{(f)_{L_3}} = \frac{1}{2f} \Rightarrow (f)_{L_3} = 2f$

Hence, $\frac{(f)_{L_1}}{(f)_{L_3}} = \frac{f}{2f}$ ....(from (1)) $= 1:2$

Question60



Two concave refracting surfaces of equal radii of curvature and refractive index 1.5 face each other in air as shown in figure. A point object O is placed midway, between P and B. The separation between the images of O, formed by each refracting surface is :

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Options:

A.

0.124R

B.

0.114R

C.

0.411R

D.

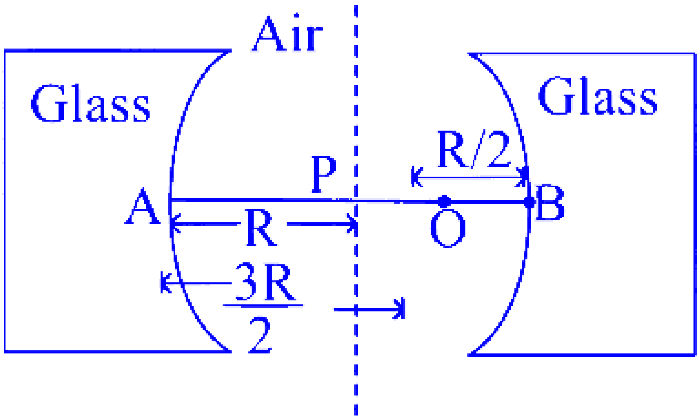
0.214R

Answer: B

Solution:

Using lens maker formula,

$\frac{\mu_2}{v}-\frac{\mu_1}{4}=\frac{\mu_2-\mu_1}{R}$



For the image formed by the surface on the right side.

$\mu_1=1,\ \mu_2=1.5,\ u=\frac{-R}{2}, R_1=-R$

so, $\frac{1.5}{v}-\frac{1}{-R/2}=\frac{1.5-1}{-R}$

$\Rightarrow \frac{1.5}{v}+\frac{2}{R}=\frac{-1}{2R}$

$\Rightarrow \frac{1.5}{v}=\frac{-5}{2R} \Rightarrow v=\frac{-3R}{5}$

Here '$-$' sign tells that image is formed left to B.

So, the distance of image from $P,=\frac{R-3R}{\Delta}=\frac{2R}{5}$

Now, for the surface on the left side,

$\mu_1=1, \mu_2=1.5, \mu=\frac{3R}{2}, R_2=R$

So, $\frac{1.5}{v}-\frac{1}{\frac{3R}{2}}=\frac{1.5-1}{R}$

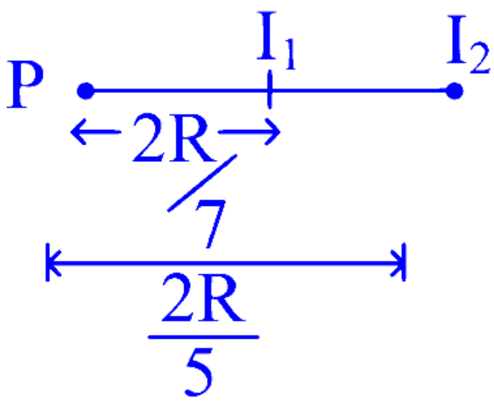
$\Rightarrow \frac{1.5}{v}-\frac{2}{3R}=\frac{1}{2R}$

$\Rightarrow \frac{1.5}{v}=\frac{1}{R}\left(\frac{1}{2}+\frac{2}{3}\right)$

$\Rightarrow \frac{1.5}{v}=\frac{7}{6R}$

$\Rightarrow v=\frac{9R}{7}$

So distance from $P=\frac{9R}{7}-R=\frac{2R}{7}$



So the distance between two images,

$= \frac{2R}{5} - \frac{2R}{7} = 0.4R - 0.2857R$

$\approx 0.4R - 0.286R = 0.114R.$

Question61

A convex lens made of glass (refractive index = 1.5) has focal length 24 cm in air. When it is totally immersed in water (refractive index = 1.33), its focal length changes to

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Options:

A.

96 cm

B.

72 cm

C.

24 cm

D.

48 cm

Answer: A

Solution:

Using lens formula,

For Ist case : $\frac{1}{f} = (a^{\mu_g} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

where, R_1, R_2 = radii of refracting surfaces

f = focal length

$a^{\mu_g} = \frac{\mu_g}{\mu_a} = \frac{1.5}{1} = 1.5$

So, $\frac{1}{24} = (1.5 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ (As f = 24 cm given)

$\Rightarrow \frac{1}{24} = 0.5\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\Rightarrow \frac{1}{12} = \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\Rightarrow \frac{1}{R_1} - \frac{1}{R_2} = \frac{1}{12}$ .... (i)

For IInd case:

$\frac{1}{f} = (w\mu_g - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\Rightarrow \frac{1}{f} = \left(\frac{1.5}{1.33} - 1\right)\left(\frac{1}{12}\right)$ ..... from (i)

$\Rightarrow \frac{1}{f} = \frac{17}{133} \times \frac{1}{12} \Rightarrow f = \frac{133 \times 12}{17}$

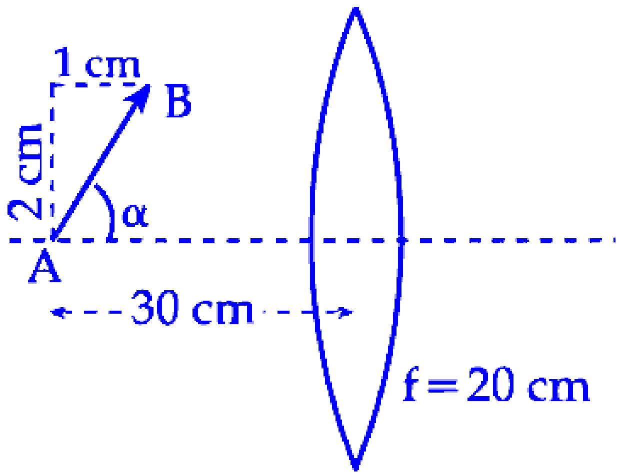
$\Rightarrow f = 93.88$ cm

≈ 94 cm

Hence, option 1 is correct.

Question62

A slanted object AB is placed on one side of convex lens as shown in the diagram. The image is formed on the opposite side. Angle made by the image with principal axis is :



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Options:

A. $-\alpha$

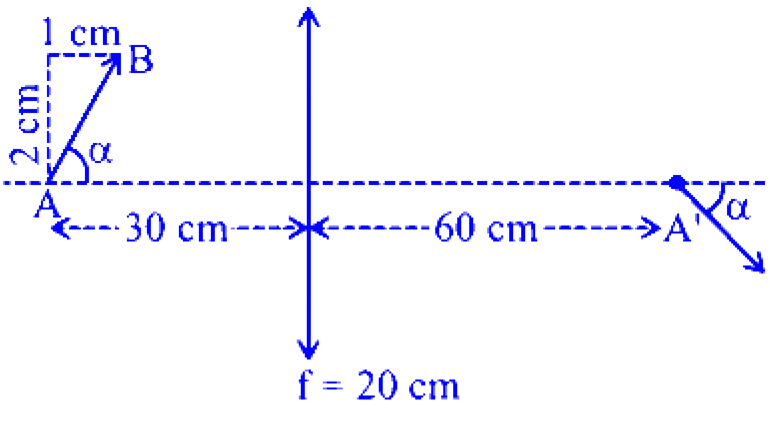
B. $-\frac{\alpha}{2}$

C. -45°

D. $+45^{\circ}$

Answer: C

Solution:



Location of image of A :-

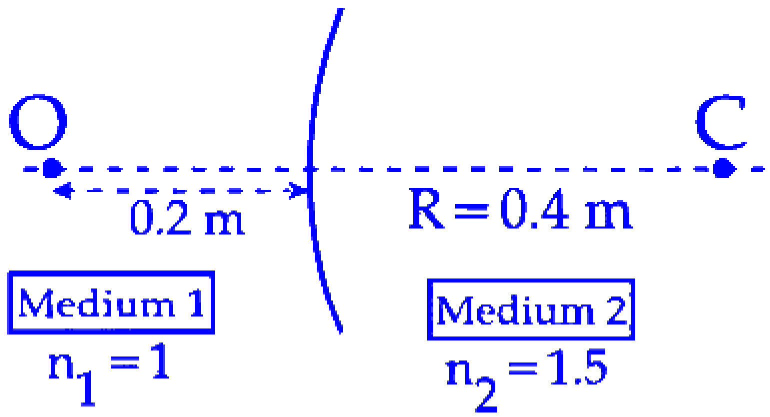
$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{1}{v}-\frac{1}{-30}=\frac{1}{20} \Rightarrow \frac{1}{v}=\frac{1}{60} \Rightarrow v=60 \text{ cm}$
$\therefore m=2$

Since size of object is small wrt the location hence

$dv=m^2 du \Rightarrow dv=4\times 1=4 \text{ cm}$
$h_i=mh_0 \Rightarrow h_i(dy)=2\times 2=4 \text{ cm}$

$\therefore$ Angle made with principle axis $=-45^{\circ}$

Question63



A spherical surface separates two media of refractive indices 1 and 1.5 as shown in figure. Distance of the image of an object ' O ', is :

(C is the center of curvature of the spherical surface and R is the radius of curvature)

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Options:

A. 0.24 m left to the spherical surface

B. 0.24 m right to the spherical surface

C. 0.4 m left to the spherical surface

D. 0.4 m right to the spherical surface

Answer: C

Solution:

To determine the image distance when a spherical surface separates two media with refractive indices of 1 and 1.5:

Use the lens maker's formula for a spherical surface:

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

Where:

μ_1 = 1 (refractive index of the first medium)

μ_2 = 1.5 (refractive index of the second medium)

u = -0.2 m (object distance, negative as per convention)

R = 0.4 m (radius of curvature)

Substituting the known values:

$\frac{1.5}{v} - \frac{1}{-0.2} = \frac{1.5-1}{0.4}$

Simplify the equation:

$\frac{1.5}{v} = \frac{0.5}{0.4} - \frac{1}{0.2}$

Calculate:

$\frac{1.5}{v} = -\frac{1.5}{0.4}$

Solve for v:

$v = -0.4\,\mathrm{m}$

The negative sign indicates that the image is 0.4 meters to the left of the spherical surface.

Question64

A bi-convex lens has radius of curvature of both the surfaces same as $1/6$ cm. If this lens is required to be replaced by another convex lens having different radii of curvatures on both sides $(R_1 \neq R_2)$, without any change in lens power then possible combination of R_1 and R_2 is :

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A. $\frac{1}{3}$ cm and $\frac{1}{7}$ cm

B. $\frac{1}{5}$ cm and $\frac{1}{7}$ cm

C. $\frac{1}{3}$ cm and $\frac{1}{3}$ cm

D. $\frac{1}{6}$ cm and $\frac{1}{9}$ cm

Answer: B

Solution:

To replace a bi-convex lens with another convex lens that has different radii of curvature on each side (i.e., $R_1 \neq R_2$), while maintaining the same lens power, the radii must satisfy a specific relationship.

For the current bi-convex lens, both radii of curvature are given as $\frac{1}{6}$ cm. The lens formula for power is tied to the radii through:

$\frac{1}{f_1} = (\mu - 1)\left(\frac{2}{R}\right)$

When this lens is replaced by a lens with different radii (R_1 and R_2), the condition that has to be met is:

$\frac{1}{f_2} = (\mu - 1)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$

For power equivalence, the expressions for $\frac{1}{f_1}$ and $\frac{1}{f_2}$ must be equal:

$(\mu - 1)\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = (\mu - 1)\left(\frac{2}{R}\right)$

This simplifies to:

$\frac{1}{R_1} + \frac{1}{R_2} = \frac{2}{R}$

Given $R = \frac{1}{6}$ cm, it follows that:

$\frac{1}{R_1} + \frac{1}{R_2} = \frac{2}{\left(\frac{1}{6}\right)}$

Thus, simplifying the equation gives:

$\frac{1}{R_1} + \frac{1}{R_2} = 12$

Therefore, any valid pair of radii R_1 and R_2 must satisfy this equation.

Question65

Two identical objects are placed in front of convex mirror and concave mirror having same radii of curvature of 12 cm , at same distance of 18 cm from the respective mirrors. The ratio of sizes of the images formed by convex mirror and by concave mirror is :

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A. 2

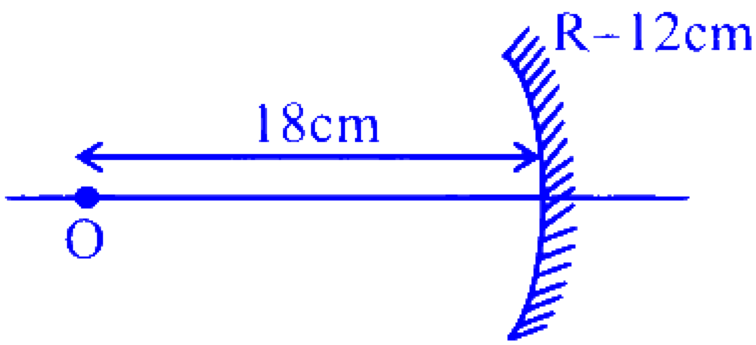
B. $1/2$

C. $1/3$

D. 3

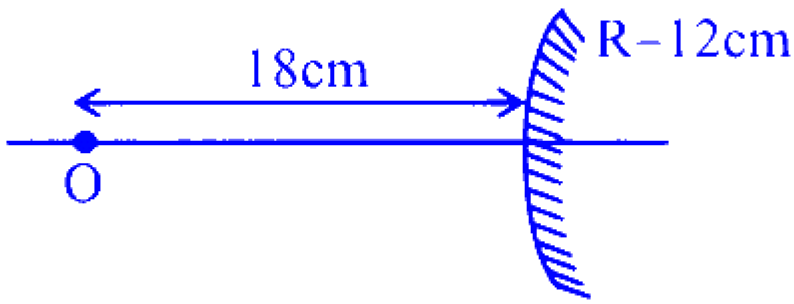
Answer: B

Solution:



Using $m = \dfrac{f}{u - f}$

$$m_1 = \frac{6}{18-6} = \frac{1}{2}$$



$m_2 = \frac{6}{18+6} = \frac{1}{4} \quad \therefore \frac{m_2}{m_1} = \frac{1}{2}$

Question66

The radii of curvature for a thin convex lens are 10 cm and 15 cm respectively. The focal length of the lens is 12 cm . The refractive index of the lens material is

JEE Main 2025 (Online) 3rd April Morning Shift

Options:

A. 1.4

B. 1.8

C. 1.5

D. 1.2

Answer: C

Solution:

To determine the refractive index (μ) of a thin convex lens, we use the lens maker's formula:

$\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

Given:

Focal length (f) = 12 cm

Radii of curvature (R_1 and R_2) = 10 cm and -15 cm, respectively

Substituting these values into the formula, we have:

$\frac{1}{12}=(\mu-1)\left(\frac{1}{10}-\frac{1}{-15}\right)$

Simplifying the equation:

$\frac{1}{12}=(\mu-1)\left(\frac{3+2}{30}\right)$

$\frac{1}{12}=(\mu-1)\times\frac{5}{30}$

$\frac{1}{12}=(\mu-1)\times\frac{1}{6}$

Solving for μ:

$\mu-1=\frac{1}{12}\times 6$

$\mu-1=\frac{1}{2}$

$\mu=\frac{3}{2}$

Thus, the refractive index of the lens material is $\mu=1.5$.

Question67

Consider following statements for refraction of light through prism, when angle of deviation is minimum.

A. The refracted ray inside prism becomes parallel to the base.

B. Larger angle prisms provide smaller angle of minimum deviation.

C. Angle of incidence and angle of emergence becomes equal.

D. There are always two sets of angle of incidence for which deviation will be same except at minimum deviation setting.

E. Angle of refraction becomes double of prism angle.

Choose the correct answer from the options given below :

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Options:

A. A, B and E Only

B. B, C and D Only

C. B, D and E Only

D. A, C and D Only

Answer: D

Solution:

When light is refracted through a prism and the angle of deviation is at its minimum, the following occurs:

The refracted ray inside the prism is parallel to the base of the prism.

The angle of incidence and the angle of emergence are equal.

There are two sets of angles of incidence that result in the same angle of deviation, except when the deviation is at its minimum.

These points are illustrated by the equation:

$\delta = \mathrm{I} + \mathrm{e} - \mathrm{A}$

For the minimum angle of deviation (δ_{min}), the condition $\mathrm{I} = \mathrm{e}$ is met, and the refracted ray becomes parallel to the base. Thus, statements A, C, and D are correct.

Question68

When an object is placed 40 cm away from a spherical mirror an image of magnification $\frac{1}{2}$ is produced. To obtain an image with magnification of $\frac{1}{3}$, the object is to be moved :

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. 20 cm away from the mirror.

B. 20 cm towards the mirror.

C. 80 cm away from the mirror.

D. 40 cm away from the mirror.

Answer: D

Solution:

To solve this problem, start by using the magnification formula for spherical mirrors:

$m = \frac{f}{f-u}$

Given that the initial magnification $m = \frac{1}{2}$, and the object distance $u = -40$ cm (the negative sign indicates distance measured against the incoming light), we can write:

$\frac{1}{2} = \frac{f}{f-(-40)}$

This simplifies to:

$\frac{1}{2} = \frac{f}{f+40}$

Cross-multiplying gives:

$f + 40 = 2f$

This leads to:

$f = 40$ cm

To find the new object distance for a magnification of $m = \frac{1}{3}$, we use the same formula:

$m = \frac{f}{f-u}$

Substitute the known values:

$\frac{1}{3} = \frac{40}{40-u}$

Cross-multiplying gives:

$40 - u = 120$

Solving for u results in:

$u = -80$ cm

Thus, the object needs to be placed 80 cm from the mirror, opposite to the initial placement of 40 cm. To achieve this repositioning, the object needs to be moved 40 cm further from its initial 40 cm distance, totaling a move of 40 cm away from the mirror.

Question69

A finite size object is placed normal to the principal axis at a distance of 30 cm from a convex mirror of focal length 30 cm . A plane mirror is now placed in such a way that the image produced by both the mirrors coincide with each other. The distance between the two mirrors is :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. 45 cm

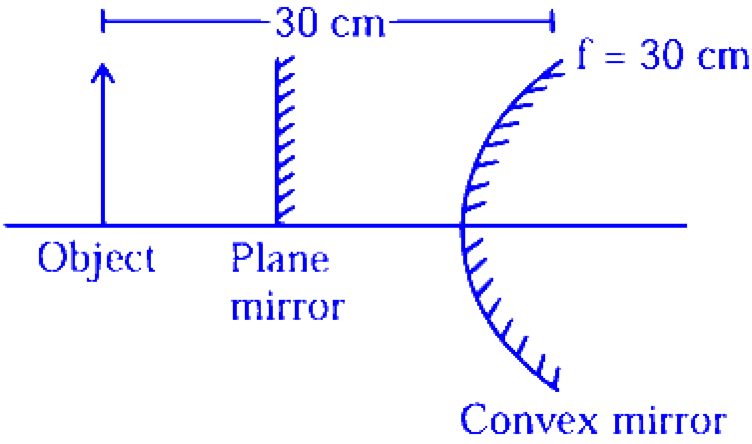
B. 15 cm

C. 22.5 cm

D. 7.5 cm

Answer: D

Solution:



For Convex mirror

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$
$$\frac{1}{v}-\frac{1}{30}=\frac{1}{30}$$
$$\frac{1}{v}=\frac{2}{30}=\frac{1}{15}\Rightarrow v=15 \text{ cm}$$

Image formed by convex mirror is at 45 cm from object so plane mirror should be placed midway at 22.5 cm from object so that both of their images may coinside,

Therefore distance between both mirrors

$=30-22.5=7.5 \text{ cm}$

Correct Answer : Option 2

Question70

A lens having refractive index 1.6 has focal length of 12 cm , when it is in air. Find the focal length of the lens when it is placed in water. (Take refractive index of water as 1.28)

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. 355 mm

B. 655 mm

C. 288 mm

D. 555 mm

Answer: C

Solution:

To find the focal length of a lens with a refractive index of 1.6 when placed in water, given that its focal length in air is 12 cm and the refractive index of water is 1.28, we can use the lens maker's formula:

$$\frac{1}{f}=\left(\frac{\mu_L}{\mu_m}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

When the lens is in air, the refractive index of the medium, μ_m, is 1. Therefore, for the lens in air:

$$\frac{1}{12}=(1.6-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

Simplifying gives:

$$\frac{1}{12}=\frac{6}{10}\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

Solving for $\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$:

$$\frac{1}{R_1}-\frac{1}{R_2}=\frac{10}{72}$$

Now, when the lens is placed in water, $\mu_m=1.28$. Substituting into the equation gives:

$$\frac{1}{f}=\left(\frac{1.6}{1.28}-1\right)\left(\frac{10}{72}\right)$$

Calculating each part:

$$\frac{1}{f}=\frac{32}{128}\times\frac{10}{72}$$

$$\frac{1}{f}=\frac{1}{4}\times\frac{10}{72}$$

Hence, solving for f:

$$f=28.8\,\text{cm}=288\,\text{mm}$$

Thus, the focal length of the lens in water is 288 mm.

Question71

Two thin convex lenses of focal lengths 30 cm and 10 cm are placed coaxially, 10 cm apart. The power of this combination is:

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. 5 D

B. 1 D

C. 20 D

D. 10 D

Answer: D

Solution:

To find the power of a combination of two thin convex lenses, we use the formula for the equivalent focal length f_{eq} of the lens system when the lenses are placed coaxially with a certain distance d between them:

$\frac{1}{f_{\text{eq}}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$

Given:

$f_1 = 30\,\text{cm}$

$f_2 = 10\,\text{cm}$

$d = 10\,\text{cm}$ (distance between the two lenses)

First, convert the focal lengths from centimeters to meters:

$f_1 = 0.3\,\text{m}$

$f_2 = 0.1\,\text{m}$

Plug in the values into the equation:

$\frac{1}{f_{\text{eq}}} = \frac{1}{0.3} + \frac{1}{0.1} - \frac{0.1}{(0.3)(0.1)}$

Calculate each term:

$\frac{1}{0.3} = \frac{10}{3} \approx 3.33$

$\frac{1}{0.1} = 10$

$\frac{0.1}{(0.3)(0.1)} = \frac{0.1}{0.03} = \frac{10}{3} \approx 3.33$

Substitute these calculated values back into the formula:

$\frac{1}{f_{\text{eq}}} = 3.33 + 10 - 3.33$

$\frac{1}{f_{\text{eq}}} = 10$

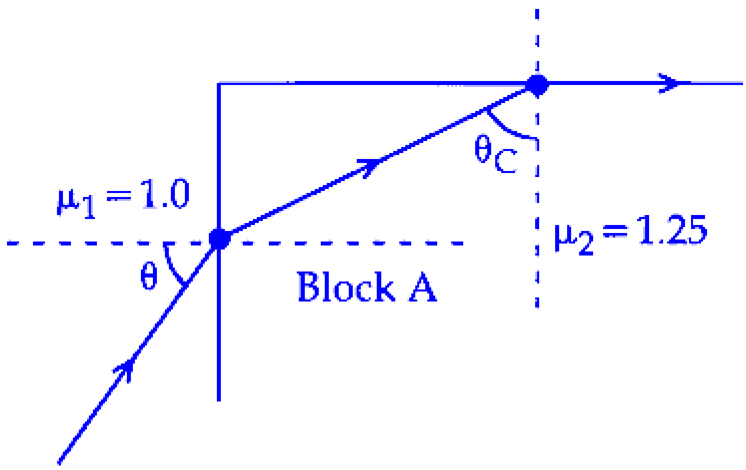
Thus, the power of the lens combination is:

$\text{Power} = \frac{1}{f_{\text{eq}}} = 10\,\text{D}$

Therefore, the power of the lens system is 10 diopters (D).

Question72

A transparent block A having refractive index $\mu = 1.25$ is surrounded by another medium of refractive index $\mu = 1.0$ as shown in the figure. A light ray is incident on the flat face of the block with incident angle θ as shown in the figure. What is the maximum value of θ for which light suffers total internal reflection at the top surface of the block?



JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

$\tan^{-1}(4/3)$

B.

$\sin^{-1}(3/4)$

C.

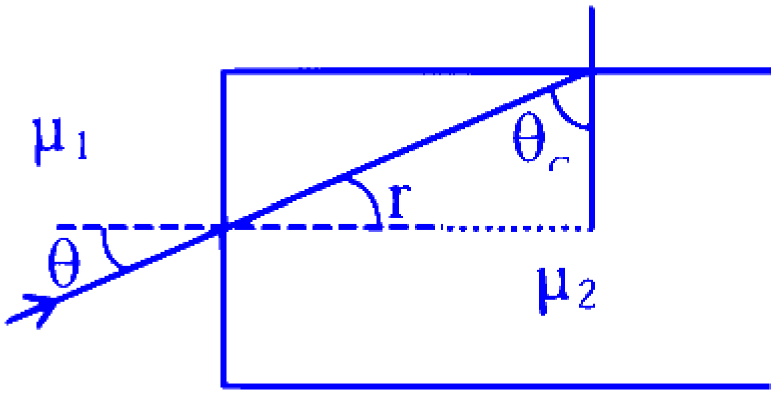
$\tan^{-1}(3/4)$

D.

$\cos^{-1}(3/4)$

Answer: B

Solution:



$$r + \theta_C = 90^\circ$$
$$\mu_1 \sin\theta = \mu_2 \, \mathrm{sinr}$$
$$\sin\theta = \frac{\mu_2}{\mu_1}\sin(90 - \theta_C)$$
$$\sin\theta = \frac{\mu_2}{\mu_1}\cos\theta_C$$
$$\sin\theta_C = \frac{\mu_1}{\mu_2}$$

$$\sin\theta = \frac{\mu_2}{\mu_1}\sqrt{1 - \frac{\mu_1^2}{\mu_2^2}}$$
$$\sin\theta = \sqrt{\frac{\mu_2^2 - \mu_1^2}{\mu_1^2}} = \sqrt{\frac{\frac{25}{16} - 1}{1}}$$
$$\sin\theta = \frac{3}{4}$$
$$\theta = \sin^{-1}\left(\frac{3}{4}\right)$$

Question73

A mirror is used to produce an image with magnification of $\frac{1}{4}$. If the distance between object and its image is 40 cm, then the focal length of the mirror is ________.

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

10 cm

B.

12.7 cm

C.

10.7 cm

D.

15 cm

Answer: C

Solution:

Magnification (m):

The magnification is given by the relationship:

$m = \frac{1}{4} = \frac{v}{u}$

Rearrange to find the object distance (u) in terms of the image distance (v):

$u = 4v$

Distance between object and image:

It is given that the sum of the object distance and the image distance is 40 cm:

$v + u = 40$

Substituting $u = 4v$ into the equation gives:

$v + 4v = 40$

$5v = 40$

Solving for v gives:

$v = 8\,\text{cm}$

Find the object distance (u):

Substitute $v = 8\,\text{cm}$ back into $u = 4v$:

$u = 4 \times 8 = 32\,\text{cm}$

Calculate the focal length (f):

Use the mirror formula:

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Substituting the values of v and u:

$\frac{1}{8} + \frac{1}{32} = \frac{1}{f}$

Simplify and solve for $\frac{1}{f}$:

$\frac{1}{8} - \frac{1}{32} = \frac{1}{f}$

$\frac{4-1}{32} = \frac{1}{f}$

$\frac{3}{32} = \frac{1}{f}$

Therefore, the focal length is:

$f = \frac{32}{3} = 10.7\,\text{cm}$

Question74

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Refractive index of glass is higher than that of air.

Reason (R): Optical density of a medium is directly proportionate to its mass density which results in a proportionate refractive index.

In the light of the above statements, choose the most appropriate answer from the options given below:

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Options:

A.

Both (A) and (R) are correct and (R) is the correct explanation of (A)

B.

(A) is correct but (R) is not correct

C.

(A) is not correct but (R) is correct

D.

Both (A) and (R) are correct but (R) is **not** the correct explanation of (A)

Answer: B

Solution:

Refractive index has no relation with mass density because both have different meaning. Hence reason is incorrect.

So (A) is correct but (R) is not correct.

Question75

A concave-convex lens of refractive index 1.5 and the radii of curvature of its surfaces are 30 cm and 20 cm, respectively. The concave surface is upwards and is filled with a liquid of refractive index 1.3. The focal length of the liquid-glass combination will be

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

$\frac{700}{11}$ cm

B.

$\frac{600}{11}$ cm

C.

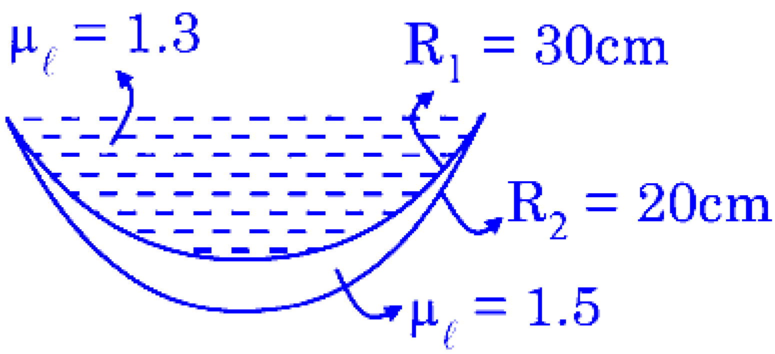
$\frac{800}{11}$ cm

D.

$\frac{500}{11}$ cm

Answer: B

Solution:



$$\begin{aligned} \frac{1}{\mathrm{f}} & =\left(\frac{1.3-1}{1}\right)\left(\frac{1}{\infty}-\frac{1}{-30}\right) \\ & =\left(\frac{1.5-1}{1}\right)\left(\frac{1}{-30}-\frac{1}{-30}\right) \\ & =\frac{0.3}{30}+\frac{0.5}{60}=\frac{1}{100}+\frac{1}{120} \\ & =\frac{6+5}{600}=\frac{11}{600} \\ \mathrm{f} & =\frac{600}{11} \mathrm{~cm} \end{aligned}$$

Question76

A convex lens of focal length 30 cm is placed in contact with a concave lens of focal length 20 cm. An object is placed at 20 cm to the left of this lens system. The distance of the image from the lens in cm is ________.

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

$\frac{60}{7}$

B.

15

C.

45

D.

30

Answer: B

Solution:

Equivalent focal length

$$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}$$
$$=\frac{1}{30}+\frac{1}{-20}=\frac{2-3}{60}=-\frac{1}{60}$$
$$f=-60\text{ cm}$$

Lens formula

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$
$$\frac{1}{v}-\frac{1}{-20}=\frac{1}{-60}$$
$$v=-15\text{ cm}$$

Question77

A convex lens of focal length 40cm forms an image of an extended source of light on a photoelectric cell. A current I is produced. The lens is replaced by another convex lens having the same diameter but focal length 20cm. The photoelectric current now is :

[27-Jan-2024 Shift 1]

Options:

A.

I/2

B.

4I

C.

2I

D.

I

Answer: D

Solution:

As amount of energy incident on cell is same so current will remain same.

Question78

If the refractive index of the material of a prism is cot (A/2), where A is the angle of prism then the angle of minimum deviation will be

[27-Jan-2024 Shift 1]

Options:

A.

$\pi - 2A$

B.

$\pi/2 - 2A$

C.

$\pi - A$

D.

$\pi/2 - A$

Answer: A

Solution:

$$\cot\frac{A}{2}=\frac{\sin\left(\frac{A+\delta_{\min}}{2}\right)}{\sin\frac{A}{2}}$$

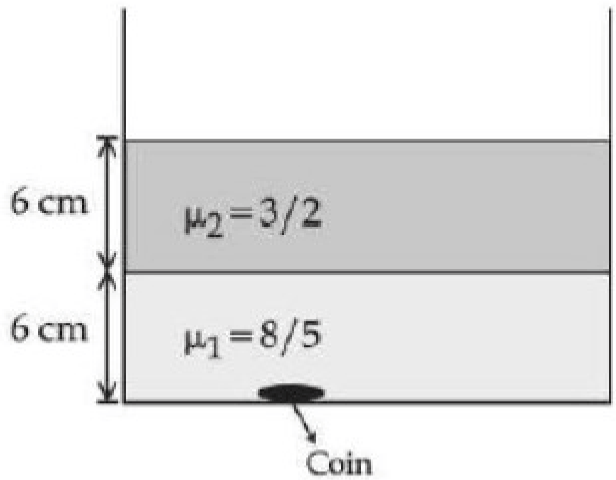
$$\Rightarrow \cos\frac{A}{2}=\sin\left(\frac{A+\delta_{\min}}{2}\right)$$

$$\frac{A+\delta_{\min}}{2}=\frac{\pi}{2}-\frac{A}{2}$$

$$\delta_{\min}=\pi-2A$$

Question79

Two immiscible liquids of refractive indices 8/5 and 3/2 respectively are put in a beaker as shown in the figure. The height of each column is 6cm. A coin is placed at the bottom of the beaker. For near normal vision, the apparent depth of the coin is α/4 cm. The value of α is_______



[27-Jan-2024 Shift 1]

Answer: 31

Solution:

$$h_{app}=\frac{h_1}{\mu_1}+\frac{h_2}{\mu_2}=\frac{6}{3/2}+\frac{6}{8/5}=4+\frac{15}{4}=\frac{31}{4}\text{ cm}$$

Question80

A convex mirror of radius of curvature 30cm forms an image that is half the size of the object. The object distance is :

[29-Jan-2024 Shift 1]

Options:

A.

−15 cm

B.

45 cm

C.

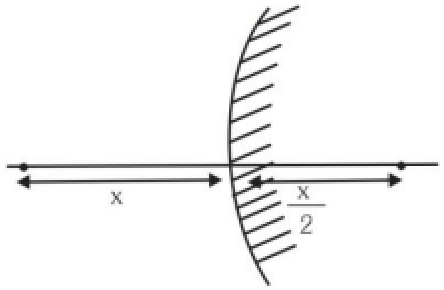
−45 cm

D.

15 cm

Answer: A

Solution:



Given $R = 30$ cm

$f = R/2 = +15$ cm

Magnification $(m) = \pm \frac{1}{2}$

For convex mirror, virtual image is formed for real object.

Therefore, m is +ve

$$\frac{1}{2} = \frac{f}{f-u}$$

$u = -15$ cm

Question81

A biconvex lens of refractive index 1.5 has a focal length of 20cm in air. Its focal length when immersed in a liquid of refractive index 1.6 will be:

[29-Jan-2024 Shift 1]

Options:

A.

−16 cm

B.

−160 cm

C.

+160 cm

D.

+16 cm

Answer: B

Solution:

$\mu_1 = 1.5$

$\mu_m = 1.6$

$f_a = 20$ cm

As $\dfrac{f_m}{f_a} = \dfrac{(\mu_1 - 1)\mu_m}{(\mu_1 - \mu_m)}$

$\dfrac{f_m}{20} = \dfrac{(1.5-1)1.6}{(1.5-1.6)}$

$f_m = -160$ cm

Question82

If the distance between object and its two times magnified virtual image produced by a curved mirror is 15cm, the focal length of the mirror must be :

[29-Jan-2024 Shift 2]

Options:

A.

15 cm

B.

-12 cm

C.

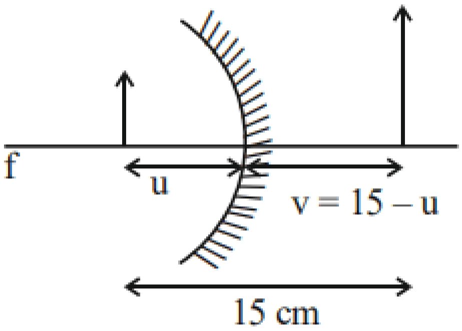
-10 cm

D.

$10/3$ cm

Answer: C

Solution:



$m = 2 = \frac{-v}{u}$

$2 = \frac{-(15-u)}{-u}$

$2u = 15 - u$

$3u = 15 \Rightarrow u = 5\text{ cm}$

$v = 15 - u = 15 - 5 = 10\text{ cm}$

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$= \frac{1}{10} + \frac{1}{(-5)} = \frac{1-2}{10} = \frac{-1}{10}$

$f = -10\text{ cm}$

Question83

The distance between object and its two times magnified real image as produced by a convex lens is 45cm. The focal length of the lens used is ____cm.

[30-Jan-2024 Shift 1]

Answer: 10

Solution:

$$\frac{v}{u} = -2$$

$v = -2u$...... (i)

$v - u = 45$.......(ii)

$\Rightarrow u = -15$ cm

$v = 30$ cm

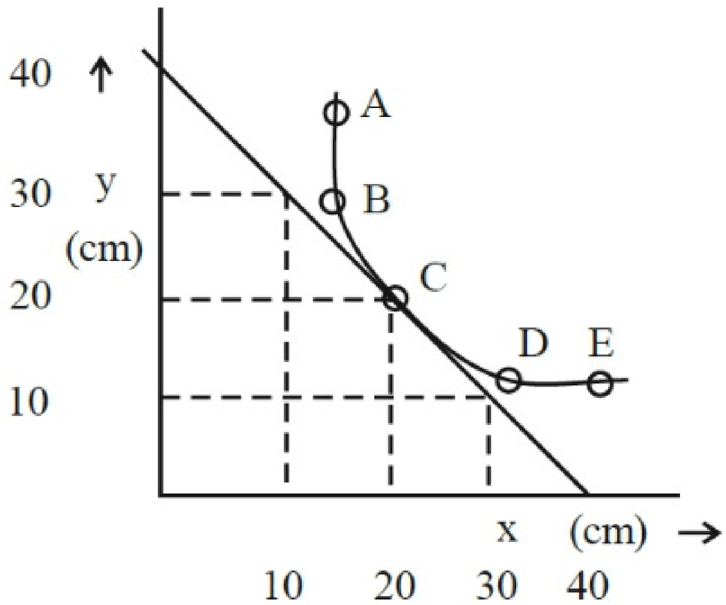
$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$f = +10$ cm

Question84

In an experiment to measure the focal length (f) of a convex lens, the magnitude of object distance (x) and the image distance (y) are measured with reference to the focal point of the lens. The y − x plot is shown in figure.

The focal length of the lens is _____cm.



[30-Jan-2024 Shift 2]

Answer: 20

Solution:

$$\frac{1}{f+20} - \frac{1}{-(f+20)} = \frac{1}{f}$$

$$\frac{2}{f+20} = \frac{1}{f} \quad f = 20\text{ cm}$$

Or $x_1 x_2 = f^2$ gives $f = 20\text{ cm}$

Question85

The refractive index of a prism with apex angle A is cotA/2. The angle of minimum deviation is :

[31-Jan-2024 Shift 1]

Options:

A.

$\delta_m = 180^\circ - A$

B.

$\delta_m = 180^\circ - 3A$

C.

$\delta_m = 180^\circ - 4A$

D.

$\delta_m = 180^\circ - 2A$

Answer: D

Solution:

$$\mu = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\frac{A}{2}}$$

$$\frac{\cos\frac{A}{2}}{\sin\frac{A}{2}} = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\frac{A}{2}}$$

$$\sin\left(\frac{\pi}{2}-\frac{A}{2}\right) = \sin\left(\frac{A+\delta_m}{2}\right)$$

$$\frac{\pi}{2}-\frac{A}{2} = \frac{A}{2}+\frac{\delta m}{2}$$

$$\delta_m = \pi - 2A$$

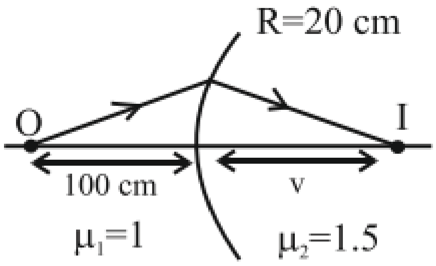
Question86

Light from a point source in air falls on a convex curved surface of radius 20cm and refractive index 1.5. If the source is located at 100cm from the convex surface, the image will be formed at ______cm from the object.

[31-Jan-2024 Shift 2]

Answer: 200

Solution:



$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.5}{v} - \frac{1}{-100} = \frac{1.5 - 1}{20}$$

$v = 100 \text{ cm}$

Distance from object

$= 100 + 100$

$= 200 \text{ cm}$

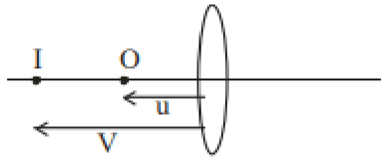
Question87

The distance between object and its 3 times magnified virtual image as produced by a convex lens is 20cm. The focal length of the lens used is ______cm.

[1-Feb-2024 Shift 1]

Answer: 15

Solution:



$v = 3u$

$v - u = 20 \text{ cm}$

$2u = 20 \text{ cm}$

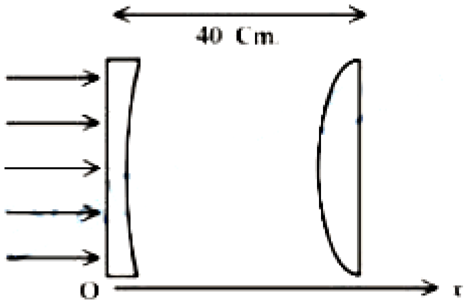
$u = 10 \text{ cm}$

$$\frac{1}{(-30)} - \frac{1}{(-10)} = \frac{1}{f}$$

$f = 15 \text{ cm}$

Question88

As shown in the figure, a combination of a thin plano concave lens and a thin plano convex lens is used to image an object placed at infinity. The radius of curvature of both the lenses is 30 cm and refraction index of the material for both the lenses is 1.75. Both the lenses are placed at distance of 40 cm from each other. Due to the combination, the image of the object is formed at distance x = ___cm, from concave lens.



[24-Jan-2023 Shift 1]

Answer: 120

Solution:

$$\frac{1}{f_1} = (1.75 - 1)\left(-\frac{1}{30}\right)$$
$$\Rightarrow f_1 = -40\,\text{cm}$$
$$\frac{1}{f_2} = (1.75 - 1)\left(\frac{1}{30}\right) \Rightarrow f_2 = 40\,\text{cm}$$
Image from L_1 will be virtual and on the left of L_1 at focal length $40\,\text{cm}$. So the object for L_2 will be $80\,\text{cm}$ from L_2 which is $2f$. Final image is formed at $80\,\text{cm}$ from L_2 on the right.
So $x = 120$

Question89

When a beam of white light is allowed to pass through convex lens parallel to principal axis, the different colours of light converge at different point on the principle axis after refraction. This is called: [24-Jan-2023 Shift 2]

Options:

A. Scattering

B. Chromatic aberration

C. Spherical aberration

D. Polarisation

Answer: B

Solution:

Based on fact.

Question90

A convex lens of refractive index 1.5 and focal length 18 cm in air is immersed in water. The change in focal length of the lens will be cm.

(. Given refractive index of water $= \frac{4}{3}$)

[24-Jan-2023 Shift 2]

Answer: 54

Solution:

$$\frac{I}{f_{H_2O}} = \left(\frac{\mu_g}{\mu_{H_2O}} - 1\right)\left(\frac{2}{R}\right)$$
$$= \frac{1}{8}\left(\frac{2}{R}\right)$$
$$= \frac{1}{(4f_{air})}$$

So, $f_{H_2O} = 4f_{air} = 72\text{ cm}$

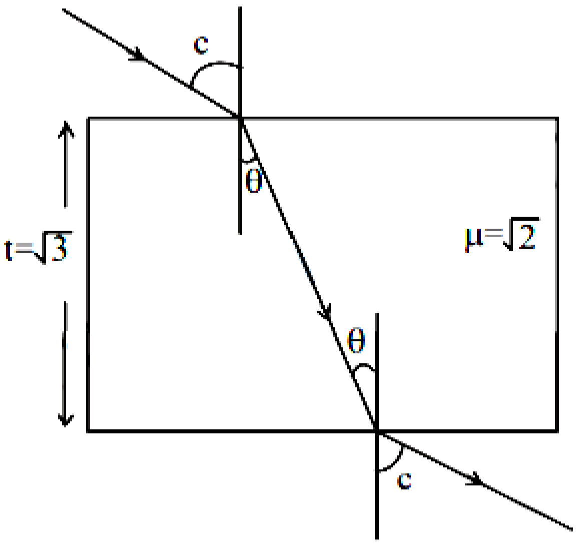
So change in focal length $= 72 - 18 = 54\text{ cm}$

Question91

A ray of light is incident from air on a glass plate having thickness $\sqrt{3}$ cm and refractive index $\sqrt{2}$. The angle of incidence of a ray is equal to the critical angle for glass-air interface. The lateral displacement of the ray when it passes through the plate is $\times 10^{-2}$ cm. (given $\sin 15^{\circ} = 0.26$)
[25-Jan-2023 Shift 1]

Answer: 52

Solution:



$\sin c = \frac{1}{\sqrt{2}}$

$c = 45^\circ$

$\sin c = \mu \sin\theta$

$\frac{1}{\sqrt{2}} = \sqrt{2}\sin\theta$

$\theta = 30^\circ$

Lateral displacement:

$x = t\sin(i - r)\sec r$

$x = \sqrt{3}\sin(45^\circ - 30^\circ)\sec 30^\circ$

$x = \sqrt{3}(0.26)\left(\frac{2}{\sqrt{3}}\right)$

$X = 0.52\text{ cm}$

$x = 52 \times 10^{-2}\text{ cm}$

Question92

The light rays from an object have been reflected towards an observer from a standard flat mirror, the image observed by the observer are :-
A. Real
B. Erect
C. Smaller in size then object
D. Laterally inverted
Choose the most appropriate answer from the options given below :
[25-Jan-2023 Shift 2]

Options:

A. B and D only

B. B and C only

C. A and D only

D. A, C and D only

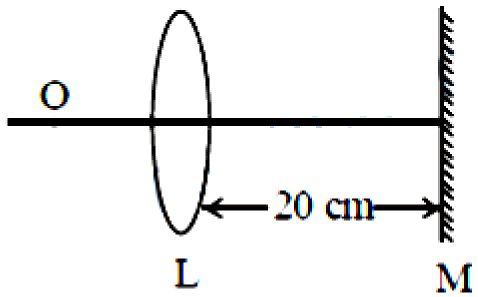
Answer: A

Solution:

Plane mirror forms erect, same sized, laterally inverted and virtual image of real object.

Question93

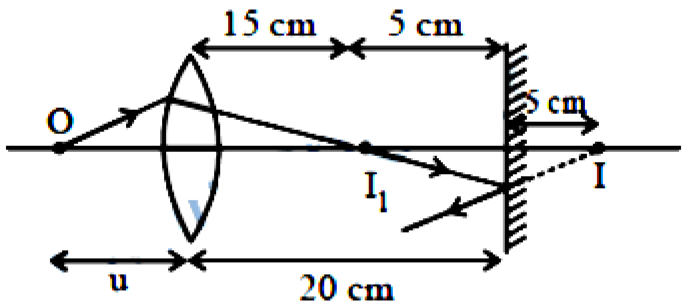
An object is placed on the principal axis of convex lens of focal length 10 cm as shown. A plane mirror is placed on the other side of lens at a distance of 20 cm. The image produced by the plane mirror is 5 cm inside the mirror. The distance of the object from the lens is ______ cm.



[25-Jan-2023 Shift 2]

Answer: 30

Solution:



$f = 10\,cm$

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{15} - \frac{1}{-u} = \frac{1}{10}$

$\Rightarrow \frac{1}{u} = \frac{1}{10} - \frac{1}{15}$

On solving we get value of u as $30\,cm$.

Question94

A scientist is observing a bacteria through a compound microscope. For better analysis and to improve its resolving power he should. (Select the best option)
[29-Jan-2023 Shift 2]

Options:

A. Increase the wave length of the light

B. Increase the refractive index of the medium between the object and objective lens

C. Decrease the focal length of the eye piece

D. Decrease the diameter of the objective lens

Answer: B

Solution:

$P = \frac{2\mu \sin \theta}{1.22\lambda}$

Question95

A person has been using spectacles of power-1.0 diopter for distant vision and a separate reading glass of power 2.0 diopters. What is the least distance of distinct vision for this person:
[30-Jan-2023 Shift 1]

Options:

A. 10 cm

B. 40 cm

C. 30 cm

D. 50 cm

Answer: D

Solution:

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$
$$P=2D=2m^{-1}$$
$$\Rightarrow \frac{1}{f}=\frac{2}{100}cm^{-1}$$
$$\frac{1}{V}-\left(-\frac{1}{25}\right)=\frac{2}{100}$$
$$\Rightarrow \frac{1}{V}=\frac{1}{50}-\frac{1}{25}$$
$$\Rightarrow V=-50\ cm$$

Question96

In an experiment for estimating the value of focal length of converging mirror, image of an object placed at 40 cm from the pole of the mirror is formed at distance 120 cm from the pole of the mirror. These distances are measured with a modified scale in which there are 20 small divisions in 1 cm. The value of error in measurement of focal length of the mirror is 1 / K cm. The value of K is ________.
[30-Jan-2023 Shift 1]

Answer: 3

Solution:

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{-1}{120}-\frac{1}{40}=\frac{1}{f},\quad f=-30\,\text{cm}$$

Now,

$$\frac{-1}{v^2}dv-\frac{1}{u^2}du=-\frac{1}{f^2}df$$

Also $dv=du=\frac{1}{20}\,\text{cm}$

$$\therefore\quad \frac{\frac{1}{20}}{(120)^2}+\frac{\frac{1}{20}}{(40)^2}=\frac{df}{(30)^2}$$

On solving

$$df=\frac{1}{32}\,\text{cm}$$

$\therefore\ k=32$

Question97

A thin prism P_1 with an angle 6° and made of glass of refractive index 1.54 is combined with another prism P_2 made from glass of refractive index 1.72 to produce dispersion without average deviation. The angle of prism P_2 is :
[30-Jan-2023 Shift 2]

Options:

A. 6°

B. 1.3°

C. 7.8°

D. 4.5°

Answer: D

Solution:

$\delta_1 = \delta_2$ [for no average deviation]

$\Rightarrow 6^\circ(1.54 - 1) = A(1.72 - 1)$

$\Rightarrow A = \dfrac{6^\circ \times 0.54}{0.72}$

$= \dfrac{18^\circ}{4} = 4.5^\circ$

Question98

In a medium the speed of light wave decreases to 0.2 times to its speed in free space The ratio of relative permittivity to the refractive index of the medium is x : 1. The value of x is _______.
(Given speed of light in free space $= 3 \times 10^8 \text{m s}^{-1}$ and for the given medium $\mu_r = 1$)
[31-Jan-2023 Shift 1]

Answer: 5

Solution:

$V = \dfrac{C}{\mu} \Rightarrow \mu = \dfrac{C}{V} = \dfrac{C}{0.2C}$

$\mu = 5$

$\mu = \sqrt{E_r \mu_r}$

$\Rightarrow E_r = \dfrac{\mu^2}{\mu_r}$

$\therefore \dfrac{E_r}{\mu} = \dfrac{\mu}{\mu_r} = 5$

Question99

A microscope is focused on an object at the bottom of a bucket. If liquid with refractive index - is poured inside the bucket, then microscope have to be raised by 30 cm to focus the object again. The height of the liquid in the bucket is :
[31-Jan-2023 Shift 2]

Options:

A. 75 cm

B. 50 cm

C. 18 cm

D. 12 cm

Answer: A

Solution:

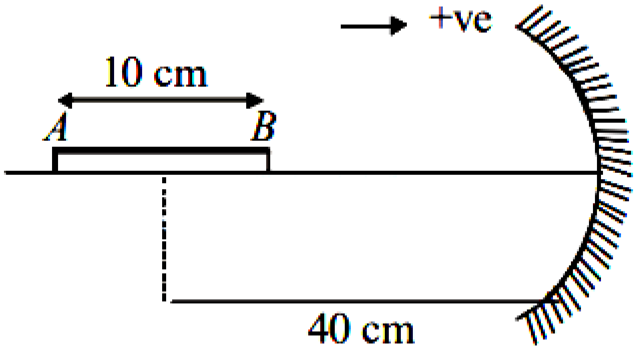
$$\text{Shift } = \left(d - \frac{d}{\mu}\right) = 30 \text{ cm}$$
$$\Rightarrow d\left[1 - \frac{1}{\frac{5}{3}}\right] = 30$$
$$\Rightarrow d = \frac{30 \times 5}{2} = 75 \text{ cm}$$

Question100

A thin cylindrical rod of length 10 cm is placed horizontally on the principle axis of a concave mirror of focal length 20 cm. The rod is placed in a such a way that mid point of the rod is at 40 cm from the pole of mirror. The length of the image formed by the mirror will be $\frac{x}{3}$ cm. The value of x is ________.
[1-Feb-2023 Shift 1]

Answer: 32

Solution:



$U_A = -45$ cm, $f = -20$ cm

$V_A = \frac{-45 \times (-20)}{-45 - (-20)} = \frac{-900}{25} = -36$ cm

And $U_B = -35$ cm

$\therefore V_B = \frac{-35 \times (-20)}{-35 - (-20)} = \frac{700}{-15}$

$\therefore V_A - V_B =$ length of image

$= \left(-36 + \frac{140}{3}\right)$ cm

$= \frac{-108 + 140}{3}$ cm

$= \frac{32}{3}$ cm

$\therefore x = 32$

Question101

Two objects A and B are placed at 15 cm and 25 cm from the pole in front of a concave mirror having radius of curvature 40 cm. The distance between images formed by the mirror is:
[1-Feb-2023 Shift 2]

Options:

A. 40 cm

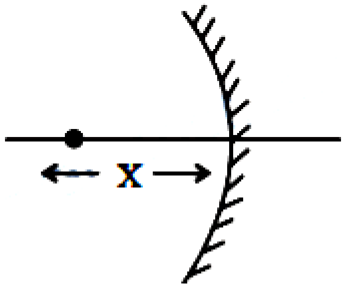
B. 60 cm

C. 160 cm

D. 100 cm

Answer: C

Solution:



by mirror formula

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v_1}+\frac{1}{-15}=\frac{1}{(-20)}$$

$$\frac{1}{v_1}=-\frac{1}{20}+\frac{1}{15}$$

$$=\frac{-3+4}{60}$$

$v_1 = 60\,\text{cm}$

$$\frac{1}{v_2}+\frac{1}{(-25)}=\frac{1}{(-20)}$$

$$\frac{1}{v_2}=\frac{-1}{20}+\frac{1}{25}$$

$$=\frac{-5+4}{100}=\frac{-1}{100}$$

$v_2 = -100\,\text{cm}$

$d = 60 + 100 = 160\,\text{cm}$

Question102

A monochromatic light wave with wavelength λ_1 and frequency v_1 in air enters another medium. If the angle of incidence and angle of refraction at the interface are 45° and 30° respectively, then the wavelength λ_2 and frequency v_2 of the refracted wave are : [6-Apr-2023 shift 1]

Options:

A. $\lambda_2 = \frac{1}{\sqrt{2}}\lambda_1,\ v_2 = v_1$

B. $\lambda_2 = \lambda_1,\ v_2 = \frac{1}{\sqrt{2}}v_1$

C. $\lambda_2 = \lambda_1, v_2 = \sqrt{2} v_1$

D. $\lambda_2 = \sqrt{2}\lambda_1, v_2 = v_1$

Answer: A

Solution:

$$1 \times \sin 45 = \mu \sin 30$$
$$\Rightarrow \frac{1}{\sqrt{2}} = \mu \times \frac{1}{2}$$
$$\Rightarrow \mu = \sqrt{2} ---- (i)$$
$$\text{Now, } \frac{\mu_1}{\mu_2} = \frac{V_2}{V_1} = \frac{\lambda_2}{\lambda_1} ---- (ii)$$
Using eq (i) and (ii),
$$\lambda_2 = \frac{1}{\sqrt{2}}\lambda_1$$
$$\text{And } V_2 = \frac{1}{\sqrt{2}} V_1$$
Now, for relation between frequencies,

Frequency, $v = \frac{v}{\lambda}$

$$\text{Or } \frac{v_1}{v_2} = \frac{v_1}{v_2} \times \frac{\lambda_2}{\lambda_1} = 1$$
$$v_1 = v_2$$

Question103

A pole is vertically submerged in swimming pool, such that it gives a length of shadow 2.15m within water when sunlight is incident at an angle of 30° with the surface of water. If swimming pool is filled to a height of 1.5m, then the height of the pole above the water surface in centimeters is ($n_w = 4/3$) ________.
[6-Apr-2023 shift 1]

Answer: 50

Solution:

$$\sin 60 = \frac{4}{3}\sin r$$

$$\Rightarrow \sin r = \frac{3}{4}\times\frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{8} --- (i)$$

$$\cos r = \sqrt{1-\frac{27}{64}} = \frac{\sqrt{37}}{8} = 0.75$$

$$\Rightarrow \tan r = \sqrt{\frac{27}{37}}$$

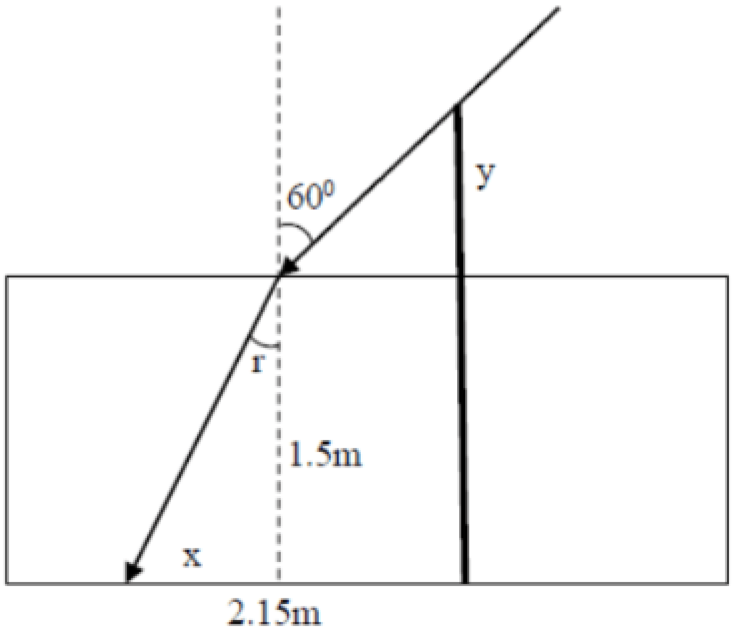
$$\Rightarrow \frac{x}{1.5} = 0.85$$

$$\Rightarrow x = 0.85 \times 1.5 = 1.275\text{m}$$

$$\tan 30 = \frac{y}{2.15-1.275} = \frac{y}{0.875}$$

$$y = \frac{0.875}{1.732} = 0.50$$

So length of pole above water surface $= 0.50\text{m} = 50\text{ cm}$



Question104

A 2 meter long scale with least count of 0.2 cm is used to measure the locations of objects on an optical bench. While measuring the focal length of a convex lens, the object pin and the convex lens are placed at 80 cm mark and 1m mark., respectively. The image of the object pin on the other side of lens coincides with image pin that is kept at 180 cm mark. The \% error in the estimation of focal length is : [6-Apr-2023 shift 2]

Options:

A. 0.51

B. 1.02

C. 0.85

D. 1.70

Answer: D

Solution:

Based on the data provided

$U = 100 - 80 = 20 \text{ cm}$

$V = 180 - 100 = 80 \text{ cm}$

Using $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ or $f = \frac{uv}{u+v} = \frac{20 \times 80}{20 + 80}$ or $f = 16 \text{ cm}$

For error analysis,

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

Differentiating

$-\frac{Df}{f^2} = -\frac{Dv}{v^2} - \frac{\Delta u}{u^2}$

To calculate $\Delta u \& \Delta v$

$U = (100 \pm 2) - (80 \pm 0.2) = (20 \pm 0.4) \text{ cm}$

Therefore $\Delta u = 0.4 \text{ cm}$,

Similarly $\Delta v = 0.4 \text{ cm}$.

Now $\frac{\Delta f}{f} = f\left[\frac{\Delta v}{v^2} + \frac{\Delta u}{u^2}\right]$

$\frac{\Delta f}{f} = 16\left[\frac{0.4}{(80)^2} + \frac{0.4}{(20)^2}\right]$

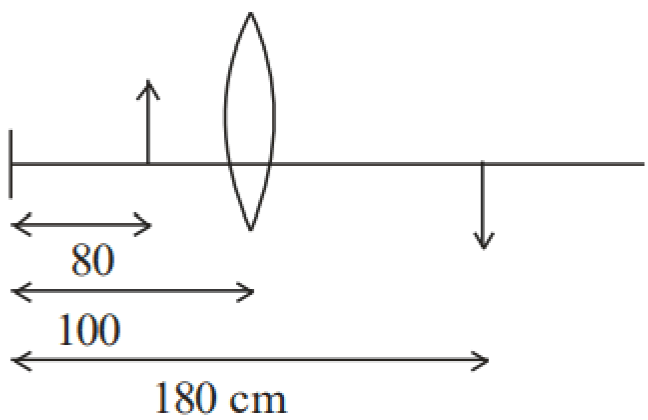
(Note: every data is in cm)

$\frac{\Delta f}{f} = \frac{16 \times 0.4}{(20)^2}\left[\frac{1}{4^2} + 1\right]$

$= \frac{16 \times 0.4}{20^2} \times \frac{17}{16} = \frac{17 \times 0.4}{400}$

$\%$ Error : $\frac{\Delta f}{f} \times 100 = \frac{17 \times 0.4}{400} \times 1000$

$= 1.7$



Question105

Given below are two statements : one is labelled as assertion A and the other is labelled as Reason R
Assertion A : The phase difference of two light wave change if they travel through different media having same thickness, but different indices of refraction
Reason R : The wavelengths of waves are different in different media.
In the light of the above statements, choose the most appropriate answer from the options given below
[6-Apr-2023 shift 2]

Options:

A. Both A and R are correct and R is the correct explanation of A

B. A is not correct but R is correct

C. A is correct but R is not correct

D. Both A and R are correct but R is NOT the correct explanation of A

Answer: A

Solution:

Both the statements are true
As we know speed of light in a medium
$v = \frac{c}{\mu}$ or $f\lambda = \frac{c}{\mu}$
therefore $\lambda \propto \frac{1}{\mu}$
when light will travel through two different mediums their phase difference will change $\Delta Q = \frac{2\pi}{\lambda} \Delta x$
and R is correction explanation

Question106

In a reflecting telescope, a secondary mirror is used to: [8-Apr-2023 shift 1]

Options:

A. Make chromatic aberration zero

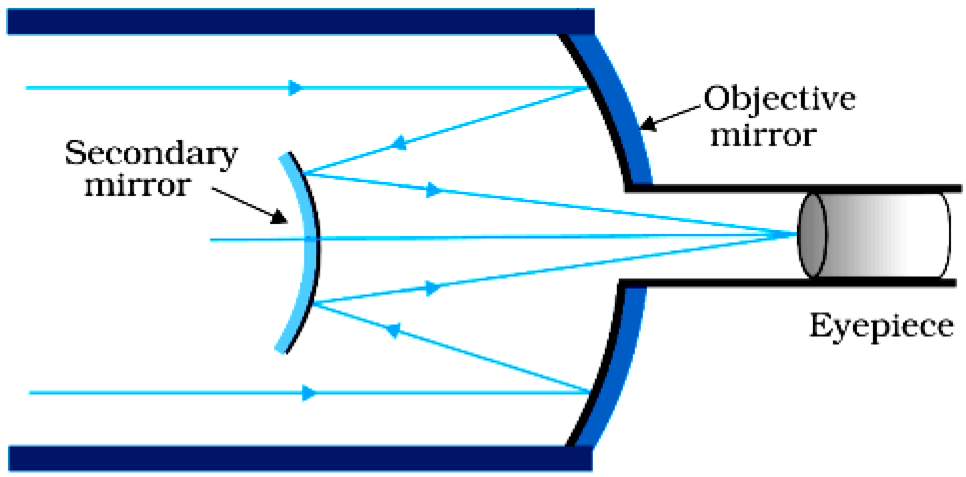
B. Reduce the problem of mechanical support

C. Move the eyepiece outside the telescopic tube

D. Remove spherical aberration

Answer: C

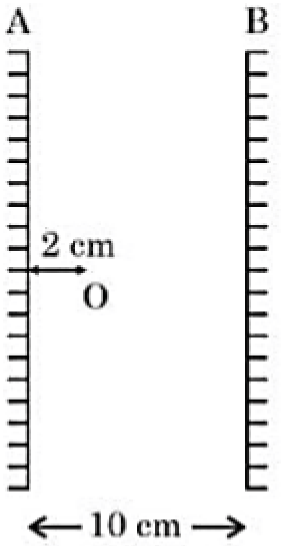
Solution:



To move the eye piece outside the telescopic tube

Question107

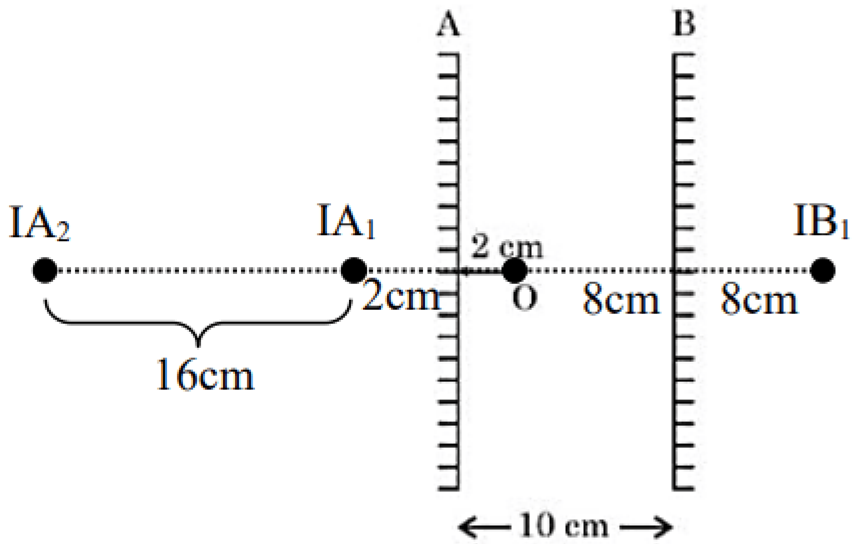
Two vertical parallel mirrors A and B are separated by 10 cm. A point object O is placed at a distance of 2 cm from mirror A. The distance of the second nearest image behind mirror A from the mirror A is ________ cm.



[8-Apr-2023 shift 1]

Answer: 18

Solution:



$$d = 2 + 16$$
$$d = 18 \text{ cm}$$

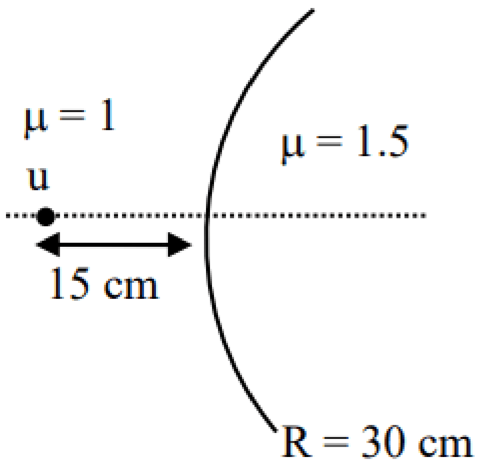
Question108

Two transparent media having refractive indices 1.0 and 1.5 are separated by a spherical refracting surface of radius of curvature 30 cm. The centre of curvature of surface is towards denser medium

and a point object is placed on the principle axis in rarer medium at a distance of 15 cm from the pole of the surface. The distance of image from the pole of the surface is ______ cm.
[8-Apr-2023 shift 2]

Answer: 30

Solution:



$$\frac{1.5}{v} - \frac{1}{(-15)} = \frac{1.5-1}{30}$$

$$\Rightarrow \frac{1.5}{v} - \frac{1}{60} - \frac{1}{15} = \frac{1-4}{60}$$

$V = -30\text{ cm}$

$= 30\text{ cm}$

Question109

An object is placed at a distance of 12 cm in front of a plane mirror. The virtual and erect image is formed by the mirror. Now the mirror is moved by 4 cm towards the stationary object. The distance by which the position of image would be shifted, will be
[10-Apr-2023 shift 1]

Options:

A. 4 cm towards mirror

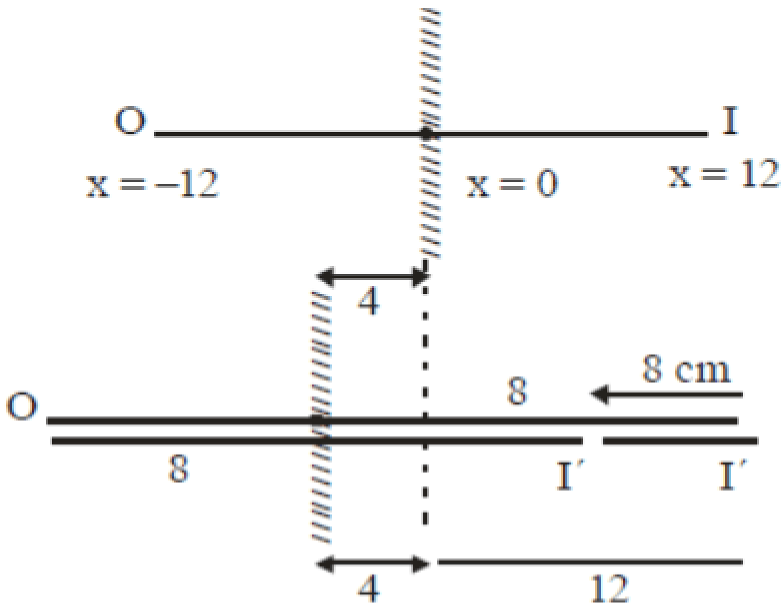
B. 8 cm away from mirror

C. 2 cm towards mirror

D. 8 cm towards mirror

Answer: D

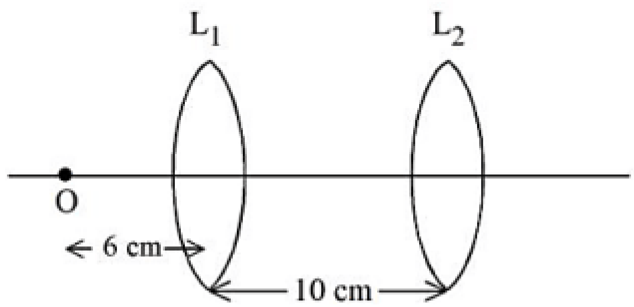
Solution:



8 cm towards mirror
Image will be shifted 8 cm towards mirror.

Question110

A point object, ' O ' is placed in front of two thin symmetrical coaxial convex lenses L_1 and L_2 with focal length 24 cm and 9 cm respectively. The distance between two lenses is 10 cm and the object is placed 6 cm away from lens L_1 as shown in the figure. The distance between the object and the image formed by the system of two lenses is ________ cm.

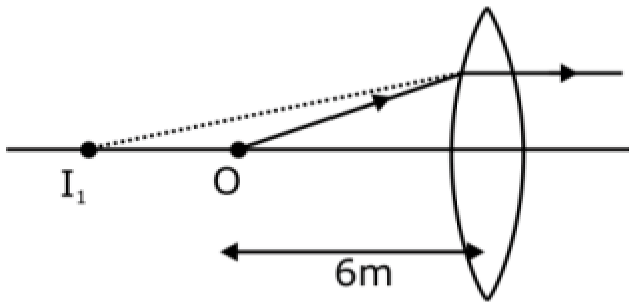


[10-Apr-2023 shift 2]

Answer: 18

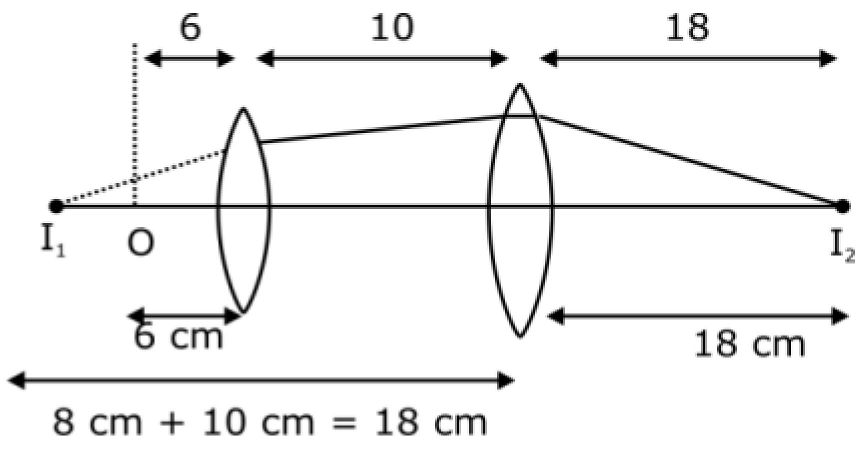
Solution:

Due to lens L_1



$u = -6m$
$f = +24m$
$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
$\frac{1}{v} = \frac{1}{24} - \frac{1}{6} = \frac{1-4}{24} \Rightarrow v = -8m$

Due to lens L_2



$U = -18m$
$F = +9m$
$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}x$

$$\frac{1}{v} - \frac{1}{9} = \frac{1}{18}$$

$$V = \frac{1}{v} = \frac{2-1}{18}$$

$$V = 18m$$

Question111

The critical angle for a denser-rarer interface is 45°. The speed of light in rarer medium is $3 \times 10^{8} m/s$. The speed of light in the denser medium is : [11-Apr-2023 shift 1]

Options:

A. $2.12 \times 10^{8} m/s$

B. $5 \times 10^{7} m/s$

C. $3.12 \times 10^{7} m/s$

D. $\sqrt{2} \times 10^{8} m/s$

Answer: A

Solution:

$$\operatorname{Sin} ic = \frac{\mu_r}{\mu_d} \Rightarrow \sin 45^{\circ} = \frac{\mu_r}{\mu_d}$$

$$\Rightarrow \frac{\mu_r}{\mu_d} = \sqrt{2} \ldots (1)$$

We know

$$V \propto \frac{1}{\mu} \Rightarrow \frac{V_d}{V_r} = \frac{\mu_r}{\mu_d}$$

$$= \frac{V_d}{3 \times 10^{8}} = \frac{1}{\sqrt{2}}$$

$$V_d = \frac{3}{\sqrt{2}} \times 10^{8} = 3 \times 0.7 \times 10^{8}$$

$$V_d = 2.12 \times 10^{8} m/sec$$

Ans. Option (1)

Question112

The radius of curvature of each surface of a convex lens having refractive index 1.8 is 20 cm. The lens is now immersed in a liquid of refractive index 1.5. The ratio of power of lens in air to its power in the liquid will be x : 1. The value of x is ________.
[11-Apr-2023 shift 1]

Answer: 4

Solution:

$$\frac{1}{f} = \left(\frac{\mu_\ell}{\mu_m} - 1\right)\left(\frac{1}{R} - \frac{1}{-R}\right)$$

$$P_1 = \frac{2}{R}\left(\frac{1.8}{1} - 1\right]$$

$$P_1 = \frac{2}{R}(0.8) = \frac{1.6}{R} \ldots (1)$$

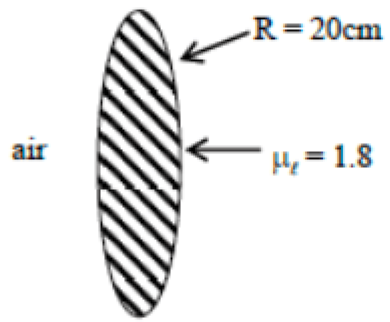
Now,

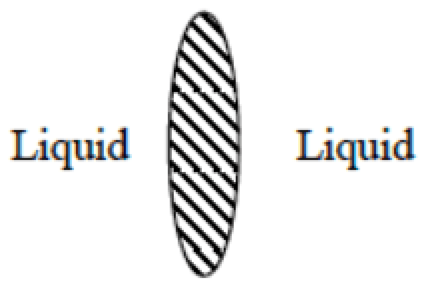
$$P_2 = \frac{2}{R}\left[\frac{1.8}{1.5} - 1\right]$$

$$P_2 = \frac{2}{R}\left[\frac{0.3}{1.5}\right] = \frac{2}{R} \times \frac{1}{5} = \frac{2}{5R}$$

$$\frac{P_{air}}{P_{liquid}} = \frac{P_1}{P_2} = \frac{\left(\frac{1.6}{R}\right)}{\left(\frac{0.4}{R}\right)} = \frac{4}{1}$$

Ans. $\rightarrow 4$





Question113

When one light ray is reflected from a plane mirror with 30° angle of reflection, the angle of deviation of the ray after reflection is: [11-Apr-2023 shift 2]

Options:

A. 140°

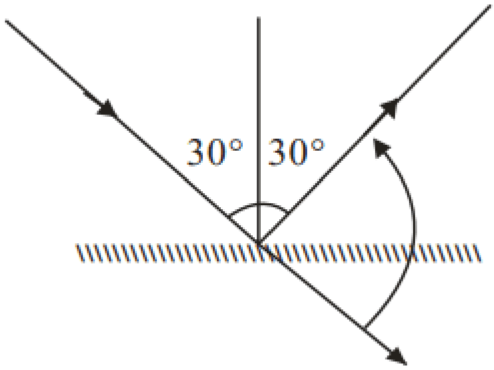
B. 130°

C. 120°

D. 110°

Answer: C

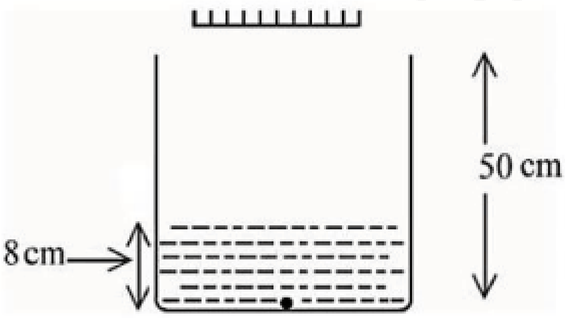
Solution:



$\delta = \Pi - 2i = \Pi - 2 \times 30$
$= 180 - 60 = 120^{\circ}$

Question114

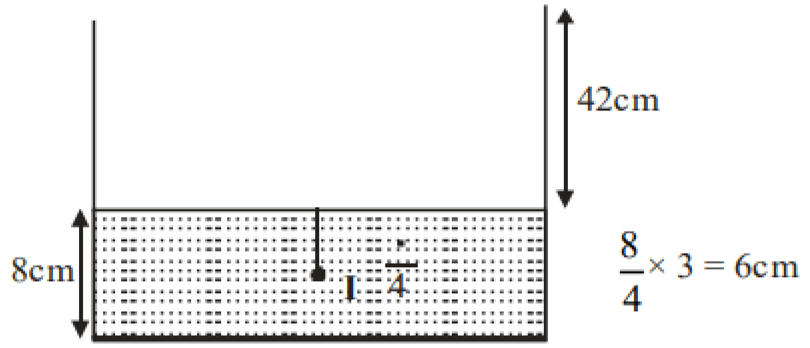
As shown in the figure, a plane mirror is fixed at a height of 50 cm from the bottom of tank containing water $\left(\mu = \frac{4}{3}\right)$. The height of water in the tank is 8 cm. A small bulb is placed at the bottom of the water tank. The distance of image of the bulb formed by mirror from the bottom of the tank is ________ cm.



[11-Apr-2023 shift 2]

Answer: 98

Solution:



Apparent depth of $O = \frac{d}{\mu} = 6$

Distance between O and $I_2 = 48 + 50 = 98$ cm

Question115

An ice cube has a bubble inside. When viewed from one side the apparent distance of the bubble is 12 cm. When viewed from the opposite side, the apparent distance of the bubble is observed as 4 cm. If the side of the ice cube is 24 cm, the refractive index of the ice cube is [12-Apr-2023 shift 1]

Options:

A. $\frac{4}{3}$

B. $\frac{3}{2}$

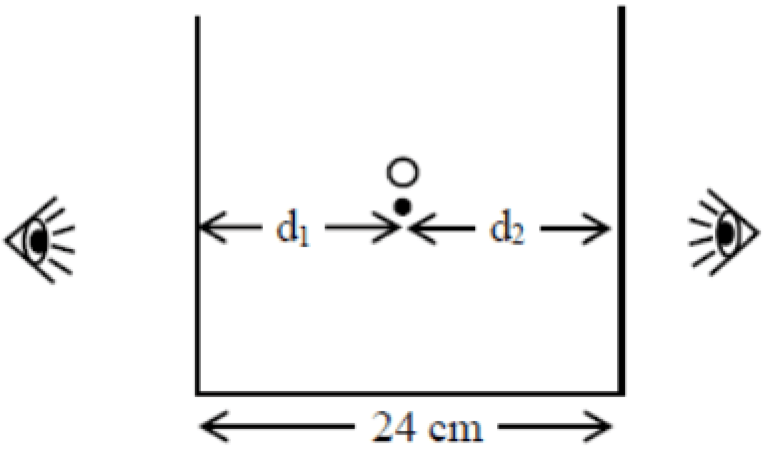
C. $\frac{2}{3}$

D. $\frac{6}{5}$

Answer: B

Solution:

Question says small bubble is trapped inside a cube of length 24 cm. If apparent distance of bubble is 12 cm and 4 cm from one and other side and find $x =$?



Using $\dfrac{\text{Apparent depth}}{\text{Realdepth}} = \dfrac{\mu_2}{\mu_1}$

$\mu_2 = 1 \quad \mu_1$ is glass

$\mu_1 = \mu \quad \mu_2$ is air

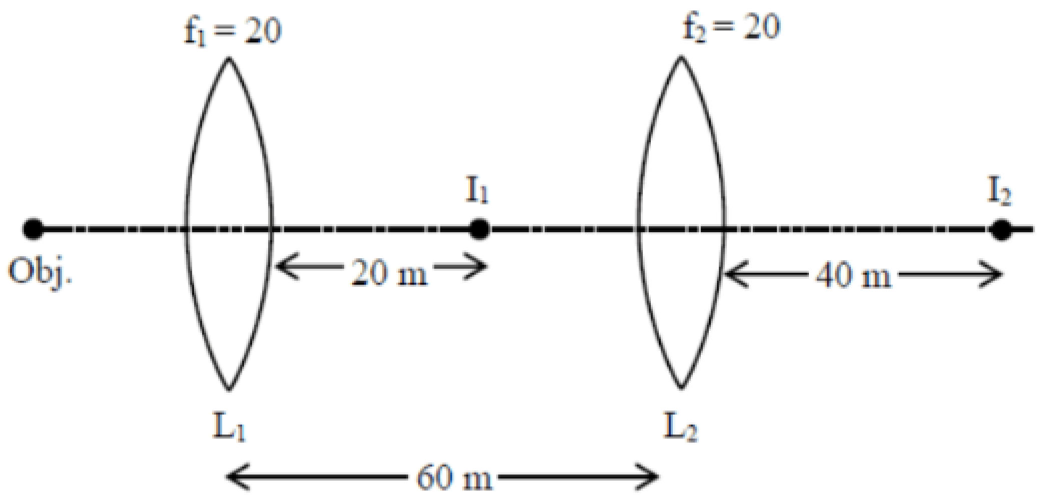
$\dfrac{12}{d_1} = \dfrac{1}{\mu}$

$\dfrac{12\mu = d_1}{\dfrac{4}{d_2}} = \dfrac{1}{\mu} \quad \Rightarrow (1)$

$d_1 + d_2 = 24m \Rightarrow \quad 12\mu + 4\mu = 24 \Rightarrow d_2$

Question116

Two convex lenses of focal length 20 cm each are placed coaxially with a separation of 60 cm between them. The image of the distant object formed by the combination is at _______ cm from the first lens.



[12-Apr-2023 shift 1]

Answer: 100

Solution:

For I st lens

$$\frac{1}{V_1} - \frac{1}{\infty} = \frac{1}{20}$$

$V_1 = 20\,\text{cm}$

For 2 nd lens

$$\frac{1}{V_2} - \frac{1}{-40} = \frac{1}{20}$$

$V_2 = 40\,\text{cm}$

So dist $= 40 + 60 = 100\,\text{cm}$

Question117

A vessel of depth ' d ' is half filled with oil of refractive index n_1 and the other half is filled with water of refractive index n_2. The apparent depth of this vessel when viewed from above will be - [13-Apr-2023 shift 1]

Options:

A. $\frac{d\,(n_1 + n_2)}{2n_1 n_2}$

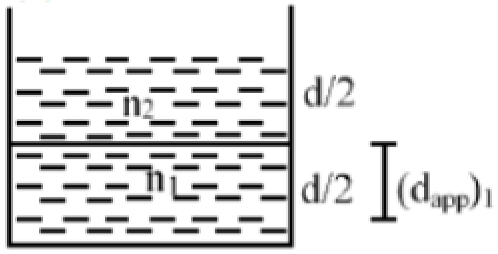
B. $\frac{dn_1 n_2}{(n_1 + n_2)}$

C. $\frac{dn_1 n_2}{2(n_1 + n_2)}$

D. $\frac{2d\,(n_1 + n_2)}{n_1 n_2}$

Answer: A

Solution:



$$(d_{spp})_1 = \frac{d}{2\left(\frac{n_1}{n_2}\right)} = \frac{n_2 d}{2n_1}$$

$$(d_{app})_2 = \frac{(d_{spp})_1 + \frac{d}{2}}{n_2}$$

$$= \frac{\left(\frac{n_2}{n_1} + 1\right)\frac{d}{2}}{n_2}$$

$$(d_{spp})_2 = \frac{(n_1 + n_2)d}{2n_1 n_2}$$

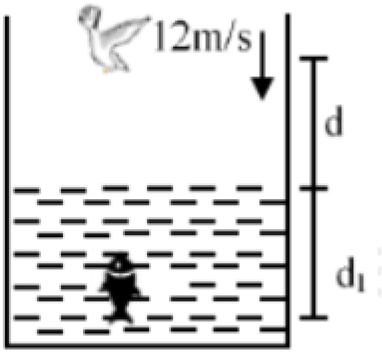
Question118

A fish rising vertically upward with a uniform velocity of $8ms^{-1}$, observes that a bird is diving vertically downward towards the fish with the velocity of $12ms^{-1}$. If the refractive index of water is $\frac{4}{3}$, then the actual velocity of the diving bird to pick the fish, will be _________ ms^{-1}.

[13-Apr-2023 shift 1]

Answer: 3

Solution:



$d_{app} = d_1 + \mu d$

$v_{app} = v_1 + \mu v$

$12 = 8 + \frac{4}{3}v$

$4 = \frac{4}{3}v$

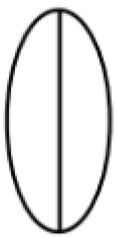
$v = 3m/s$

Question119

A bi convex lens of focal length 10 cm is cut in two identical parts along a plane perpendicular to the principal axis. The power of each lens after cut is ________ D.
[13-Apr-2023 shift 2]

Answer: 5

Solution:



$P_1 + P_1 = P = \frac{1}{f}$

$2P_1 = \frac{1}{0.1}$

$P_1 = 5D$

Question120

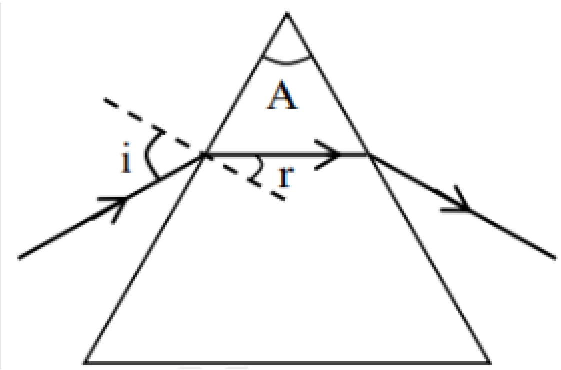
The refractive index of a transparent liquid filled in an equilateral hollow prism is $\sqrt{2}$. The angle of minimum deviation for the liquid will be ________.
[15-Apr-2023 shift 1]

Answer: 30

Solution:

For minimum deviation

$r = \frac{A}{2} = \frac{60}{2} = 30^{\circ}$



$1 \sin i = \sqrt{2} \sin r$

$\sin i = \sqrt{2} \times \sin 30^{\circ}$

$\sin i = \frac{1}{\sqrt{2}}$

$i = 45^{\circ}$

$\delta_{min} = 2i - A = 90 - 60 = 30^{\circ}$

$\delta_{min} = 30^{\circ}$

Question121

Two identical thin biconvex lens of focal length 15 cm and refractive index 1.5 are in contact with each other. The space between the lenses is filled with a liquid of refractive index 1.25. The focal length of the combination is _____cm.
[24-Jun-2022-Shift-1]

Answer: 10

Solution:

$$\frac{1}{f_1} = \left(\frac{\mu_e}{\mu_m} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

here $|R_1| = |R_2| = R$

$$\Rightarrow \frac{1}{f_{1_1}} = (1.5 - 1)\left(\frac{2}{R} \right) = \frac{1}{15}$$

$$\Rightarrow \frac{1}{R} = \frac{1}{15} \text{ or } R = 15 \text{ cm}$$

for the concave lens made up of liquid

$$\frac{1}{f_{1_2}} = (1.25 - 1)\left(-\frac{2}{R} \right) = -\frac{1}{30} \text{ cm}$$

now for equivalent lens

$$\frac{1}{f_e} = \frac{2}{f_{1_1}} + \frac{1}{f_{1_2}}$$

$$= \frac{2}{15} - \frac{1}{30} = \frac{3}{30} = \frac{1}{10}$$

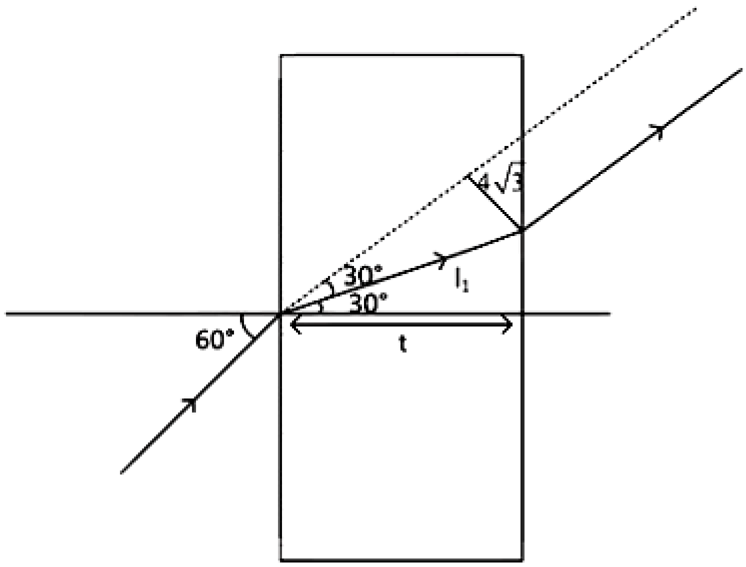
or $f_e = 10$ cm

Question122

A ray of light is incident at an angle of incidence 60° on the glass slab of refractive index $\sqrt{3}$. After refraction, the light ray emerges out from other parallel faces and lateral shift between incident ray and emergent ray is $4\sqrt{3}$ cm. The thickness of the glass slab is_____ cm.
[24-Jun-2022-Shift-2]

Answer: 12

Solution:



$1 \times \sin 60^\circ = \sqrt{3} \times \sin r$

$\Rightarrow r = 30^\circ$

$\therefore l_1 = 4\sqrt{3} \times 2$

$= 8\sqrt{3}$ cm

$\therefore$ Thickness, $t = l_1 \cos 30^\circ$

$= 8\sqrt{3} \times \frac{\sqrt{3}}{2}$

$= 4 \times 3$

$= 12$ cm

Question123

A light wave travelling linearly in a medium of dielectric constant 4, incidents on the horizontal interface separating medium with air. The angle of incidence for which the total intensity of incident wave will be reflected back into the same medium will be :
(Given : relative permeability of medium $\mu_r = 1$)

[25-Jun-2022-Shift-1]

Options:

A. 10°

B. 20°

C. 30°

D. 60°

Answer: D

Solution:

$n = \sqrt{K\mu} = 2$ ($n \Rightarrow$ refractive index)
So for TIR

$\theta > \sin^{-1}\left(\frac{1}{n}\right)$

$\theta > 30$

Only option is 60°

Question124

The difference of speed of light in the two media A and B$(v_A - v_B)$ is $2.6 \times 10^7 m/s$. If the refractive index of medium B is 1.47, then the ratio of refractive index of medium B to medium A is : (Given : speed of light in vacuum $.c = 3 \times 10^8 ms^{-1}$) [25-Jun-2022-Shift-1]

Options:

A. 1.303

B. 1.318

C. 1.13

D. 0.12

Answer: C

Solution:

Speed of light in a medium $= \frac{c}{n}$

$\Rightarrow$ According to given information,

$\frac{c}{n_A} - \frac{c}{n_B} = 2.6 \times 10^7$

$$\Rightarrow \frac{n_B}{n_A} - 1 = \frac{2.6 \times 10^7}{3 \times 10^8} \times n_B$$

$$\Rightarrow \frac{n_B}{n_A} \simeq 1.13$$

Question125

A light whose electric field vectors are completely removed by using a good polaroid, allowed to incident on the surface of the prism at Brewster's angle. Choose the most suitable option for the phenomenon related to the prism.
[25-Jun-2022-Shift-2]

Options:

A. Reflected and refracted rays will be perpendicular to each other.

B. Wave will propagate along the surface of prism.

C. No refraction, and there will be total reflection of light.

D. No refraction, and there will be total transmission of light.

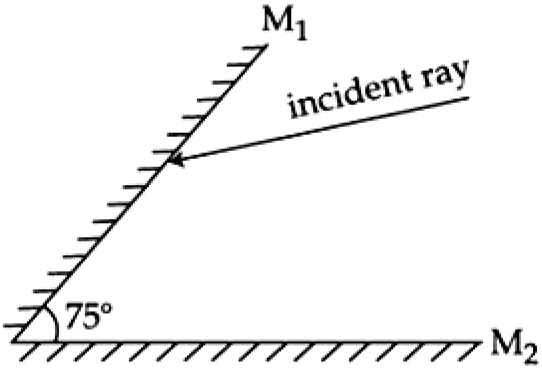
Answer: D

Solution:

When electric field vector is completely removed and incident on Brewster's angle then only refraction takes place.

Question126

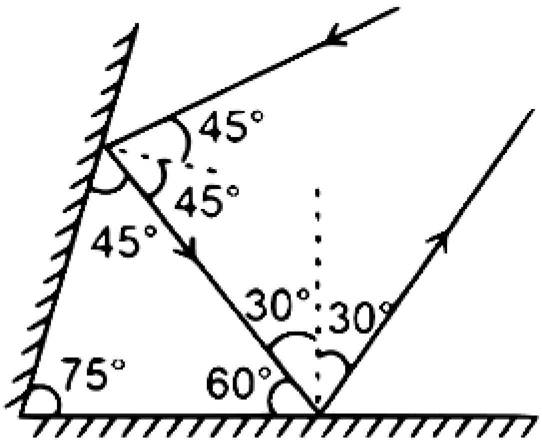
A light ray is incident, at an incident angle θ_1, on the system of tow plane mirrors M_1 and M_2 having an inclination angle 75° between them (as shown in figure). After reflecting from mirror M_1 it gets reflected back by the mirror M_2 with an angle of reflection 30°. The total deviation of the ray will be_____ degree.



[26-Jun-2022-Shift-1]

Answer: 210

Solution:



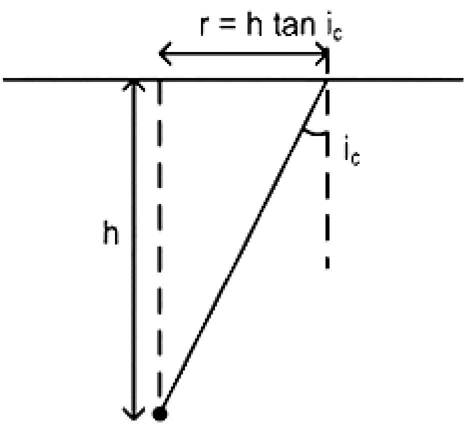
On first reflection angel of deviation is 90° and on second reflection angle of deviation is 120°
So total deviation is $\delta = 90^{\circ} + 120^{\circ} = 210^{\circ}$

Question127

A small bulb is placed at the bottom of a tank containing water to a depth of $\sqrt{7}$m. The refractive index of water is $\frac{4}{3}$. The area of the surface of water through which light from the bulb can emerge out is $x\pi m^2$. The value of x is____
[26-Jun-2022-Shift-2]

Answer: 9

Solution:



So $r = h \dfrac{\sin i_c}{\sqrt{1-\sin^2 i_c}}$

So $A = \pi r^2$

$= \dfrac{\pi h^2 \sin^2 i_c}{1-\sin^2 i_c}$

$= \dfrac{\pi 7 \times \frac{9}{16}}{1-\frac{9}{16}} = \dfrac{\pi \times 7 \times 9}{7} = 9\pi$

Question128

Consider a light ray travelling in air is incident into a medium of refractive index $\sqrt{2n}$. The incident angle is twice that of refracting angle. Then, the angle of incidence will be : [27-Jun-2022-Shift-1]

Options:

A. $\sin^{-1}(\sqrt{n})$

B. $\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

C. $\sin^{-1}(\sqrt{2n})$

D. $2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

Answer: D

Solution:

According to the law,

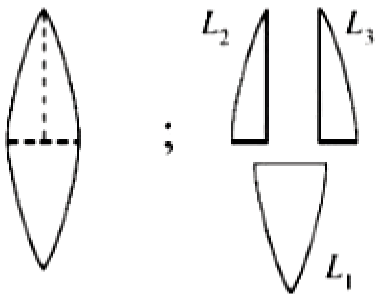
$1 \times \sin\theta = \sqrt{2n} \times \sin\left(\frac{\theta}{2}\right)$

$\Rightarrow \cos\frac{\theta}{2} = \sqrt{\frac{n}{2}}$

$\Rightarrow \theta = 2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

Question129

A convex lens has power P. It is cut into two halves along its principal axis. Further one piece (out of the two halves) is cut into two halves perpendicular to the principal axis (as shown in figures). Choose the incorrect option for the reported pieces.



[27-Jun-2022-Shift-2]

Options:

A. Power of $L_1 = \frac{P}{2}$

B. Power of $L_2 = \frac{P}{2}$

C. Power of $L_3 = \frac{P}{2}$

D. Power of $L_1 = P$

Answer: A

Solution:

We know $P = \frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$L_1 : \frac{1}{f_1} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = P_1 = (\mu - 1)\left(\frac{2}{R}\right) = P$

$L_2 : \frac{1}{f_2} = (\mu - 1)\left(\frac{1}{R_1}\right) = P_2 = \frac{\mu - 1}{R}$

$L_3 : \frac{1}{f_3} = (\mu - 1)\left(-\frac{1}{R_2}\right) = P_3 = \frac{\mu - 1}{R}$

Question130

The refracting angle of a prism is A and refractive index of the material of the prism is cot(A/2). Then the angle of minimum deviation will be -
[28-Jun-2022-Shift-1]

Options:

A. $180 - 2A$

B. $90 - A$

C. $180 + 2A$

D. $180 - 3A$

Answer: A

Solution:

$$\mu = \frac{\sin\left(\frac{\delta_m + A}{2}\right)}{\sin(A/2)} = \cot A/2$$

$$\Rightarrow \cos A/2 = \sin\left(\frac{\delta_m + A}{2}\right)$$

$$\Rightarrow \frac{\pi}{2} - \frac{A}{2} = \frac{\delta_m + A}{2}$$

$$\Rightarrow \pi - 2A = \delta_m$$

Question131

The aperture of the objective is 24.4 cm. The resolving power of this telescope, if a light of wavelength 2440$\overset{\circ}{A}$ is used to see th object will be :
[28-Jun-2022-Shift-1]

Options:

A. 8.1×10^{6}

B. 10.0×10^{7}

C. 8.2×10^{5}

D. 1.0×10^{-8}

Answer: C

Solution:

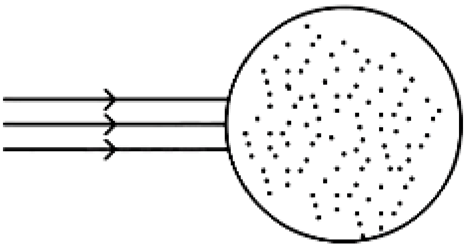
$$\text{R.P.} = \frac{1}{1.22\lambda / a}$$
$$= \frac{24.4 \times 10^{-2}}{1.22 \times 2440 \times 10^{-10}}$$
$$= 8.2 \times 10^{5}$$

Question132

A parallel beam of light is allowed to fall on a transparent spherical globe of diameter 30 cm and refractive index 1.5. The distance from the centre of the globe at which the beam of light can converge is_____ mm.
[29-Jun-2022-Shift-1]

Answer: 225

Solution:



1st refraction: $\frac{1.5}{v_1} - 0 = \frac{0.5}{15}$

$\Rightarrow v_1 = 45\text{ cm}$

2nd refraction : $\frac{1}{v_2} - \frac{1.5}{15} = \frac{-0.5}{-15}$

$\Rightarrow \frac{1}{v_2} = \frac{1}{30} + \frac{1}{10}$

$= \frac{4}{30}$

$\Rightarrow v_2 = +7.5\text{ cm}$

$\Rightarrow$ Distance from centre $= 22.5\text{ cm} = 225\text{ mm}$

Question133

The speed of light in media 'A' and 'B' are 2.0×10^{10} cm/s and 1.5×10^{10} cm/s respectively. A ray of light enters from the medium B to A at an incident angle ' θ '. If the ray suffers total internal reflection, then
[29-Jun-2022-Shift-2]

Options:

A. $\theta = \sin^{-1}\left(\frac{3}{4}\right)$

B. $\theta > \sin^{-1}\left(\frac{2}{3}\right)$

C. $\theta < \sin^{-1}\left(\frac{3}{4}\right)$

D. $\theta > \sin^{-1}\left(\frac{3}{4}\right)$

Answer: D

Solution:

$\mu_A = \frac{3 \times 10^8}{2 \times 10^8} = 1.5$
$\mu_B = \frac{3 \times 10^8}{1.5 \times 10^8} = 2$
For TIR
$\theta > i_c$
$\theta > \sin^{-1}\left(\frac{1.5}{2}\right)$
$\theta > \sin^{-1}\left(\frac{3}{4}\right)$

Question134

Which of the following statement is correct? [25-Jul-2022-Shift-1]

Options:

A. In primary rainbow, observer sees red colour on the top and violet on the bottom

B. In primary rainbow, observer sees violet colour on the top and red on the bottom

C. In primary rainbow, light wave suffers total internal reflection twice before coming out of water drops.

D. Primary rainbow is less bright than secondary rainbow.

Answer: A

Solution:

In primary rainbow, observer sees red colour on the top and violet on the bottom.

Question135

Time taken by light to travel in two different materials A and B of refractive indices μ_A and μ_B of same thickness is t_1 and t_2 respectively. If $t_2 - t_1 = 5 \times 10^{-10}$ s and the ratio of μ_A to μ_B is 1 : 2. Then, the thickness of material, in meter is: (Given v_A and v_B are velocities of light in A and B materials respectively.)
[25-Jul-2022-Shift-1]

Options:

A. $5 \times 10^{-10} v_A m$

B. $5 \times 10^{-10} m$

C. $1.5 \times 10^{-10} m$

D. $5 \times 10^{-10} v_B m$

Answer: A

Solution:

$t_2 - t_1 = 5 \times 10^{-10}$

$\Rightarrow \frac{d}{v_B} - \frac{d}{v_A} = 5 \times 10^{-10}$

and, $\frac{v_B}{v_A} = \frac{\mu_A}{\mu_B} = \frac{1}{2}$

$\Rightarrow d\left(1 - \frac{v_B}{v_A}\right) = 5 \times 10^{-10} \times v_B$

$\Rightarrow d\left(1 - \frac{1}{2}\right) = 5 \times 10^{-10} \times v_B$

$\Rightarrow d = 10 \times 10^{-10} \times v_B m$

$\Rightarrow d = 5 \times 10^{-10} \times v_A m$

Question136

For an object placed at a distance 2.4m from a lens, a sharp focused image is observed on a screen placed at a distance 12cm from the lens. A glass plate of refractive index 1.5 and thickness 1cm is

introduced between lens and screen such that the glass plate plane faces parallel to the screen. By what distance should the object be shifted so that a sharp focused image is observed again on the screen?
[25-Jul-2022-Shift-2]

Options:

A. 0.8m

B. 3.2m

C. 1.2m

D. 5.6m

Answer: B

Solution:

The shift produced by the glass plate is
$d = t\left(1 - \frac{1}{\mu}\right) = 1 \times \left(1 - \frac{1}{1.5}\right) = \frac{1}{3}\text{cm}$

So final image must be produced at $\left(12 - \frac{1}{3}\right)\text{cm} = \frac{35}{3}\text{cm}$ from lens so that glass plate must shift it to produce image at screen. So

$\frac{1}{12} - \frac{1}{-240} = \frac{1}{f} = \frac{1}{35/3} - \frac{1}{u}$

$\frac{1}{u} = \frac{3}{35} - \frac{1}{12} - \frac{1}{240}$

or $u = -560\text{cm}$

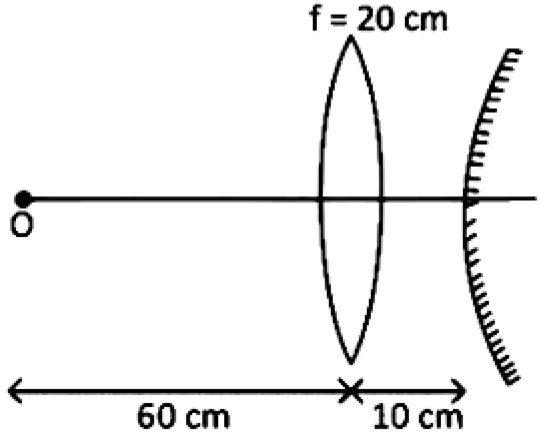
so shift $= 5.6 - 2.4 = 3.2\text{m}$

Question137

A convex lens of focal length 20cm is placed in front of a convex mirror with principal axis coinciding each other. The distance between the lens and mirror is 10cm. A point object is placed on principal axis at a distance of 60cm from the convex lens. The image formed by combination coincides the object itself. The focal length of the convex mirror is ___cm.
[25-Jul-2022-Shift-2]

Answer: 10

Solution:



$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} - \frac{1}{-60} = \frac{1}{20}$

$\frac{1}{v} = -\frac{1}{60} + \frac{1}{20} = \frac{-1+3}{60} = \frac{2}{60}$

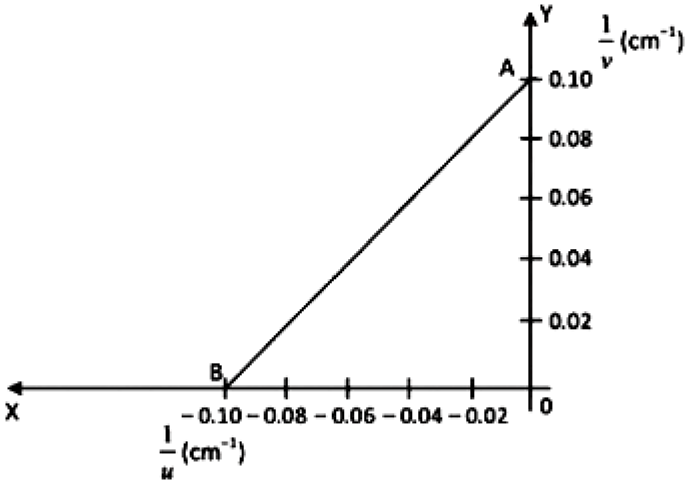
$\Rightarrow v = +30\text{cm}$

$\therefore$ Radius of curvature of mirror $= 30 - 10 = 20\text{cm}$

$\Rightarrow f_{\text{mirror}} = \frac{20}{2} = 10\text{cm}$

Question138

The graph between $\frac{1}{u}$ and $\frac{1}{v}$ for a thin convex lens in order to determine its focal length is plotted as shown in the figure. The refractive index of lens is 1.5 and its both the surfaces have same radius of curvature R. The value of R will be _________ cm. (where u = object distance, v = image distance)



[26-Jul-2022-Shift-1]

Answer: 10

Solution:

$$f = 10\,\text{cm}$$
$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R} - \frac{1}{-R}\right)$$
$$\frac{1}{10} = \frac{1.5 - 1}{1} \times \frac{2}{R}$$
$$\frac{1}{10} = \frac{1}{R}$$
$$R = 10\,\text{cm}$$

Question139

Light travels in two media M_1 and M_2 with speeds $1.5 \times 10^8 ms^{-1}$ and $2.0 \times 10^8 ms^{-1}$ respectively. The critical angle between them is : [26-Jul-2022-Shift-2]

Options:

A. $\tan^{-1}\left(\frac{3}{\sqrt{7}}\right)$

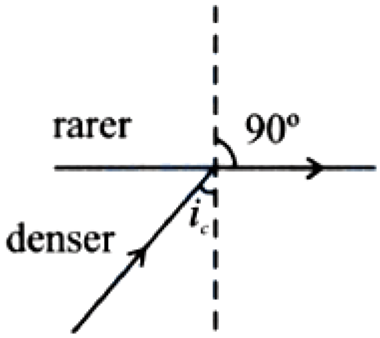
B. $\tan^{-1}\left(\frac{2}{3}\right)$

C. $\cos^{-1}\left(\frac{3}{4}\right)$

D. $\sin^{-1}\left(\frac{2}{3}\right)$

Answer: A

Solution:

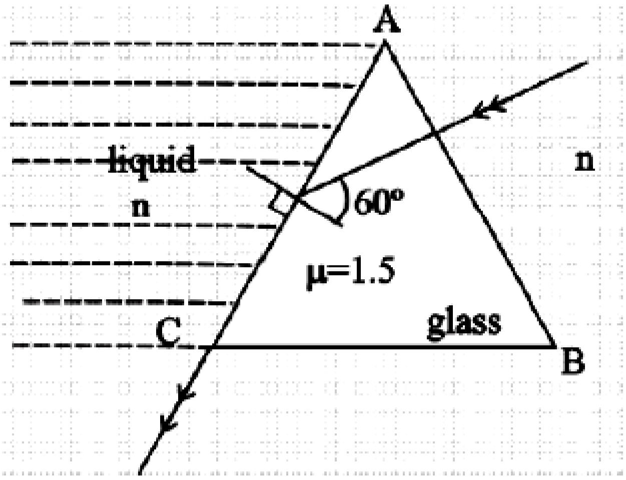


$$v = \frac{c}{n}$$
$$n_d \sin i_c = n_r \sin 90^{\circ}$$
$$\sin i_c = \frac{n_r}{n_d} = \frac{V_d}{V_r}$$
$$\sin i_c = \frac{1.5 \times 10^8}{2 \times 10^8} = \frac{1.5}{2}$$
$$\sin i_c = \frac{3}{4}$$
$$\tan i_c = \frac{3}{\sqrt{4^2 - 3^2}} \Rightarrow \frac{3}{\sqrt{7}}$$
$$i_c = \tan^{-1}\left(\frac{3}{\sqrt{7}}\right)$$

Question140

In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index ' n '. For incident angle 60° at the side AC, the refracteel light beam just grazes along face AC. The refractive index of the liquid n = $\frac{\sqrt{x}}{4}$. The value of x is ________.

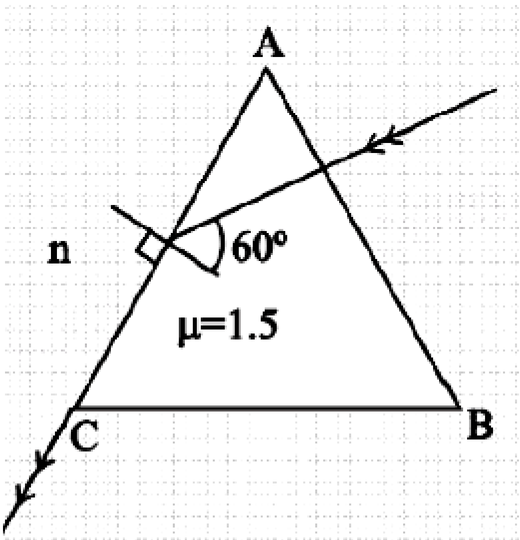
(Given refractive index of glass = 1.5)



[26-Jul-2022-Shift-2]

Answer: 27

Solution:



Using snell's law at face AC

$1.5 \sin 60^{\circ} = n \times \sin 90^{\circ}$

$1.5 \times \frac{\sqrt{3}}{2} = n = \frac{\sqrt{x}}{4}$

$3\sqrt{3} = \sqrt{x}$

$x = 27$

Question141

A microscope was initially placed in air (refractive index 1). It is then immersed in oil (refractive index 2). For a light whose wavelength in air is λ, calculate the change of microscope's resolving power due to oil and choose the correct option.
[27-Jul-2022-Shift-1]

Options:

A. Resolving power will be $\frac{1}{4}$ in the oil than it was in the air.

B. Resolving power will be twice in the oil than it was in the air.

C. Resolving power will be four times in the oil than it was in the air.

D. Resolving power will be $\frac{1}{2}$ in the oil than it was in the air.

Answer: B

Solution:

$\because$ Resolving power $= \frac{2\mu \sin\theta}{1.22\lambda}$

$\frac{P_1}{P_2} = \frac{\mu_1}{\mu_2} \times \frac{\mu_1}{\mu_2}$

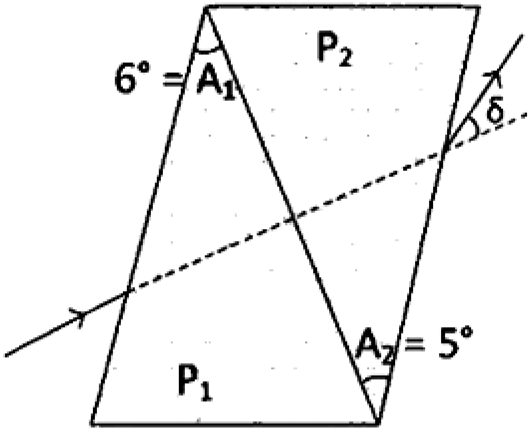
$= \left(\frac{\mu_1}{\mu_2}\right)^2$

$\Rightarrow \frac{P_1}{P_2} = \frac{1}{4}$

$\Rightarrow P_2 = 4P_1$

Question142

A thin prism of angle 6° and refractive index for yellow light (n_Y)1.5 is combined with another prism of angle 5° and n_Y = 1.55. The combination produces no dispersion. The net average deviation (δ) produced by the combination is $\left(\frac{1}{x}\right)^0$. The value of x is ________.



[27-Jul-2022-Shift-2]

Answer: 4

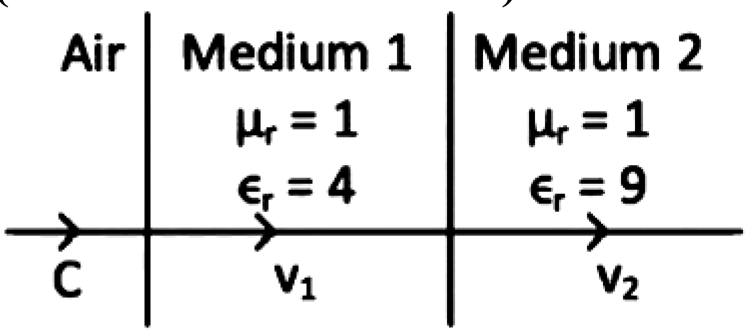
Solution:

$$\delta_{net} = \delta_1 + \delta_2$$
$$= |(\mu_1 - 1)A_1 - (\mu_2 - 1)A_2|$$
$$= |3^\circ - 2.75^\circ|$$
$$\delta_{net} = \frac{1^\circ}{4}$$
$$\Rightarrow x = 4$$

Question143

As shown in the figure, after passing through the medium 1 . The speed of light v_2 in medium 2 will be:

(. Given .c = $3 \times 10^8 ms^{-1}$)



[28-Jul-2022-Shift-1]

Options:

A. $1.0 \times 10^{8} \mathrm{~ms}^{-1}$

B. $0.5 \times 10^{8} \mathrm{~ms}^{-1}$

C. $1.5 \times 10^{8} \mathrm{~ms}^{-1}$

D. $3.0 \times 10^{8} \mathrm{~ms}^{-1}$

Answer: A

Solution:

$$V = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_r \varepsilon r \mu_0 \varepsilon_0}}$$
$$\Rightarrow V_2 = \frac{c}{\sqrt{9}} = 10^{8} \mathrm{m/s}$$

Question144

In normal adujstment, for a refracting telescope, the distance between objective and eye piece is 30 cm. The focal length of the objective, when the angular magnification of the telescope is 2 , will be
[28-Jul-2022-Shift-1]

Options:

A. 20 cm

B. 30 cm

C. 10 cm

D. 15 cm

Answer: A

Solution:

$\because m = \frac{f_o}{f_e}$
$\Rightarrow 2 = \frac{f_o}{f_e}$...... (i)
and, $l = f_o + f_e$
$\Rightarrow 30 = f_o + f_e$..... (ii)
$\Rightarrow 30 = f_o + \frac{f_o}{2}$
$\Rightarrow 30 \times \frac{2}{3} = f_o$
$\Rightarrow f_o = 20\,\text{cm}$

Question145

The power of a lens (biconvex) is $1.25m^{-1}$ in particular medium. Refractive index of the lens is 1.5 and radii of curvature are 20 cm and 40 cm respectively. The refractive index of surrounding medium:
[28-Jul-2022-Shift-2]

Options:

A. 1.0

B. $\frac{9}{7}$

C. $\frac{3}{2}$

D. $\frac{4}{3}$

Answer: B

Solution:

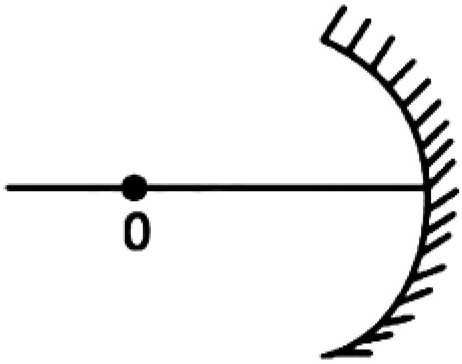
$P = \frac{\mu_2}{f} = (\mu_1 - \mu_2)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ (For this formula refer to NCERT Part-2, Chapter-9, solved example 8)
($\mu_{1.}$ is refractive index of lens and μ_2 is of surrounding medium)
$1.25 = (1.5 - \mu_2)\left(\frac{1}{0.2} + \frac{1}{0.4}\right)$

$\frac{1.25 \times 0.08}{0.6} = (1.5 - \mu_2)$

$\Rightarrow \mu_2 = \frac{4}{3}$

Question146

An object ' O ' is placed at a distance of 100 cm in front of a concave mirror of radius of curvature 200 cm as shown in the figure. The object starts moving towards the mirror at a speed 2 cm/ s. The position of the image from the mirror after 10 s will be at ____cm.



[28-Jul-2022-Shift-2]

Answer: 400

Solution:

After 10 sec.

$u = -80$ cm

$f = -100$ cm

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$v = 400$ cm

Question147

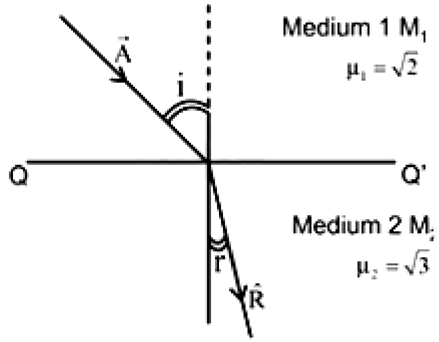
The X − Y plane be taken as the boundary between two transparent media M_1 and $M_2 \cdot M_1$ in $Z \geq 0$ has a refractive index of $\sqrt{2}$ and M_2

with Z < 0 has a refractive index of $\sqrt{3}$. A ray of light travelling in M_1 along the direction given by the vector $\vec{P} = 4\sqrt{3}\hat{i} - 3\sqrt{3}\hat{j} - 5\hat{k}$, is incident on the plane of separation. The value of difference between the angle of incident in M_1 and the angle of refraction in M_2 will be____ degree.
[29-Jul-2022-Shift-1]

Answer: 15

Solution:

$\vec{A} = 4\sqrt{3}\hat{i} - 3\sqrt{3}\hat{j} - 5\hat{k}$



$\mu_1 \sin i = \mu_2 \sin r$

As incident vector A makes i angle with normal z-axis \& refracted vector R makes r angle with normal z - axis with help of direction cosine

$i = \cos^{-1}\left(\frac{A_z}{A}\right) = \cos^{-1}\left(\frac{5}{\sqrt{(4\sqrt{3})^2 + (3\sqrt{3})^2 + 5^2}}\right)$

$= \cos^{-1}\left(\frac{5}{10}\right) \Rightarrow i = 60^{\circ}$

$\sqrt{2}\sin 60 = \sqrt{3} \times \sin r$

$r = 45^{\circ}$

Difference between $i \& r = 60 - 45 = 15$

Question148

Light enters from air into a given medium at an angle of 45° with interface of the air-medium surface. After refraction, the light ray is

deviated through an angle of 15° from its original direction. The refractive index of the medium is:
[29-Jul-2022-Shift-2]

Options:

A. 1.732

B. 1.333

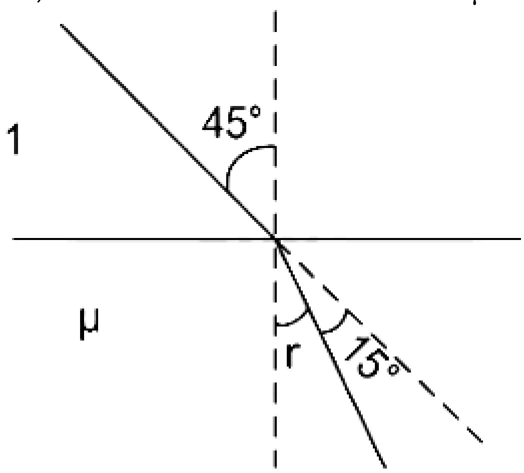
C. 1.414

D. 2.732

Answer: C

Solution:

Let, refractive index of medium $= \mu$



$\therefore r + 15^{\circ} = 45^{\circ}$
$\Rightarrow r = 30^{\circ}$
Using Snell's law,
1. $\sin 45^{\circ} = \sin 30^{\circ} \times \mu$

$\Rightarrow \frac{1}{\sqrt{2}} = \frac{1}{2} \times \mu$

$\Rightarrow \mu = \frac{2}{\sqrt{2}} = \sqrt{2} = 1.141$

Question149

The focal length f is related to the radius of curvature r of the spherical convex mirror by :

[24feb2021shift1]

Options:

A. $f = +frac\ 1\ 2\ r$

B. $f = -r$

C. $f = -frac\ 1\ 2\ r$

D. $f = r$

Answer: A

Solution:

For convex mirror, focus is behind the mirror. So, its focal length (f) is positive. $\therefore\ f = +\frac{r}{2}$

Question150

The incident ray, reflected ray and the outward drawn normal are denoted by the unit vectors a, b and c, respectively. Then, choose the correct relation for these vectors. [26 Feb 2021 Shift 2]

Options:

A. $b = a + 2c$

B. $b = 2a + c$

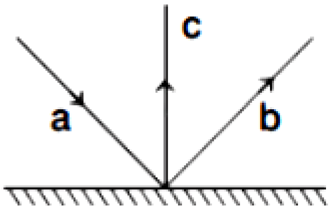
C. $b = a - 2(a \cdot c)c$

D. $b = a - c$

Answer: C

Solution:

Taking component of a, b and c along $\hat{i}$, $\hat{j}$ and $\hat{k}$.



$a = \sin\theta \hat{i} - \cos\theta \hat{j}$

$b = \sin\theta \hat{i} + \cos\theta \hat{j}$

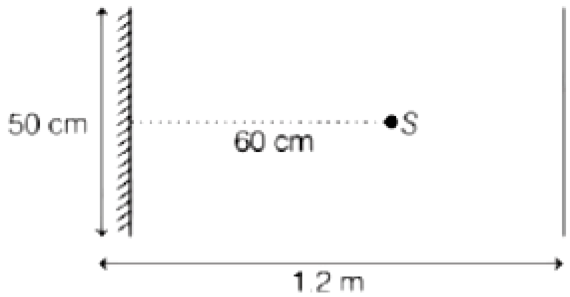
$c = \hat{j}$

Now, on solving option (c), we get

$$a - 2(a \cdot c)c = (\sin\theta \hat{i} - \cos\theta \hat{j}) - 2[(\sin\theta \hat{i} - \cos\theta \hat{j}) \cdot \hat{j}]\hat{j}$$
$$= \sin\theta \hat{i} + \cos\theta \hat{j}$$
$$= b$$

Question151

A point source of light S, placed at a distance 60cm in front of the centre of a plane mirror of width 50cm, hangs vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 1.2m from it (see in the figure). The distance between the extreme points, where he can see the image of the light source in the mirror is .... cm.



[26 Feb 2021 Shift 2]

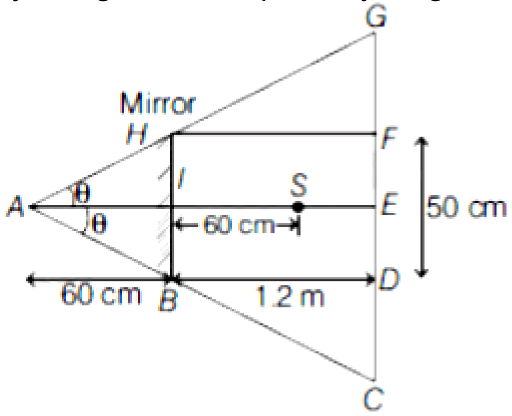
Answer: 150

Solution:

Given, length of mirror, $m = 50\text{cm} = 50 \times 10^{-2}\text{m}$

Distance of source from mirror, $d = 60\text{cm}$

$= 60 \times 10^{-2}$m Distance of man from mirror, $d_m = 1.2$m
By using the concept of ray diagram of plane mirror shown below



Now, using the concept of similar triangle,
$\Delta HAl \sim \triangle GAE$ and $\Delta BAl \sim \Delta CAE$

$\therefore \frac{AI}{AE} = \frac{HI}{EG}$

$\Rightarrow \frac{0.60}{1.8} = \frac{0.25}{EG}$

$\Rightarrow EG = 0.25 \times \frac{1.8}{0.6} = 0.25 \times 3 = 0.75\text{m}$

As, $CG = 2EG$

$\Rightarrow CG = 0.75 \times 2 = 1.50\text{m}$

$\frac{AI}{AE} = \frac{HI}{EG}$

$\Rightarrow \frac{0.60}{1.8} = \frac{0.25}{EG}$

$\Rightarrow EG = 0.25 \times \frac{1.8}{0.6} = 0.25 \times 3 = 0.75\text{m}$

As, $CG = 2EG$

$\Rightarrow CG = 0.75 \times 2 = 1.50\text{m}$

Hence, distance between the extreme points, where he can see image of light source in mirror is 150cm.

Question152

A short straight object of height 100cm lies before the central axis of a spherical mirror, whose focal length has absolute value f = 40cm. The image of object produced by the mirror is of height 25cm and has the same orientation of the object. One may conclude from the information.
[26 Feb 2021 Shift 1]

Options:

A. Image is real, same side of concave mirror

B. Image is virtual, opposite side of concave mirror

C. Image is real, same side of convex mirror

D. Image is virtual, opposite side of convex mirror

Answer: D

Solution:

Given, height of object, $h_o = 100\text{cm}$
Focal length of mirror, $f = 40\text{cm}$
Height of image, $h_i = 25\text{cm}$
Nature of image is erect means virtual.
As, h_i is less than h_o,
so mirror used is convex mirror.
Hence, (d) is correct option, i.e. image is virtual, opposite and is convex.

Question153

The same size images are formed by a convex lens when the object is placed at 20cm or at 10cm from the lens. The focal length of convex lens is .... cm.
[25 Feb 2021 Shift 1]

Answer: 15

Solution:

Let v be the position of image, h be the height of image and h_o be the height of object.
Given, $h_i = h_o$
Since, magnification, $m = h_i / h_o = h_i / h$ . . . (i)
By using lens formula,
$1/f = 1/v - 1/u \Rightarrow 1/v = 1/f + 1/u$

$$v = \frac{fu}{u+f}$$

$$m = \frac{v}{u} = \frac{f}{f+u}$$

Now, from Eq. (i), m can be ± 1.
For, $m = +1 = \dfrac{f}{-10+f}$ . . . (ii)

For $m=-1=\dfrac{f}{-20+f}$ . . . (iii)
On dividing Eq. (iii) by Eq. (ii), we get
$-1=\dfrac{-10+f}{-20+f}$
$\Rightarrow\ 20-f=-10+f \Rightarrow 30=2f$
$\Rightarrow\ f=15cm$

Question154

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) For a simple microscope, the angular size of the object equals the angular size of the image.
Reason (R) Magnification is achieved as the small object can be kept much closer to the eye than 25cm and hence, it subtends a large angle.
In the light of the above statements, choose the most appropriate answer from the options given below.
[26 Feb 2021 Shift 2]

Options:

A. A is true but R is false.

B. Both A and R are true but R is not the correct explanation of A.

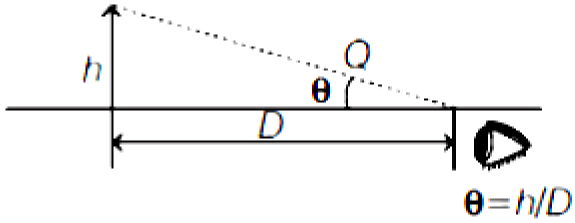
C. Both A and R are true and R is the correct explanation of A.

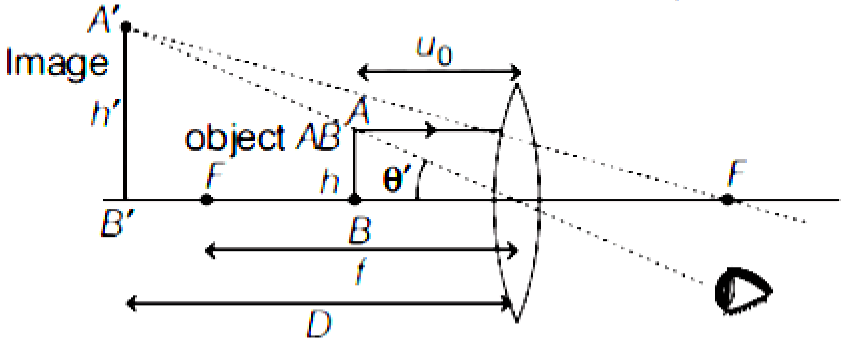
D. A is false but R is true.

Answer: C

Solution:

The formation of image with simple microscope is shown below.





Here, $\theta' = \frac{h}{u_0} = \frac{h'}{D} = \frac{h'}{25}$

where, $D = 25\text{cm}$ (least distance of distinct vision)

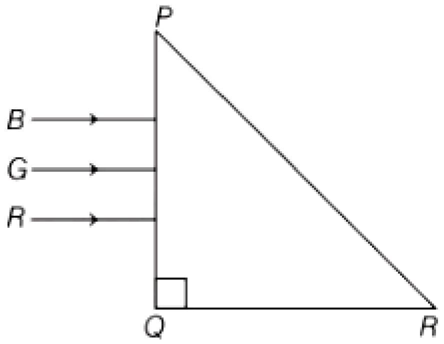
Here, θ' is same for both object and image, hence Assertion is true.

Magnification, $m = \frac{\theta'}{\theta} = \frac{D}{u_0}$

Hence, if $u_0 < D(25\text{cm})$, hence the value of θ' will obtain large. So, option (c) is the correct.

Question155

Three rays of light, namely red (R), green (G) and blue (B) are incident on the face PQ of a right angled prism PQR as shown in figure



The refractive indices of the material of the prism for red, green and blue wavelength are 1.27, 1.42 and 1.49, respectively. The colour of the ray(s) emerging out of the face PR is [18 Mar 2021 Shift 2]

Options:

A. green

B. red

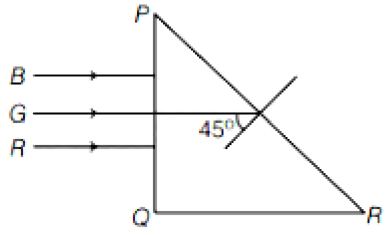
C. blue and green

D. blue

Answer: B

Solution:

From the given figure,



We know that,

$\mu = \dfrac{1}{\sin i_C}$

Here, i_c is the critical angle of incidence and μ is the refractive index.

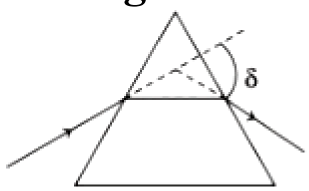
$\mu = \dfrac{1}{\sin 45^{\circ}} \Rightarrow \mu = 1.414$

The rays will emerge out when angle of incidence is less than the angle of critical angle of glass-air interface PR.

As $\mu_R < \mu$ while μ_C and $\mu_B > \mu$, so only red colour will be transmitted through face PR while green and blue rays will suffer total internal reflection.

Question156

The angle of deviation through a prism is minimum when



A. incident ray and emergent ray are symmetric to the prism
B. the refracted ray inside the prism becomes parallel to its base
C. angle of incidence is equal to that of the angle of emergence
D. angle of emergence is double the angle of incidence
Choose the correct answer from the options given below.
[16 Mar 2021 Shift 1]

Options:

A. Statements (A), (B) and (C) are true.

B. Only statement (D) is true.

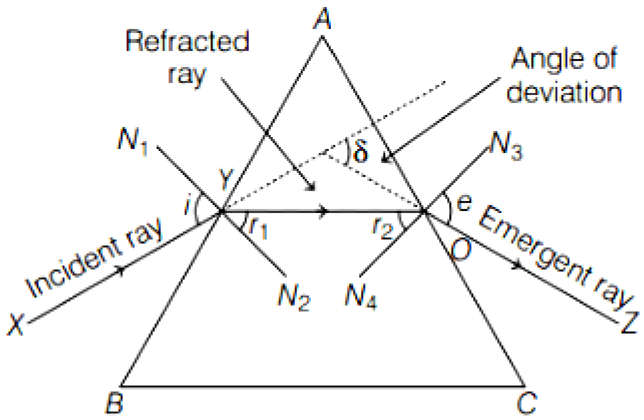
C. Only statements (A) and (B) are true.

D. Statements (B) and (C) are true.

Answer: A

Solution:

The propagation of light ray from a prism is shown below



From the above figure, we can say that minimum value of angle of deviation can only be achieved when
A. incident ray and emergent ray are symmetric to the prism.
B. the refracted ray inside the prism becomes parallel to its base.
C. angle of incidence is equal to angle of emergence.
∴Statement (A), (B) and (C) are true.
Note Refracted ray inside the prism is parallel to the base only for equilateral and isosceles prism.

Question157

The image of an object placed in air formed by a convex refracting surface is at a distance of 10m behind the surface. The image is real and is at $\frac{2}{3}$ of the distance of the object from the surface .The wavelength of light inside the surface is $\frac{2}{3}$ times the wavelength in air. The radius of the curved surface is $\frac{x}{13}$m. The value of x is
[17 Mar 2021 Shift 2]

Answer: 30

Solution:

Given,
Image distance, $v = 10m$

Object distance, $u = \left(\frac{3}{2}\right) \times 10 - 15m$

and $n_2 = \frac{3}{2}n_1 \quad \left(\because n \propto \frac{1}{\lambda}\right)$

Using the lens Maker's formula,

$\frac{n_2}{v} - \frac{n_1}{u} = \left(\frac{n_2 - n_1}{R}\right)$

Substituting the values in the above equation, we get

$\frac{\frac{3}{2}n_1}{10} - \frac{n_1}{-15} = \left(\frac{\frac{3}{2}n_1 - n_1}{R}\right) \Rightarrow R = \frac{30}{13}m$

The radius of the curved surface is $\frac{30}{13}m$.

Comparing with $\frac{x}{13}$, we get

$x = 30$

Question158

Red light differs from blue light as they have [16 Mar 2021 Shift 2]

Options:

A. different frequencies and different wavelengths

B. different frequencies and same wavelengths

C. same frequencies and same wavelengths

D. same frequencies and different wavelengths

Answer: A

Solution:

Since, $\lambda v = c =$ constant
where, $\lambda =$ wavelength of light

and $v =$ frequency of light.
Red light and blue light have different wavelengths and different frequencies but same speed.

Question159

A deviation of 2° is produced in the yellow ray when prism of crown and flint glass are achromatically combined. Taking dispersive powers of crown and flint glass are 0.02 and 0.03 respectively and refractive index for yellow light for these glasses are 1.5 and 1.6, respectively. The refracting angles for crown glass prism will be ...... (in degree).
(Round off to the nearest integer)
[16 Mar 2021 Shift 2]

Answer: 12

Solution:

Given
Dispersive power of crown glass, $\omega_1 = 0.02$
Dispersive power of flint glass, $\omega_2 = 0.03$
Refractive index of yellow light,
for crown glass, $\mu_1 = 1.5$ and
for flint glass, $\mu_2 = 1.6$
This is a case of achromatic combination.
$\therefore\ \theta_{\text{net}} = 0$
$\Rightarrow\ \theta_1 - \theta_2 = 0 \Rightarrow \theta_1 = \theta_2$
$\Rightarrow\ \omega_1\delta_1 = \omega_2\delta_2\ \ [\because \theta = \omega\delta]$... (i)
and $\delta_{\text{net}} = \delta_1 - \delta_2 = 2^{\circ}$......(ii)
From Eqs. (i) and (ii), we get
$$\delta_1 - \frac{\omega_1\delta_1}{\omega_2} = 2^{\circ}$$
$$\Rightarrow\ \delta_1\left(1 - \frac{\omega_1}{\omega_2}\right) = 2^{\circ} \Rightarrow \delta_1\left(1 - \frac{2}{3}\right) = 2^{\circ}$$
$\Rightarrow\ \delta_1 = 6^{\circ}$
Also, $\delta_1 = (\mu_1 - 1)A_1$

$\Rightarrow\ 6^{\circ} = (1.5 - 1)A_1 \Rightarrow A_1 = 12^{\circ}$

$\therefore$ The refracting angle for crown glass prism will be 12°.

Question160

The thickness at the centre of a plano convex lens is 3mm and the diameter is 6cm. If the speed of light in the material of the lens is $2 \times 10^{8} ms^{-1}$, then the focal length of the lens is [17 Mar 2021 Shift 1]

Options:

A. 0.30cm

B. 15cm

C. 1.5cm

D. 30cm

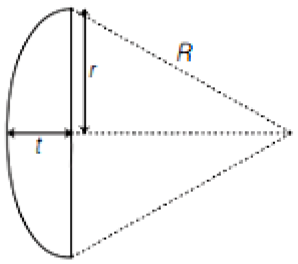
Answer: D

Solution:

Given, thickness at the centre of plano-convex lens, $t = 3mm = 3 \times 10^{-3} m$
Diameter of plano-convex lens, $d = 6cm$
$\therefore$ Radius of plano-convex lens, $r = 3cm = 3 \times 10^{-2} m$
Speed of light in lens material, $v = 2 \times 10^{8} ms^{-1}$
Refractive index $= \dfrac{\text{Speed of light in air}}{\text{Speed of light in medium}}$

$\mu = \dfrac{3 \times 10^{8}}{2 \times 10^{8}} = \dfrac{3}{2} = 1.5$

$\Rightarrow\ \mu = 1.5$



We know that, $R^2 = (R - t)^2 + r^2$
where, R is the radius of curvature of plano convex lens.
$\Rightarrow\ R^2 = R^2 + t^2 - 2Rt + r^2 \Rightarrow 2Rt = r^2 + t^2$

As, t is small, then t^2 will be very very small, so it can be neglected.

$\Rightarrow\ 2Rt = r^2$

$\Rightarrow\ R = r^2/2t$....(i)

From lens Maker's formula, we have

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Here $R_1 = R \& R_2 = \infty$

$$\Rightarrow\ \frac{1}{f} = (\mu - 1)\left(\frac{1}{R} - \frac{1}{\infty}\right)$$

$$\Rightarrow\ \frac{1}{f} = \frac{\mu - 1}{R} \text{...(ii)}$$

From Eqs. (i) and (ii), we get

$$\frac{1}{f} = \frac{\mu - 1(2t)}{r^2} \Rightarrow f = \frac{r^2}{(\mu - 1)2t}$$

Putting the given values in above equation, we get

$$f = \frac{(3 \times 10^{-2})^2}{(1.5 - 1)2 \times 3 \times 10^{-3}} \Rightarrow f = 0.3\text{m}$$

$f = 30\text{cm}$

Question161

The refractive index of a converging lens is 1.4. What will be the focal length of this lens if it is placed in a medium of same refractive index? (Assume the radii of curvature of the faces of lens are R_1 and R_2 respectively)
[16 Mar 2021 Shift 2]

Options:

A. 1

B. Infinite

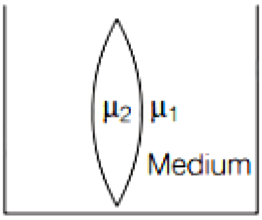
C. $\frac{R_1 R_2}{R_1 - R_2}$

D. Zero

Answer: B

Solution:

Consider a convex lens of refractive index μ_2 and μ_1 is the refractive index of the medium in which it is placed.



$\Rightarrow \mu_2 = \mu_1$.....(i)

According to the lens Maker's formula,

$$\frac{1}{f} = \left[\frac{\mu_1}{\mu_2} - 1\right]\left[\frac{1}{R_1} - \frac{1}{R_2}\right] \text{....(ii)}$$

where, f is the focal length of the lens,

R_1 and R_2 are the radii of curvature of respective faces of lens.

From Eqs. (i) and (ii), we can write

$$\frac{1}{f} = [1 - 1]\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = 0$$

$\Rightarrow \frac{1}{f} = 0 \Rightarrow f =$ Infinite

Question162

Your friend is having eye sight problem. She is not able to see clearly a distant uniform window mesh and it appears to her as non-uniform and distorted. The doctor diagnosed the problem as [18 Mar 2021 Shift 1]

Options:

A. astigmatism

B. myopia with astigmatism

C. presbyopia with astigmatism

D. myopia and hypermetropia

Answer: B

Solution:

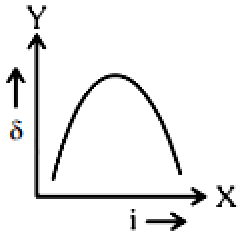
A friend is not seen clearly the distant object, then its diagnosis is myopia because in myopia the distant object is blurry and it also appear non-uniform and distorted images of the object, then its diagnosis is astigmatism also.

Question163

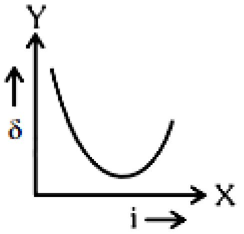
The expected graphical representation of the variation of angle of deviation 'δ' with angle of incidence 'i' in a prism is :
[27 Jul 2021 Shift 2]

Options:

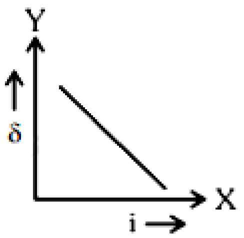
A.



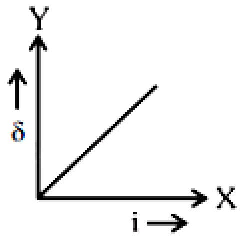
B.



C.



D.



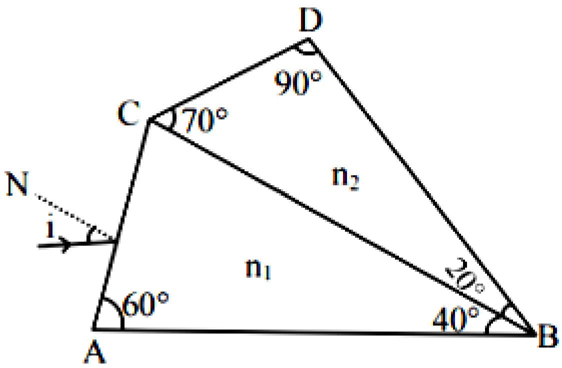
Answer: B

Solution:

Standard graph between angle of deviation and incident angle.

Question164

A prism of refractive index n_1 and another prism of refractive index n_2 are stuck together (as shown in the figure). n_1 and n_2 depend on λ, the wavelength of light, according to the relation
$n_1 = 1.2 + \frac{10.8 \times 10^{-14}}{\lambda^2}$ and $n_2 = 1.45 + \frac{1.8 \times 10^{-14}}{\lambda^2}$
The wavelength for which rays incident at any angle on the interface BC pass through without bending at that interface will be ________nm.



[27 Jul 2021 Shift 1]

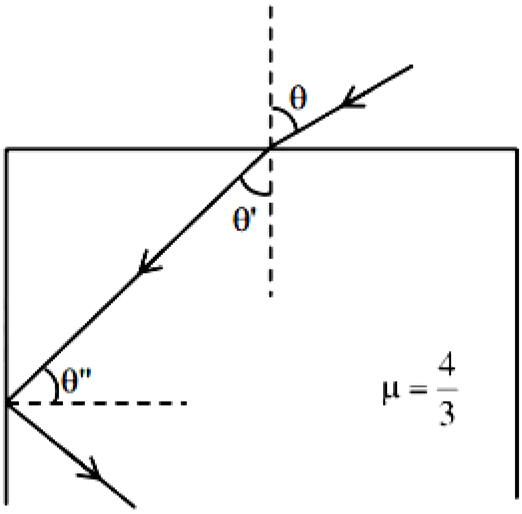
Answer: 600

Solution:

For no bending, $n_1 = n_2$
$1.2 + \frac{10.8 \times 10^{-14}}{\lambda^2} = 1.45 + \frac{1.8 \times 10^{-4}}{\lambda^2}$
On solving,
$9 \times 10^{-14} = 25\lambda^2$
$\lambda = 6 \times 10^{-7}$
$\lambda = 600\text{nm}$

Question165

A ray of light entering from air into a denser medium of refractive index $\frac{4}{3}$, as shown in figure. The light ray suffers total internal reflection at the adjacent surface as shown. The maximum value of angle theta should be equal to :



[25 Jul 2021 Shift 2]

Options:

A. $\sin^{-1}\frac{\sqrt{5}}{4}$

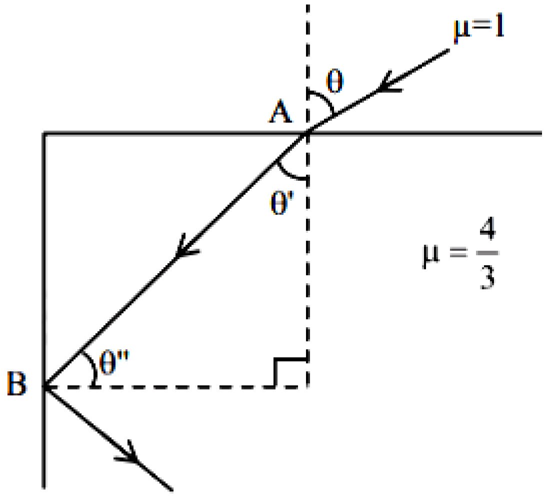
B. $\sin^{-1}\frac{\sqrt{7}}{3}$

C. $\sin^{-1}\frac{\sqrt{5}}{3}$

D. $\sin^{-1}\frac{\sqrt{7}}{4}$

Answer: A

Solution:



At maximum angle θ ray at point B goes in gazing emergence, at all less values of θ, TIR occurs.
At point B

$$\frac{4}{3} \times \sin\theta'' = 1 \times \sin 90^\circ$$

$$\theta'' = \sin^{-1}\left(\frac{3}{4}\right)$$

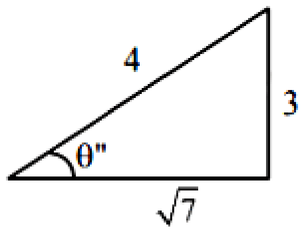
$$\theta' = \left(\frac{\pi}{2} - \theta''\right)$$

At point A

$$1 \times \sin\theta = \frac{4}{3} \times \sin\theta'$$

$$\sin\theta = \frac{4}{3} \times \sin\left(\frac{\pi}{2} - \theta''\right)$$

$$\sin\theta = \frac{4}{3}\cos\left[\cos^{-1}\frac{\sqrt{7}}{4}\right]$$



$$\sin\theta = \frac{4}{3} \times \frac{\sqrt{7}}{4}$$

$$\theta = \sin^{-1}\left(\frac{\sqrt{7}}{3}\right)$$

Question166

A ray of laser of a wavelength 630 nm is incident at an angle of 30° at the diamond-air interface. It is going from diamond to air. The refractive index of diamond is 2.42 and that of air is 1. Choose the correct option.
[25 Jul 2021 Shift 1]

Options:

A. angle of refraction is 24.41°

B. angle of refraction is 30°

C. refraction is not possible

D. angle of refraction is 53.4°

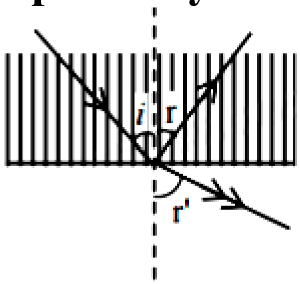
Answer: C

Solution:

$\sin \theta_C = \frac{1}{\mu} = \frac{1}{2\mu_2} < \sin \theta_C$
$\sin \theta > \sin \theta_C$
$\theta > \theta_C$
Total internal reflection will happen

Question167

A ray of light passes from a denser medium to a rarer medium at an angle of incidence i. The reflected and refracted rays make an angle of 90° with each other. The angle of reflection and refraction are respectively r and r'. The critical angle is given by :



[22 Jul 2021 Shift 2]

Options:

A. $\sin^{-1}$ (cotr)

B. $\tan^{-1}(\sin i)$

C. $\sin^{-1}$(tanr')

D. $\sin^{-1}(\tan r)$

Answer: D

Solution:

$r + r' + 90° = 180° \Rightarrow r' = 90 - r = 90 - i$
$n_1 \sin i = n_2 \sin r' = n_2 \sin(90 - i)$
$n_1 \sin i = n_2 \cos i \Rightarrow \tan i = \frac{n_2}{n_1}$
Now $\sin C = \frac{n_2}{n_1} = \tan i$
$\Rightarrow C = \sin^{-1}(\tan i) = \sin^{-1}(\tan r)$

Question168

A ray of light passing through a prism ($\mu = \sqrt{3}$)suffers minimum deviation. It is found that the angle of incidence is double the angle of refraction within the prism. Then, the angle of prism is ________ (in degrees)
[22 Jul 2021 Shift 2]

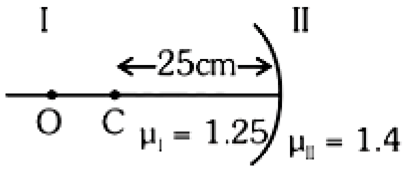
Answer: 60

Solution:

At minimum deviation $r_1 = r_2 = \frac{A}{2}$
Also given $i = 2r_1 = A$
Now $1 . \sin i = \sqrt{3} \sin r_1$
$1 \sin A = \sqrt{3} \sin \frac{A}{2}$
$\Rightarrow 2 \sin \frac{A}{2} \cos \frac{A}{2} = \sqrt{3} \sin \frac{A}{2}$
$\Rightarrow \cos \frac{A}{2} = \frac{\sqrt{3}}{2} \Rightarrow \frac{A}{2} = 30°$
$\Rightarrow A = 60°$

Question169

Region I and II are separated by a spherical surface of radius 25 cm. An object is kept in region I at a distance of 40 cm from the surface. The distance of the image from the surface is :



[20 Jul 2021 Shift 1]

Options:

A. 55.44 cm

B. 9.52 cm

C. 18.23 cm

D. 37.58 cm

Answer: D

Solution:

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$
$$\frac{1.4}{v} - \frac{1.25}{-40} = \frac{1.4 - 1.25}{-25}$$
$$\frac{1.4}{v} = -\frac{0.15}{25} - \frac{1.25}{40}$$
$$v = -37.58\text{cm}$$

Hence option (4)

Question170

An object viewed from a near point distance of 25 cm, using a microscopic lens with magnification '6', gives an unresolved image. A resolved image is observed at infinite distance with a total magnification double the earlier using an eyepiece along with the given lens and a tube of length 0.6 m, if the focal length of the

eyepiece is equal to ______ cm.
[20 Jul 2021 Shift 1]

Answer: 25

Solution:

For simple microscope,

$m = 1 + \frac{D}{f_0}$

$6 = 1 + \frac{D}{f_0}$

$5 = \frac{25}{f_0}$

$f_0 = 5\text{cm}$

For compound microscope,

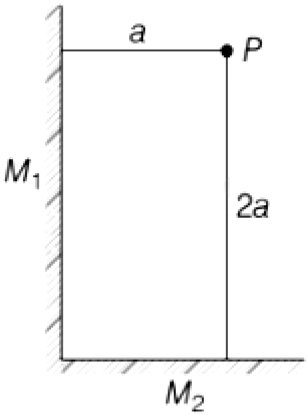
$m = \frac{1 \cdot D}{f_0 \cdot f_e}$

$12 = \frac{60 \times 25}{5 \cdot f_e}$

$f_e = 25\text{cm}$

Question171

Two plane mirrors M_1 and M_2 are at right angle to each other shown. A point source P is placed at a and 2a meter away from M_1 and M_2, respectively. The shortest distance between the images thus formed is
(Take $\sqrt{5} = 2.3$)



[31 Aug 2021 Shift 1]

Options:

A. 3a

B. 4.6 a

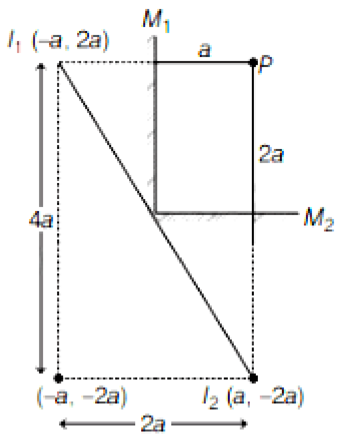
C. 2.3 a

D. $2\sqrt{10}a$

Answer: B

Solution:

According to the given figure, we have two plane mirror, placed at 90º to each other.
Since, image formed by plane mirror are virtual, erect, same size and at same distance behind mirror.
Let I_1 is the image of P due to mirror M_1 and I_2 is the image of P due to mirror M_2



Therefore, the shortest distance between images i.e., between I_1 and I_2

$$I_1I_2 = \sqrt{(4a)^2 + (2a)^2}$$
$$= \sqrt{16a^2 + 4a^2}$$
$$= a\sqrt{20} = a2\sqrt{5}$$
$$= a \times 2 \times 2.3 = 4.6a$$

Question172

An object is placed beyond the centre of curvature C of the given concave mirror. If the distance of the object is d_1 from C and the distance of the image formed is d_2 from C, the radius of curvature of this mirror is
[27 Aug 2021 Shift 1]

Options:

A. $\frac{2d_1d_2}{d_1 - d_2}$

B. $\frac{2d_1d_2}{d_1 + d_2}$

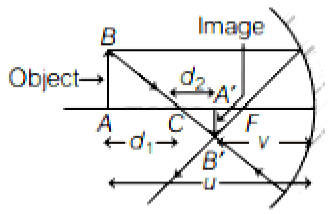
C. $\frac{d_1d_2}{d_1 + d_2}$

D. $\frac{d_1d_2}{d_1 - d_2}$

Answer: A

Solution:

The given situation is shown in the following ray diagram.



In concave mirror, when object is placed beyond C the image is formed between C and focus of mirror.
Consider radius of curvature of mirror is R.
Object distance, $u = -(R + d_1)$
Image distance, $v = -(R - d_2)$
Focal length of mirror, $f = \frac{-R}{2}$
Now, from mirror formula, we have
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{u + v}{uv}$
$\Rightarrow uv = f(u + v)$
Substituting the values in above expression, we get

$[-(R+d_1)][-(R-d_2)] = -\frac{R}{2}(R+d_1+R-d_2)$
$\Rightarrow 2(R^2-d_2R+d_1R-d_1d_2) = 2R^2+d_1R-d_2R$
$\Rightarrow 2R^2-2d_2R+2d_1R-2R^2-d_1R+d_2R = 2d_1d_2$
$\Rightarrow d_1R-d_2R = 2d_1d_2$
$\Rightarrow R = \frac{2d_1d_2}{d_1-d_2}$
Thus, radius of curvature of mirror is $\frac{2d_1d_2}{d_1-d_2}$.

Question173

Car B overtakes another car A at a relative speed of $40ms^{-1}$. How fast will the image of car B appear to move in the mirror of focal length 10 cm fitted in car A, when the car B is 1.9m away from the car A ?
[26 Aug 2021 Shift 1]

Options:

A. $4ms^{-1}$

B. $0.2ms^{-1}$

C. $40ms^{-1}$

D. $0.1ms^{-1}$

Answer: D

Solution:

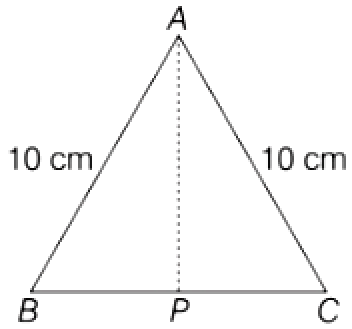
According to the question,
Velocity of car B w.r.t. mirror of car A, $v_{BM} = 40m/s$
Distance of car B from mirror of car A, $u = 1.9m = 190\,cm$
Focal length of mirror, f = 10 cm
We have to find the velocity of image of car B in the mirror, i.e. v_{IM}
We know that,
$v_{IM} = -m^2 v_{BM}$ ....(I)
where, m is the magnification produced by mirror of car A,
$\Rightarrow m = \frac{f}{f-u} = \frac{10}{10-(-190)}$

$=\frac{10}{200}=\frac{1}{20}$
Substituting the values in Eq. (i), we get
$v_{IM}=-\left(\frac{1}{20}\right)^2\times 40$
$=-\frac{1}{400}\times 40=-0.1m/s$
Here, negative sign shows that, the speed of image w.r.t. mirror ofcar A is in opposite direction.

Question174

Cross-section view of a prism is the equilateral triangle ABC in the figure. The minimum deviation is observed using this prism when the angle of incidence is equal to the prism angle. The time taken by light to travel from P (mid-point of BC) to A is..... $\times 10^{-10}$s.

(Given, speed of light in vacuum $=3\times 10^8 m/s$ and $\cos 30°=\frac{\sqrt{3}}{2}$)



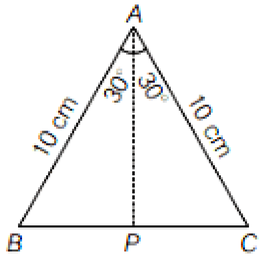
[31 Aug 2021 Shift 2]

Answer: 5

Solution:

Given, the base of a prism is equilateral triangle of side 10 cm as shown in figure.
At minimum deviation, Angle of incident = Angle of prism

i.e. i = A = 60°



Angle of refraction, $r = \frac{A}{2} = 60°2 = 30°$

Let μ be the refractive index.
Then, by Snell's law of refraction,

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin 60°}{\sin 30°} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$$

Now, in ΔABP, AP = AB cos 30°

$$= \frac{10\sqrt{3}}{2} = 5\sqrt{3}\text{ cm}$$

$$= 5\sqrt{3} \times 10^{-2}\text{m}$$

i.e. optical distance travelled by light along AP,

$$d = \mu \times AP = \sqrt{3} \times 5\sqrt{3} \times 10^{-2} = 15 \times 10^{-2}\text{m}$$

Now, the time taken by light $(c = 3 \times 10^{8}\text{m/s})$ to travel from P to A,

$$t = \frac{d}{C} = \frac{15 \times 10^{-2}}{3 \times 10^{8}}\text{s} = 5 \times 10^{-10}\text{s}$$

Thus, correct answer is 5.

Question175

An object is placed at the focus of concave lens having focal length f. What is the magnification and distance of the image from the optical centre of the lens?
[31 Aug 2021 Shift 1]

Options:

A. 1, ∞

B. Very high, ∞

C. $\frac{1}{2}, \frac{f}{2}$

D. $\frac{1}{4}, \frac{f}{4}$

Answer: C

Solution:

Given, focal length of concave lens, f' = -f
Object distance from lens, u = -f
Image distance from lens = v
Magnification = m
By using lens formula,

$$\frac{1}{f'} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{f'} + \frac{1}{u}$$

$$= -\frac{1}{f} - \frac{1}{f} = -\frac{2}{f}$$

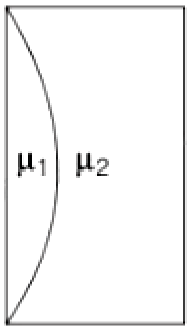
$$\Rightarrow v = -\frac{f}{2}$$

$$\therefore |v| = \frac{f}{2}$$

$$\therefore m = \frac{v}{u} = \frac{-\frac{f}{2}}{-f} = \frac{1}{2}$$

Question176

Curved surfaces of a plano-convex lens of refractive index μ_1 and a plano-concave lens of refractive index μ_2 have equal radius of curvature as shown in figure. Find the ratio of radius of curvature to the focal length of the combined lenses.



[27 Aug 2021 Shift 2]

Options:

A. $\frac{1}{\mu_2 - \mu_1}$

B. $\mu_1 - \mu_2$

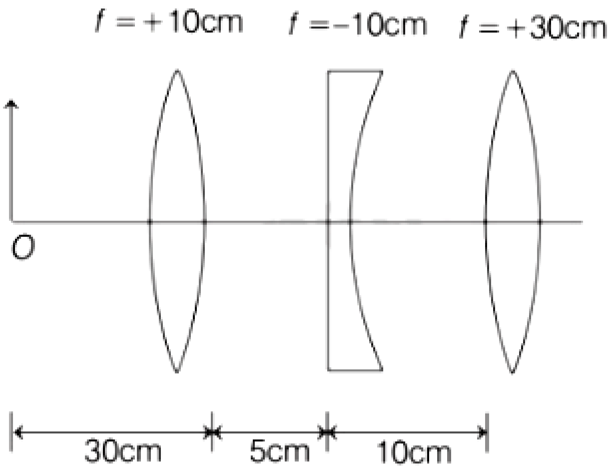
C. $\frac{1}{\mu_1 - \mu_2}$

D. $\mu_2 - \mu_1$

Answer: B

Question177

Find the distance of the image from object O, formed by the combination of lenses in the figure.



[27 Aug 2021 Shift 1]

Options:

A. 75 cm

B. 10 cm

C. 20 cm

D. infinity

Answer: A

Solution:

Given, focal length of first lens, $f_1 = +10\,\text{cm}$

Focal length of second lens, $f_2 = -10\,\text{cm}$

Focal length of third lens, $f_3 = +30\,\text{cm}$

Object distance from first lens, $u_1 = -30\,\text{cm}$

Using lens formula for first lens, we get

$$\frac{1}{f_1} = \frac{1}{v_1} - \frac{1}{u_1}$$

$$\Rightarrow \frac{1}{-10} = \frac{1}{v_1} - \frac{1}{-30}$$

$$\Rightarrow \frac{1}{v_1} = \frac{1}{10} - \frac{1}{30} = \frac{2}{30}$$

$$v_1 = 15\,\text{cm}$$

Thus, the image formed by first lens is 15 cm to the right of first lens.
The object distance for second lens will be

$$u_2 = 15\,\text{cm} - 5\,\text{cm} = 10\,\text{cm}$$

Using lens formula for first lens, we get

$$\frac{1}{f_2} = \frac{1}{v_2} - \frac{1}{u_2}$$

$$\Rightarrow \frac{1}{-10} = \frac{1}{v_2} - \frac{1}{10}$$

$$\Rightarrow \frac{1}{v_2} = \frac{1}{-10} + \frac{1}{10} = 0$$

$$\Rightarrow v_2 = \infty$$

Thus, the object distance for third lens, $u_3 = \infty$.

Using lens formula for third lens, we get

$$\frac{1}{f_3} = \frac{1}{v_3} - \frac{1}{u_3}$$

$$\Rightarrow \frac{1}{30} = \frac{1}{v_3} - \frac{1}{\infty}$$

$$\frac{1}{v_3} = \frac{1}{30}$$

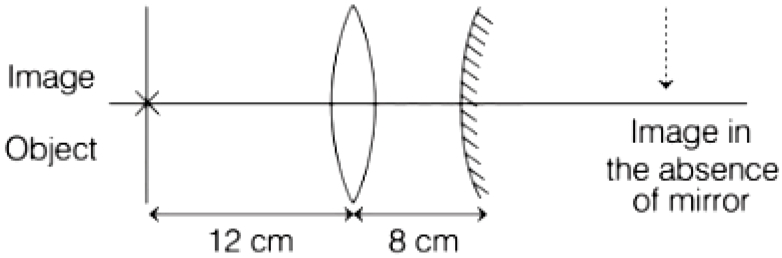
$$v_3 = 30\,\text{cm}$$

The final image will form 30 cm from the third lens.
The distance of final image from object will be $30 + 10 + 5 + 30 = 75\,\text{cm}$
Thus, the distance of final image formed from object O is 75 cm.

Question178

An object is placed at a distance of 12 cm from a convex lens. A convex mirror of focal length 15 cm is placed on other side of lens at 8 cm as shown in the figure. Image of object coincides with the

object.

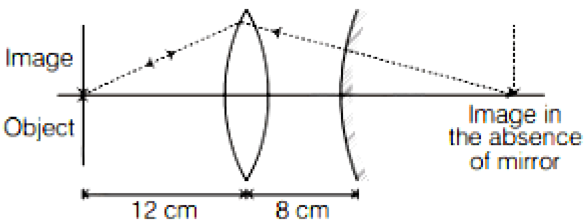


When the convex mirror is removed, a real and inverted image is formed at a position. The distance of the image from the object will be............ cm.
[26 Aug 2021 Shift 2]

Answer: 50

Solution:

It is given that image coincides with the object. In the arrangement given, the image will only coincide with the object when the ray on mirror falls directly perpendicular on the surface of it.



The light will converge at centre of curvature of mirror and due to that after removing the mirror the light will converge at same position.
The focal length of mirror, f = 15 cm. So, radius of curvature of mirror, 2f = 30 cm
The image distance from the object after convex mirror is removed will be calculated as, 12 cm + 8 cm + 30 cm = 50 cm.
Thus, the image will formed at a distance of 50 cm from object.

Question179

A glass tumbler having inner depth of 17.5 cm is kept on a table. A student starts pouring water $\left(\mu = \frac{4}{3}\right)$ into it while looking at the surface of water from the above. When he feels that the tumbler is half filled, he stops pouring water. Up to what height, the tumbler is

actually filled?
[1 Sep 2021 Shift 2]

Options:

A. 11.7 cm

B. 10 cm

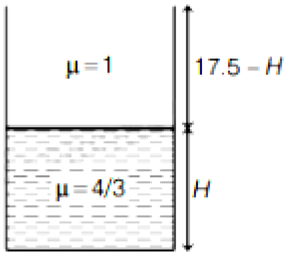
C. 7.5 cm

D. 8.75 cm

Answer: B

Solution:

Let us draw the diagram of glass tumbler.



Consider the actual height of the tumbler be H.
The refractive index of the water, $\mu = \frac{4}{3}$
The refractive index of the air, $\mu = 1$
As we know that,

$$\mu_{water} = \frac{H_{real}}{H_{appearent}}$$

$$\frac{4}{3} = \frac{H}{H_{appearent}}$$

$$\Rightarrow H_{appearent} = \frac{3H}{4}$$

The height of air observed by observer = 17.5 –H
According to the above figure,

$$\frac{3H}{4} = 17.5 - H$$

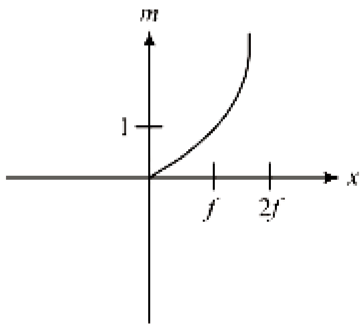
⇒ H = 10.11 cm ≃ 10 cm
The actual height of the water in tumbler is 10 cm.

Question180

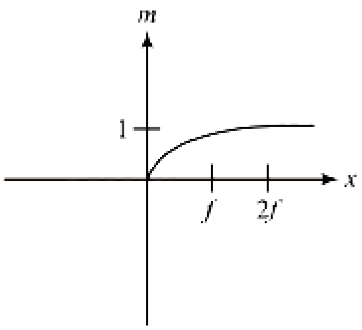
An object is gradually moving away from the focal point of a concave mirror along the axis of the mirror. The graphical representation of the magnitude of linear magnification (m) versus distance of the object from the mirror (x) is correctly given by (Graphs are drawn schematically and are not to scale)
[8 Jan. 2020 II]

Options:

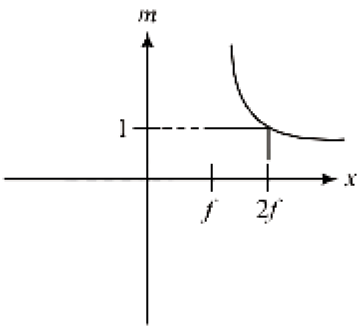
A.



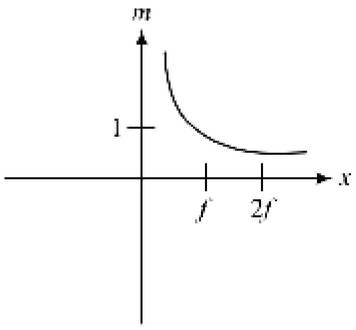
B.



C.



D.

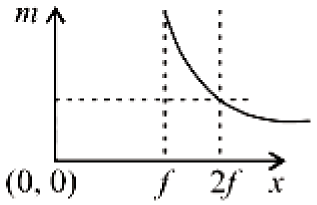


Answer: C

Solution:

Using mirror formula, magnification is given by $m = \frac{f}{u-f} = \frac{-1}{1-\frac{u}{f}}$

At focus magnification is ∞
And at $u = 2f$, magnification is 1 .
Hence graph (d) correctly depicts ' m ' versus distance of object ' x ' graph.



Question181

A vessel of depth 2h is half filled with a liquid of refractive index $2\sqrt{2}$ and the upper half with another liquid of refractive index $\sqrt{2}$. The liquids are immiscible. The apparent depth of the inner surface of the bottom of vessel will be:
[9 Jan. 2020 I]

Options:

A. $\frac{h}{\sqrt{2}}$

B. $\frac{h}{2(\sqrt{2}+1)}$

C. $\frac{h}{3\sqrt{2}}$

D. $\frac{3}{4}h\sqrt{2}$

Answer: D

Solution:

Apparent depth,
$h_{app} =$ apperent depth

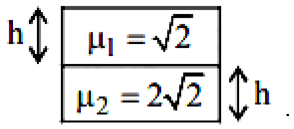
$$h_{\text{app}} = 1\left|\frac{h_1}{\mu_1} + \frac{h_2}{\mu_2}\right|$$

$$= \frac{h}{2\sqrt{2}} + \frac{h}{\sqrt{2}}$$

$$= \frac{h}{\sqrt{2}}\left(\frac{1}{2} + 1\right) = \frac{3h}{2\sqrt{2}}$$

$$h_{\text{app}} = \frac{3h}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{3h}{4} \times \sqrt{2}$$

$$h_{app} = \frac{3h}{4} \times \sqrt{2}$$



Question182

There is a small source of light at some depth below the surface of water (refractive index $= \frac{4}{3}$) in a tank of large cross sectional surface area. Neglecting any reflection from the bottom and absorption by water, percentage of light that emerges out of surface is (nearly):
[Use the fact that surface area of a spherical cap of height h and radius of curvature r is $2\pi rh$]
[9 Jan. 2020 II]

Options:

A. 21%

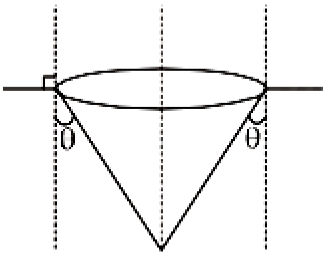
B. 34%

C. 17%

D. 50%

Answer: C

Solution:



Given,

Refractive index, $\mu = \frac{4}{3}$

$\frac{4}{3} \sin\theta = 1 \sin 90°$

$\Rightarrow \sin\theta = \frac{3}{4}$

$\cos\theta = \frac{\sqrt{7}}{4}$

Solid angle, $\Omega = 2\pi(1-\cos\theta) = 2\pi(1-\sqrt{7}/4)$

Fraction of energy transmitted

$= \frac{2\pi(1-\cos\theta)}{4\pi} = \frac{1-\sqrt{7}/4}{2} = 0.17$

Percentage of light emerges out of surface

$= 0.17 \times 100 = 17\%$

Question183

The critical angle of a medium for a specific wavelength, if the medium has relative permittivity 3 and relative permeability $\frac{4}{3}$ for this wavelength, will be:
[8 Jan. 2020 I]

Options:

A. 15°

B. 30°

C. 45°

D. 60°

Answer: B

Solution:

Here, from question, relative permittivity

$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0} = 3 \Rightarrow \varepsilon = 3\varepsilon_0$

Relative permeability $\mu_r = \frac{\mu}{\mu_0} = \frac{4}{3} \Rightarrow \mu = \frac{4}{3}\mu_0$

$\therefore \mu\varepsilon = 4\mu_0\varepsilon_0$

$\sqrt{\frac{\mu_0\varepsilon_0}{\mu\varepsilon}} = \frac{v}{c} = \frac{1}{2} \left(\because c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}\right)$

$n = \sqrt{\mu_r\varepsilon_r} = \sqrt{\frac{4}{3} \times 3} = 2$

And $n = \frac{1}{\sin\theta_c}$

$\Rightarrow \sin\theta_c = \frac{1}{n} = \frac{1}{2}$

$\therefore$ Critical angle, $\theta_c = 30°$

Question184

A point object in air is in front of the curved surface of a plano-convex lens. The radius of curvature of the curved surface is 30 cm and the refractive index of the lens material is 1.5, then the focal length of the lens (in cm) is__________.
[NA 8 Jan. 2020 I]

Answer: 60

Solution:

Given :$\mu = 1.5$; $R_{curved} = 30cm$

Using, Lens-maker formula

$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - 1R_2\right)$

For plano-convex lens $R_1 \rightarrow \infty$ then $R_2 = -R$

$\therefore f = \frac{R}{\mu - 1} = \frac{30}{1.5 - 1} = 60\text{cm}$

Question185

A thin lens made of glass (refractive index = 1.5) of focal length f = 16 cm is immersed in a liquid of refractive index 1.42. If its focal length in liquid is f_l ,then the ratio f_l / f is closest to the integer: [7 Jan. 2020 II]

Options:

A. 1

B. 9

C. 5

D. 17

Answer: B

Solution:

Using lens maker's formula

$\frac{1}{f} = \left(\frac{\mu_g}{\mu_a} - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$

Here, μ_g and μ_a are the refractive index of glass and air respectively

$\Rightarrow \frac{1}{f} = (1.5 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ....(i)

When immersed in liquid

$\frac{1}{f_l} = \left(\frac{\mu_g}{\mu_l} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

[Here, $\mu_l =$ refractive index of liquid]

$\Rightarrow \frac{1}{f_l} = \left(\frac{1.5}{1.42} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ....(ii)

Dividing (i) by(ii)

$\Rightarrow \frac{f_l}{f} = \frac{(1.5 - 1)1.42}{0.08} = \frac{1.42}{0.16} = \frac{142}{16} \approx 9$

Question186

The magnifying power of a telescope with tube length 60 cm is 5. What is the focal length of its eye piece?
[8 Jan. 2020 I]

Options:

A. 20 cm

B. 40 cm

C. 30 cm

D. 10 cm

Answer: D

Solution:

For telescope
Tube length $(L) = f_o + f_e = 60$
and magnification $(m) = \frac{f_o}{f_e} = 5 \Rightarrow f_0 = 5f_e$
$\therefore f_o = 50cm$ and $f_e = 10cm$
Hence focal length of eye-piece, $f_e = 10cm$

Question187

If we need a magnification of 375 from a compound microscope of tube length 150 mm and an objective of focal length 5 mm, the focal length of the eye-piece, should be close to:
[7 Jan. 2020 I]

Options:

A. 22 mm

B. 12 mm

C. 2 mm

D. 33 mm

Answer: A

Solution:

According question, $M = 375$
$L = 150\text{mm}, f_0 = 5\text{mm}$ and $f_e = ?$

Using, magnification, $M \simeq \frac{L}{f_0}\left(1 + \frac{D}{f_e}\right)$

$\Rightarrow 375 = \frac{150}{5}\left(1 + \frac{250}{f_e}\right)\ (\because D = 25\text{cm} = 250\text{mm})$

$\Rightarrow 12.5 = 1 + \frac{250}{f_e}$

$\Rightarrow f_e = \frac{250}{11.5} = 21.7 \approx 22\text{mm}$

Question188

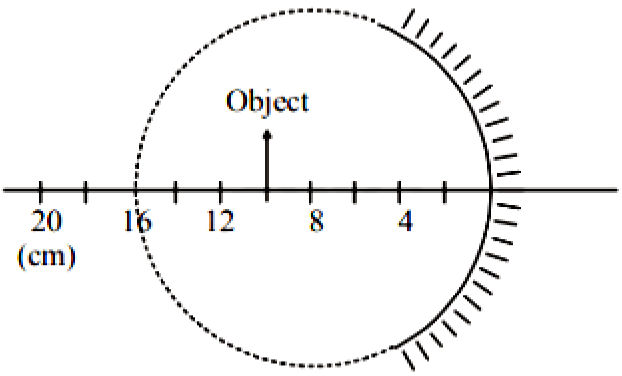
TOPIC 1 -Plane Mirror, Spherical Mirror and Reflection of Light
When an object is kept at a distance of 30cm from a concave mirror, the image is formed at a distance of 10cm from the mirror. If the object is moved with a speed of 9 cms^{-1}, the speed (in cms^{-1}) with which image moves at that instant is ________.
[NA Sep. 03,2020 (II)]

Answer: 1

Solution:

Distance of object, $u = -30\text{cm}$
Distance of image, $v = 10\text{cm}$
Magnification, $m = \frac{-v}{u} = \frac{(-10)}{-30} = \frac{1}{3}$
Speed of image $= m^2 \times$ speed of object $= \frac{1}{9} \times 9 = 1\text{cms}^{-1}$

Question189



A spherical mirror is obtained as shown in the figure from a hollow glass sphere. If an object is positioned in front of the mirror, what will be the nature and magnification of the image of the object? (Figure drawn as schematic and not to scale)
[Sep. 02, 2020 (I)]

Options:

A. Inverted, real and magnified

B. Erect, virtual and magnified

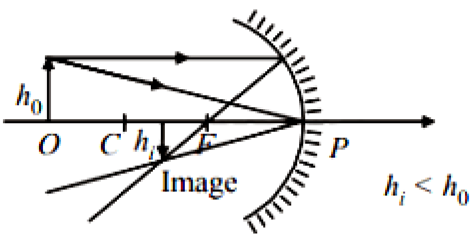
C. Erect, virtual and unmagnified

D. Inverted, real and unmagnified

Answer: D

Solution:

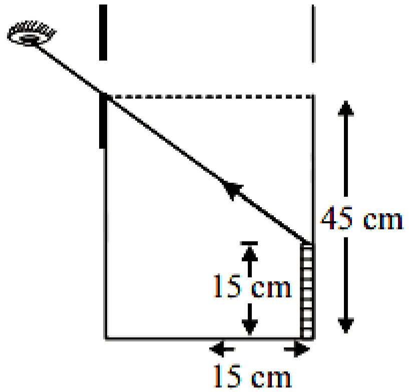
Object is placed beyond radius of curvature (R) of concave mirror hence image formed is real, inverted and diminished or unmagnified.



Question190

TOPIC 2 - Refraction of Light at Plane Surface and Total Internal Reflection

An observer can see through a small hole on the side of a jar (radius 15 cm) at a point at height of 15 cm from the bottom (see figure). The hole is at a height of 45 cm. When the jar is filled with a liquid up to a height of 30 cm the same observer can see the edge at the bottom of the jar. If the refractive index of the liquid is N/100, where N is an integer, the value of N is ___________.



[NA Sep. 03, 2020 (I)]

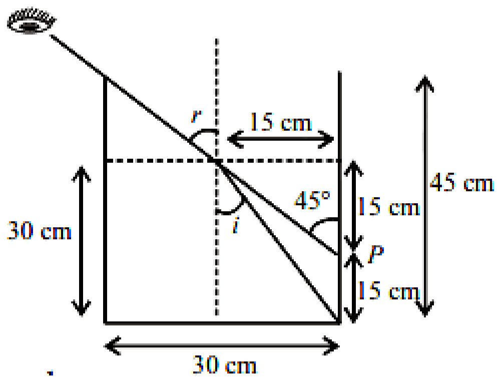
Answer: 158

Solution:

From figure, $\sin i = \frac{15}{\sqrt{15^2 + 30^2}}$ and $\sin r = \sin 45°$

From Snell's law, $\mu \times \sin i = 1 \times \sin r$

$\Rightarrow \mu \times \frac{15}{\sqrt{15^2 + 30^2}} = 1 \times \sin 45° = \frac{1}{\sqrt{2}}$



$$\therefore \mu = \frac{\frac{1}{\sqrt{2}}}{\frac{15}{\sqrt{1125}}} = 158 \times 10^{-2} = \frac{N}{100}$$

Hence, value of $N \simeq 158$

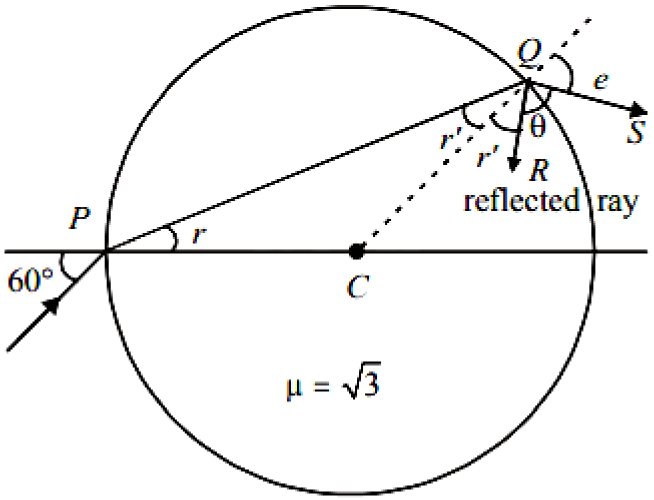
Question191

A light ray enters a solid glass sphere of refractive index $\mu = \sqrt{3}$ at an angle of incidence 60°. The ray is both reflected and refracted at the farther surface of the sphere. The angle (in degrees) between the reflected and refracted rays at this surface is _______.
[NA Sep. 02, 2020 (II)]

Answer: 90

Solution:

In the figure, QR is the reflected ray and QS is refracted ray. CQ is normal.



Apply Snell's law at P

$1 \sin 60° = \sqrt{3} \sin r$

$\Rightarrow \sin r = \frac{1}{2} \Rightarrow r = 30°$

From geometry, $CP = CQ$

$\therefore r' = 30°$

Again apply snell's law at Q,

$\sqrt{3} \sin r' = 1 \sin e$

$\Rightarrow \frac{\sqrt{3}}{2} = \sin e \Rightarrow e = 60°$
From geometry
$r' + \theta + e = 180°$ (As angles lies on a straight line)
$\Rightarrow 30° + \theta + 60° = 180° \Rightarrow \theta = 90°$

Question192

TOPIC 3 - Refraction at Curved Surface Lenses and Power of Lens A point like object is placed at a distance of 1 m in front of a convex lens of focal length 0.5 m. A plane mirror is placed at a distance of 2 m behind the lens. The position and nature of the final image formed by the system is : [Sep. 06, 2020 (I)]

Options:

A. 2.6 m from the mirror, real

B. 1 m from the mirror, virtual

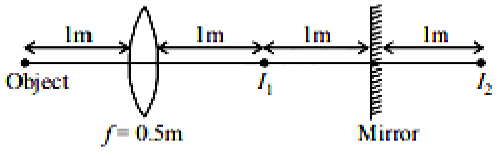
C. 1 m from the mirror, real

D. 2.6 m from the mirror, virtual

Answer: D

Solution:

Focal length of the convex lens, $f = 0.5m$
Object is at $2f$ so, image (I_1) will also be at $2f$.
Image of I_1 i.e., I_2 will be $1m$ behind mirror.
Now I_2 will be object for lens.



$\therefore u = (-1) + (-1) + (-1) = -3m$
Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{+0.5} + \frac{1}{-3}$ or $v = \frac{3}{5} = 0.6m$
Hence, distance of image from mirror
$= 2 + 0.6 = 2.6m$ and real.

Question193

A double convex lens has power P and same radii of curvature R of both the surfaces. The radius of curvature of a surface of a plano-convex lens made of the same material with power 1.5P is : [Sep. 06, 2020 (II)]

Options:

A. 2R

B. $\frac{R}{2}$

C. $\frac{3R}{2}$

D. $\frac{R}{3}$

Answer: D

Solution:

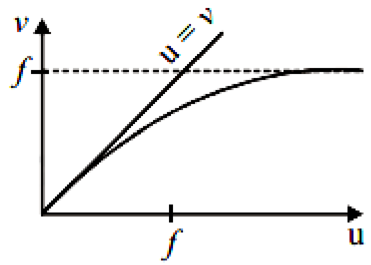
Given, using lens maker's formula
$\frac{1}{f} = (k-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$
Here, $R_1 = R_2 = R$ (For double convex lens)
$\therefore \frac{1}{f} = (\mu - 1)\left(\frac{1}{R} - \frac{1}{-R}\right)$
$\Rightarrow P = \frac{1}{f} = (\mu - 1)\frac{2}{R}$ ....(i)
For plano convex lens,
$R_1 = R', R_2 = \infty$
Using lens maker's formula again, we have
$1.5P = (\mu - 1)\left(\frac{1}{R'} - \frac{1}{\infty}\right)$ ....(ii)
$\Rightarrow \frac{3}{2}P = \frac{\mu - 1}{R'}$
From (i) and (ii),
$\frac{3}{2} = \frac{R'}{2R} \Rightarrow R' = \frac{R}{3}$

Question194

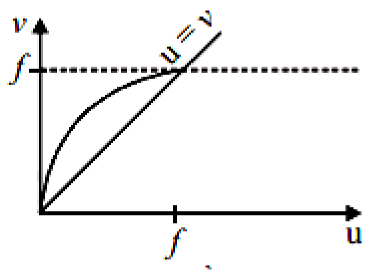
For a concave lens of focal length f, the relation between object and image distances u and v, respectively, from its pole can best be represented by (u = v is the reference line) :
[Sep. 05, 2020 (I)]

Options:

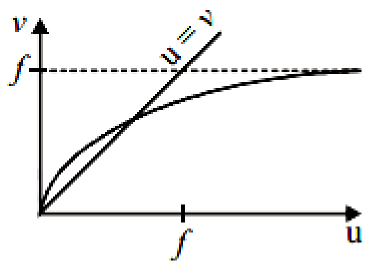
A.



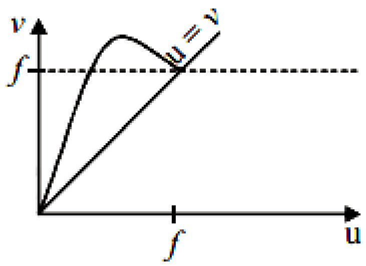
B.



C.



D.



Answer: D

Solution:

From lens formula,
$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \Rightarrow v=\frac{uf}{u+f}$
Case-I : If $v=u \Rightarrow f+u=f \Rightarrow u=0$
Case-II : If $u=\infty$ then $v=f$.
Hence, correct u versus v graph, that satisfies this condition is (a).

Question195

The distance between an object and a screen is 100cm. A lens can produce real image of the object on the screen for two different positions between the screen and the object. The distance between these two positions is 40cm. If the power of the lens is close to $\left(\frac{N}{100}\right)$D where N is an integer, the value of N is _______.
[NA Sep. 04, 2020 (II)]

Answer: 476.19

Solution:

Given,
Distance between an object and screen, D = 100 cm
Distance between the two position of lens, d = 40 cm
Focal length of lens,
$f=\frac{D^2-d^2}{4D}=\frac{100^2-40^2}{4(100)}=\frac{(100+40)(100-40)}{4(100)}=21\text{cm}$
Power, $P=\frac{1}{f}=\frac{100}{21}=\frac{N}{100}$
$\therefore N=476.19$

Question196

TOPIC 4 - Prism and Dispersion of Light

The surface of a metal is illuminated alternately with photons of energies $E_1 = 4eV$ and $E_2 = 2.5eV$ respectively. The ratio of maximum speeds of the photoelectrons emitted in the two cases is 2 . The work function of the metal in (eV) is _______.

[NA Sep. 05, 2020 (II)]

Answer: 2

Solution:

From the Einstein's photoelectric equation
Energy of photon = Kinetic energy of photoelectrons + Work function
⇒ Kinetic energy = Energy of Photon − Work Function Let ϕ_0 be the work function of metal and v_1 and v_2 be the velocity of photoelectrons. Using Einstein's photoelectric equation we have

$\frac{1}{2}mv_1{}^2 = 4 - \phi_0$ ....(i)

$\frac{1}{2}mv_2{}^2 = 2.5 - \phi_0$ ....(ii)

$$\Rightarrow \frac{\frac{1}{2}mv_1{}^2}{\frac{1}{2}mv_2{}^2} = \frac{4-\phi_0}{2.5-\phi_0}$$

$$\Rightarrow (2)^2 = \frac{4-\phi_0}{2.5-\phi_0} \Rightarrow 10 - 4\phi_0 = 4 - \phi_0$$

$\phi_0 = 2eV$

Question197

. A compound microscope consists of an objective lens of focal length 1 cm and an eye piece of focal length 5 cm with a separation of 10 cm.

The distance between an object and the objective lens, at which the strain on the eye is minimum is $\frac{n}{40}$ cm.

The value of n is ______.

[NA Sep. 05, 2020 (I)]

Answer: 50

Solution:

Given : Length of compound microscope, $L = 10cm$
Focal length of objective $f_0 = 1cm$ and of eye-piece,

$f_e = 5cm$

$u_0 = f_e = 5cm$

Final image formed at infinity (∞), $v_e = \infty$

$v_0 = 10 - 5 = 5$

Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0} \Rightarrow \frac{1}{5} - \frac{1}{u_0} = \frac{1}{1} \Rightarrow u_0 = -\frac{5}{4}cm$

or, $\frac{5}{4} = \frac{N}{40}$

$\therefore N = \frac{200}{4} = 50cm$

Question198

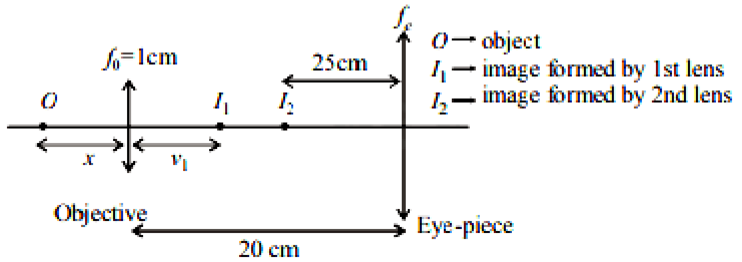
In a compound microscope, the magnified virtual image is formed at a distance of 25 cm from the eye-piece. The focal length of its objective lens is 1 cm. If the magnification is 100 and the tube length of the microscope is 20 cm, then the focal length of the eye-piece lens (in cm) is __________.
[NA Sep. 04, 2020 (I)]

Answer: 4.48

Solution:

According to question, final image i.e., $v_2 = 25cm$,

$f_0 = 1cm$, magnification, $m = m_1 m_2 = 100$



Using lens formula, For first lens or objective $= \frac{1}{v_1} - \frac{1}{-x} = \frac{1}{1} \Rightarrow v_1 = \frac{x}{x-1}$

Also magnification $|m_1| = \left|\frac{v_1}{u_1}\right| = \frac{1}{x-1}$

For 2 nd lens or eye-piece, this is acting as object

$\therefore u_2 = -(20 - v_1) = -\left(20 - \frac{x}{x-1}\right)$ and $v_2 = -25\text{cm}$

Angular magnification $|m_A| = \left|\frac{D}{u_2}\right| = \frac{25}{|u_2|}$

Total magnification $m = m_1 m_A = 100$

$$\left(\frac{1}{x-1}\right)\left(\frac{25}{20 - \frac{x}{x-1}}\right) = 100$$

$$\Rightarrow \frac{25}{20(x-1) - x} = 100 \Rightarrow 1 = 80(x-1) - 4x$$

$$\Rightarrow 76x = 81 \Rightarrow x = \frac{81}{76}$$

$$\Rightarrow u_2 = -\left(20 - \frac{\frac{81}{76}}{\frac{81}{76} - 1}\right) = \frac{-19}{5}$$

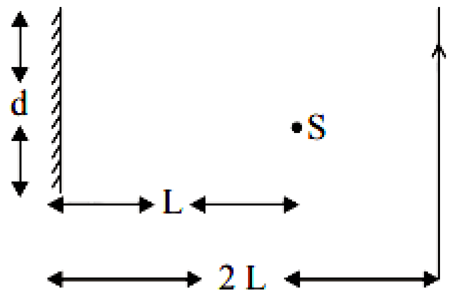
Again using lens formula for eye-piece

$$\frac{1}{-25} - \frac{1}{-\frac{19}{5}} = \frac{1}{f_e} \Rightarrow f_e = \frac{25 \times 19}{106} \approx 4.48\text{cm}$$

Question199

A point source of light, S is placed at a distance L in front of the centre of plane mirror of width d which is hanging vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror, at a distance 2L as show below. The distance over which the

man can see the image of the light source in the mirror is:



[12 Jan. 2019 I]

Options:

A. d

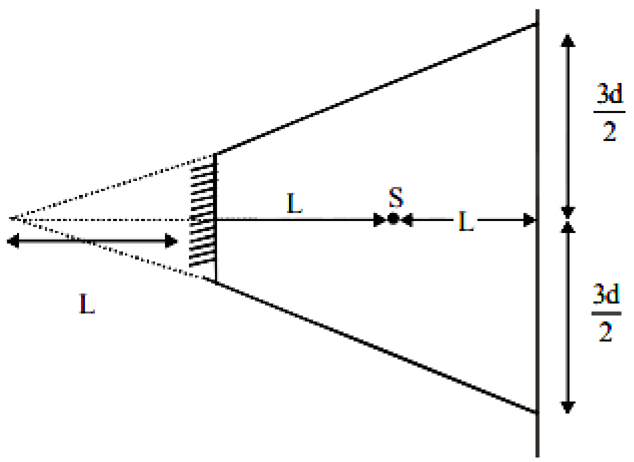
B. 2d

C. 3d

D. $\frac{d}{2}$

Answer: C

Solution:



Total distance $= \frac{3d}{2} + \frac{3d}{2} = 3d$

Question200

Two plane mirrors are inclined to each other such that a ray of light incident on the first mirror (M_1) and parallel to the second mirror

(M_2) is finally reflected from the second mirror (M_2) parallel to the first mirror (M_1). The angle between the two mirrors will be: [9 Jan. 2019 II]

Options:

A. 45°

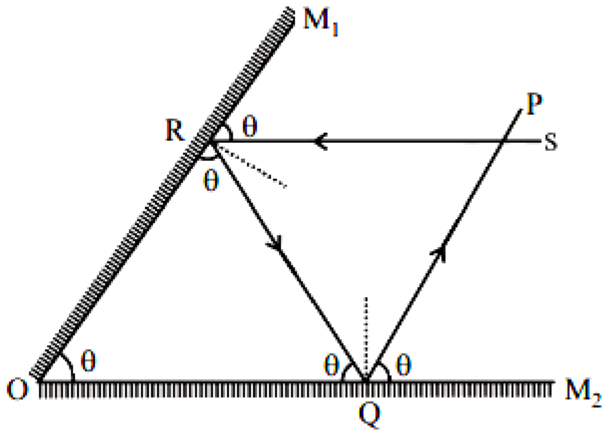
B. 60°

C. 75°

D. 90°

Answer: B

Solution:



Let angle between the two mirrors be θ.
Ray PQ || mirror M_1 and Rs || mirror M_2
$\therefore M_1Rs = \angle ORQ = \angle M_1OM_2 = \theta$
Similarly, $\angle M_2QP = \angle OQR = \angle M_2OM_1 = \theta$
In ΔORQ, $3\theta = 180° \Rightarrow \theta = \dfrac{180°}{3} = 60°$

Question201

A light wave is incident normally on a glass slab of refractive index 1.5. If 4% of light gets reflected and the amplitude of the electric field of the incident light is 30 V/m, then the amplitude of the electric

field for the wave propogating in the glass medium will be: [12 Jan. 2019 I]

Options:

A. 30 V/m

B. 10 V/m

C. 24 V/ m

D. 6 V/m

Answer: C

Solution:

As 4% of light gets reflected, so only $(100-4=96\%)$ of light comes after refraction so,

$P_{\text{refracted}} = \frac{96}{100} P_I$

$\Rightarrow K_2 A_t^2 = \frac{96}{100} K_1 A_i^2$

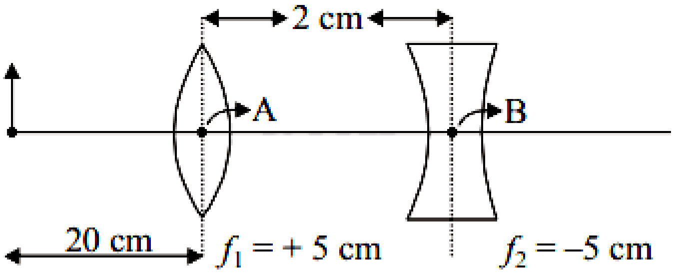
$\Rightarrow r_2 A_t^2 = \frac{96}{100} r_1 A_i^2$

$\Rightarrow A_t^2 = \frac{96}{100} \times \frac{1}{3} \times (30)^2$

$A_t \sqrt{\frac{64}{100} \times (30)^2} = 24$

Question202

What is the position and nature of image formed by lens combination shown in figure? (f_1, f_2 are focal lengths)



[12 Jan. 2019 I]

Options:

A. 70 cm from point B at left; virtual

B. 40 cm from point B at right; real

C. $\frac{20}{3}$ cm from point B at right, real

D. 70 cm from point B at right; real

Answer: D

Solution:

By lens's formula, $\frac{1}{V}-\frac{1}{u}=\frac{1}{f}$

For first lens, $[u_1=-20]$

$\frac{1}{V_1}-\frac{1}{-20}=\frac{1}{5} \Rightarrow V_I=\frac{20}{3}$

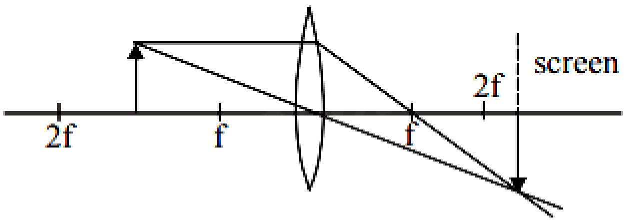
Image formed by first lense will behave as an object for second lens

so, $u_2=\frac{20}{3}-2=\frac{14}{3}$

$\frac{1}{V_2}-\frac{1}{\frac{14}{3}}=\frac{1}{-5} \Rightarrow V_2=70\text{cm}$

Question203

Formation of real image using a biconvex lens is shown below :



If the whole set up is immersed in water without disturbing the object and the screen positions, what will one observe on the screen ?
[12 Jan. 2019 II]

Options:

A. Image disappears

B. Magnified image

C. Erect real image

D. No change

Answer: A

Solution:

According to lens maker's formula,

$$\frac{1}{f}=(\mu_{rel}-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

Focal length of lens will change due to change in refrective index μ_{rel}. So, image will be formed at new position. Hence image disappears

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Question204

A plano-convex lens (focal length f_2, refractive index μ_2, radius of curvature R) fits exactly into a plano-concave lens (focal length f_1, refractive index μ_1, radius of curvature R). Their plane surfaces are parallel to each other. Then, the focal length of the combination will be : [12 Jan. 2019 II]

Options:

A. f_1-f_2

B. $\frac{R}{\mu_2-\mu_1}$

C. $\frac{2f_1f_2}{f_1+f_2}$

D. f_1+f_2

Answer: B

Solution:

$\frac{1}{f_2} = (\mu_2 - 1)\left(\frac{+1}{R}\right)$
$\frac{1}{f_1} = (\mu_1 - 1)\frac{(-1)}{R}$
Now when combined the focal length is given by
$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$
$= (\mu_1 - 1)\frac{(-1)}{R} + (\mu_2 - 1)\frac{+1}{R}$
$= \frac{1}{R}[\mu_2 - 1 - \mu_1 + 1]$
$= \frac{\mu_2 - \mu_1}{R}$
$\Rightarrow f = \frac{R}{\mu_2 - \mu_1}$

Question205

An object is at a distance of 20 m from a convex lens of focal length 0.3 m. The lens forms an image of the object. If the object moves away from the lens at a speed of 5m/s, the speed and direction of the image will be :
[11 Jan. 2019 I]

Options:

A. 2.26×10^{-3} m/s away from the lens

B. 0.92×10^{-3} m/s away from the lens

C. 3.22×10^{-3} m/s towards the lens

D. 1.16×10^{-3} m/s towards the lens

Answer: D

Solution:

By lens formula
$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
$\frac{1}{v} - \frac{1}{(-20)} = \frac{10}{3}$
$\frac{1}{V} = \frac{10}{3} - \frac{1}{20}$

$\frac{1}{v}=\frac{197}{60}; v=\frac{60}{197}$
Magnification of lens (m) is given by
$m=\left(\frac{v}{u}\right)=\frac{\left(\frac{60}{197}\right)}{20}$
velocity of image wrt. to lens is given by
$v_{I/L}=m^2 v_{O/L}$
direction of velocity of image is same as that of object
$v_{O/L}=5m/s$
$v_{I/L}=\left(\frac{60\times 1}{197\times 20}\right)^2$
$=1.16\times 10^{-3} m/s$ towards the lens

Question206

A plano convex lens of refractive index μ_1 and focal length f_1 is kept in contact with another plano concave lens of refractive index μ_2 and focal length f_2 If the radius of curvature of their spherical faces is R each and $f_1 = 2f_2$, then μ_1 and μ_2 are related as: [10 Jan. 2019 I]

Options:

A. $\mu_1+\mu_2=3$

B. $2\mu_1-\mu_2=1$

C. $3\mu_2-2\mu_1=1$

D. $2\mu_2-\mu_1=1$

Answer: B

Solution:

From lens maker's formula,
$\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$
$\frac{1}{f_1}=(\mu_1-1)\left(\frac{1}{\infty}-\frac{1}{-R}\right)=\frac{1}{2f_2}$

Similarly, for plano-concave lens

$\frac{1}{f_2} = (\mu_2 - 1)\left(\frac{1}{-R} - \frac{1}{\infty}\right)$

Dividing $\frac{1}{f_1}$ by $\frac{1}{f_2}$ we get,

$\frac{(\mu_1 - 1)}{R} = \frac{(\mu_2 - 1)}{2R}$

or, $2\mu_1 - \mu_2 = 1$

Question207

The eye can be regarded as a single refracting surface. The radius of curvature of this surface is equal to that of cornea (7.8 mm). This surface separates two media of refractive indices 1 and 1.34. Calculate the distance from the refracting surface at which a parallel beam of light will come to focus. [10 Jan. 2019 II]

Options:

A. 1 cm

B. 2 cm

C. 4.0 cm

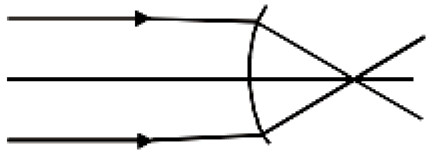
D. 3.1 cm

Answer: D

Solution:

using, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$R = 7.8mm$



$\mu_1 = 1 \mu_2 = 1.34$

$\Rightarrow \frac{1.34}{V} - \frac{1}{\infty} = \frac{1.34 - 1}{7.8} [\because u = \infty]$

$\therefore V = 30.7mm = 3.07cm \simeq 3.1cm$

Question208

A convex lens is put 10 cm from a light source and it makes a sharp image on a screen, kept 10 cm from the lens. Now a glass block (refractive index 1.5) of 1.5 cm thickness is placed in contact with the light source. To get the sharp image again, the scream is shifted by a distance d. Then d is: [9 Jan. 2019 I]

Options:

A. 1.1 cm away from the lens

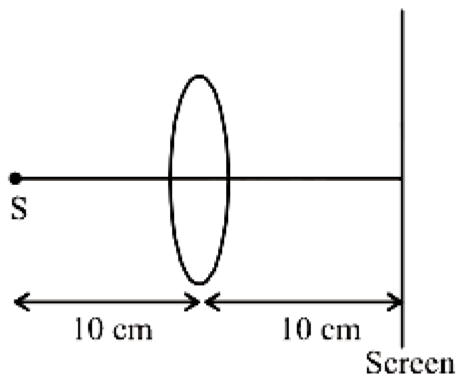
B. 0

C. 0.55 cm towards the lens

D. 0.55 cm away from the lens

Answer: D

Solution:



Using lens formula

$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{1}{10}-\frac{1}{-10}=\frac{1}{f} \Rightarrow f=5\mathrm{cm}$

Shift due to slab, $=t\left(1-\frac{1}{\mu}\right)$ in the direction of incident ray

or, $d=1.5\left(1-\frac{2}{3}\right)=0.5$

Now, $u=-9.5$

Again using lens formulas $\frac{1}{v}-\frac{1}{-9.5}=\frac{1}{5}$

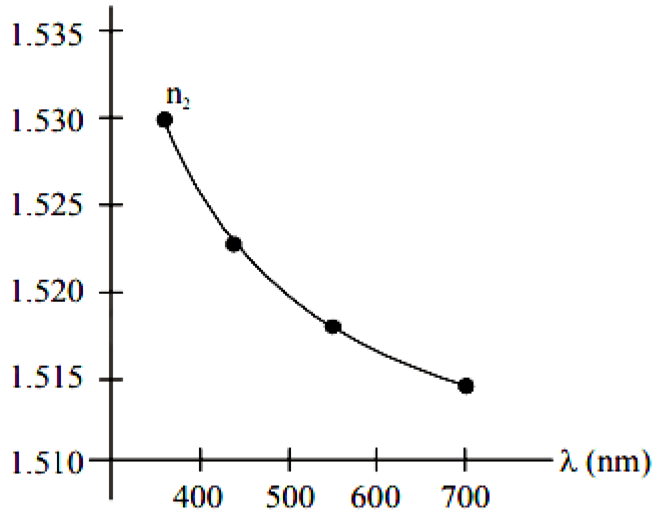
$\Rightarrow \frac{1}{v}=\frac{1}{5}-\frac{2}{19}=\frac{9}{95}$

or, $v=\frac{95}{9}=10.55\mathrm{cm}$

Thus, screen is shifted by a distance
$d = 10.55 - 10 = 0.55 \mathrm{cm}$ away from the lens.

Question209

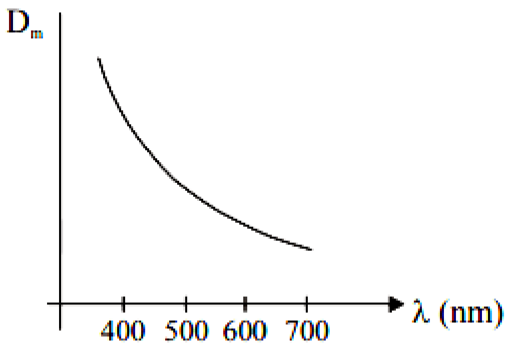
The variation of refractive index of a crown glass thin prism with wavelength of the incident light is shown. Which of the following graphs is the correct one, if D_m is the angle of minimum deviation ?



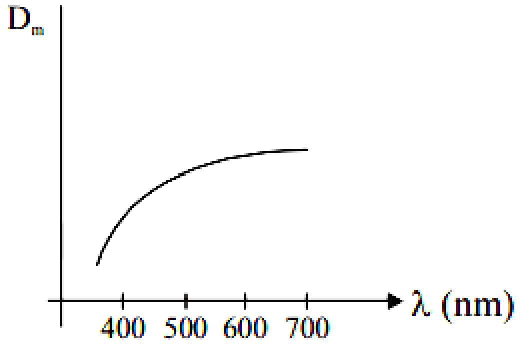
[11 Jan. 2019, I]

Options:

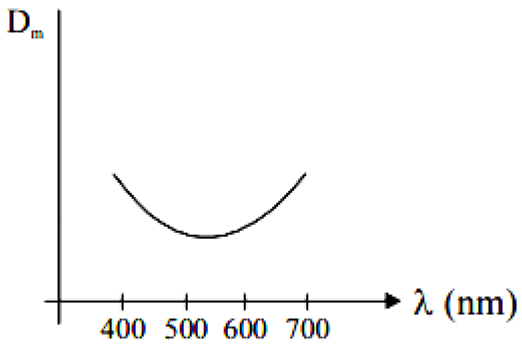
A.



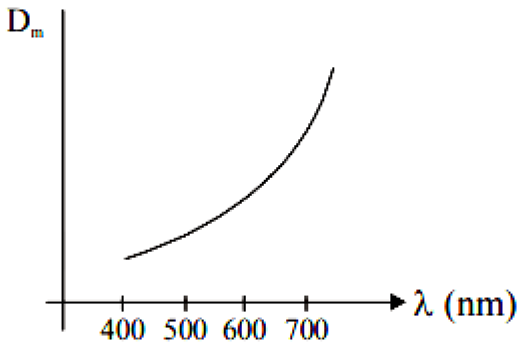
B.



C.



D.



Answer: A

Solution:

When angle of prism is small, then angle of deviation is given by $D_m = (\mu - 1)A$
So, if wavelength of incident light is increased, mu decreases and hence D_m decreases.

Question210

A monochromatic light is incident at a certain angle on an equilateral triangular prism and suffers minimum deviation. If the refractive index of the material of the prism is $\sqrt{3}$, then the angle of incidence is :
[11 Jan. 2019 II]

Options:

A. 90°

B. 30°

C. 60°

D. 45°

Answer: C

Solution:

For minimum deviation:
$r_1 = r_2 = \frac{A}{2} = 30°$
by Snell's law $\mu_1 \sin i = \mu_2 \sin r$
$1 \times \sin i = \sqrt{3} \times \frac{1}{2} = \frac{\sqrt{3}}{2} \Rightarrow i = 60$

Question211

A concave mirror for face viewing has focal length of 0.4 m. The distance at which you hold the mirror from your face in order to see your image upright with a magnification of 5 is:
[9 April 2019 I]

Options:

A. 0.24 m

B. 1.60 m

C. 0.32 m

D. 0.16 m

Answer: C

Solution:

$+5 = -\frac{v}{u} \Rightarrow v = -5u$

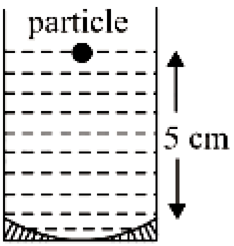
Using $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

or $\frac{1}{-5u} + \frac{1}{u} = \frac{1}{0.4}$

$\therefore u = 0.32m$

Question212

A concave mirror has radius of curvature of 40 cm. It is at the bottom of a glass that has water filled up to 5 cm (see figure). If a small particle is floating on the surface of water, its image as seen, from directly above the glass, is at a distance d from the surface of water. The value of d is close to :
(Refractive index of water = 1.33)



[12 Apr. 2019 I]

Options:

A. 6.7 cm

B. 13.4 cm

C. 8.8 cm

D. 11.7 cm

Answer: C

Solution:

If v is the distance of image formed by mirror, then

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$$ or $\frac{1}{v}+\frac{1}{-5}=\frac{1}{-20}$

$\therefore v=\frac{20}{3}\text{cm}$

Distance of this image from water surface

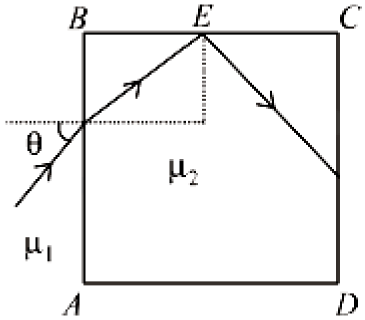
$=\frac{20}{3}+5=\frac{35}{3}\text{cm}$

Using, $\frac{RD}{AD}=\mu$

$\therefore AD=d=\frac{RD}{\mu}=\frac{(35/3)}{1.33}=8.8\text{cm}$

Question213

A transparent cube of side d, made of a material of refractive index μ_2, is immersed in a liquid of refractive index $\mu_1(\mu_1<\mu_2)$. Aray is incident on the face AB at an angle θ (shown in the figure). Total internal reflection takes place at point E on the face BC.



Then θ must satisfy :
[12 Apr. 2019 II]

Options:

A. $\theta<\sin^{-1}\frac{\mu_1}{\mu_2}$

B. $\theta>\sin^{-1}\sqrt{\frac{\mu_2^2}{\mu_1^2}-1}$

C. $\theta<\sin^{-1}\sqrt{\frac{\mu_2^2}{\mu_1^2}-1}$

D. $\theta > \sin^{-1}\frac{\mu_1}{\mu_2}$

Answer: C

Solution:

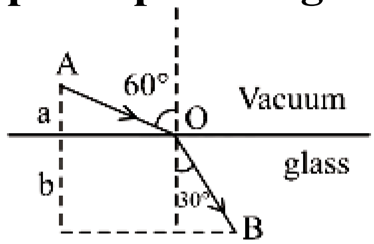
Using, $\sin\theta_{max} = \mu_1\sqrt{\mu_2^2 - \mu_1^2} = \sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}$

or $\theta_{max} = \sin^{-1}\left(\sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}\right)$

For T_1R, $\theta < \sin^{-1}\left(\sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}\right)$

Question214

A ray of light AO in vacuum is incident on a glass slab at angle 60° and refracted at angle 30° along OB as shown in the figure. The optical path length of light ray from A to B is :



[10 Apr. 2019 I]

Options:

A. $\frac{2\sqrt{3}}{a} + 2b$

B. $2a\frac{2b}{3}$

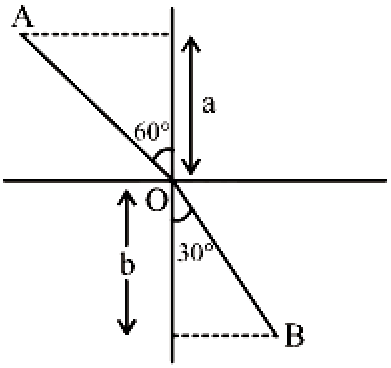
C. $2a + \frac{2b}{\sqrt{3}}$

D. $2a + 2b$

Answer: D

Solution:

From the given figure
As $\sin 60° = \mu \sin 30°$



$\Rightarrow \mu = \dfrac{\sin 60°}{\sin 30°} = \sqrt{3}$

$\dfrac{a}{AO} = \cos 60° \Rightarrow AO = 2a$

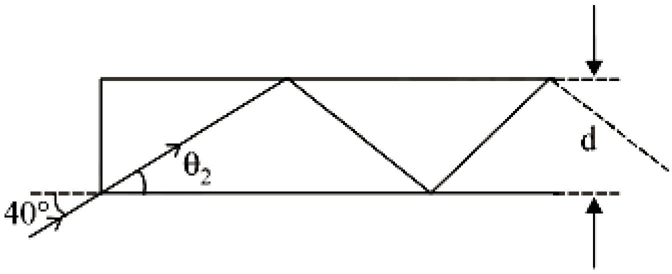
$\dfrac{b}{BO} = \cos 30° \Rightarrow BO = \dfrac{2b}{\sqrt{3}}$

Optical path length $= AO + \mu BO$

$= 2a + (\sqrt{3})\dfrac{2b}{\sqrt{3}} = 2a + 2b$

Question215

In figure, the optical fiber is $l = 2m$ long and has a diameter of $d = 20\mu m$. If a ray of light is incident on one end of the fiber at angle $\theta_1 = 40°$, the number of reflections it makes before emerging from the other end is close to :
(refractive index of fiber is 1.31 and $\sin 40° = 0.64$)



[8 April 2019 I]

Options:

A. 55000

B. 66000

C. 45000

D. 57000

Answer: D

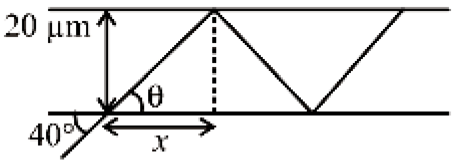
Solution:

Using Snell's law of refraction,

$1 \times \sin 40° = 1.31 \sin \theta$

$\Rightarrow \sin \theta = \frac{0.64}{1.31} = 0.49 \approx 0.5$

$\Rightarrow \theta = 30°$



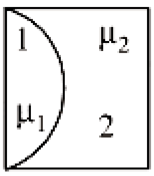
$x = 20\mu m \times \cot \theta$

$\therefore$ Number of reflections $= \frac{2}{20 \times 10^{-6} \times \cot \theta}$

$= \frac{2 \times 10^6}{20 \times \sqrt{3}} = 57735 \approx 57000$

Question216

One plano-convex and one plano-concave lens of same radius of curvature 'R' but of different materials are joined side by side as shown in the figure. If the refractive index of the material of 1 is μ_1 and that of 2 is μ_2, then the focal length of the combination is :



[10 Apr. 2019 I]

Options:

A. $\frac{R}{\mu_1 - \mu_2}$

B. $\frac{2R}{\mu_1 - \mu_2}$

C. $\frac{2R}{2(\mu_1 - \mu_2)}$

D. $\frac{R}{2-(\mu_1 - \mu_2)}$

Answer: A

Solution:

Focal length of plano-convex lens-

$\frac{1}{f_1} = (\mu_1 - 1)\left(\frac{1}{\infty} - \frac{1}{-R}\right) = \frac{\mu_1 - 1}{R}$

$\Rightarrow f_1 = \frac{R}{(\mu_1 - 1)}$

Focal length of plano-concave lens -

$\frac{1}{f_2} = (\mu_2 - 1)\left(\frac{1}{-R} - \frac{1}{\infty}\right) = \frac{\mu_2 - 1}{-R}$

$\Rightarrow f_2 = \frac{-R}{(\mu_2 - 1)}$

For the combination of two lens-

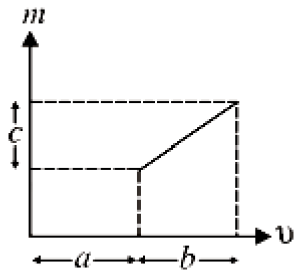
$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{\mu_1 - 1}{R} - \frac{\mu_2 - 1}{R}$

$= \frac{\mu_1 - \mu_2}{R}$

$\Rightarrow f_{eq} = \frac{R}{\mu_1 - \mu_2}$

Question217

The graph shows how the magnification m produced by a thin lens varies with image distance v. What is the focal length of the lens used ?



[10 Apr. 2019 II]

Options:

A. $\frac{b^2}{ac}$

B. $\frac{b^2c}{a}$

C. $\frac{a}{c}$

D. $\frac{b}{c}$

Answer: D

Solution:

From the equation of line
$m = k_1v + k_2 (\because y = mx + c)$
$\Rightarrow \frac{v}{u} = k_1v + k_2 \left(\because m = \frac{v}{u}\right)$
$\Rightarrow \frac{1}{u} = k_1 + \frac{k_2}{v}$ (Dividing both sides by v)
$\Rightarrow \frac{k_2}{v} - \frac{1}{u} - k_1$
Comparing with lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get
$k_1 = \frac{1}{-f}$ and $k_2 = 1$
$\therefore f = \frac{1}{\text{slope of } m - v \text{ graph}} = -\frac{b}{c}$

Question218

A convex lens of focal length 20cm produces images of the same magnification 2 when an object is kept at two distances x_1 and $x_2 (x_1 > x_2)$ from the lens. The ratio of x_1 and x_2 is: [9 Apr. 2019 II]

Options:

A. 2: 1

B. 3: 1

C. 5: 3

D. 4: 3

Answer: B

Solution:

Using, $M = \frac{v}{u}$

or $-2 = \frac{v_1}{x_1} \Rightarrow v_1 = -2x_1$

We have $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

or $\frac{1}{-2x_1} - \frac{1}{x_1} = \frac{1}{20}$

$x_1 = 30\text{cm}$

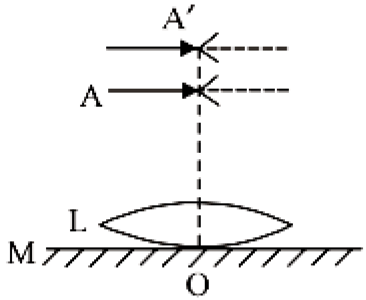
And $\frac{1}{2x_2} - \frac{1}{x_2} = \frac{1}{20}$

or $x_2 = -10\text{cm}$

So, $\frac{x_1}{x_2} = \frac{30}{10} = 3$

Question219

A thin convex lens L (refractive index = 1.5) is placed on a plane mirror M . When a pin is placed at A, such that OA = 18cm, its real inverted image is formed at A itself, as shown in figure. When a liquid of refractive index μ_i is put between the lens and the mirror, the pin has to be moved to A′, such that OA′ = 27cm, to get its inverted real image at A′ itself. The value of μ_i will be:



[9 Apr. 2019 II]

Options:

A. $\frac{4}{3}$

B. $\frac{3}{2}$

C. $\sqrt{3}$

D. $\sqrt{2}$

Answer: A

Solution:

$\frac{1}{f_1} = \frac{2}{f_1}$
Here $2f_1 = 18cm$ or $f_1 = 9cm$
So, $\frac{1}{9} = \frac{2}{f_1}$ or $f_1 = 18cm$
Using, $\frac{1}{f_1} = (\mu - 1)\left(\frac{2}{R}\right)$
or $\frac{1}{18} = (1.5 - 1)\left(\frac{2}{R}\right)$
$\therefore R = 18cm$
when liquid is put between, then
$\frac{1}{f_2} = \frac{2}{f_1} + \frac{2}{f}$
or $\frac{1}{(27/2)} = \frac{2}{18} + \frac{2}{f}$
or $f = -54cm$
Now $-\frac{1}{54} = (\mu_1 - 1) \times \frac{1}{R}$
$= (\mu_1 - 1) \times \left(\frac{1}{-18}\right)$
$\therefore \mu_1 = \frac{1}{3} + 1 = \frac{4}{3}$

Question220

An upright object is placed at a distance of 40 cm in front of a convergent lens of focal length 20 cm. A convergent mirror of focal length 10 cm is placed at a distance of 60 cm on the other side of the lens. The position and size of the final image will be : [8 April 2019 I]

Options:

A. 20 cm from the convergent mirror, same size as the object

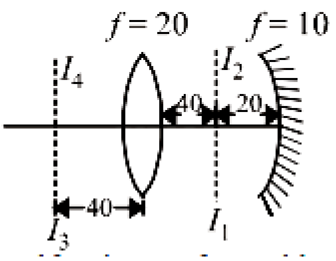
B. 40 cm from the convergent mirror, same size as the object

C. 40 cm from the convergent lens, twice the size of the object

D. (Bouns)

Answer: D

Solution:



$v_1 = \frac{40 \times 20}{(40-20)} = 40\text{cm}$

$u_2 = 60 - 40 = 20\text{cm}$

$\therefore v_2 = \frac{20 \times 10}{(20-10)} = 20\text{cm}$

$\therefore$ Image traces back to object itself as image formed by lens is a centre of curvature of mirror.

Question221

A convex lens (of focal length 20 cm) and a concave mirror, having their principal axes along the same lines, are kept 80 cm apart from each other. The concave mirror is to the right of the convex lens. When an object is kept at a distance of 30 cm to the left of the convex lens, its image remains at the same position even if the concave mirror is removed. The maximum distance of the object for which this concave mirror, by itself would produce a virtual image would be : [8 Apr. 2019 II]

Options:

A. 30 cm

B. 25 cm

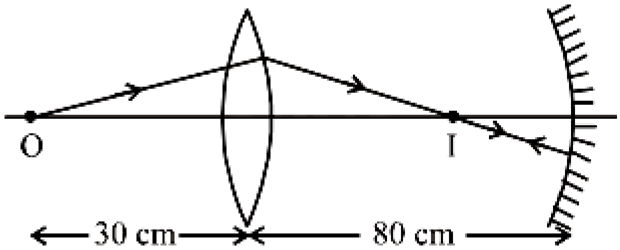
C. 10 cm

D. 20 cm

Answer: C

Solution:

For lens



$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$

or $\frac{1}{v}-\frac{1}{-30}=\frac{1}{20}$

$\therefore v=+60cm$

According to the condition, image formed by lens should be the centre of curvature of the mirror, and so $2f'=20$ or $f'=10cm$

Question222

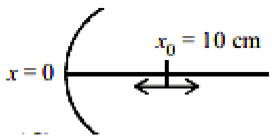
A particle is oscillating on the X-axis with an amplitude 2cm about the point x_0 = 10cm with a frequency ω. A concave mirror of focal length 5cm is placed at the origin (see figure) Identify the correct statements:

(A) The image executes periodic motion

(B) The image executes non-periodic motion

(C) The turning points of the image are asymmetric w.r.t the image of the point at x = 10cm

(D) The distance between the turning points of the oscillation of the image is $\frac{100}{21}=0$



[Online April 15, 2018]

Options:

A. (B), (D)

B. (B), (C)

C. (A), (C), (D)

D. (A), (D)

Answer: C

Solution:

When object is at 8cm

Image $V_1 = \frac{f \times u}{u - f} = \frac{5 \times 8}{8 - 5} = -\frac{40}{3}\text{cm}$

When object is at 12cm

Image $V_2 = \frac{f \times u}{u - f} = \frac{5 \times 12}{12 - 5} = -\frac{60}{7}\text{cm}$

Separation $= \left| V_1 - V_2 \right| = \frac{40}{3} - \frac{60}{7} = \frac{100}{21}\text{cm}$

So A, C and D are correct statements.

Question223

A planoconvex lens becomes an optical system of 28 cm focal length when its plane surface is silvered and illuminated from left to right as shown in Fig-A. If the same lens is instead silvered on the curved surface and illuminated from other side as in Fig. B, it acts like an optical system of focal length 10 cm. The refractive index of the material of lens if:

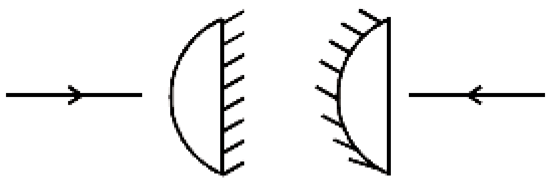


Fig. A Fig. B

[Online April 15, 2018]

Options:

A. 1.50

B. 1.55

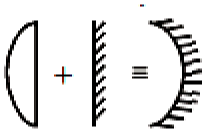
C. 1.75

D. 1

Answer: B

Solution:

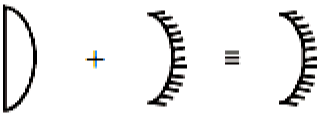
Case-1



$$\frac{1}{f_1}=\left(\frac{\mu-1}{R}\right) \quad f=-28$$

$$P=2P_1+P_2 \Rightarrow \frac{1}{28}=2\left(\frac{\mu-1}{R}\right) \quad \left(\because \text{Power, } P=\frac{1}{f} \;\&\; f_{\text{plane mirror}}=\infty\right)$$

Case-2



$$\frac{1}{f_1}=\left(\frac{\mu-1}{R}\right) \quad f_2=-\frac{R}{2} \quad f=-10\text{ cm}$$

$$P=2P_1+P_2 \Rightarrow \frac{1}{10}=2\left(\frac{\mu-1}{2}\right)+\frac{2}{R}$$

or, $\frac{1}{10}=\frac{1}{28}+\frac{2}{R} \Rightarrow \frac{2}{R}=\frac{1}{10}-\frac{2}{28}=\frac{18}{280}$

or, $R=\frac{280}{9}cm$

or, $\frac{1}{28}=2\left(\frac{\mu-1}{280}\right)9 \Rightarrow \mu-1=\frac{5}{9}$

$\therefore \mu=1+\frac{5}{9}=\frac{14}{9}=1.55$

Question224

A convergent doublet of separated lenses, corrected for spherical aberration, has resultant focal length of 10cm. The separation between the two lenses is 2cm. The focal lengths of the component lenses
[Online April 15, 2018]

Options:

A. 18cm, 20cm

B. 10cm, 12cm

C. 12cm, 14cm

D. 16cm, 18cm

Answer: A

Solution:

For minimum spherical aberration separation,
$d = f_1 - f_2 = 2\text{cm}$
Resultant focal length $= F = 10\text{cm}$
Using $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$ and solving, we get $f_1, f_2 18$ cm and 20cm respectively.

Question225

A ray of light is incident at an angle of 60° on one face of a prism of angle 30°. The emergent ray of light makes an angle of 30° with incident ray. The angle made by the emergent ray with second face of prism will be:
[Online April 16, 2018]

Options:

A. 30°

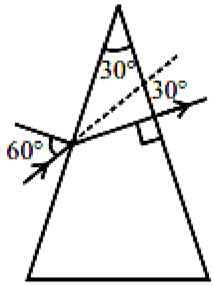
B. 90°

C. 0°

D. 45°

Answer: C

Solution:



Angle of prism, $A = 30°$, $i = 60°$,
angle of deviation, $\delta = 30°$
Using formula, $\delta = i + e - A$
$\Rightarrow e = \delta + A - i$
$= 30° + 30° - 60° = 0°$
∴ Emergent ray will be perpendicular to the face
So it will make angle 90° with the force through which it emerges.

Question226

Let the refractive index of a denser medium with respect to a rarer medium be n_{12} and its critical angle be θ_C. At an angle of incidence A when light is travelling from denser medium to rarer medium, a part of the light is reflected and the rest is refracted and the angle between reflected and refracted rays is 90°. Angle A is given by: [Online April 8, 2017]

Options:

A. $\frac{1}{\cos^{-1}(\sin\theta_C)}$

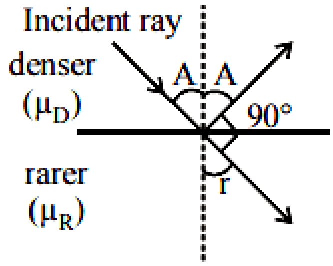
B. $\frac{1}{\tan^{-1}(\sin\theta_C)}$

C. $\cos^{-1}(\sin\theta_C)$

D. $\tan^{-1}(\sin\theta_C)$

Answer: D

Solution:



From Snell's law, $\frac{\mu_R}{\mu_D} = \frac{\sin i}{\sin r}$

$\because \angle i = A$ and $\angle r = (90° - A)$

We also know that, $\sin\theta_C = \frac{\mu_R}{\mu_D}$

From eq $^n(i)$, $\sin\theta_C = \frac{\sin A}{\sin(90° - A)}$

$\sin\theta_C = \frac{\sin A}{\cos A}$

$\sin\theta_C = \tan A$

or $A = \tan^{-1}(\sin\theta_C)$

Question227

In an experiment a convex lens of focal length 15 cm is placed coaxially on an optical bench in front of a convex mirror at a distance of 5 cm from it. It is found that an object and its image coincide, if the object is placed at a distance of 20 cm from the lens. The focal length of the convex mirror is :
[Online April 9, 2017]

Options:

A. 27.5 cm

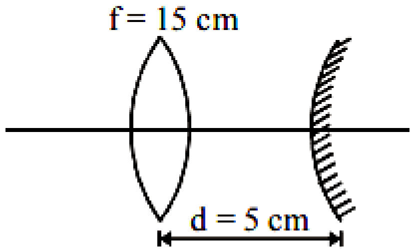
B. 20.0 cm

C. 25.0 cm

D. 30.5 c

Answer: A

Solution:



Given, focal length of lens (f) = 15 cm
object is placed at a distance (u) = – 20 cm
By lens formula,

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

$$\frac{1}{v}=\frac{1}{f}+\frac{1}{u}=\frac{1}{15}-\frac{1}{20}$$

$$\frac{1}{v}=\frac{4-3}{60}$$

$v = 60cm$

The image I gets formed at 60 cm to the right of the lens and it will be inverted.
The rays from the image (I) formed further falls on the convex mirror forms another image. This image should formed in such a way that it coincide with object at the same point due to reflection takes place by convex mirror.
Distance between lens and mirror will be
d = image distance (v) – radius of curvature of convex mirror

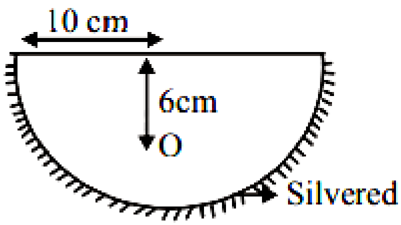
$5 = 60 - 2f$

$2f = 60 - 5$

$f = \frac{55}{2} = 27.5cm$ (convex mirror)

Question228

A hemispherical glass body of radius 10 cm and refractive index 1.5 is silvered on its curved surface. A small air bubble is 6 cm below the flat surface inside it along the axis. The position of the image of the air bubble made by the mirror is seen :



[Online April 10, 2016]

Options:

A. 14 cm below flat surface

B. 20 cm below flat surface

C. 16 cm below flat surface

D. 30 cm below flat surface

Answer: B

Solution:

Given, radius of hemispherical glass $R = 10cm$

$\therefore$ Focal length $f = \frac{10}{2} = -5cm$

$u = (10 - 6) = -4cm$

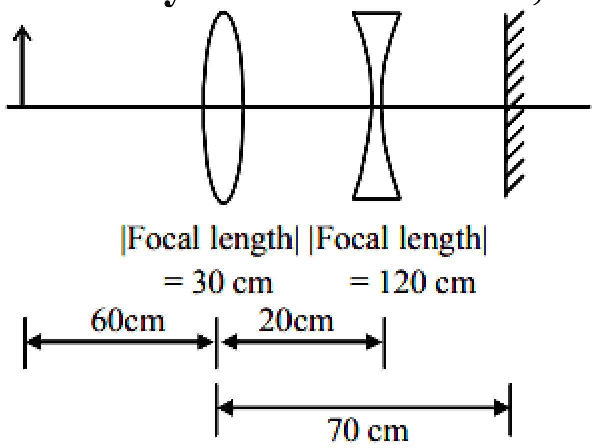
By using mirror formula,

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{-4} = \frac{1}{-5} \Rightarrow v = 20cm$

Apparent height, $h_a = h_r \frac{\mu_1}{\mu_2} = 30 \times \frac{1}{1.5} = 20cm$ below flat surface.

Question229

A convex lens, of focal length 30 cm, a concave lens of focal length 120 cm, and a plane mirror are arranged as shown. For an object kept at a distance of 60 cm from the convex lens, the final image, formed by the combination, is a real image, at a distance of



[Online April 9, 2016]

Options:

A. 60 cm from the convex lens

B. 60 cm from the concave lens

C. 70 cm from the convex lens

D. 70 cm from the concave lens

Answer: A

Solution:

Len's formula is given by
$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$
For convex lens,
$\frac{1}{30}=\frac{1}{v}+\frac{1}{60} \Rightarrow \frac{1}{60}=\frac{1}{v}$
Similarly for concave lens
$\frac{1}{-120}=\frac{1}{v}-\frac{1}{40} \Rightarrow \frac{1}{v}=\frac{1}{60}$
Virtual object $10\mathrm{cm}$ behind plane mirror.
Hence real image $10\mathrm{cm}$ infront of mirror or, $60\mathrm{cm}$ from convex lens.

Question230

To find the focal length of a convex mirror, a student records the following data :

Object Pin	Convex Lens	Convex Mirror	Image Pin
22.2cm	32.2cm	45.8cm	71.2cm

The focal length of the convex lens is f_1 and that of mirror is f_2. Then taking index correction to be negligibly small, f_1 and f_2 are close to :
[Online April 9, 2016]

Options:

A. $f_1 = 7.8\mathrm{cm}, f_2 = 12.7\mathrm{cm}$

B. $f_1 = 12.7\mathrm{cm}, f_2 = 7.8\mathrm{cm}$

C. $f_1 = 15.6\mathrm{cm}, f_2 = 25.4\mathrm{cm}$

D. $f_1 = 7.8\mathrm{cm}, f_2 = 25.4\mathrm{cm}$

Answer: A

Solution:

Taking $f_2 = 12.07$

Using Mirror's formula

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$\Rightarrow \frac{1}{12.7} = \frac{1}{25.4} + \frac{1}{u} \Rightarrow \frac{1}{12.7} - \frac{1}{25.4} = \frac{1}{u}$

$u = 25.4 = v'$

Now using Len's formula

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{f_1} = \frac{1}{25.4 + 13.6} + \frac{1}{10}$

$\Rightarrow \frac{1}{f_1} = \frac{1}{39} + \frac{1}{10} \Rightarrow f_1 = \frac{390}{49} = 7.96$

The closest answers is (a) as option (c) and (d) are not possible.

Question231

In an experiment for determination of refractive index of glass of a prism by $i - \delta$, plot it was found that a ray incident at angle 35°, suffers a deviation of 40° and that it emerges at angle 79°. In that case which of the following is closest to the maximum possible value of the refractive index? [2016]

Options:

A. 1.7

B. 1.8

C. 1.5

D. 1.6

Answer: C

Solution:

We know that $i + e - A = \delta$

$35° + 79° - A = 40° \therefore A = 74°$

But $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin A/2} = \frac{\sin\left(\frac{74+\delta_m}{2}\right)}{\sin\frac{74}{2}}$

$= \frac{5}{3}\sin\left(37° + \frac{\delta_m}{2}\right)$

μ_{max} can be $\frac{5}{3}$. That is μ_{max} is less than $\frac{5}{3} = 1.67$

But δ_m will be less than $40°$ so

$\mu < \frac{5}{3}\sin 57° < \frac{5}{3}\sin 60° \Rightarrow \mu = 1.5$

Question232

An observer looks at a distant tree of height 10 m with a telescope of magnifying power of 20. To the observer the tree appears : [2016]

Options:

A. 20 times taller

B. 20 times nearer

C. 10 times taller

D. 10 times nearer

Answer: B

Solution:

A telescope magnifies by making the object appearing closer

Question233

To determine refractive index of glass slab using a travelling microscope, minimum number of readings required are : [Online April 10, 2016]

Options:

A. Two

B. Four

C. Three

D. Five

Answer: C

Solution:

Reading one ⇒ without slab
Reading two ⇒ with slab
Reading three ⇒ with saw dust
Minimum three readings are required to determine refractive index of glass slab using a travelling microscope.

Question234

You are asked to design a shaving mirror assuming that a person keeps it 10 cm from his face and views the magnified image of the face at the closest comfortable distance of 25 cm. The radius of curvature of the mirror would then be :
[Online April 10, 2015]

Options:

A. 60 cm

B. –24 cm

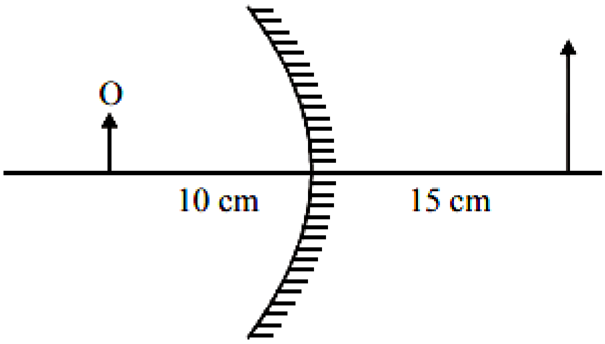
C. – 60 cm

D. 24 cm

Answer: C

Solution:

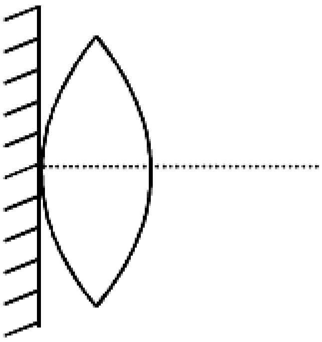
Convex mirror is used as a shaving mirror.



From question : $v = 15cm, u = -10cm$
Radius of curvature, $R = 2f = ?$
Using mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$
$\frac{1}{15} + \frac{1}{(-10)} = \frac{1}{f} \Rightarrow f = -30cm$
Therefore radius of curvature, $R = 2f = -60cm$

Question235

A thin convex lens of focal length ' f ′ is put on a plane mirror as shown in the figure. When an object is kept at a distance 'a' from the lens - mirror combination, its image is formed at a distance $\frac{a}{3}$ in front of the combination. The value of ' a ' is :



[Online April 11, 2015

Options:

A. 3f

B. $\frac{3}{2}$f

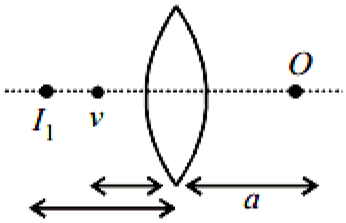
C. f

D. 2f

Answer: D

Solution:

When object is kept at a distance ‘a’ from thin covex lens

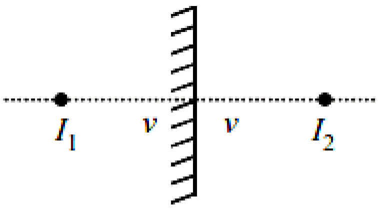


By lens formula : $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

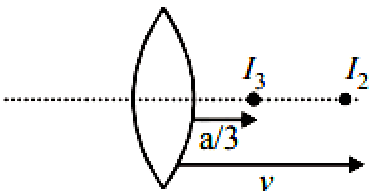
$$\frac{1}{V} - \frac{1}{(-a)} = \frac{1}{f}$$

or, $\frac{1}{v} = \frac{1}{f} - \frac{1}{a}$ .....(i)

Mirror forms image at equal distance from mirror



Now, again from lens formula



$$\frac{3}{a} - \frac{1}{V} = \frac{1}{f}$$

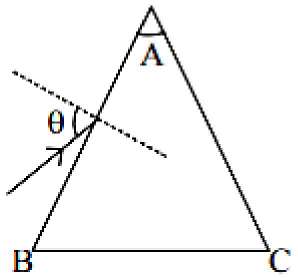
$\frac{3}{a} - \frac{1}{f} + \frac{1}{a} = \frac{1}{f}$ [From eqn. (i)]

Hence, $a = 2f$

Question236

Monochromatic light is incident on a glass prism of angle A. If the refractive index of the material of the prism is μ, a ray, incident at an angle θ, on the face AB would get transmitted through the face

AC of the prism provided :



[2015]

Options:

A. $\theta > \cos^{-1}\left[\mu \sin\left(A + \sin^{-1}\left(\frac{1}{\mu}\right)\right]\right.$

B. $\theta < \cos^{-1}\left[\mu \sin\left(A + \sin^{-1}\left(\frac{1}{\mu}\right)\right]\right.$

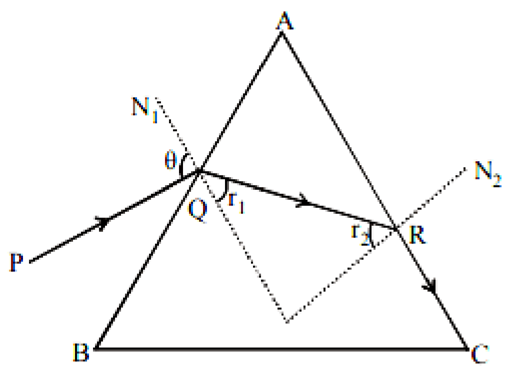
C. $\theta > \sin^{-1}\left[\mu \sin\left(A - \sin^{-1}\left(\frac{1}{\mu}\right)\right]\right.$

D. $\theta < \sin^{-1}\left[\mu \sin\left(A - \sin^{-1}\left(\frac{1}{\mu}\right)\right]\right.$

Answer: C

Solution:

When $r_2 = C, \angle N_2RC = 90°$
Where $C =$ critical angle
As $\sin C = \frac{1}{\mu} = \sin r_2$



Applying snell's law at 'R'
$\mu \sin r_2 = 1 \sin 90°$ ......(i)
Applying snell's law at 'Q'
$1 \times \sin\theta = \mu \sin r_1$ ......(ii)
But $r_1 = A - r_2$

So, $\sin\theta = \mu\sin(A - r_2)$

$\sin\theta = \mu\sin A\cos r_2 - \cos A$ ....(iii) [using (i)]

From (1)

$\cos r_2 = \sqrt{1 - \sin^2 r_2} = \sqrt{1 - \frac{1}{\mu^2}}$ ....(iv)

By eq. (iii) and (iv)

$\sin\theta = \mu\sin A\sqrt{1 - \frac{1}{\mu^2}} - \cos A$

on further solving we can show for ray not to transmitted through face AC

$\theta = \sin^{-1}\left[\mu\sin\left(A - \sin^{-1}\left(\frac{1}{\mu}\right)\right]\right.$

So, for transmission through face AC

$\theta > \sin^{-1}\left[\mu\sin\left(A - \sin^{-1}\left(\frac{1}{\mu}\right)\right]\right.$

Question237

A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. If a 50 m tall tower at a distance of 1 km is observed through this telescope in normal setting, the angle formed by the image of the tower is θ , then θ is close to : [Online April 10, 2015]

Options:

A. 30°

B. 15°

C. 60°

D. 1°

Answer: C

Solution:

Magnifying power of telescope,

$M\,P = \frac{\beta\text{ (angle subtended by image at eye piece)}}{\alpha\text{(angle subtended by object on objective)}}$

Also, $M\,P = \frac{f_o}{f_e} = \frac{150}{5} = 30$

$\alpha = \frac{50}{1000} = \frac{1}{20}\text{rad}$

$\therefore \beta = \theta = MP \times \alpha = 30 \times \frac{1}{20} = \frac{3}{2} = 1.5 rad$

or, $\beta = 1.5 \times \frac{180°}{\pi} \simeq 84°$

Question238

A diver looking up through the water sees the outside world contained in a circular horizon. The refractive index of water is $\frac{4}{3}$, and the diver's eyes are 15cm below the surface of water. Then the radius of the circle is:
[Online April 9, 2014]

Options:

A. $15 \times 3 \times \sqrt{5}$cm

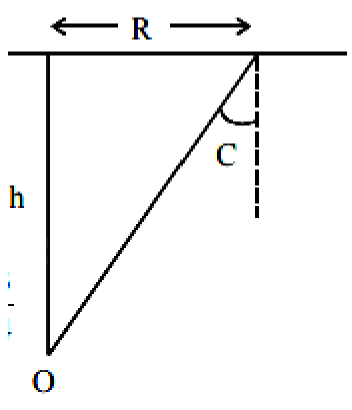
B. $15 \times 3\sqrt{7}$cm

C. $\frac{15 \times \sqrt{7}}{3}$cm

D. $\frac{15 \times 3}{\sqrt{7}}$cm

Answer: D

Solution:



Given, $\mu = \frac{4}{3}$

$h = 15cm$
$R = ?$

$\frac{\sin 90°}{\sin C} = \mu$

$\Rightarrow \sin C = \frac{1}{\mu} = \frac{R}{\sqrt{R^2 + h^2}} = \frac{3}{4}$

$\Rightarrow 16R^2 = 9R^2 + 9h^2$

or, $7R^2 = 9h^2$

or, $R = \frac{3}{\sqrt{7}}h = \frac{3}{\sqrt{7}} \times 15\text{cm}$

Question239

A thin convex lens made from crown glass $\left(\mu = \frac{3}{2}\right)$ has focal length f. When it is measured in two different liquids having refractive indices $\frac{4}{3}$ and $\frac{5}{3}$, it has the focal lengths f_1 and f_2 respectively. The correct relation between the focal lengths is:
[2014]

Options:

A. $f_1 = f_2 < f$

B. $f_1 > f$ and f_2 becomes negative

C. $f_2 > f$ and f_1 becomes negative

D. f_1 and f_2 both become negative

Answer: B

Solution:

By Lens maker's formula for convex lens

$\frac{1}{f} = \left(\frac{\mu}{\mu_L} - 1\right)\left(\frac{2}{R}\right)$

for, $\mu_{L_1} = \frac{4}{3}, f_1 = 4R$

for $\mu_{L_2} = \frac{5}{3}, f_2 = -5R$

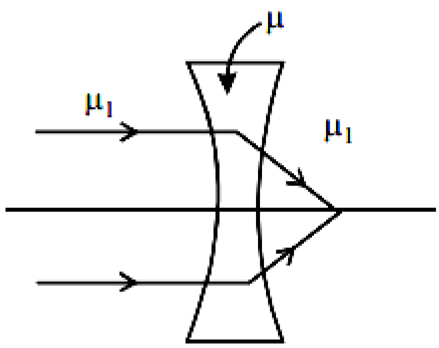
$\Rightarrow f_2 = (-)$ ve

Question240

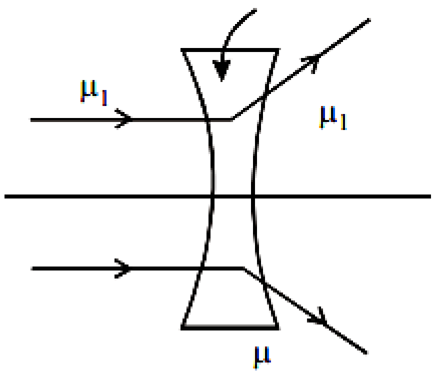
The refractive index of the material of a concave lens is μ. It is immersed in a medium of refractive index μ_1. A parallel beam of light is incident on the lens. The path of the emergent rays when $\mu_1 > \mu$ is:
[Online April 12, 2014]

Options:

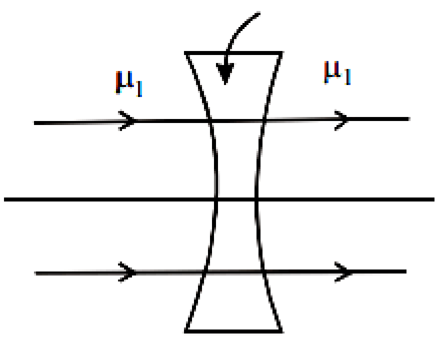
A.



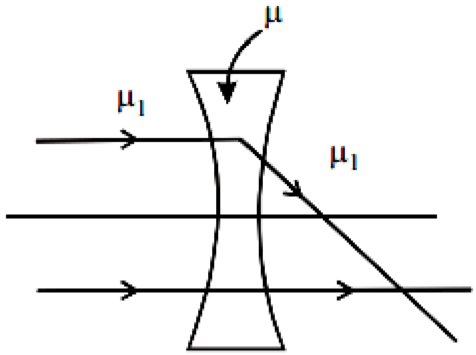
B.



C.



D.



Answer: A

Solution:

If a lens of refractive index μ is immersed in a medium of refractive index μ_1, then its focal length in medium is given by

$$\frac{1}{f_m} = ({}_m\mu_1 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

If f_a is the focal length of lens in air, then

$$\frac{1}{f_a} = ({}_a\mu_1 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\Rightarrow \frac{f_m}{f_a} = \frac{({}_a\mu_1 - 1)}{({}_m\mu_1 - 1)}$$

If $\mu_1 > \mu$, then f_m and f_a have opposite signs and the nature of lens changes i.e. a convex lens diverges the light rays and concave lens converges the light rays. Thus given option (a) is correct.

Question241

An object is located in a fixed position in front of a screen. Sharp image is obtained on the screen for two positions of a thin lens separated by 10 cm. The size of the images in two situations are in the ratio 3 : 3. What is the distance between the screen and the object? [Online April 11, 2014]

Options:

A. 124.5 cm

B. 144.5 cm

C. 65.0 cm

D. 99.0 cm

Answer: D

Solution:

Given: Separation of lens for two of its position, $d = 10\text{cm}$
Ratio of size of the images in two positions
$\frac{I_1}{I_2} = \frac{3}{2}$
Distance of object from the screen, $D = ?$
Applying formula,
$\frac{I_1}{I_2} = \frac{(D+d)^2}{(D-d)^2}$
$\Rightarrow \frac{3}{2} = \frac{(D+10)^2}{(D-10)^2}$
$\Rightarrow \frac{3}{2} = \frac{D^2+100+20D}{D^2+100-20D}$
$\Rightarrow 3D^2+300-60D = 2D^2+200+40D$
$\Rightarrow D^2-100D+100=0$
On solving, we get $D = 99\text{cm}$
Hence the distance between the screen and the object is 99cm.

Question242

In a compound microscope, the focal length of objective lens is 1.2 cm and focal length of eye piece is 3.0 cm. When object is kept at 1.25 cm in front of objective, final image is formed at infinity. Magnifying power of the compound microscope should be: [Online April 11, 2014]

Options:

A. 200

B. 100

C. 400

D. 150

Answer: A

Solution:

Given : $f_0 = 1.2cm; f_e = 3.0cm$
$u_0 = 1.25cm; M_\infty = ?$
From $\frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{u_0}$
$\Rightarrow \frac{1}{1.2} = \frac{1}{v_0} - \frac{1}{(-1.25)}$
$\Rightarrow \frac{1}{v_0} = \frac{1}{1.2} - \frac{1}{1.25}$
$\Rightarrow v_0 = 30cm$
Magnification at infinity,
$M_\infty = -\frac{v_0}{u_0} \times \frac{D}{f_e}$
$= \frac{30}{1.25} \times \frac{25}{3}$
$(\because D = 25cm$ least distance of distinct vision $) = 200$
Hence the magnifying power of the compound microscope is 200

Question243

The focal lengths of objective lens and eye lens of a Galilean telescope are respectively 30 cm and 3.0 cm. telescope produces virtual, erect image of an object situated far away from it at least distance of distinct vision from the eye lens. In this condition, the magnifying power of the Galilean telescope should be: [Online April 9, 2014]

Options:

A. + 11.2

B. – 11.2

C. – 8.8

D. + 8.8

Answer: D

Solution:

Given, Focal length of objective, $f_0 = 30cm$
focal length of eye lens, $f_e = 3.0cm$
Magnifying power, $M = ?$

Magnifying power of the Galilean telescope,

$$M_D = \frac{f_0}{f_e}\left(1 - \frac{f_e}{D}\right)$$

$$= \frac{30}{3}\left(1 - \frac{3}{25}\right) [\because D = 25\text{cm}]$$

$$= 10 \times \frac{22}{25} = 8.8\text{cm}$$

Question244

A printed page is pressed by a glass of water. The refractive index of the glass and water is 1.5 and 1.33, respectively. If the thickness of the bottom of glass is 1 cm and depth of water is 5 cm, how much the page will appear to be shifted if viewed from the top ? [Online April 25, 2013]

Options:

A. 1.033 cm

B. 3.581 cm

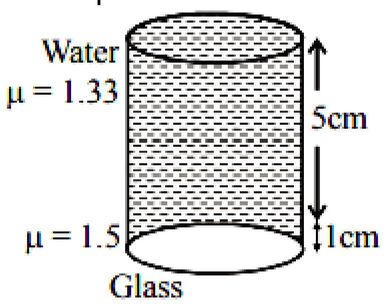
C. 1.3533 cm

D. 1.90 cm

Answer: C

Solution:

Real depth = 5 cm + 1cm = 6 cm



Apparent depth $= \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2} + \ldots..$

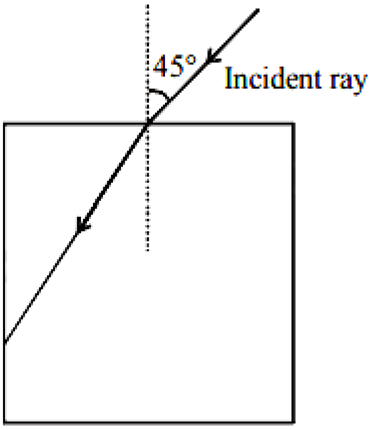
$$= \frac{5}{1.33} + \frac{1}{1.5}$$

$\simeq 3.8 + 0.7 \simeq 4.5\text{cm}$

$\therefore$ Shift $= 6\text{cm} - 4.5\text{cm} \cong 1.5\text{cm}$

Question245

A light ray falls on a square glass slab as shown in the diagram. The index of refraction of the glass, if total internal reflection is to occur at the vertical face, is equal to :



[Online April 23, 2013]

Options:

A. $\frac{(\sqrt{2}+1)}{2}$

B. $\sqrt{\frac{5}{2}}$

C. $\frac{3}{2}$

D. $\sqrt{\frac{3}{2}}$

Answer: D

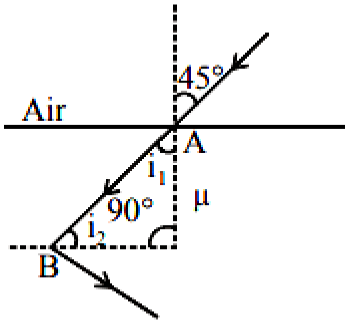
Solution:

At point A by Snell's law

$\mu = \frac{\sin 45°}{\sin r} \Rightarrow \sin r = \frac{1}{\mu\sqrt{2}}$ ....(i)

At point B, for total internal reflection,

$\sin i_1 = \frac{1}{\mu}$



From figure, $i_1 = 90° - r$

$\therefore (\sin 90° - r) = \frac{1}{\mu}$

$\Rightarrow \cos r = \frac{1}{\mu}$ ....(ii)

Now $\cos r = \sqrt{1 - \sin^2 r} = \sqrt{1 - \frac{1}{2\mu^2}}$

$= \sqrt{\frac{2\mu^2 - 1}{2\mu^2}}$ ....(iii)

From eqs (ii) and (iii)

$\frac{1}{\mu} = \sqrt{\frac{2\mu^2 - 1}{2\mu^2}}$

Squaring both sides and then solving, we get

$\mu = \sqrt{\frac{3}{2}}$

Question246

Light is incident from a medium into air at two possible angles of incidence (A) 20° and (B) 40°. In the medium light travels 3.0 cm in 0.2 ns. The ray will :
[Online April 9, 2013]

Options:

A. suffer total internal reflection in both cases (A) and (B)

B. suffer total internal reflection in case (B) only

C. have partial reflection and partial transmission in case (B)

D. have 100% transmission in case (A)

Answer: B

Solution:

Velocity of light in medium

$$V_{med} = \frac{3cm}{0.2ns} = \frac{3 \times 10^{-2}m}{0.2 \times 10^{-9}s} = 1.5m/s$$

Refractive index of the medium

$$\mu = \frac{V_{air}}{V_{med}} = \frac{3 \times 10^{8}}{1.5} = 2m/s$$

As $\mu = \frac{1}{\sin C}$

$$\therefore \sin C = \frac{1}{\mu} = \frac{1}{2} = 30°$$

Condition of TIR is angle of incidence i must be greater than critical angle. Hence raywill suffer TIR in case of (B) $(i = 40° > 30°)$ only.

Question247

Diameter of a plano-convex lens is 6 cm and thickness at the centre is 3 mm. If speed of light in material of lens is 2×10^{8} m/s, the focal length of the lens is
[2013]

Options:

A. 15 cm

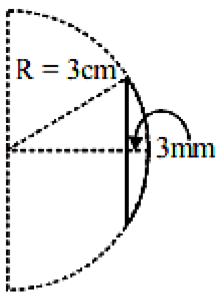
B. 20 cm

C. 30 cm

D. 10 cm

Answer: C

Solution:



$\therefore n=\frac{\text{Velocity of light in vacuum}}{\text{Velocity of light in medium}}$
$\therefore n=\frac{3}{2}$
$3^2+(R-3mm)^2=R^2$
$\Rightarrow 3^2+R^2-2R(3mm)+(3mm)^2=R^2$
$\Rightarrow R\approx 15cm$
$\frac{1}{f}=\left(\frac{3}{2}-1\right)\left(\frac{1}{15}\right)\Rightarrow f=30cm$

Question248

The image of an illuminated square is obtained on a screen with the help of a converging lens. The distance of the square from the lens is 40 cm. The area of the image is 9 times that of the square. The focal length of the lens is : [Online April 22, 2013]

Options:

A. 36 cm

B. 27 cm

C. 60 cm

D. 30 cm

Answer: D

Solution:

If side of object square $=l$
and side of image square $=l'$
From question, $\frac{l'^2}{l}=9$
or $\frac{l'}{l}=3$
i.e., magnification $m=3$
$u=-40cm$
$v=3\times 40=120cm$
$f=?$
From formula , $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$

$$\frac{1}{120} - \frac{1}{-40} = \frac{1}{f}$$

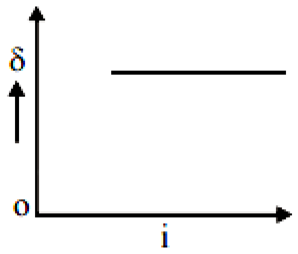
or, $\frac{1}{f} = \frac{1}{120} + \frac{1}{40} = \frac{1+3}{120}$ $\therefore f = 30cm$

Question249

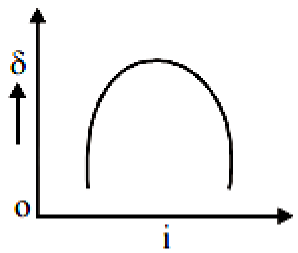
The graph between angle of deviation (δ) and angle of incidence (i) for a triangular prism is represented by [2013]

Options:

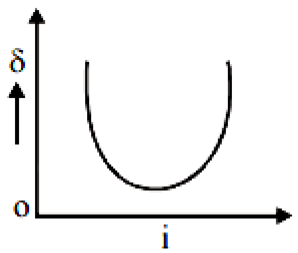
A.



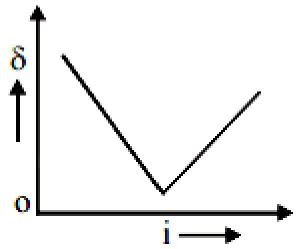
B.



C.



D.



Answer: C

Solution:

For the prism as the angle of incidence (i) increases, the angle of deviation (δ) first decreases goes to minimum value and then increases.

Question250

This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: Very large size telescopes are reflecting telescopes instead of refracting telescopes.
Statement 2: It is easier to provide mechanical support to large size mirrors than large size lenses.
[Online April 23, 2013]

Options:

A. Statement-1 is true and Statement-2 is false.

B. Statement-1 is false and Statement-2 is true.

C. Statement-1 and statement-2 are true and Statement - 2 is correct explanation for statement-1.

D. Statements-1 and statement-2 are true and Statement-2 is not the correct explanation for statement-1.

Answer: C

Solution:

One side of mirror is opaque and another side is reflecting this is not in case of lens hence, it is easier to provide mechanical support to large size mirrors than large size lenses. Reflecting telescopes are based on the same principle except that the formation of images takes place by reflection instead of refraction.

Question251

The focal length of the objective and the eyepiece of a telescope are 50 cm and 5 cm respectively. If the telescope is focussed for distinct vision on a scale distant 2 m from its objective, then its magnifying power will be:
[Online April 22, 2013]

Options:

A. – 4

B. – 8

C. + 8

D. – 2

Answer: D

Solution:

Given: $f_0 = 50\text{cm}, f_e = 5\text{cm}$
$d = 25\text{cm}, u_0 = -200\text{cm}$
Magnification $M = ?$
As $\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0}$
$\Rightarrow \frac{1}{v_0} = \frac{1}{f_0} + \frac{1}{u_0} = \frac{1}{50} - \frac{1}{200} = \frac{4-1}{200} = \frac{3}{200}$
or $v_0 = \frac{200}{3}\text{cm}$
Now $v_e = d = -25\text{cm}$
From, $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$
$-\frac{1}{u_e} = \frac{1}{f_e} - \frac{1}{v_e}$
$= \frac{1}{5} + \frac{1}{25} = \frac{6}{25}$
or, $v_e = \frac{-25}{6}\text{cm}$

Magnification $M = M_0 \times M_e$

$$= \frac{v_0}{u_0} \times \frac{v_e}{u_e} = \frac{-200/3}{200} \times \frac{-25}{-25/6}$$

$$= -\frac{1}{3} \times 6 = -2$$

Question252

An object at 2.4 m in front of a lens forms a sharp image on a film 12 cm behind the lens. A glass plate 1cm thick, of refractive index 1.50 is interposed between lens and film with its plane faces parallel to film. At what distance (from lens) should object shifted to be in sharp focus of film? [2012]

Options:

A. 7.2 m

B. 2.4 m

C. 3.2 m

D. 5.6 m

Answer: D

Solution:

The focal length of the lens

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{1}{12} + \frac{1}{240}$$

$$= \frac{20+1}{240} = \frac{21}{240}$$

$$f = \frac{240}{21} \text{cm}$$

When glass plate is interposed between lens and film, so shift produced will be

Shift $= t\left(1 - \frac{1}{\mu}\right)$

$$1\left(1 - \frac{1}{3/2}\right) = 1 \times \frac{1}{3}$$

Now image should be form at

$v' = 12 - \frac{1}{3} = \frac{35}{3}\text{cm}$

Now the object distance u.

Using lens formula again

$\frac{1}{f} = \frac{1}{v'} - \frac{1}{u}$

$\Rightarrow \frac{1}{u} = \frac{1}{v'} - \frac{1}{f}$

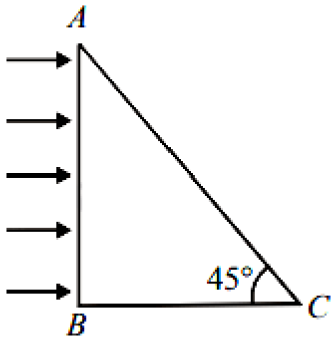
$\Rightarrow \frac{1}{u} = \frac{3}{35} - \frac{21}{240} = \frac{1}{5}\left[\frac{3}{7} - \frac{21}{48}\right]$

$\Rightarrow \frac{1}{u} = \frac{1}{5}\left[\frac{48-49}{7 \times 16}\right]$

$\Rightarrow u = -7 \times 16 \times 5 = -560\text{cm} = -5.6\text{m}$

Question253

A beam of light consisting of red, green and blue colours is incident on a right-angled prism on face AB. The refractive indices of the material for the above red, green and blue colours are 1.39, 1.44 and 1.47 respectively. A person looking on surface AC of the prism will see



[Online May 26, 2012]

Options:

A. no light

B. green and blue colours

C. red and green colours

D. red colour only

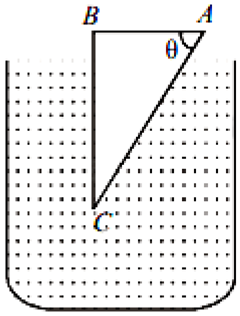
Answer: D

Solution:

For light to come out through face 'AC', total internal reflection must not take place.
i.e., $\theta < c \Rightarrow \sin\theta < \sin c$
$\Rightarrow \sin\theta < \frac{1}{\mu}$
or $\mu < \frac{1}{\sin\theta} \Rightarrow \mu < \frac{1}{\sin 45°}$
$\Rightarrow \mu < \sqrt{2} \Rightarrow \mu < 1.414$

Question254

A glass prism of refractive index 1.5 is immersed in water (refractive index $\frac{4}{3}$) as shown in figure. A light beam incident normally on the face AB is totally reflected to reach the face BC, if



[Online May 19, 2012]

Options:

A. $\sin\theta > \frac{5}{9}$

B. $\sin\theta > \frac{2}{3}$

C. $\sin\theta > \frac{8}{9}$

D. $\sin\theta > \frac{1}{3}$

Answer: C

Solution:

For total internal reflection on face AC
$\theta >$ critical angle (C)
and $\sin\theta \geq \sin C$

$\sin\theta \geq \frac{1}{{}^{w}\mu_g}$

$\sin\theta \geq \frac{\mu_w}{\mu_g} \Rightarrow \sin\theta \geq \frac{\frac{4}{3}}{\frac{3}{2}}$

$\therefore \sin\theta \geq \frac{8}{9}$

Question255

Which of the following processes play a part in the formation of a rainbow?
(i) Refraction
(ii) Total internal reflection
(iii) Dispersion
(iv) Interference
[Online May 7, 2012]

Options:

A. (i), (ii) and (iii)

B. (i) and (ii)

C. (i), (ii) and (iv)

D. (iii) and (iv)

Answer: A

Solution:

Rainbow is formed due to the dispersion of light suffering refraction and total internal reflection (TIR) in the droplets present in the atmosphere

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Question256

A telescope of aperture 3×10^{-2}m diameter is focused on a window at 80m distance fitted with a wiremesh of spacing 2×10^{-3}m. Given:

$\lambda = 5.5 \times 10^{-7}$m, which of the following is true for observing the mesh through the telescope?
[Online May 26, 2012]

Options:

A. Yes, it is possible with the same aperture size.

B. Possible also with an aperture half the present diameter.

C. No, it is not possible.

D. Given data is not sufficient.

Answer: A

Solution:

Given : $d = 3 \times 10^{-2}$m
$\lambda = 5.5 \times 10^{-7}$m
Limit of resolution, $\Delta\theta = \dfrac{1.22\lambda}{d}$
$= \dfrac{1.22 \times 5.5 \times 10^{-7}}{3 \times 10^{-2}} = 2.23 \times 10^{-5}$rad
At a distance of 80m, the telescope is able to resolve between two points which are separated by
$2.23 \times 10^{-5} \times 80$m
$= 1.78 \times 10^{-3}$m

Question257

We wish to make a microscope with the help of two positive lenses both with a focal length of 20 mm each and the object is positioned 25 mm from the objective lens. How far apart the lenses should be so that the final image is formed at infinity?
[Online May 12, 2012]

Options:

A. 20 mm

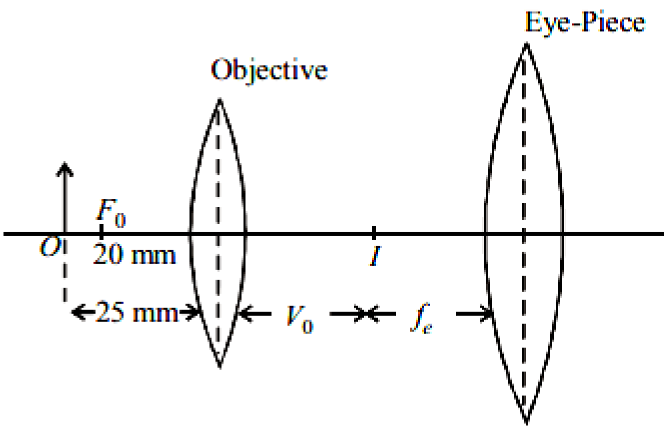
B. 100 mm

C. 120 mm

D. 80 mm

Answer: C

Solution:



To obtain final image at infinity, object which is the image formed by objective should be at focal distance of eyepiece.
By lens formula (for objective)

$$\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0}$$

or, $\frac{1}{v_0} - \frac{1}{-25} = \frac{1}{20}$

$$\Rightarrow \frac{1}{v_0} = \frac{1}{20} - \frac{1}{25} = \frac{5-4}{100} = \frac{1}{100}\text{mm}$$

$\therefore v_0 = 100\text{mm}$

Therefore the distance between the lenses

$= v_0 + f_e = 100\text{mm} + 20\text{mm} = 120\text{mm}$

Question258

A car is fitted with a convex side-view mirror of focal length 20 cm. A second car 2.8 m behind the first car is overtaking the first car at a relative speed of 15 m/s. The speed of the image of the second car as seen in the mirror of the first one is : [2011]

Options:

A. $\frac{1}{15}$ m/s

B. 10 m/s

C. 15 m/s

D. $\frac{1}{10}$ m/s

Answer: A

Solution:

From mirror formula

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

Differentiating the above equation, we get

$$\frac{dv}{dt}=-\frac{v^2}{u^2}\left(\frac{du}{dt}\right)$$

Also,

$$\frac{v}{u}=\frac{f}{u-f}$$

$$\Rightarrow\frac{dv}{dt}=-\left(\frac{f}{u-f}\right)^2\frac{du}{dt}$$

$$\Rightarrow\frac{dv}{dt}=\left(\frac{0.2}{2.8-0.2}\right)^2\times 15$$

$$\Rightarrow\frac{dv}{dt}=\frac{1}{15}m/s$$

Question259

Let the $x-z$ plane be the boundary between two transparent media. Medium 1 in $z\geq 0$ has a refractive index of $\sqrt{2}$ and medium 2 with $z<0$ has a refractive index of $\sqrt{3}$. Aray of light in medium 1 given by the vector $\vec{A}=6\sqrt{3}\hat{i}+8\sqrt{3}\hat{j}-10\hat{k}$ is incident on the plane of separation. The angle of refraction in medium 2 is:
[2011]

Options:

A. 45°

B. 60°

C. 75°

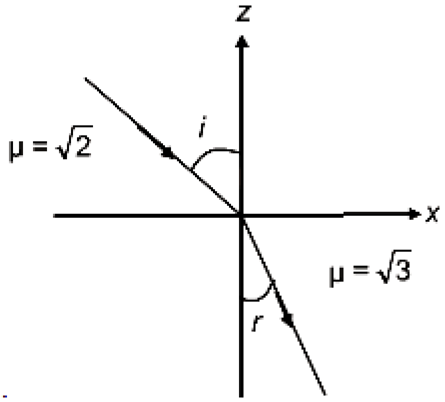
D. 30°

Answer: A

Solution:

As refractive index for $z > 0$ and $z \leq 0$ is different xy plane should be the boundry between two media.
Angle of incidence is given by

$$\cos(\pi - i) = \frac{\left(6\sqrt{3}\,\hat{i} + 8\sqrt{3}\,\hat{j} - 10\hat{k}\right) . \hat{k}}{20}$$



$-\cos i = -\frac{1}{2}$
$\Rightarrow \angle i = 60°$
From Snell's law, $\mu = \sqrt{2}$
$\frac{\sin i}{\sin r} = \frac{u_2}{u_1}$
$\Rightarrow \frac{\sin i}{\sin r} = \frac{\sqrt{3}}{\sqrt{2}}$
$\Rightarrow \sqrt{2}\sin i = \sqrt{3}\sin r$
$\Rightarrow \sqrt{2}\sin 60° = \sqrt{3}$
$\Rightarrow \sqrt{2} \times \frac{\sqrt{3}}{2} = \sqrt{3}\sin r$
$\Rightarrow \angle r = 45°$

Question260

A beaker contains water up to a height h_1 and kerosene of height h_2 above water so that the total height of (water + kerosene) is ($h_1 + h_2$). Refractive index of water is μ_1 and that of kerosene is μ_2. The apparent shift in the position of the bottom of the beaker when viewed from above is
[2011 RS]

Options:

A. $\left(1+\frac{1}{\mu_1}\right)h_1-\left(1+\frac{1}{\mu_2}\right)h_2$

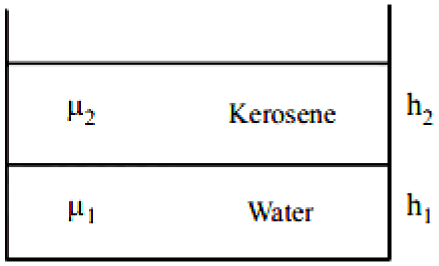
B. $\left(1-\frac{1}{\mu_1}\right)h_1+\left(1-\frac{1}{\mu_2}\right)h_2$

C. $\left(1+\frac{1}{\mu_1}\right)h_2-\left(1+\frac{1}{\mu_2}\right)h_1$

D. $\left(1-\frac{1}{\mu_1}\right)h_2+\left(1-\frac{1}{\mu_2}\right)h_1$

Answer: B

Solution:



Apparent shift of the bottom due to water,

$\Delta h_1 = h_1\left[1-\frac{1}{\mu_1}\right]$

Apparent shift of the bottom due to kerosene, Δh_2

$= h_2\left[1-\frac{1}{\mu_2}\right]$

Thus, total apparent shift :

$= \Delta h_1 + \Delta h_2$

$= \left(1-\frac{1}{\mu_1}\right)h_1+\left(1-\frac{1}{\mu_2}\right)h_2$

Question261

When monochromatic red light is used instead of blue light in a convex lens, its focal length will
[2011 RS]

Options:

A. increase

B. decrease

C. remain same

D. does not depend on colour of light

Answer: A

Solution:

From the Cauchy
Formula, $\mu = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^1}$

$\therefore \mu \propto \frac{1}{\lambda}$

As, $\lambda_{blue} < \lambda_{red}$

$\therefore \lambda_{blue} > \mu_{red}$

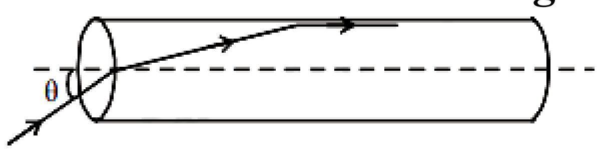
From lens maker's formula

and $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\Rightarrow \frac{1}{f_B} > \frac{1}{f_R} \Rightarrow f_R > f_B$

Question262

A transparent solid cylindrical rod has a refractive index of $\frac{2}{\sqrt{3}}$. It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure.



The incident angle θ for which the light ray grazes along the wall of the rod is:
[2009]

Options:

A. $\sin^{-1}(\sqrt{3}/2)$

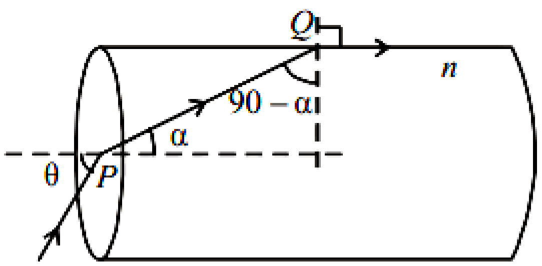
B. $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

C. $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

D. $\sin^{-1}(1/2)$

Answer: C

Solution:



Applying Snell's law for medium inside the cylinder and air at Q we get

$$n = \frac{\sin 90^\circ}{\sin(90^\circ - \alpha)} = \frac{1}{\cos\alpha}$$

$$\therefore \cos\alpha = \frac{1}{n}$$

$$\therefore \sin\alpha = \sqrt{1-\cos^2\alpha} = \sqrt{1-\frac{1}{n^2}} = \frac{\sqrt{n^2-1}}{n} \text{ ......(i)}$$

Applying Snell's Law for air and medium inside the cylinder at P we get

$$n = \frac{\sin\theta}{\sin\alpha}$$

$\Rightarrow \sin\theta = n \times \sin\alpha = \sqrt{n^2-1}$; [from (i)]

$$\therefore \sin\theta = \sqrt{\left(\frac{2}{\sqrt{3}}\right)^2 - 1} = \sqrt{\frac{4}{3}-1} = \frac{1}{\sqrt{3}}$$

or $\theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Question263

In an optics experiment, with the position of the object fixed, a student varies the position of a convex lens and for each position, the screen is adjusted to get a clear image of the object. A graph between the object distance u and the image distance v, from the lens, is plotted using the same scale for the two axes. A straight line passing

through the origin and making an angle of 45° with the x-axis meets the experimental curve at P. The coordinates of P will be [2009]

Options:

A. $\left(\frac{f}{2}, \frac{f}{2}\right)$

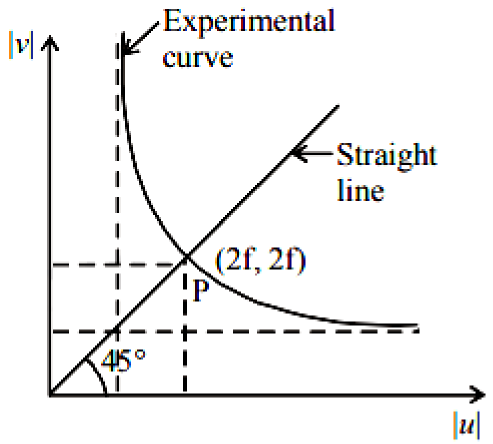
B. (f, f)

C. $(4f, 4f)$

D. $(2f, 2f)$

Answer: D

Solution:



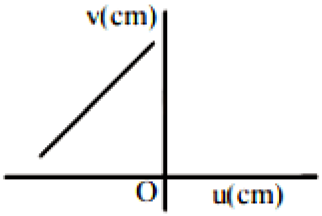
For the graph to intersect $y = x$ line. The value of $|v|$ and $|u|$ must be equal.
From lens formula
$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
When $u = -2f$, $v = 2f$
Also $v = \frac{f}{1 + \frac{f}{u}}$
As $|u|$ increases, v decreases for $|u| > f$. The graph between $|v|$ and $|u|$ is shown in the figure. A straight line passing through the origin and making an angle of $45°$ with the X-axis meets the experimental curve at $P(2f, 2f)$.

Question264

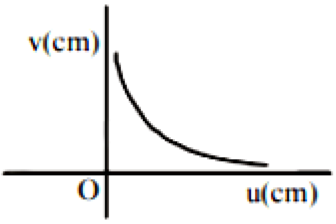
A student measures the focal length of a convex lens by putting an object pin at a distance ‘u’ from the lens and measuring the distance ‘v’ of the image pin. The graph between ‘u’ and ‘v’ plotted by the student should look like
[2008]

Options:

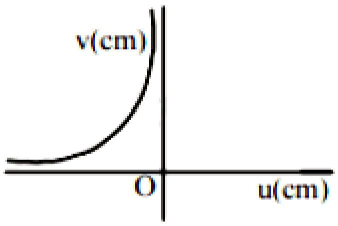
A.



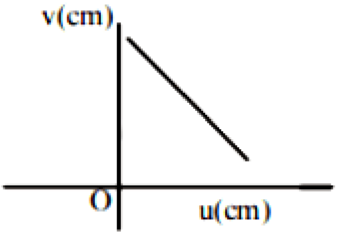
B.



C.



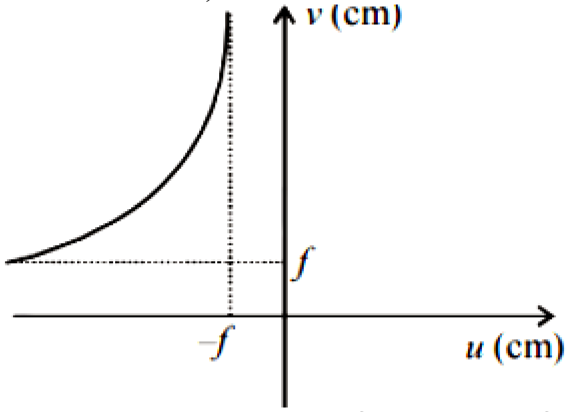
D.



Answer: C

Solution:

From the lens formula $\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$
This graph suggest that when
$u=-f, v=+\infty$
When u is at $-\infty$, $v=f$.



When the object is moved further away from the lens, v decreases but remains positive.

Question265

An experiment is performed to find the refractive index of glass using a travelling microscope. In this experiment distances are measured by [2008]

Options:

A. a vernier scale provided on the microscope

B. a standard laboratory scale

C. a meter scale provided on the microscope

D. a screw gauge provided on the microscope

Answer: A

Solution:

To find the refractive index of glass using a travelling microscope, a vernier scale is provided on the microscope

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Question266

Two lenses of power –15 D and +5 D are in contact with each other. The focal length of the combination is
[2007]

Options:

A. + 10 cm

B. – 20 cm

C. – 10 cm

D. + 20 cm

Answer: C

Solution:

When two thin lenses are in contact coaxially, power of combination is given by
$P = P_1 + P_2$
$= (-15 + 5)D$
$= -10D.$
Also, $P = \frac{1}{f}$
$\Rightarrow f = \frac{1}{P} = \frac{1}{-10}$ metre
$\therefore f = -\left(\frac{1}{10} \times 100\right) cm = -10cm$

Question267

The refractive index of a glass is 1.520 for red light and 1.525 for blue light. Let D_1 and D_2 be angles of minimum deviation for red and blue light respectively in a prism of this glass. Then,
[2006]

Options:

A. $D_{1}

B. $D_1 = D_2$

C. D_1 can be less than or greater than D_2 depending upon the angle of prism

D. $D_1 > D_2$

Answer: A

Solution:

When angle of prism is small,
Angle of deviation, $D = (\mu - 1)A$
Since $\lambda_b < \lambda_r$
$\Rightarrow \mu_r < \mu_b \quad \Rightarrow D_1 < D_2$

Question268

A fish looking up through the water sees the outside world contained in a circular horizon. If the refractive index of water is $\frac{4}{3}$ and the fish is 12cm below the surface, the radius of this circle in cm is [2005]

Options:

A. $\frac{36}{\sqrt{7}}$

B. $36\sqrt{7}$

C. $4\sqrt{5}$

D. $36\sqrt{5}$

Answer: A

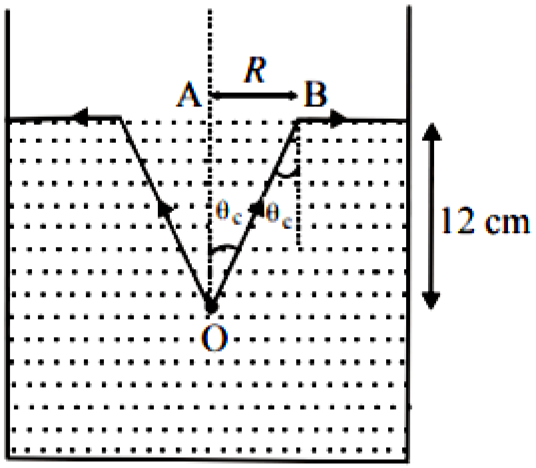
Solution:

From the figure it is clear that
$\tan\theta_c = \frac{AB}{OA}$
$\Rightarrow R = OA\tan\theta_c$
$\Rightarrow R = \frac{OA\sin\theta_c}{\cos\theta_c}$
$\Rightarrow R = \frac{OA\sin\theta_c}{\sqrt{1 - \sin^2\theta_c}}$

$$\Rightarrow \tan\theta_c = \frac{R}{12} = \frac{\sin\theta_c}{\sqrt{1-\sin^2\theta_c}}$$

$$\because \sin\theta_c = \frac{1}{\mu} = \frac{3}{4}$$

$$\Rightarrow \tan\theta_c = \frac{3}{\sqrt{16-9}} = \frac{3}{\sqrt{7}} = \frac{R}{12}$$



$$\Rightarrow R = \frac{36}{\sqrt{7}} \text{cm}$$

Question269

A thin glass (refractive index 1.5) lens has optical power of – 5 D in air. Its optical power in a liquid medium with refractive index 1.6 will be
[2005]

Options:

A. – 1D

B. 1 D

C. – 25 D

D. 25 D

Answer: B

Solution:

According to lens maker's formula in air

$$\frac{1}{f_a} = ({}_a\mu_g - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\Rightarrow \frac{1}{f_a} = \left(\frac{1.5}{1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \text{ ......(i)}$$

Using lens maker's formula in liquid medium,

$$\frac{1}{f_m} = \left(\frac{\mu_g}{\mu_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\Rightarrow \frac{1}{f_m} = \left(\frac{1.5}{1.6} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \text{ ....(ii)}$$

Dividing (i) by (ii),

$$\frac{f_m}{f_a} = \left(\frac{1.5 - 1}{\frac{1.5}{1.6} - 1}\right) = -8$$

$$P_a = -5 = \frac{1}{f_a}$$

$$\Rightarrow f_a = -\frac{1}{5}$$

$$\Rightarrow f_m = -8 \times f_a = -8 \times -\frac{1}{5} = \frac{8}{5}$$

$$P_m = \frac{\mu}{f_m} = \frac{1.6}{8} \times 5 = 1\text{D}$$

Question270

A plano convex lens of refractive index 1.5 and radius of curvature 30 cm. Is silvered at the curved surface. Now this lens has been used to form the image of an object. At what distance from this lens an object be placed in order to have a real image of size of the object [2004]

Options:

A. 60 cm

B. 30 cm

C. 20 cm

D. 80 cm

Answer: C

Solution:

The focal length (F) of the final mirror is
$$\frac{1}{F} = \frac{2}{f_1} + \frac{1}{f_m}$$
Using lens maker's formula

Here $$\frac{1}{f_1} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$
Here, $R_1 = \infty$

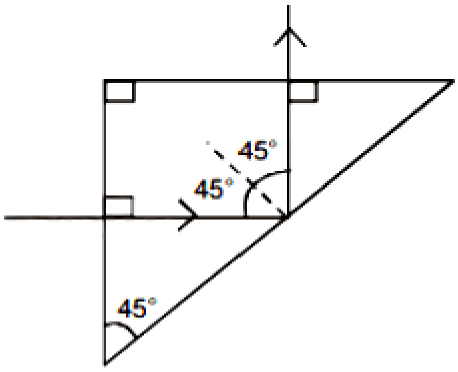
$R_2 = 30\text{cm}$
$$= (1.5 - 1)\left[\frac{1}{\infty} - \frac{1}{-30}\right] = \frac{1}{60}$$
$$\therefore \frac{1}{F} = 2 \times \frac{1}{60} + \frac{1}{30/2} = \frac{1}{10}$$
$\therefore F = 10\text{cm}$

Real image will be equal to the size of the object if the object distance

$u = 2F = 20\text{cm}$

Question271

A light ray is incident perpendicularly to one face of a 90° prism and is totally internally reflected at the glass-air interface. If the angle of reflection is 45°, we conclude that the refractive index n



[2004]

Options:

A. $n > \frac{1}{\sqrt{2}}$

B. $n > \sqrt{2}$

C. $n < \frac{1}{\sqrt{2}}$

D. $n < \sqrt{2}$

Answer: B

Solution:

For total internal reflection
Incident angle $(i)>$ critical angle (i_c),

$\therefore \sin i > \sin i_c$

$\Rightarrow \sin 45° > \sin i_c \Rightarrow \sin i_c = \frac{1}{n}$

$\therefore \sin 45° > \frac{1}{n}$

$\Rightarrow \frac{1}{\sqrt{2}} > \frac{1}{n} \Rightarrow n > \sqrt{2}$

Question272

To get three images of a single object, one should have two plane mirrors at an angle of
[2003]

Options:

A. 60º

B. 90º

C. 120º

D. 30º

Answer: B

Solution:

The number of images formed is given by

$n = \frac{360}{\theta} - 1$

$\Rightarrow \frac{360}{\theta} - 1 = 3$

$\Rightarrow \theta = \frac{360°}{4} = 90°$

Question273

Consider telecommunication through optical fibres. Which of the following statements is not true?
[2003]

Options:

A. Optical fibres can be of graded refractive index

B. Optical fibres are subject to electromagnetic interference from outside

C. Optical fibres have extremely low transmission loss

D. Optical fibres may have homogeneous core with a suitable cladding.

Answer: B

Solution:

Optical fibres form a dielectric wave guide and are free from electromagnetic interference or radio frequency interference. There is extremely low transmission loss in optical fibre

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Question274

The image formed by an objective of a compound microscope is
[2003]

Options:

A. virtual and diminished

B. real and diminished

C. real and enlarged

D. virtual and enlarged

Answer: C

Solution:

A real, inverted and enlarged image of the object is formed by the objective lens of a compound microscope.

Question275

If two plane mirrors are kept at 60° to each other, then the number of images formed by them is
[2002]

Options:

A. 5

B. 6

C. 7

D. 8

Answer: A

Solution:

When two plane mirrors are inclined at each other at an angle θ then the number of the images (n) of a point object kept between the plane mirrors is

$n = \frac{360°}{\theta} - 1$

(if $\frac{360°}{\theta}$ is even integer)

∴ Number of images $d = \frac{360°}{60°} - 1 = 5$

Question276

Which of the following is used in optical fibres?
[2002]

Options:

A. total internal reflection

B. scattering

C. diffraction

D. refraction.

Answer: A

Solution:

In an optical fibre, light is sent through the fibre without any loss by the phenomenon of total internal reflection. Total internal reflection of light waves confine the light rays inside the optical fiber.

Question277

An astronomical telescope has a large aperture to [2002]

Options:

A. reduce spherical aberration

B. have high resolution

C. increase span of observation

D. have low dispersion

Answer: B

Solution:

The resolving power of a telescope is
$R.P = \frac{D}{1.22\lambda}$
where $D =$ diameter of the objective lens
$\lambda =$ wavelength of light.
Clearly, $R.P \propto \frac{D}{\lambda}$
Resolving power of telescope resolution will be high if its objective is of large aperture.
